DNV

WHEN TRUST MATTERS

ENERGY TRANSITION OUTLOOK 2026

A global and regional forecast to 2060

10TH EDITION

FOREWORD

Welcome to the 10th edition of our annual Energy Transition Outlook.

10 years ago, we set out to deliver an independent, 'most likely' forecast of the world's energy future. We established a core team of system dynamics experts with the idea of grounding our forecasts in the experience of thousands of DNV engineers and professionals working with customers building the energy infrastructure of tomorrow.

Now is a good time to take stock of our forecasting work. In the opening chapter of this Outlook, we list some of the things we got right (so far) and other areas where we have had to course-correct. Most important to me is what has stayed the same. 10 years ago, we predicted that the present-day primary energy mix, with an 80% fossil share, would shift to a 50/50 split between fossils and non-fossils by 2050. That is still our forecast.

Why is the transition not accelerating? Or, rather, why are we not bringing down GHG emissions at the speed required? We need to look critically at developments over the last decade for answers.

We began work on our first ETO in 2016, the year after the *Paris Agreement* was adopted. A no-nonsense forecast was needed amid both the optimism that accompanied the Agreement and the scepticism it gave rise to about a 'disruptive' energy transition.

Neither sentiment has since prevailed. The 'ratcheting' mechanism of ever-stronger commitments to decarbonization by the parties to **the Agreement has not resulted in a fast-enough transition.** Some parts of the transition have accelerated, especially solar, wind, and batteries. Other aspects have made more limited progress, with many countries failing to address grid constraints. And hard-to-electrify sectors such as heavy industry, aviation, and shipping are still struggling to address their emissions with cleaner fuels, including hydrogen and advanced biofuels.

I should emphasize, however, that while **the transition may not be accelerating, in sum it is not slowing.** By historical standards, it remains rapid and consequential. What is changing is its geography. Transition strategies are now more heavily linked to national interests. Fossil-importing nations are accelerating the adoption of clean energy technologies to reduce exposure to volatile fuel markets, while many fossil-exporting countries are transitioning more slowly. Intensified by Russia's invasion of Ukraine and more recently the Iran war, energy security is shaping the pace and direction of change, not just decarbonization.

One theme has remained constant over the past 10 years: we have argued consistently that faster **electrification of society is the single most powerful lever available** to improve prosperity, strengthen energy security, and accelerate decarbonization. It should therefore remain a priority for decision-makers across the public and private sectors.

The defining challenge of the coming decade will be turning the accelerating deployment of renewable generation into economic advantage. That means extracting more value from every renewable megawatt we build by applying a **new level of systems thinking** to how energy assets are deployed and in how policy shapes the value they deliver.

The data centres that hyperscalers are building illustrate the challenge. Their concentrated demand can strain local networks and force expensive grid upgrades. This makes them unpopular when the costs are borne by local communities, but the real issue is decades of underinvestment in grid infrastructure. Expanding and modernizing the grid is essential with or without data centres. The countries that prosper in an increasingly electrified economy will be those that can attract new electricity-intensive industries while continuing to provide **reliable and affordable power** for households and businesses alike.

In our Outlook, you will also find **good news:** total spending on energy is falling as a share of the global economy. As energy becomes more efficient, affordable, and productive, the opportunities to create value across the industry continue to expand. However, so too do risks if the public and private sectors cannot find new ways to cooperate, invest, and scale with a long-term perspective.

I conclude with a reminder that what DNV has done consistently over the last 10 years is to call things as we see them, not as we wish them to be. Emissions associated with our 'most likely' forecast mean the world is currently heading for **global warming of 2.3°C** by 2100. That is a deeply concerning outcome, and although we do not talk so much about climate in 2026, we must not forget that accelerating decarbonization remains an urgent priority for all.

On behalf of DNV, I thank you for reading this report and I invite you to challenge us and work with us across the value chain to find new solutions. I hope this year's report will inspire new thinking about how all of us can contribute to the much-needed energy transition so that it can move even faster than we predict.

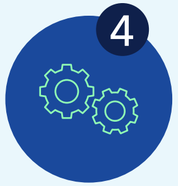
Knut Ørbeck-Nilsen
Group President & CEO
DNV

HIGHLIGHTS

Energy security is driving increasingly divergent transition pathways for energy importers and exporters

Short-term supply shocks are accelerating oil's long-term decline

The rate of electrification over the next 20 years will be more than double that of the previous 20 years



The coming decade will be about system integration, not just renewable generation

The gap is widening between competitive mature technologies and emerging technologies that are reliant on increasingly constrained policy support

By 2050, the world will have reduced emissions by 44% from today's levels – but will not even be halfway to net-zero emissions

HIGHLIGHTS

1. Energy security is driving increasingly divergent transition pathways for energy importers and exporters

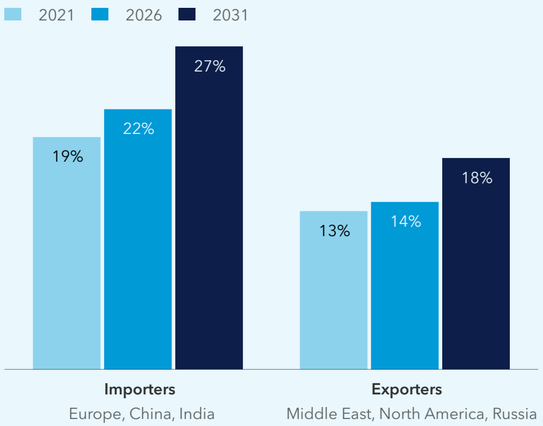
– Energy exporters are maximizing their production to mitigate the shortfall of oil and gas from the Middle East, but most of the world is seeking to reduce or avoid fossil fuel dependence and prioritize strengthened supply security through renewables, nuclear, and energy efficiency.

– Economic and job security considerations are reinforcing these trends, with countries increasingly prioritizing domestic energy, industrial preservation, and development. Energy exporters are supporting oil majors, national oil companies, and the associated supplier industry, while importers are building clean technology industries for domestic deployment and export.

– Over the last five years, the share of non-fossil energy in the primary energy mix has increased more than three times as much in energy-importing regions (2.2 percentage points across China, India, and Europe) as in energy-exporting regions (0.7 percentage points across the Middle East, North America, and Russia).

– Global investments in renewables are surging with annual expenditure doubling over the last five years from USD 860bn in 2021 to USD 1,770bn in 2026.

Share of non-fossil in primary energy



HIGHLIGHT 1 | Non-fossil includes renewable and nuclear energy.

2. Short-term supply shocks are accelerating oil's long-term decline

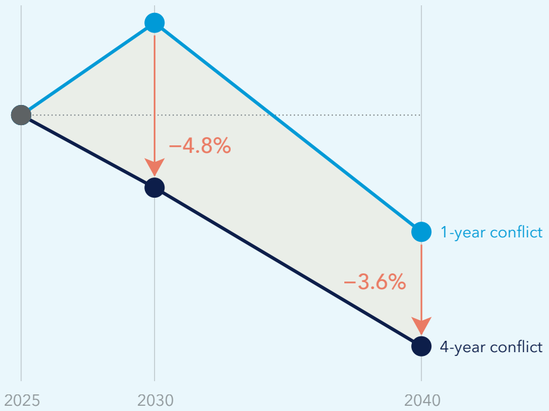
– The Iran war has further weakened the case for oil as a global export commodity and is causing permanent demand destruction. The longer the conflict persists and energy prices remain elevated, the greater the resulting demand destruction for both oil and gas.

– EV uptake in markets outside China, Europe, and the US has doubled over the past year.

– Middle Eastern oil producers will see their future market share shrink. We forecast the region supplying around 40% of global oil production in 2050, down from 50% in our 2025 Outlook projection.

– In a sensitivity study assuming the Middle East conflict persists until 2030 and oil and gas prices remain moderately elevated for those four years, we find a short-term reduction in demand of 4–6% while the conflict lasts. We also find a permanent destruction of global oil and gas demand of 2–5% relative to our main forecast through the remainder of our forecast period.

Oil and gas primary energy supply (index, 2025 = 100)



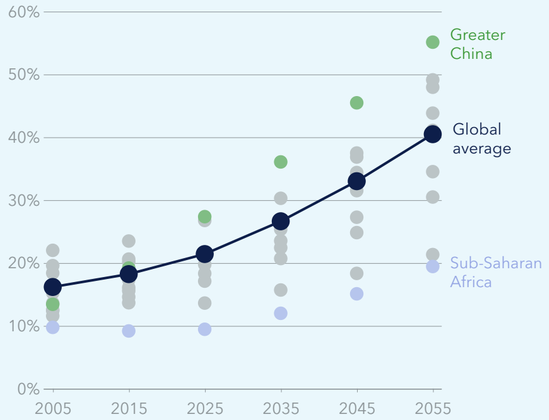
HIGHLIGHT 2 | Scale shown spans roughly 92–103, to highlight differences.

HIGHLIGHTS

3. The rate of electrification over the next 20 years will be more than double that of the previous 20 years

– In 2005, electricity represented 16% of global final energy demand. Today the share is 21%, and in 2045 we find the share will be 33%.

Share of electricity in final energy demand



HIGHLIGHT 3 | Includes self-generated production.

– Electrification is driven by fossil-importing countries. Thus, in Europe, China, and India, absolute electricity demand grows on average by 62% over the coming 20 years. However, fossil exporters are also electrifying due to its efficiency gains and cost advantages. In North America, the Middle East, and Russia, electricity increases by an average of 47% over the same period.

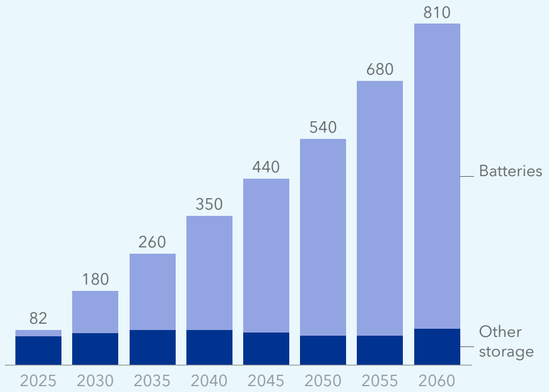
– China is setting the pace for electrification, with electricity supplying 27% of its final energy demand today and exceeding 50% by 2060. It also has unrivalled manufacturing scale, producing more than 80% of solar modules and 95% of solar wafers (IEA, 2026).

– In the next decade, North America accounts for 40% of global AI energy demand, while China follows as the second major growth centre. However, the main driver of growing electricity demand from today to 2040 will be space cooling (22%) and EV charging (21%), followed by data centres, including AI (18%).

4. The coming decade will be about system integration, not just renewable generation

– Improved pricing, tariffs, infrastructure, and market design are main priorities to integrate variable renewables in the energy system.

Installed storage capacity per electricity produced (kWh/GWh)



HIGHLIGHT 4 | Batteries include EV storage, standalone batteries, batteries co-located with solar PV, and long duration storage.

– Batteries are consistently outperforming all mainstream forecasts, with installed capacity increasing 14-fold over the past five years.

– Neither transmission capacity nor demand flexibility are growing fast enough. While expanding transmission capacity takes time, demand flexibility can deliver significant benefits in the near term.

– Gas remains critical for balancing the grid, even as its share of generation falls from 23% to 8% by 2050. Its average capacity factor (utilization) declines throughout our forecast period.

– Hybrid threats, including cyber attacks, are emerging risks for energy systems. Greater reliance on electricity increases vulnerability to disruptions, but expanding interconnection, storage, and distributed generation can also enhance system resilience.

– Renewable energy is seen as low risk with lower cost of capital (CoC) than fossil sources in all regions except the Middle East and Russia. The CoC advantage for renewables over fossil fuels widens from 1 to 5 percentage points relative to oil and gas, and from 4 to 8 percentage points relative to coal.

HIGHLIGHTS

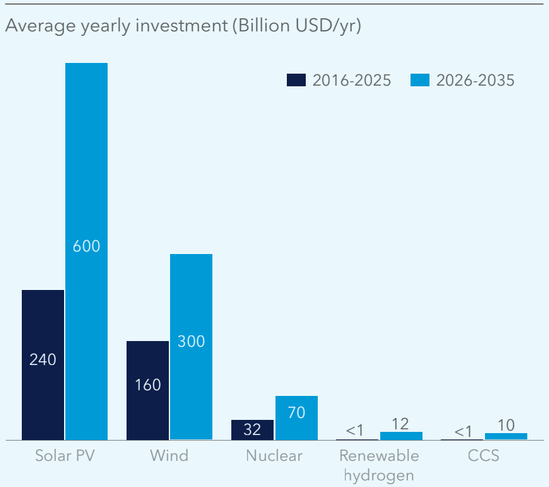
5. The gap is widening between competitive mature technologies and emerging technologies that are reliant on increasingly constrained policy support

– Solar PV and onshore wind are low-cost sources displacing fossil fuels from power generation in a growing number of countries. Fossil-fired power generation peaked globally last year and has now started to decline in absolute terms.

– EVs are increasingly competitive with internal combustion engines. Commercial EVs are advancing rapidly and are now only five years behind passenger EVs in terms of share of new vehicle sales.

– Early-stage technologies essential to decarbonizing the hard-to-electrify sectors are being given lower priority due to high costs and concerns about industrial competitiveness. Compared with ETO 2025, we have reduced the amount of hydrogen and CCS in our long-term (2060) forecast by 29% and 15% respectively. Policy support is needed to overcome cost gaps.

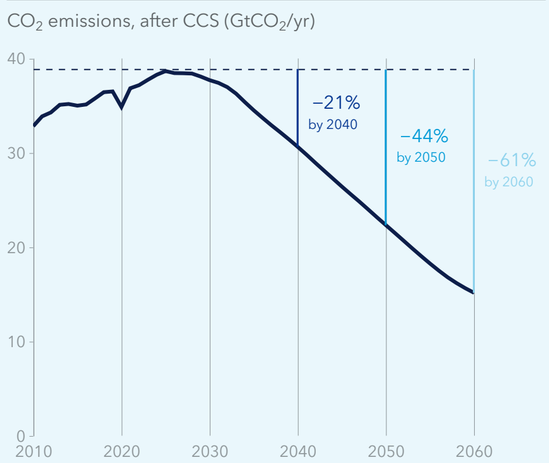
– Nuclear is an exception to this cost-driven divide. It retains its present share of power production as electrification accelerates despite high costs and supply chain risks. Capacity will grow 30% in the next decade and 170% to 2060. Small modular reactors only become competitive with large-scale nuclear around 2040 in most regions; their share of nuclear generation grows rapidly thereafter.



HIGHLIGHT 5 | Renewable hydrogen: produced from water electrolysis.

6. By 2050, the world will have reduced emissions by 44% from today's levels – but will not even be halfway to net-zero emissions

– Energy- and process-related CO_2 emissions reduce 44% from 39 Gt today to 22 Gt in 2050. Power emissions lead with a 60% reduction, followed by manufacturing (45%) and transport (34%), while buildings and process emissions are lagging.



HIGHLIGHT 6 | Includes energy- and process-related emissions.

– Net-zero emissions are delayed until the 2090s, and this milestone is very uncertain and based on technologies like direct air capture that have yet to be deployed at scale, and on uncertain developments in land use changes.

– The world is heading for 2.3°C of warming based on current IPCC climate sensitivity. New global warming records increasingly indicate that this climate sensitivity is on the low side and that there is a heightened risk of greater warming. Still, neither the most optimistic nor the most pessimistic IPCC scenario now appear likely.

– Renewed climate ambitions are needed as limiting global warming to 1.5°C is now effectively unattainable. The goal of keeping warming well below 2°C is still attainable but requires stronger commitments.

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10 YEARS OF FORECASTING

Why did we start?

It is difficult to recapture the atmosphere of 2015. The adoption of the *Sustainable Development Goals* (SDGs) and, just months later, the *Paris Agreement* created an unprecedented sense of global alignment around a shared vision of sustainable development and climate action. Yet, at the time, **the central question was not how to manage the energy transition, but whether it could happen at all.** Could the world continue to improve living standards while reducing emissions? Would clean energy technologies become competitive at scale?

DNV's contribution to that debate began not with energy forecasting, but with sustainable development. Together with the UN Global Compact, we set out, in a report titled *The Future of Spaceship Earth* (2016), to assess the likelihood of achieving the SDGs. The Goals themselves acknowledge their interconnectedness – you cannot pull one apart from the others without affecting the whole system. Consequently, assessing the likely success of the SDGs favoured a systems-dynamics methodology. That is a modelling approach designed to capture complex interactions, feedback loops, and delays that shape long-term outcomes – as described more fully in Chapter 2 of this report. It quickly became apparent to us that the same systems perspective was essential for understanding the future of energy.

The energy transition as a source of risk – and opportunity

At the time, we saw the energy transition emerging as one of the greatest sources of both risk and opportunity facing our customers. Energy companies, industrial firms, shipowners, investors, and policy-makers were making decisions with consequences stretching many decades into the future, yet the forecasting landscape offered limited guidance. 10 years ago, many mainstream outlooks were rooted in assumptions and modelling approaches developed for a fossil-dominated energy system and repeatedly underestimated the pace of change in renewable technologies. Other outlooks were more along the lines of wishful thinking: describing pathways that were desirable while underplaying the strength of energy technologies already in existence. Our customers needed a trusted, independent view of where the energy system was actually heading.

We believed we could fill that role. For decades, our experts and engineers have worked with energy companies (both oil and gas and renewables), developers, utilities, and governments, to help design, test, certify, and deploy the infrastructure that fuels and powers the world. Combined with our independence and research-based approach, **we believed we could offer something different:** not a range of possible futures, but a single forecast of the most likely future. From that conviction, our *Energy Transition Outlook (ETO)* was born. Each subsequent ETO has attracted thousands of industry professionals and stakeholders to our subscriber base. Every year we have sought to answer not, 'what possible energy futures are there?', but 'what is the most likely energy future?' given our best assessment of how technology, economy, and policy will develop in all countries and regions.

So, how have we done so far?

Our knowledge is deeper and our models are more sophisticated than they were 10 years ago, but forecasting the future remains a demanding exercise. The pandemic of 2020, the war in Ukraine, and the war in Iran were not in our forecast. More broadly, geopolitical rivalry and concerns over energy security have become far more prominent than we anticipated in 2017. At the same time, although climate impacts are becoming more visible and severe, political and societal responses have often been more muted than many expected.

Looking at the energy system today, we can clearly see where we were right and where we were wrong. We underestimated the pace of progress for solar PV and battery storage, while overestimating the speed of development for hydrogen and improvements in energy intensity. Many other developments were captured well. Our forecast for EV uptake was considered unusually ambitious in 2017, yet it closely matches the reality of today's market. Likewise, we correctly anticipated the overall pace of electrification.

The purpose of a forecast is not to predict every event correctly. Rather, it is to identify the structural forces that shape long-term development. Judged against that standard, our record gives us confidence. While individual technologies have developed faster or slower than expected, the overall direction of the energy transition, and many of the forces driving it, have unfolded broadly as we anticipated.

Why forecasting still matters

The relevance of forecasting has not diminished over the past decade. If anything, it has increased. Better data availability, greater computing power, and advances in AI have improved the tools available to forecasters. Yet the world itself has become more complex. The energy transition now unfolds alongside geopolitical fragmentation, industrial policy competition, energy security concerns, supply-chain vulnerabilities, and rapidly evolving technologies.

AI makes it easier than ever to generate outlooks and scenarios, but that elevates questions of trust and makes transparency, methodology, and credibility even more important. Decision-makers need more than numbers; they need confidence in the assumptions, evidence, and reasoning behind them.

A perfect forecast would never need updating. Reality, of course, has other plans. New information constantly emerges, policies change, and unexpected events (like the supply shock that the Iran war has set in train) occur. Every year, decision-makers across the energy industry make choices based on their best available understanding of the future. Our objective remains the same as it was 10 years ago: to provide an independent, science-based assessment of the most likely path ahead, balancing security, affordability, and sustainability.

On the question of sustainability, we remind readers that each year for the last 10 years we have said that our forecast does not see the world on track to achieve the goals of the *Paris Agreement*. In fact, we have calculated the global warming associated with our forecast at between 2.2 and 2.5°C of warming by 2100. Perhaps, therefore, the most important contribution from our sustained forecasting efforts has been to show that the world is not accelerating its reduction of GHG emissions. That task is becoming ever more urgent.

A stable forecast for 2050 non-fossil share
Non-fossil share of primary energy, 2050

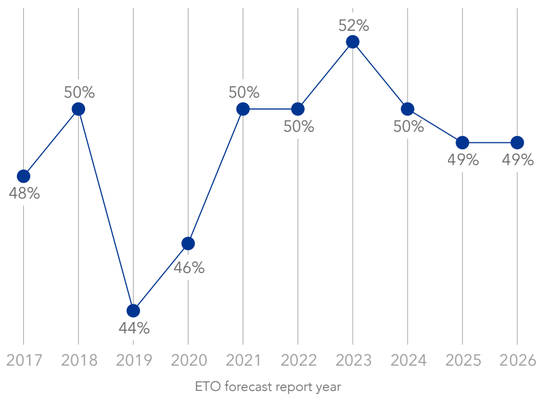

FIGURE 0.1 | Non-fossil energy includes renewables and nuclear.

Looking back at our first forecast: close for oil and EV uptake, but underestimating renewables growth and persisting coal
2017 ETO forecast for year 2025 vs actual 2025 values, for selected global metrics

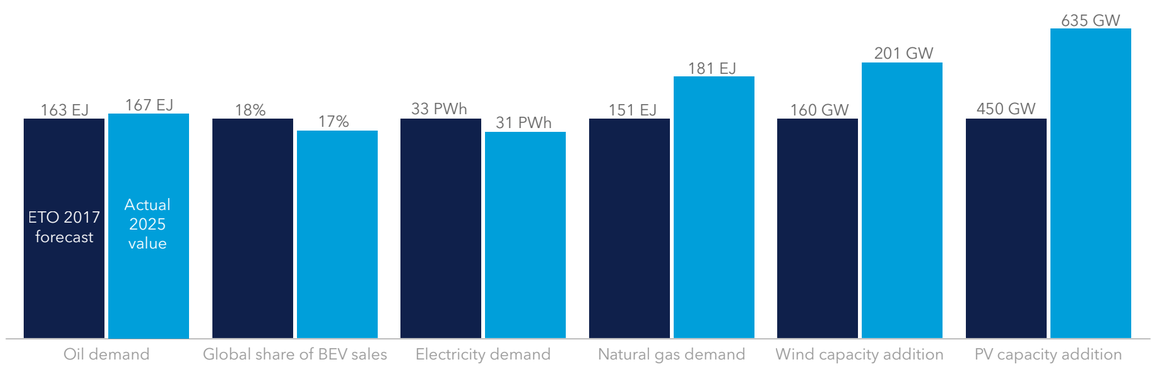

FIGURE 0.2 | BEV: battery electric vehicle only, excluding hybrid.

INTRODUCTION

About this Outlook: This is the 10th edition of our annual *Energy Transition Outlook* (ETO), which presents the results of our independent model of the world's energy system. It covers the period through to 2060 and forecasts the energy transition globally and in 10 world regions.

Our **best estimate**, not the future we want

A single forecast, not scenarios

Long-term dynamics, not short-term imbalances

Continued development of proven **technology**, not uncertain breakthroughs

Main **policy** trends included; caution on untested commitments, e.g. NDCs, etc.

Behavioural changes: some assumptions made, e.g. linked to a changing environment

Web resources – www.dnv.com/eto

More detailed descriptions of the regional transitions can be found on our main ETO web hub, along with more specialized reports on the energy transition, country-level reports, and downloadable versions of most of the charts in this and other reports. Our forecast data may be freely accessed at www.dnv.com/eto-data.

For our readers in maritime and related industries, our annual Maritime Forecast is a widely consulted source on the maritime energy transition.

Our approach

Our guiding principles in producing this forecast include:

- Producing a best estimate of the energy future, which differs in many ways from the future we want to see unfold
- Publishing a single 'most likely' forecast, and not a range of scenarios, which often serve to confuse rather than inform
- Emphasizing and exploring long-term dynamics rather than short-term imbalances
- Focusing on proven technologies and excluding unproven or future potentially 'breakthrough' technologies
- Incorporating main policy trends; treating untested policy commitments with caution
- Modelling effects of behavioural changes – e.g. in relation to energy efficiency.

Chapter 11 contains more details on our modelling methodology and presents an overview of the updates to our model over the last 12 months.

Independent view

DNV was founded 162 years ago to safeguard life, property, and the environment. We are owned by a foundation and are trusted by a wide range of customers, including the world's leading energy companies across fossil and non-fossil energy value chains, to advance the safety and sustainability of their businesses.

Developing an independent understanding of, and forecasting, the energy transition is of strategic importance to both us and our customers. This Outlook draws on the expertise of over 150 professionals in DNV. In addition, we are very grateful for the assistance provided by external experts. All contributors are listed on the last page of this report.

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1.0 KEY DEVELOPMENTS THE LAST 12 MONTHS

War in Iran

- The oil and gas sector is highly exposed to geopolitical instability; around half of global oil production and more than a third of global gas production is traded internationally. This vulnerability has been dramatically demonstrated by the closure of the Strait of Hormuz, threats to the Bab al-Mandeb maritime chokepoint, and Ukraine's attacks on Russian refineries.
- Global oil inventories have mitigated the immediate effect of the supply crunch on consumption. However, long-term sustained high oil and gas prices and high uncertainty result in permanent oil demand destruction.
- We assume normalization during Q1 2027, i.e. the conflict lasts for one year. We have also modelled a prolonged-war sensitivity (the conflict lasting for four years) and substantially larger demand destruction.

Geopolitical tension and energy security

- Energy security has deteriorated across multiple dimensions, including infrastructure resilience, security of supply, and economic security. This has prompted diverging responses from energy importers and exporters.
- Energy security policy responses produce cumulative effects that outlast short-term price shocks. A security premium (additional cost incurred for resilience and domestic capability) persists throughout our forecast period. Our model includes factors for the realignment of global trade, the reshoring of manufacturing, and the willingness to pay higher costs for domestically-produced or controlled energy.

Policy debrief

- The evolving global energy landscape is increasingly shaped by divergent state interests and policy strategies. Import-dependent economies pursue diversification and renewables to align decarbonization with energy security and resilience goals. In contrast, fossil fuel exporters seek to safeguard hydrocarbon production and consumption.
- Electrification is increasingly a shared global policy priority, driven by efforts to strengthen energy security, capitalize on the favourable economics of renewable power and electro-technologies, and make greater use of domestic energy resources.
- Strategic energy priorities and industrial policy tactics include expanding local content requirements to build domestic manufacturing bases that supply deployment needs.

AI deployment expands

- We forecast data-centre electricity demand will nearly triple to 1,100 TWh by 2030, driven by AI inference. Capabilities are advancing faster than adoption, while energy demand depends on actual use.
- Capital expenditure by the five largest technology companies exceeded USD 400bn in 2025 and could reach USD 715bn in 2026. Growing reliance on debt makes the buildout increasingly sensitive to financing conditions and capital-market sentiment.
- Data centres can be built faster than grid connections. This is pushing developers to build on-site generation, with increasing public opposition and likely stronger regulation to follow.

1.1 THE WAR IN IRAN

The Iran war demonstrates, once again, the continued vulnerability of the energy industry to geopolitical instability. While all parts of the industry are exposed to such risks, the oil and gas sector is particularly sensitive because almost half of global oil and over a third of gas production is traded across borders. Our main forecast assumes that conditions gradually normalize during Q4 2026 and Q1 2027 following a relatively short conflict. This time horizon is broadly congruent with the IEA's view and market signals.

At the time of writing, the conflict remains far from resolved and the disruption is extending beyond the Strait of Hormuz to other critical energy corridors. Given the growing uncertainty around the conflict's duration, we have also assessed a prolonged-war sensitivity in which instability in Iran and the wider Gulf region persists significantly longer than currently expected and results in substantially greater demand destruction.

Over the last four years, the *Energy Transition Outlook* (ETO) has provided detailed commentary on the consequences of the war in Ukraine (Section 1.2). Recent Ukraine attacks on Russian energy production, transport, and refining have added further complexity to the global energy transition. However, the Iran war has larger implications for the global energy transition than the war in Ukraine. The main impact is the closing of the Strait of Hormuz, which, prior to February 2026, served as the gateway for approximately 20% of global oil and LNG exports.

In this section, we assess the implications of these developments for the energy transition across four key dimensions:

- **Flow of commodities:** production, transport, supply
- **Price** effects
- **Demand sectors:** manufacturing, feedstock, transport, buildings
- **Geographies:** regional consequences

Most energy shocks are on the demand side, while the Hormuz crisis originated as a supply shock. Demand and supply are, however, closely related and there is a clear link between supply constraints and demand response. In our modelling, demand destruction is not simply a consequence of higher energy prices. It also emerges from the adaptive responses of households, industries, and governments and includes reduced consumption, fuel switching, greater efficiency, changes in mobility patterns, and accelerated investment in alternative energy sources. These responses can outlast the crisis itself, permanently reshaping patterns of energy demand. The 'Modelling a crisis' factbox summarizes how this is reflected in our forecast.

Flow of commodities

The wider Middle East region produces one-third of the world's oil and one-quarter of its natural gas. Before the outbreak of the war, more than 20% of the global oil export and almost 20% of refined products and LNG exports passed through the Strait of Hormuz.

So far, the conflict has focused more on energy transport than energy production. Both sides have shown relative restraint in targeting oil and gas production wells, refineries, and LNG facilities. Oil and gas transport has been affected far more significantly, as most export from the region was traditionally transported by ships through Hormuz. Disruption to the East-West Pipeline demonstrates that infrastructure previously viewed as a strategic workaround to Hormuz is itself vulnerable. The crisis has therefore evolved from a threat to a single chokepoint into a broader challenge to the resilience of the region's entire energy export system. For LNG and refined products, alternatives are even more constrained, as these commodities rely heavily on specialised export terminals and shipping infrastructure.

Global strategic oil inventories have buffered much of the shortfall and are being drawn down to mitigate

Modelling a crisis

The present Iran war and wider Middle East crisis impact a wide range of parameters in the ETO Model. The most important of these are:

- We set the **oil price** to USD 95 in 2026 and USD 79 in 2027. If the conflict continues, it will be USD 100.
- We set **gas prices** regionally based on a supplier-change factor plus a global gas-supply scarcity premium of up to 55%. Prices are modelled based on unavailable volumes of LNG from the Middle East partly replaced by LNG from other regions.
- We adjust **GDP** to higher energy costs and changes in export volumes; Middle East GDP is adjusted further down due to trade disruptions and lower investments.
- Increased **EV adoption**, mainly due to higher oil (gasoline, diesel) prices, but also power prices and other factors.
- Reduced **road travel** distance in South East Asia and North America for a short-duration crisis, extending to other regions for a longer-duration crisis.
- Reduced **aviation demand** based on elasticity of −0.6 to −0.8 on estimated increased aviation fares of 6–10%.
- Temporary cut in **feedstock** use, and some change in feedstock mix due to high oil and gas prices.
- A 2–3% temporary increase in **military oil and gas consumption**.

the impact of reduced production and transport. China in particular holds huge inventories, while many high-income countries also maintain strategic and commercial stocks that help support supply. Refined products and natural gas have benefitted from inventories that now are increasingly drained.

Nevertheless, both oil supply and demand (consumption) have declined significantly. The demand reduction amounts to 5.3 Mbpd in Q2 and 3.4 Mbpd in Q3 (IEA, 2026d). In our main case, oil demand declines by 1.9% for the full year 2026, with most of the lost demand recovering in 2027 when demand grows by 2.5%. Natural gas demand falls by 1.1% in 2026 before recovering by 1.9% in 2027. However, oil demand destruction persists because of the conflict (DNV, 2026a). For example, EV sales in 'new markets' (i.e. outside China, Europe, and North America) jumped 97% in July this year compared with the same month last year. The response becomes structural in road transport (new equipment) and chemical feedstocks (new plant configuration). Shipping and aviation have few large-scale alternatives at present, so the reduction there is behavioural and is much more likely to reverse with the price.

Price

The price of oil has spiked at USD 120/bbl but mostly fluctuates between USD 80 and 110/bbl, compared with a pre-war level of USD 70/bbl. In our model, we use an average oil price for 2026 of USD 95/bbl declining to USD 79/bbl in 2027. Refined products are harder pressed (e.g. diesel and aviation jet fuel prices are almost double pre-war levels). Natural gas prices tend to be more regional, but Asian LNG prices are typically up by 50-100% since February. However, the duration of the disruption matters more than the severity of price spikes, as investment and policy decisions depend on expected prices over an asset's lifetime rather than on short-term spot prices. A vehicle replaced, a boiler converted, or an efficiency standard legislated is not likely to revert when the price comes down.

Demand sectors

Petrochemicals and non-energy use appear to be experiencing the largest demand reduction, with both oil and gas consumption declining by about 7% in 2026. Most of this decrease is in China. Road transport, which accounts for the largest share of global oil demand, appears to be more resilient, with consumption falling by around 1% for the full year. In regions such as India and South East Asia, the lack of availability of oil, diesel, and petrol has curbed demand. In North America, higher petrol prices are impacting road travel. However, rising demand for EVs is a greater motivator for the energy transition than an immediate reduction in kilometres driven. Aviation demand declines by around 3% as cancelled flights, higher fuel costs, and partly limited availability of refined products reduce activity. However, we note again that this is a temporary behavioural response: there are no indications of a faster switching to sustainable aviation fuels (SAFs). In the maritime sector, some ships have been trapped, but longer sailing distances offset these effects so overall demand is broadly unchanged. In the manufacturing sector, oil demand falls by about 3% globally, driven by both higher prices and supply constraints.

Regional effects

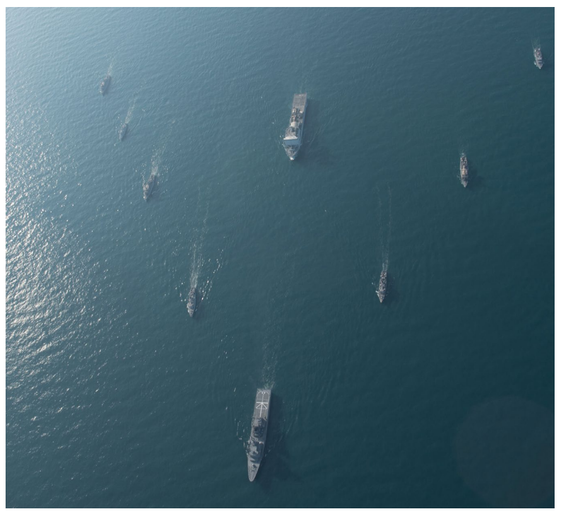
Asia is the main market for oil and gas export passing through the Strait of Hormuz and has been hardest hit by the shortfall, especially in low- and middle-income countries without sufficient inventories to draw down. On the other hand, China has the world's largest inventories and could theoretically maintain previous consumption levels for an extended period. However, that has not been the case. We expect the region to see a 4% decline in demand in 2026, the largest of any in our model. Production of feedstock like petrochemicals has been reduced particularly sharply, and some consumers have begun to substitute oil (and to some extent gas) with coal.

Global energy markets have partially adapted to supply disruptions through increased output for export and rapid shifts in trade flows. Increased US exports, especially LNG exports, have offset a significant share of the lost supply from the Middle East, while other energy-producing countries have also sought to maximize export. In Europe, US deliveries have softened the impact but not fully replaced LNG from the Middle East. Exigent relaxation of sanctions has allowed the return of some Russian oil and LNG volumes, but recent Ukraine attacks on refining and export infrastructure have curbed the throughput and made Russia a net importer of certain refined products.

Growing concern over energy security is driving energy-importing and energy-exporting nations towards increasingly divergent strategies and transition pathways. The Iran war has highlighted importing nations' vulnerability to supply disruption and exposed the vulnerability of the exporters themselves. Exporters can no longer assume unhindered access to markets and, in the worst case, the conflict could establish a precedent in which maritime energy trade can no longer be taken for granted. This would have profound implications for energy security, trade patterns and investment decisions worldwide.

Time Horizon

Restoring trust is likely to be the most time-consuming aspect of the recovery process (DNV, 2026b). Confidence that vessels can transit the Strait of Hormuz without being attacked or detained is essential for resuming normal shipping activity and, in turn, normalizing exports and production.



The consequences of a prolonged Middle East crisis

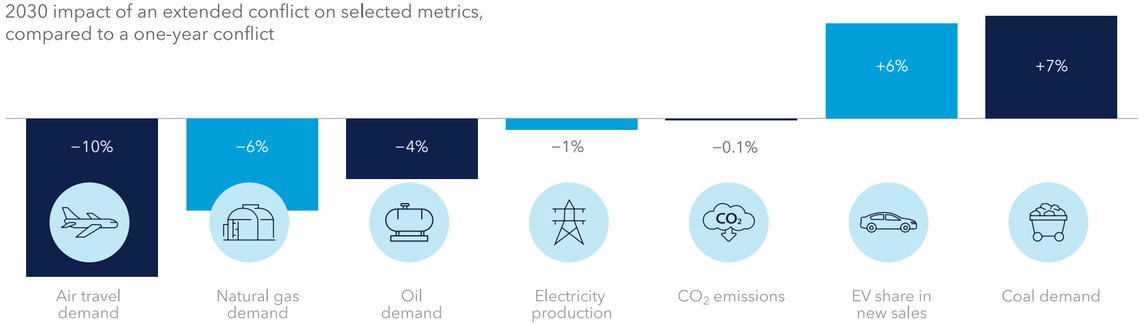
This infographic shows the differences in our model results between the most likely future (a one-year conflict with normalization in Q1 2027) and an extended war sensitivity case (a four-year conflict with normalization only in 2030).

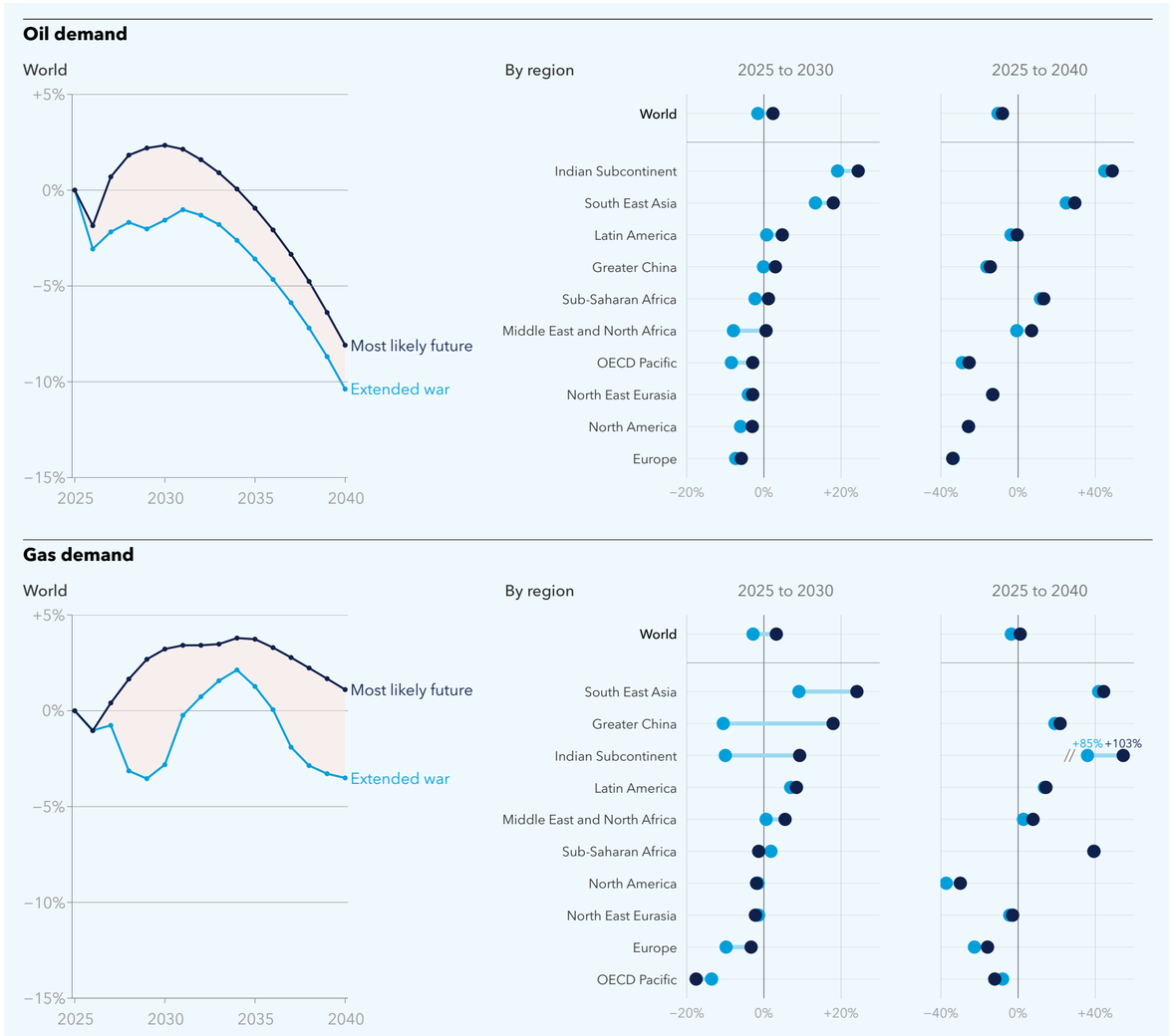
Key assumptions of the four-year crisis:

- Oil prices just above USD 100/bbl
- 75 Mtpa of Middle East LNG liquefaction capacity assumed unavailable through 2030

- Regional gas-price premiums peak at 55% (India), 50% (OPA), 45% (Europe and SEA), and 40% (China)
- Coal prices increase by around 25% by 2030, uranium by around 25%, and biomass by around 15%
- GDP reduced by around 1–2% in most importing regions and up to −5% in the Middle East by 2030
- Building retrofit activity increases by up to 7%
- Coal-to-gas switching in Chinese industry reduced by around 20%
- Petrochemical feedstock demand reduced by up to 6% for oil feedstocks and 10% for gas feedstocks
- Stronger energy-security preferences incorporated into power-sector investment decisions

2030 impact of an extended conflict on selected metrics, compared to a one-year conflict





1.2 GEOPOLITICS AND THE ENERGY TRANSITION

Energy security is the most powerful geopolitical force shaping today's energy decisions. The Iran war brought that reality into sharp focus, but it did not emerge in isolation. A decade of fraying trade relationships, fragmented supply chains, and rising geopolitical tensions had already heightened concerns about energy dependence. Russia's invasion of Ukraine exposed how energy could be weaponized. The conflict with Iran has now reinforced the vulnerability of global energy systems and cemented energy security as a defining policy priority.

The consequence of these conflicts is a growing divide between energy importers and exporters. Countries that rely on imported energy are accelerating investment in domestic, low-carbon sources to reduce strategic dependence, while exporters face a different set of economic and geopolitical pressures. This divergence is one of the clearest findings from this year's research and modelling.

Geopolitical influence over energy is here to stay

Before the present crisis in the Middle East, at least three energy disruptions produced measures that outlasted the price shocks themselves: the pandemic reduced demand and exposed the concentration of manufacturing capacity; Russia's invasion of Ukraine removed a large pipeline gas supplier from the European market; and attacks on Red Sea shipping from late 2023 rerouted a substantial share of Europe-to-Asia trade. Countries responded with measures that include strategic stockpiling, supply chain screening, domestic content requirements, and non-price resilience criteria in renewables auctions. They are written into law, tariff schedules, and auction rules that stay in place when prices fall back.

The European response to the 2022 Russian gas cutoff supports the institutional reading. The *REPowerEU* gas regulation entered into force in February 2026, converting a crisis objective into binding law four years after the shock and well after prices had normalized (EC, 2026a).

Responses to energy security in 2026 are following this pattern. The European Commission launched the *AccelerateEU* strategy in April 2026, with around half of its 44 actions concerned with scaling clean energy and electrification (Evans, 2026). Global energy investment now directs USD 2.2trn to clean energy against USD 1.2trn to fossil fuels (IEA, 2026a). The International Energy Agency (IEA) reports that security, trust, and diversity have entered project and partner selection alongside cost and environmental performance (IEA, 2026a).

AccelerateEU is not an isolated response to rising energy security concerns. Rather, it is one manifestation of a broader trend visible across many advanced economies. From Europe's *AccelerateEU* strategy to Australia's *Future Made in Australia* programme, Japan's *Green Transformation Strategy*, and South Korea's strategic industry support, governments are increasingly using energy and industrial policy to reduce strategic dependencies, strengthen resilience, and enhance national competitiveness.

We view these collectively as an institutional layer affecting the energy transition that will outlast short-term price perturbations. Indeed, we go further in treating a security premium as persistent across the forecast period. We use this term to describe the additional cost incurred when energy and industrial policy prioritize resilience, domestic capability, and reduced geopolitical exposure alongside economic efficiency.

In previous Outlooks, we have explained how this security premium boosts willingness to invest in otherwise relatively expensive nuclear generation. More broadly, the security premium boosts domestic renewable generation, which is set to grow by more than 8% in 2026 and overtake coal-fired output. The IEA attributes part of that expansion to supply diversification during LNG market disruption (IEA, 2026d). Short-run substitution can still run in the opposite direction, with coal generation rising on higher gas prices in several importing countries (IEA, 2026d; WEF, 2026a).

Importers and exporters are diverging in policy and investment

Importers and exporters are responding to the same rise in energy security concerns with opposing policy and investment choices. The divergence is shown in policy priorities and investment allocation rather than in trade itself, which continues.

Energy security has many components, but the more prominent risks are:

- **Energy infrastructure resilience** towards disruptions, including attacks and hybrid threats
- **Energy supply security**, including diversification of type of energy, where it is sourced from, and how it is transported
- **Price risk**, including vulnerability to fluctuations beyond control
- **Economic security** to safeguard the jobs, innovation, and capital formation that underpin the clean energy economy and reduce excessive reliance on single sources of supply

The first two of these risks have escalated dramatically with the weaponization of energy in the last five years. Exporters and importers alike are exposed to the first of these risks (infrastructure

and supply disruption), but it is energy importers who suffer disproportionately from energy security and price risks for as long as they remain significantly dependent on imports to cover their energy needs. To the extent that importers opt to invest in domestic cleantech in response to import supply and price shocks, they invariably grapple with economic security.

Exporters have shifted toward pursuing individual interests, rather than collective management. Producers with high fiscal breakevens depend on defending price through output restraint, while producers with lower fiscal exposure and expanding capacity earn more from selling volume. The two therefore hold different tolerances for price volatility and the result is a looser alignment between exporter interests than in earlier periods.

Importers are now almost all directing investments towards domestic supply. China's *15th Five-Year Plan* for 2026-2030 sets non-fossil energy at 25% of primary energy by 2030 and raises nuclear capacity ambitions to 110 GW by 2030 from 62 GW at the end of 2025 (CREA, 2026). The European Commission puts nuclear investment needs at USD 281bn to 2050 for lifetime extensions and new large reactors, aiming to grow installed capacity from 98 GWe in 2025 to around 109 GWe, close to the ETO forecast of 113 GW including the UK, by 2050 (EC, 2026b).

Countries pursuing reshoring accept a cost premium for doing so and run into physical limitations.

The importer shift to domestic supply is by no means uniform. China, for example, is expanding non-fossil supply while continuing to rely on coal. Lower-income importers have less room to invest, so they face a harder choice between energy access, fiscal stability, and transition spending.

Despite the plethora of policies and vast budgets earmarked for energy security, the risks have intensified. Energy security scores fell in 2026, with both the resilience and supply diversification components declining (WEF, 2026a). The loss of external supply or reliable transport routes is felt immediately, while the domestic capacity that responds to it takes years to build, so the security payoff arrives later than the cost. Even so, we expect the exporter/importer divergence to outlast the present crisis. Nations are adopting through-cycle strategies which give rise to infrastructure that will not be dismantled if the prices fall.

Effectively, we are seeing an energy transition that is diverging along importer and exporter lines. For energy importers, the pursuit of security is accelerating investment in domestic clean energy, electrification, and nuclear power as a means of reducing dependence on external suppliers. Exporters are also expanding clean energy, but alongside continued investment in fossil fuel production and export capacity. The result is a transition that is becoming more geographically differentiated, but not slower overall.

Domestic manufacturing comes at a premium while trade realigns

Reshoring policies aim to address concerns over dependencies, the need for diversification, and a desire for domestic value creation. The countries pursuing reshoring accept a cost premium for doing so and run into physical limitations. China holds around 85% of global solar PV supply chain capacity and processes more than 70% of the world's lithium, cobalt, graphite, and rare earths (IEA, 2026c). Production outside the largest exporter is insufficient to meet demand at several steps in the solar, wind, battery, and heat pump supply chains.

The US has proposed a 15% tariff and minimum import prices on polysilicon and related solar products, while existing legislation increases domestic-content requirements and restricts tax-credit eligibility for certain foreign-linked entities (Pao, 2026). The cost is visible in deployment, where many renewable energy projects have seen rising costs. Credit phase-outs increased solar costs by between 36% and 55% within a year (Deloitte, 2025). Other countries have followed broadly similar policies. For example, India has required domestically manufactured cells in government-supported projects since June 2026, combined with tariffs on imports (TimesTech, 2026).

However, the reshoring push should not be seen as a retreat from trade. Global goods trade grew by

6.5% in 2025, faster than the world economy. At the same time, geopolitical distance of that trade fell by 1.2%, against an annualized 0.9% a year over 2017 to 2024 (MGI, 2026) – meaning that leading national economies are trading somewhat more with politically aligned partners. The more substantial shifts are along the lines of exigent realignment. The US replaced roughly two-thirds of the imports it lost from China through suppliers in ASEAN and India (MGI, 2026). We do not foresee a full decoupling of manufacturing value chains. Rather, supply chains are realigning somewhat along political lines while remaining globally dispersed.

Incorporating geopolitical trends into the forecast

Geopolitics enters our modelling through four mechanisms:

- We adjust the economic growth path that drives energy demand
- We reduce the demand for maritime trade
- We apply an energy security cost factor built on each region's dependence on imported fuel and imported energy technology
- We reduce technology learning rates to reflect reshored manufacturing

Two of these mechanisms pull against each other since the push for energy independence favours renewables and nuclear, while the push for domestic clean technology supply chains raises their cost.

The ETO Model also carries short-term parameters for the impact of the war in Iran on energy prices, fossil fuel supply and energy demand, and for the effect of recent economic developments on the cost of capital across regions. Each factor is a parameter adjustment against an otherwise largely cost-driven forecast.

Restructuring the global economy

GDP per capita enters our model as an exogenous driver of energy demand. For the short term, we follow the growth rates implied by the International Monetary Fund (IMF) *World Economic Outlook*, which projects global growth of 3.0% in 2026, with global disinflation stalled and the war shock weighing on energy importers (IMF, 2026). UN Trade and Development (UNCTAD) foresees growth falling further to 2.6% in 2026 (UNCTAD, 2026). We treat this pathway as a potential further downside that reflects a deceleration in global trade directly linked to the scale and duration of the conflict in the Middle East. The economic consequences depend strongly on both the extent of the conflict and how long it persists.

For the long term, we base GDP per capita growth on projections from the Organisation for Economic Co-operation and Development (OECD) and the International Institute for Applied Systems Analysis (IIASA). Rather than following the central Shared Socioeconomic Pathway 2 (SSP2 – 'middle of the road'), we average SSP2 with SSP3 (the 'regional rivalry pathway' in which economic nationalism, weak institutions, and lack of coordination produce slower and more uneven development, particularly in lower-income regions). The average reflects that we no longer assume economic development continues along historical trends, while not assuming a full transition to regional rivalry either. The trajectory includes a modest productivity dividend from artificial intelligence, in line with IMF and OECD expectations (IMF, 2026; OECD, 2026). On this path, world GDP nearly doubles from USD 193trn per year in 2025 to USD 378trn per year in 2060, with annual growth slowing to 2% during the 2040s.

Demand for maritime trade

Recent conflicts have further disrupted maritime trade patterns leading to changes in both volumes and distance. The effect on oil and gas is direct and has both short- and long-term impacts. For other commodities and freight of goods, insurance premiums for Strait of Hormuz transits rose from 1%–3% of hull value to 7.5%–10% (Shamim, 2026). Shipping lanes are being rerouted to avoid the Red Sea and Suez. In addition, expanding trade tariffs have introduced uncertainty for global maritime trade, further accelerating the shift towards reshoring, friend-shoring, and supplier diversification.

The reduction in trade extends beyond hydrocarbons. We observe that merchandise trade growth has already started decelerating, from 4.7% between 2024 and 2025 to between 1.5% and 2.5% expected in 2026 (UNCTAD, 2026). Half of the global clean energy technology trade passes through the Strait of Malacca (IEA, 2026c). With some regions diversifying or relying on reshoring, further decline is likely. Thus, we apply a reduction to container, bulk, and other cargo trade to reflect reshoring, tariffs, and supplier diversification. This reduction applies to volumes rather than to tonne-miles because rerouting raises distance while these policies lower volume and a single adjustment cannot represent both. The volume and distance effects are therefore specified separately, with the distance effect following the rerouting evidence. This results in a 5% to 10% overall reduction of global trade in these segments by 2040.

> The economic consequences of the conflict in the Middle East depend strongly on both its extent and duration.

Prioritizing domestic sources for energy security

Governments prioritize domestic energy sources to reduce import dependence, build immunity to supply disruptions, and mitigate the risk of energy being used as a geopolitical weapon. Our power model represents this with an energy security cost factor for new investments applied to the levelized cost of each generation technology in each region. The factor combines two relative scores. The first scores dependence on imported fuel (such as gas, oil, or coal) and raises the levelized cost of the affected generation by up to 30%. The second scores dependence on imported technology (such as solar PV modules and battery storage) and raises it by up to 10%. The factor expresses the additional cost a region's government accepts, or the support it extends, to favour power generation under domestic production or control.

The two scores separate regions, primary (direct imported fuels) and secondary technology-dependent measures to enable a combined effect of both. China imports much of its gas and oil but manufactures its own solar, battery, and wind equipment, so its cost factor penalizes gas- and oil-fired generation and leaves renewables unaffected. North America

shows the inverse pattern, with no fuel penalty and a premium of around 5% on solar and storage. Europe carries both, with gas- and oil-fired generation at the 30% cap and around 5% on solar. The factor also works in the opposite direction; where a technology strengthens a region's energy security, it falls below the neutral level and lowers the levelized cost to represent government support for secure domestic supply. Nuclear carries such an advantage in the regions that treat it as a strategic domestic source (at 5% in North America, Europe, China, and the Indian Subcontinent, 10% in the OECD Pacific, and 20% in North East Eurasia). Domestically abundant fuels are treated in the same way, with gas-fired generation in the Middle East and North Africa and coal-fired generation in China carrying advantages of 8% and 5% respectively. Nuclear's remaining geopolitical exposure enters through the bloc constraint on technology adoption.

The resulting preferences vary between the regions. Europe combines nuclear lifetime extension and new build with accelerated renewables and electrification. China combines renewables and nuclear expansion with continued coal use in power and chemicals. North America favours oil and gas on its resource base and the technology dependence score carries that cost effect.

Technology deployment and reshoring of manufacturing

We represent energy technology manufacturing reshoring as a long-lasting but not permanent reduction in learning rates for solar, wind, and lithium-ion batteries in the regions rebuilding domestic supply chains. The reduction applies in North America, Europe, the Indian Subcontinent, and the OECD Pacific. It reflects the scale and experience disadvantages of building new supply chains relative to established suppliers. At its peak, the adjustment reduces the learning rates by between 10% and 30%, depending on technology and region. The adjustment peaks around 2030 and then gradually declines, with costs converging towards global levels during the 2040s. Despite the cost disadvantages, governments continue to reinforce domestic manufacturing objectives through trade, procurement, and local-content policies. Examples include US foreign entity of concern (FEOC) rules and solar tariffs, EU auction resilience criteria, and Indian domestic cell requirements (Pao, 2026; TimesTech, 2026). Local production is succeeding unevenly. For example, the US has expanded module assembly capacity much faster than upstream manufacturing capacity with continuing shortfalls in domestic cell production and parts of the wind supply chain (Deloitte, 2025). The cost adjustment is regional and we expect its effect on global technology costs and the overall pace of the transition to be small.

We assume that adoption of nuclear technology is bloc constrained. North America, Europe, Japan, and South Korea do not adopt Russian or Chinese nuclear technology, and the reverse holds true as well. We assume all other regions will source nuclear technology in roughly equal measure from both blocs. This constraint reduces learning curve effects and raises costs in supply chains for spare parts and fuel.

1.3 POLICY DEBRIEF 2026

Global energy policy developments are marked by diverging state interests and energy pursuits, but new alliances among like-minded states are forming to solve climate change problems and achieve the energy transition.

With great-power competition accelerating, alliance formation is centred on shared values and synergistic interests in clean-energy ambitions, supply chain resilience, and climate priorities. This direction was captured by Canadian Prime Minister Mark Carney in a special Davos address in early 2026 where he urged middle powers to pursue strategic coalitions and partnerships, emphasizing that "collective investments in resilience are cheaper than everyone building their own fortress" (WEF, 2026b). The 2025 EU-Canada partnerships exemplify this deepening cooperation on clean energy, critical minerals, and defence. Both are also expanding trade ties across Asia and Latin America. Meanwhile, Global South cooperation is also deepening, as seen with the BRICS expansion in 2025 to include Indonesia as a full bloc member, and its New Development Bank financing renewable energy and infrastructure projects (SDG News, 2026).

With fossil resources at the crux of conflicts and energy price surges, polarization on fossil and non-fossil energy policy is high. This is compounded by the opposite directions in US and Chinese policies on energy transition and climate; the US is moving backward while China is forging ahead. Most import-dependent nations see the energy transition, trade diversification, and renewable energy in particular as the pragmatic step towards rebuilding national energy control and insurance against energy disruption and climate catastrophe. Other countries, notably fossil fuel exporters, adamantly apply political pressure to preserve fossil energy exploitation if not dominance.

Key global developments since last year's ETO signify the stark fronts on fossil energy and climate action

The IMO's *Net-Zero Framework* (NZF), the cornerstone of shipping decarbonization, was delayed in October 2025 amid opposition from oil-exporting countries and US pressure. Negotiations were postponed for a year, delaying implementation until at least 2028 and increasing regulatory fragmentation as regional measures such as the EU ETS, *FuelEU* regulation remain in force.

At COP30, in Belém, Brazil in November 2025, the *Mutirão Decision* (UNFCCC, 2025) recognized the likely overshoot of the 1.5°C target but omitted mention of the pledge to transition away from fossil fuels – the outcome agreed on at COP28 in Dubai, 2023 – as blocked by opposition from fossil fuel producers and like-minded developing countries, despite a push for a transition roadmap from more than 80 countries.

In April 2026, the 'Santa Marta' conference, co-hosted by Colombia and the Netherlands and sparked by the lack of mention in the final COP30 text, convened more than 50 countries to discuss practical steps to transitioning away from fossil fuels for economic security, cost and competitiveness, and climate reasons (Dunne, 2026).

In May 2026, 141 countries in the UN General Assembly supported text turning the International Court of Justice's 2025 advisory opinion on state obligations in respect of climate change into tangible action and accountability, despite opposing votes from 8 countries, including Iran, Russia, the US, and Saudi Arabia (UN, 2026).

In June 2026, electrification solidified as a shared global policy priority. Turkey and Australia, co-hosts of COP31, advanced a '35% by 2035' electrification target mantra on the Action Agenda to accelerate the shift from direct fossil fuel use to clean electricity across demand sectors (COP31 Presidency, 2026).

Key policy developments in select ETO regions signify diversity in policy stances

North America: US climate repeal

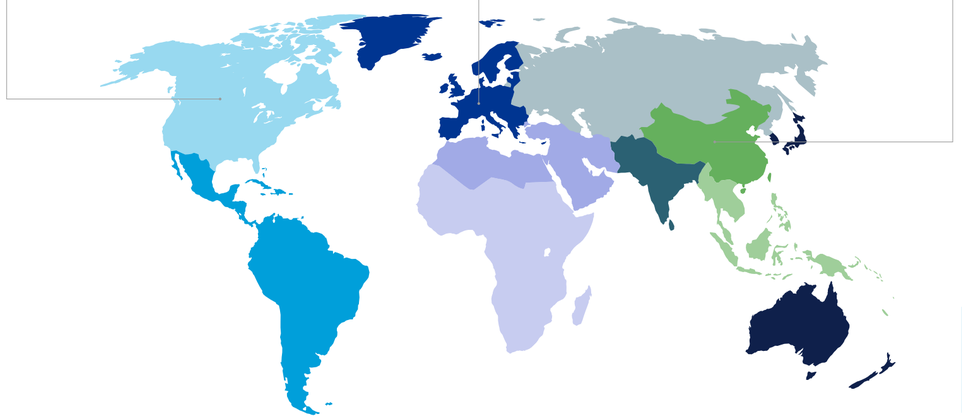
- In February 2026, the US Environmental Protection Agency repealed the 2009 Endangerment Finding removing the legal basis for emissions regulation. US federal policy refocuses on energy dominance favouring nuclear and fossil fuels.
- Canada's decarbonization agenda remains intact, while increasingly balancing climate ambition with economic security and resource development priorities.

Europe: Stays the course on clean competition and climate

- In January 2026, CBAM comes into effect to address carbon leakage, marking the first cross-border extension of a jurisdiction's carbon price.
- In March 2026, the EU adopted a legally binding 2040 climate target of a 90% GHG reduction versus 1990 levels, but with increasing flexibilities.

Greater China: Poised for economy-wide emissions reduction

- China adopted its *15th Five-Year Plan* in March 2026,
- In July 2026, the State Council released the *Action Plan for Carbon Peaking During the 15th Five-Year Plan Period* (2026–2030) translating objectives into sector-specific measures for energy, industry, transport, buildings, and carbon markets.



See Chapter 7 for more details on energy policy.

Strategic priorities and industrial policy tactics under geopolitical pressures

We focus on geopolitical objectives, where states combine industrial policy (IP) instruments to enhance energy and economic security and competitiveness. Policy choices depend on the stage of economic development where Landesmann (2026) outlines three broad development states and corresponding IP orientations, paraphrased below:

1. Economies lagging in the adoption of state-of-the-art technologies direct IP towards attracting investment and facilitating technology transfer from more advanced economies.
2. Economies close to the technological frontier direct IP towards reaping the benefits by advancing and commercializing new technologies.
3. Economies challenged in their technological leadership direct IP to include broader geo-political considerations, moving beyond traditional economic, market-failure correction.

These categories broadly reflect patterns observed across ETO regions. Low-income regions base energy and technology decisions primarily on development priorities and economic factors (State 1). Middle-income regions practice active non-alignment (Spektor, 2025), strategically hedging among partners to advance national interests (State 2). High-income regions use IP to respond to competitive pressures and reduce strategic dependencies (State 3).

Resource endowments and climate priorities further shape IP tactics (see right, 1–5), frequently creating trade-offs between supply chain efficiency, costs and decarbonization.

North America

- Strategic priorities: energy sovereignty, resource control, and exports.
- US pursues energy dominance through fossil fuels and selected clean-energy tech, using energy for strategic leverage and trade instruments in economic coercion tactics.
- Canada repositions towards global clean energy partnerships, critical minerals, and maintaining hydrocarbon exports.

IP tactics 1 2 3 4

Europe

- Strategic priorities: accelerating decarbonization, energy security, and competitiveness in parallel.
- Electrification pivot with domestic renewables/manufacturing preference to reduce hydrocarbon imports and preserve industry leadership in established strengths.
- Vendor diversification and reducing reliance on non-European tech providers, notably in critical infrastructure.

IP tactics 1 2 3 4

North East Eurasia

- Strategic priorities: maximizing fossil fuel production and export revenues, with Russia redirecting trade toward growing Asian markets as Europe plans phase out.
- Ukraine advances European integration, while Kazakhstan balances major powers, pursuing transit independence and clean energy partnerships to strengthen economic resilience.

IP tactics 1 5

Greater China

- Strategic priorities: self-reliance, tech sovereignty, decarbonization. Domestic fossil and non-fossil power is incentivized for energy security.
- China supports strategic tech sectors and leads global cleantech manufacturing, applies countermeasures where market access is infringed, and is a key financier of energy and raw materials globally.

IP tactics 1 2 3 4

Latin America

- Strategic priorities: hydrocarbon expansion and pursuit of green competitive advantages in domestic renewables, hydrogen production and mineral reserves, looking to boost domestic manufacturing.
- The region's distance from major conflict zones creates opportunities to leverage its diverse energy and resource base amid importers' diversification efforts.

IP tactics 1 4 5

Middle East and North Africa

- Strategic priorities: maximizing fossil fuel production, export revenues (Gulf states), economic diversification through renewable deployment, and cleantech industry investments.
- Energy importers, such as Turkey and Egypt, pursue pragmatic energy strategies, combining gas infrastructure and transit ambitions with renewable deployment while deepening clean-energy cooperation with Europe.

IP tactics 1 4 5

Indian Subcontinent

- Strategic priorities: development and energy security, prioritizing affordable energy from both fossil and non-fossil sources to support growth.
- India pursues strategic autonomy through diversified oil and gas imports while leveraging international partnerships to strengthen domestic cleantech industries and supply chains.

IP tactics 1 4 5

South East Asia

- Strategic priorities: energy security amid rapid industrialization and rising demand. Import dependence and chokepoint exposure spur diversification across fuels and technologies, prioritizing renewable deployment and continued coal use.
- Within global supply chain reconfiguration, countries pursue strategic neutrality, attracting manufacturing and mineral investment from competing geopolitical blocs while advancing local-content requirements.

IP tactics 1 4 5

Sub-Saharan Africa

- Strategic priorities: development through expanded energy access, industrialization, and resource development across hydrocarbons, renewables and critical minerals.
- The region leverages diverse partnerships across great powers and multilateral lenders to secure favourable financing, infrastructure investment, and resource development.

IP tactics 1 4 5

OECD Pacific

- Strategic priorities: decarbonization, clean growth through established strengths, energy security through lower import dependence. Non-fossil power is incentivized to bolster domestic production.
- Australia, Japan, and South Korea deepen clean-energy, overseas investments, and supply cooperation with allies in balance with economic ties to China.

IP tactics 1 2 3 4 5

1. Domestic energy prioritization to leverage domestic energy sources (fossil and non-fossil) and technological knowhow for energy security reasons.

2. Trade policy to protect domestic industry: tariffs, duties, export subsidies, and border carbon adjustments.

3. Trade and controls to restrict market access and deny strategic capabilities: sanctions, embargos, export controls, investment screening, and dual-use regulation.

4. Financial support and procurement for domestic supply chains and markets: R&D support, state aid, preferential subsidies, grants, tax credits, and local content requirements.

5. Exchange governed by economics, but conditioned by funding availability and partnership conditions.

Regional policy examples emphasizing developments since last year's ETO

Europe
- The EU is targeting reductions in gas (70%) and crude oil (40%) imports by 2040 through electrification and incentivizing non-fossil power.
- The *Net-Zero Industry Act* aims for domestic manufacturing (40% of annual deployment needs by 2030). The *Critical Raw Materials Act* targets capabilities in strategic materials, supported by procurement, and funding instruments.
- Unfair competition tariffs and investment screening rules restrict high-risk technology suppliers
- The *Industrial Accelerator Act* establishes 'Made in EU' preferences, and the *Clean Industrial Deal State Aid Framework* underpins European industry support, alongside deeper strategic partnership.

Latin America
- Nova Indústria Brazil (2024) policy pursues industrialization through sector missions, including electrotechnology and biofuels. Brazil faces 25% US tariffs on most imports (USTR, 2026).
- Mexico pursues clean-tech industry and public-private renewable investments while remaining dependent on US natural gas. It raised vehicle tariffs (on non-free agreement countries) while the US-Mexico-Canada Agreement (USMCA) is under renegotiation.

OECD Pacific
- *Future Made in Australia* targets market diversification and domestic production and processing capabilities. Japan's *Green Transformation Strategy* and South Korea's Advanced Strategic Industry Fund support strategic industry sectors for technological sovereignty, backed by government co-financing.
- Japan's *Second Offshore Wind Industry Vision* (2025) sets a 65% domestic procurement ratio, supported by the Green Innovation Fund.
- South Korea's *Offshore Wind Promotion Act* (2025) and tenders have non-price criterion favouring domestic manufacturing in winning auction contracts.

Greater China
- China applies export restrictions on low-carbon technologies and critical materials to counter tariffs and market-access barriers, and allows it to influence global supply chains and foreign manufacturing ambitions.
- The *Belt and Road Initiative* targets Global South economies with cleantech exports shifting to emerging markets. China's overseas manufacturing investments exceed USD 220bn since 2022 (Xue and Larsen, 2025) with industrial park projects concentrated in Sub-Saharan Africa and South East Asia (Liu, 2026).

North America
- The US has expanded tariff protections citing national security, balance-of-payments concerns, and forced labour provisions.
- In February 2026, screening began for projects' ownership, financing, technology, and supply chain links to foreign entities of concern. The US OBBBA preserves support for domestic manufacturing.
- 'Building Canada' accelerates nation-building projects with manufacturing tax credits supporting renewable energy, nuclear power, and EV supply chains.

South East Asia
- Factory relocation from China is driving cleantech manufacturing expansions. Countries provide tax breaks to incentivize foreign direct investments and industry localization.
- Singapore restricts high-risk technology suppliers in critical infrastructure, while countries in the region broadly balance security concerns, costs, and local content preferences.

Middle East and North Africa
- GCC countries have China as their main hydrocarbon energy customer and clean-tech partner for long-term economic diversification and pursuing renewable energy ambitions.
- Cleantech manufacturing is incentivized, including in Saudi Arabia and the UAE. Turkey has a domestic-content bonus in renewable projects (YEKDEM mechanism) and its HIT-30 High Technology Investment Program attracts international investments in priority areas like mobility and green energy.

Indian Subcontinent
- Reducing energy imports is prioritized through non-fossil power expansion and coal. India is diversifying coal imports and supporting investment in foreign coal assets to strengthen energy security (S&P Global, 2025).
- India's 2026 *Blueprint for Cleantech Manufacturing* targets 50% domestic value addition by 2030 and has imposed tariffs on Chinese solar products.
- India's *Production-Linked Incentive* (PLI) mandates local-content requirements for cells and modules in government-backed projects and solar tenders.

North East Eurasia
- Russia is redirecting hydrocarbon exports from Europe to China and India (Direskeneli, 2025). Gas power is favoured in the region, disincentivizing import-dependent resources.
- Kazakhstan balances state cooperation with partnerships built on assets of importance, including exporting critical minerals to Europe, continuing oil and uranium extraction, and participating in China's Belt and Road Initiative (Kassenova, 2026).

Sub-Saharan Africa
- Vast natural resources have global and middle powers, especially China, interested in expanding economic involvement in resource extraction, energy, and infrastructure (S&P Global, 2025). Partnerships and investment are vital for development.
- Several countries have localization policies, South Africa most explicitly with local-content requirements in its renewable energy procurement programme (REIPPPP).

1.4 DATA CENTRES AND AI

Data centres consumed approximately 400 TWh of electricity in 2025, 1.2% of global supply. Investment has shifted from software to physical assets such as semiconductors, buildings, cooling systems, and grid connections. These are commissioned on timescales set by power system investment. We forecast electricity demand almost tripling to 1,100 TWh by 2030 and reaching approximately 2,000 TWh by 2035, with AI workloads overtaking all conventional data centre use in 2031.

Growth is primarily constrained by grids, equipment supply chains, and public acceptance. Together, these define the maximum pace at which capacity can be added over the next decade. Our AI energy demand forecast is among the most uncertain in our Outlook because efficiency gains, growing use of AI, and rising computation per task pull demand in different directions. New computing architectures or algorithms could significantly alter the energy needed to deliver a given outcome, a possibility that lies outside the range of assumptions captured in our model.

While physical constraints determine the ceiling for data centre expansion, financing conditions influence how much of that potential is realized. Capital expenditure by the five largest technology companies exceeded USD 400bn in 2025 and we expect it to rise by a further 75% in 2026 to approximately USD 700bn. For comparison, global upstream oil and gas investment was USD 550bn in 2025. Going forward, we expect operating cash flows to fund about half of this spending, with debt markets covering the remainder. This makes the pace of growth sensitive to market sentiment, expected returns, and macroeconomic conditions. Financing conditions can change much faster than grids or supply chains, so a reassessment of AI revenues, tenant creditworthiness, or refinancing risk could slow investment even where power, land, and equipment remain available.

Market data shows a buildout operating against its limits. North America had 6 GW under construction at the end of 2025, within a record development pipeline of 38 GW across the Americas, of which 90% was leased before delivery. US capacity forecasts have been revised upward repeatedly as new projects are announced. We forecast global IT capacity rising from 82 GW in 2025 to 202 GW in 2030 and 342 GW in 2035, with electricity demand growing 22% annually over the first five years. North American data centre demand becomes a major driver of electricity consumption, accounting for 35% of total power generation growth to 2030. Fleet load factors rise from an estimated 42% today towards just over 50% within a decade as newly commissioned facilities complete their multi-year occupancy and equipment ramp-up.

Frontier model capability is advancing quickly, with the length of tasks they can complete doubling approximately every seven months. Evidence on actual productivity is more mixed. Use still remains concentrated in a few information-intensive occupations. Energy demand follows economically productive use rather than capability, and our forecast reflects that lag. Even so, AI workloads grow from 12% of data centre electricity demand in 2025 to 48% in 2030 and 61% in 2035. Inference rather than training carries this growth, comprising 55% of AI energy demand by 2030, as revenue-generating deployment expands across analysis, synthesis, customer interaction, and creative work.

Grid connection remains the dominant supply strategy, at about 80% of global capacity additions, because it lets operators buy electricity as a service rather than build, permit, and fuel their own generation. However, US grid interconnection timelines of three to seven years now stand against 18 to 24 months to build a data centre, so behind-the-meter generation has become a speed-to-power strategy and grid-only supply falls to around 60% of North American additions in 2025. The alternatives are themselves constrained, with gas turbine delivery slots tightening while access to gas pipeline capacity can prove as challenging as securing electricity-grid connections. Further up the supply chain, key semiconductor components remain constrained, transformers and switchgear carry multi-year lead times, and skilled electricians are scarce. No single constraint halts the buildout, but together they mean capacity arrives at the rate supply chains and grids permit rather than the rate demand would support.

Data-centre expansion is also becoming a political issue, with growing scrutiny of who bears the associated costs and impacts. Communities and regulators are responding with measures ranging from permitting restrictions and location-specific construction bans to stricter requirements on water use, grid access, and infrastructure funding. Who pays for the infrastructure buildout will influence where and how quickly capacity gets built. The cost of accommodating data centre demand is already visible on household bills and in wholesale capacity prices in parts of North America (Kunkel, 2025; Paullin, 2026), and other regions are likely to follow the same pattern. Regulators are responding with dedicated rules for large loads, and continued grid access increasingly depends on public acceptance of the resulting costs. We expect cost allocation between data centres and other consumers will be a defining regulatory question of the buildout. Future growth will depend not only on investment and technology, but on whether projects can secure a socially and technically acceptable claim on local power, water, and land.

2 THE DNV FORECAST

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2.0 SUMMARY OF ENERGY SUPPLY AND DEMAND

1 Energy insecurity is driving energy independence

- Energy importers are turning to domestic renewables and nuclear energy to reduce dependence on imported fuels. Fossil producers are relying more heavily on their own resources, but are still adding large volumes of renewable power because of increasingly favourable economics.
- For generations, fossil energy has provided a constant 80% share of primary energy globally. By 2050, the fossil share falls to 51% of global primary energy, and to 40% in energy-importing regions like Greater China, Europe, and OECD Pacific as they move fastest to replace fuel imports with domestic renewables.

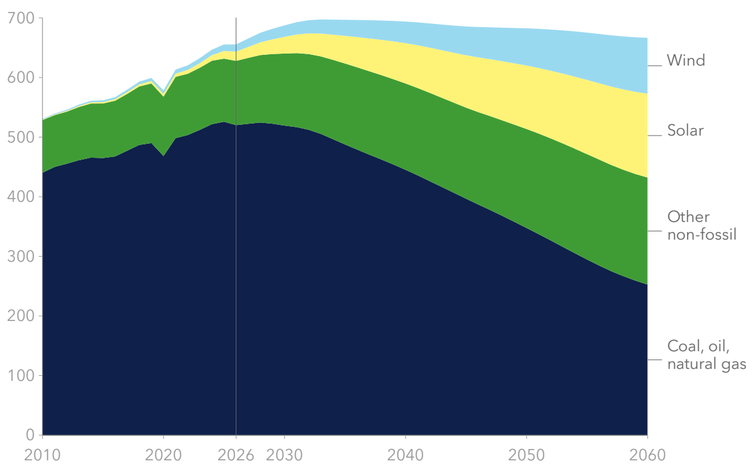
2 Infrastructure and system integration are critical enablers

- Final energy demand for electricity will double by 2060, with wind and solar expanding ten-fold to deliver 77% of electricity by then.
- Market designs need to continue to boost low-cost generation while increasingly rewarding the capacity, flexibility, and system services needed to balance supply and demand (Chapter 4).
- Energy security during the transition also depends on maintaining and protecting the fossil-fuel infrastructure and trade routes on which the world will continue to rely (Chapters 1 and 5).

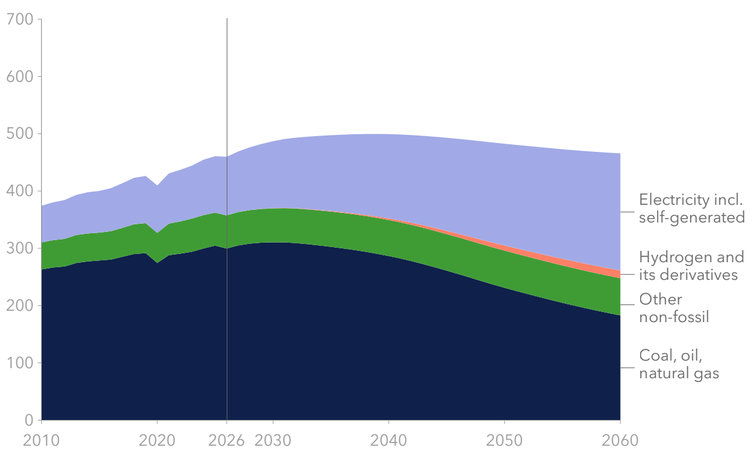
3 Electrification brings vast efficiencies, but is not a solution for all

- As efficient electrification replaces inefficient combustion, consumers get more value out of every unit of energy delivered. Final energy demand thus plateaus and then falls while useful energy grows in aggregate.
- Electricity increasingly dominates the transition because it is efficient, affordable, and increasingly domestic. Electrification cuts waste on an unprecedented scale, with EVs and heat pumps delivering the largest reductions in energy losses across end-use sectors.
- Where direct electrification cannot penetrate deeply, the transition is much slower (e.g. aviation, maritime transport, and heavy industries). These sectors depend on policy-driven uptake of biofuels, hydrogen derivatives, and CCS for decarbonization (Chapter 3).

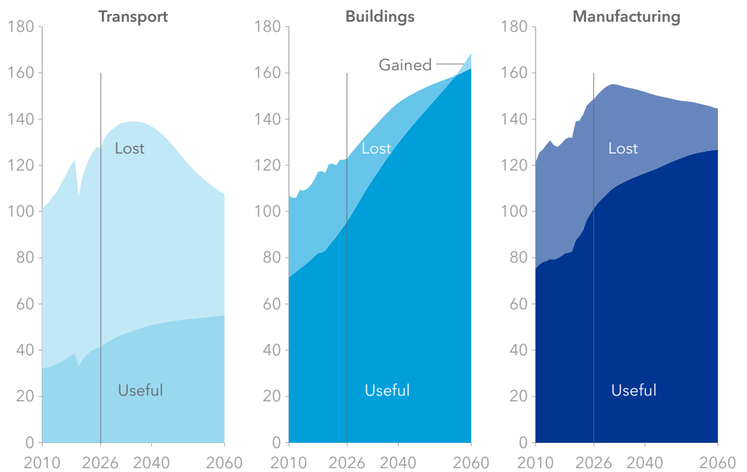
Primary energy supply (EJ/yr)



Final energy demand by carrier (EJ/yr)

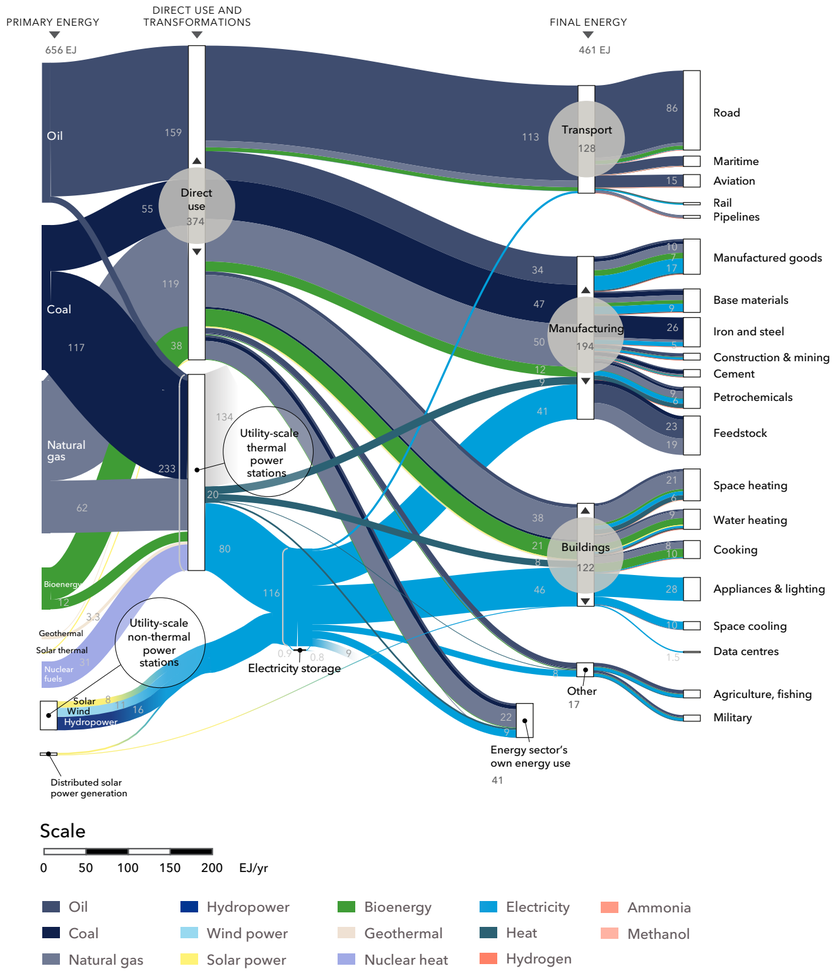


Final energy demand by sector (EJ/yr), split in useful, lost, and gained energy

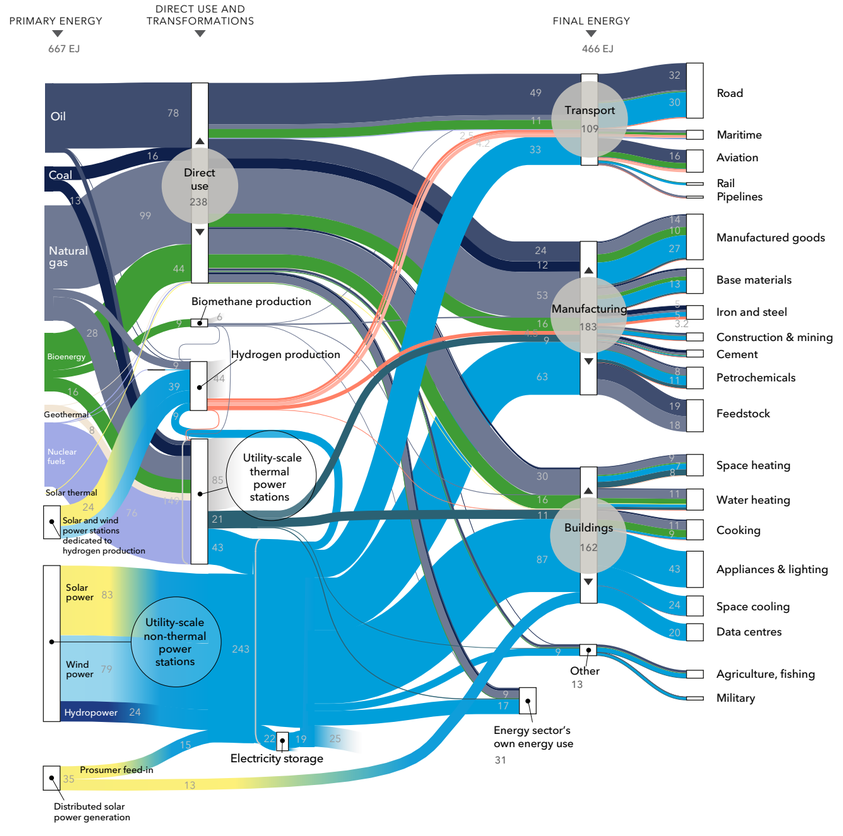


Comparison of energy flows

2025



2060



2.1 ELECTRIFICATION DRIVES PRIMARY ENERGY PEAK

The world is approaching peak primary energy, not because prosperity is reaching its limits, but because energy productivity is rising. Electrification's attendant efficiencies allow economies to create more value from each unit of energy supplied. This means global GDP can almost double by 2060 while primary energy demand remains broadly flat. At the same time, renewables will steadily erode the fossil fuel dominance that has characterized the energy system for more than a century, though not quickly enough to achieve global climate goals.

We forecast that primary energy supply will shortly break a 200-year-old trend and decouple from population and economic growth. After five more years of slowing growth, primary energy will level off and stay virtually constant for the rest of our forecast period to 2060. This major change in a well-established trend is due to the emergence of a more efficient energy system with fewer energy losses. Thus, the world consumes ever more useful energy (i.e. the energy that directly delivers heat, lighting, and motion) despite no further growth in primary energy supply.

The most critical aspect of the global energy transition is non-fossil energy displacing fossil energy. The share of fossil fuels in the present primary energy mix is 80%. This has remained near-constant for more than 60 years. However, we forecast a drop of about one percentage point per year in the coming five years. This is an unprecedented change by historical standards. This displacement will continue and deepen, reducing the fossil share to under 40% in 2060. Coal is experiencing the fastest reduction with a decline of more than 80% to 2060, followed by oil at 50%, and gas at almost 25%. Today, these three fossil fuels have about equal share, but in a few decades natural gas will have a bigger share than oil and coal combined.

Primary energy mix decarbonization is a disruptively rapid shift. Even so, it is not fast enough to reduce emissions in line with climate ambitions; global net-zero emissions targets for mid-century and associated temperature thresholds are unattainable. (Chapter 9). Consensus on a feasible global net-zero target has yet to be reached.

All the renewable energy sources grow in both share of primary energy and in absolute numbers. Solar primary energy grows ten-fold to become the largest source of primary energy by the end of our forecast period. Wind energy grows eight-fold to 2060, while nuclear more than doubles. Interestingly, nuclear's share of primary energy is larger than its share of electricity because primary energy statistics count reactor heat rather than electricity output. The thermal energy produced in reactors is typically three times the electricity ultimately generated.

Primary energy: the raw energy supplied
Global primary energy supply represents all energy entering the energy system in its raw form measured as the chemical energy of fossil fuels and biomass, the thermal energy from nuclear fission, and the electricity generated by hydro, wind, solar, other non-combustible renewables. It also includes the energy sector's own energy use and the fossil fuels used for feedstock production.

Non-fossil grows quickly and overtakes fossil in the 2050s
Primary energy supply (EJ/yr)

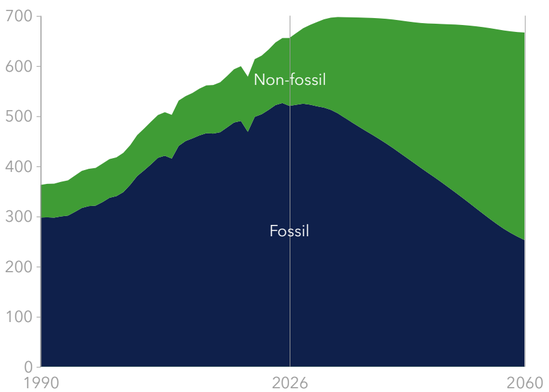


FIGURE 2.1 | Historical data source: IEA WEB (2026).

World primary energy by source (EJ/yr)

Source	2025	2030	2040	2050	2060
Wind	11	19	36	62	94
Solar	13	28	68	106	141
Hydropower	16	18	23	25	24
Bioenergy	55	60	70	77	71
Geothermal	4	6	8	9	8
Nuclear	31	37	44	55	76
Sum non-fossil fuel	**130**	**168**	**249**	**334**	**414**
Natural gas	176	181	178	166	135
Oil	178	182	163	122	88
Coal	173	156	104	60	29
Sum fossil fuel	**527**	**519**	**445**	**348**	**252**
Total supply (non-fossil + fossil)	**657**	**687**	**694**	**682**	**666**

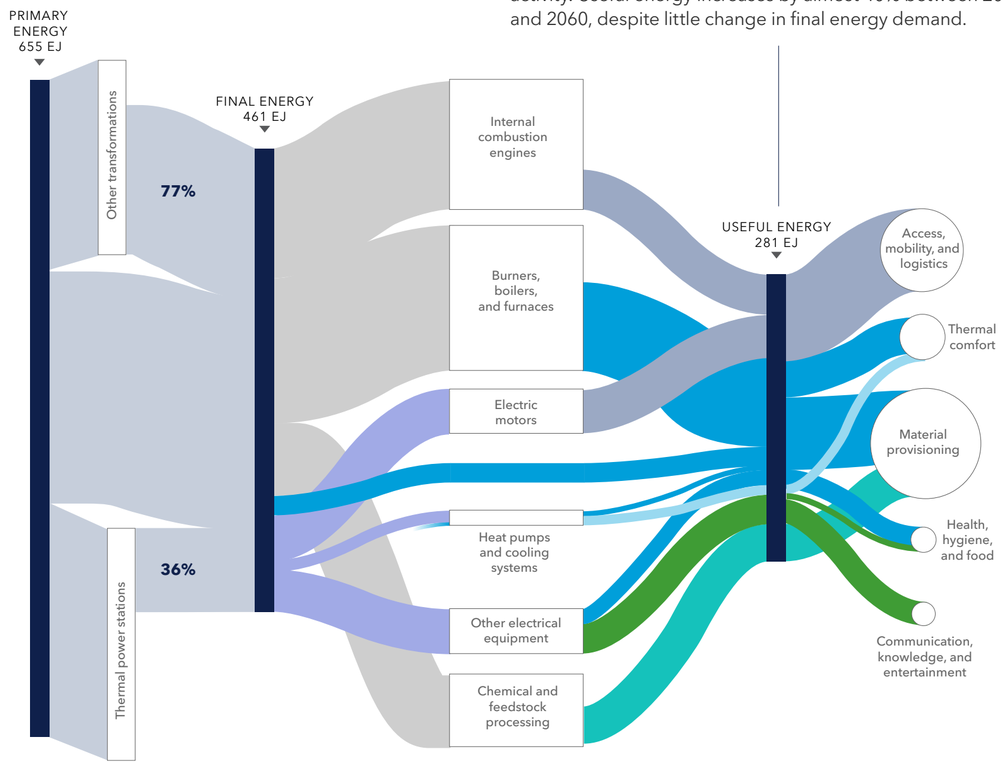
TABLE 2.1

2.2 ENERGY EFFICIENCY

The next wave of efficiency gains occurs where energy is used, not where it is produced.

2025

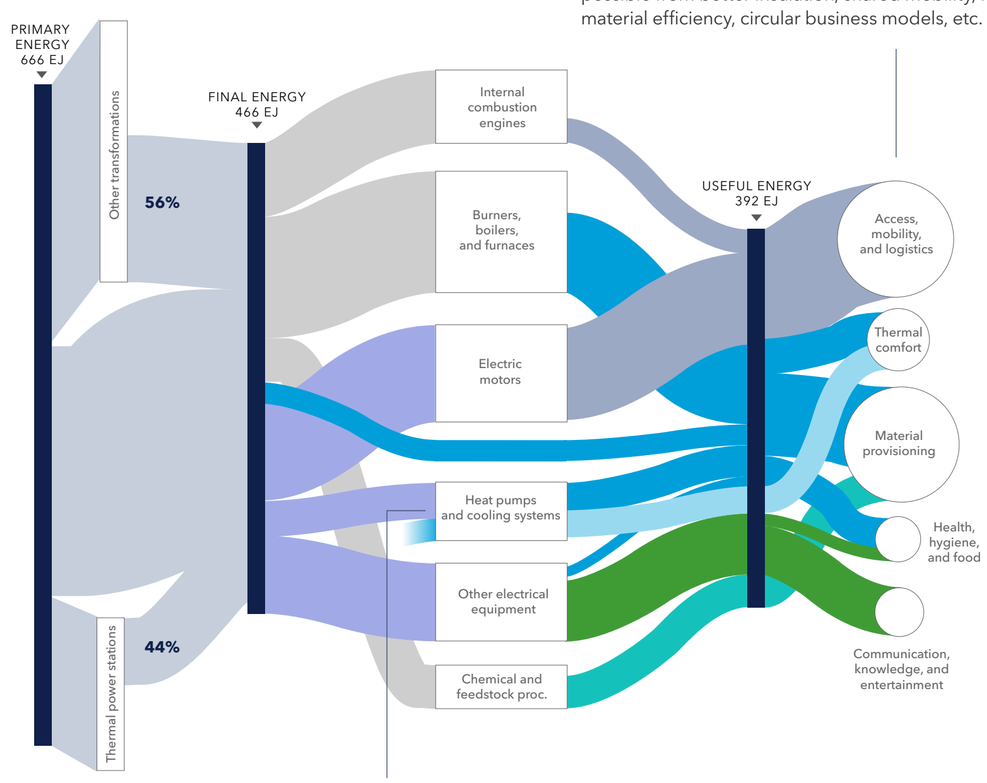
Useful energy is the portion of final energy converted into heat, motion, cooling, and other forms that directly enable human activity. Useful energy increases by almost 40% between 2025 and 2060, despite little change in final energy demand.



In 2025, thermal power stations convert just **36%** of their primary energy input into electricity and heat. These losses decline towards 2060, but losses under 'Other transformations' increase as hydrogen and synthetic-fuel production expands. **Final energy**, delivered to end users before conversion into useful energy, therefore remains broadly unchanged.

2060

Even after losses in energy conversion and end-use technologies are reduced, **further efficiency gains** are possible from better insulation, shared mobility, recycling, material efficiency, circular business models, etc.



Heat pumps use electricity to move heat rather than generate it. The difference between useful heat output and electricity input is supplied by ambient heat from air, water, and the ground.

Fuels · Electricity · Electrical services · Heat · Motion · Cooling · Feedstock

Efficiency beyond energy

The largest efficiency opportunities to 2060 increasingly lie downstream (in end-use equipment, electrification, and the organization of energy services), rather than in the transformation of primary energy into final energy. While early progress came mainly from changes in energy supply, future efficiency gains will increasingly come from how energy is used. Further out, AI will focus attention on where energy use is necessary in the first place.

Most energy analyses stop at final energy – the energy supplied to end users, analogous to the electricity arriving at a building or the fuel dispensed into a vehicle's tank. That provides only a partial picture of efficiency. It measures the electricity, fuels, and heat supplied to consumers, but not how much is converted into useful outputs or lost in end-use equipment. *Our ETO goes further by quantifying useful energy: the motion, heat, cooling, and other outputs delivered by engines, boilers, furnaces, and other equipment*.

Between 2025 and 2060, primary and final energy remain broadly similar, but useful energy will increase substantially. This reflects a major improvement in end-use efficiency as combustion technologies are progressively replaced by electric motors, heat pumps, and other more efficient equipment. By contrast, the ratio of final to primary energy changes little. Falling losses from thermal power generation are largely offset by increasing losses from hydrogen production and other energy transformations.

Pathways to greater efficiency

Different forms of efficiency influence the primary energy required to deliver projected energy services (Figure 2.2). The red line shows the central trajectory for primary energy in our forecast. The upper trajectories explore how primary energy demand would evolve if successive efficiency improvements – each less likely than the previous – were frozen at the current levels. The lower trajectories show how rapidly demand could fall if technologies and practices expected later in the century were adopted with minimal delay.

Together, the trajectories bound the contribution of four cumulative efficiency mechanisms: transformation efficiency, equipment efficiency, electrification, and structural efficiency.

Transformation efficiency determines how much primary energy reaches consumers as final energy. It covers electricity and heat generation, fuel transformation, energy-industry own use, and transport, transmission, and distribution losses. Although renewable electricity avoids the thermal conversion losses associated with fossil-fuel power stations, hydrogen and other secondary fuels production introduces new losses. Consequently, transformation efficiency contributes relatively little to the overall change in total primary energy; by 2060, accelerated improvements reduce primary energy by only 3%, while freezing efficiency at its current level changes the total by less than 1%.

Equipment efficiency determines how effectively end-use equipment converts final energy into useful energy without changing its underlying operating principle. In other words, this category is concerned with efficiency improvements to things like an internal combustion engine or an electric drivetrain, rather than replacing one with the other. Examples include improvements in these engines and drivetrains, but also boilers, furnaces, turbines, cooling equipment, and so on. These incremental gains remain important, but mature technologies typically improve only gradually. Modern condensing gas boilers, for example, already convert around 90% to 95% of fuel energy into heat, leaving limited scope for further improvement without changing technology. Improvements also spread slowly through the installed stock, as older, less-efficient equipment remains in operation until it is replaced.

Electrification involves replacing one technological pathway with another rather than simply improving existing equipment. Electric drivetrains convert electricity into motion far more efficiently than combustion engines convert fuels. Heat pumps move heat rather than generating it through combustion and can therefore deliver several units of useful heat for each unit of electricity consumed. Electrification consequently produces much larger savings than incremental equipment improvements alone. By 2060, halting further electrification would increase primary energy demand by 23%, compared with 13% if incremental equipment-efficiency improvements ceased. However, the cumulative energy savings due to electrification are constrained by the gradual turnover of the existing stock using other energy sources.

Structural efficiency reduces the useful energy required to deliver economic activity and human needs. Better-insulated buildings require less heat to maintain thermal comfort; public transport, higher vehicle occupancy, and improved logistics can provide mobility with less motion; recycling, material efficiency, and circular production reduce the need for energy-intensive primary material production; and structural efficiency also includes organizational and behavioural changes that improve the utilization of buildings, vehicles, infrastructure, and materials.

The largest efficiency gains lie beyond energy conversion, in electrification and structural change

Global primary energy consumption under various efficiency pathways (EJ/yr)

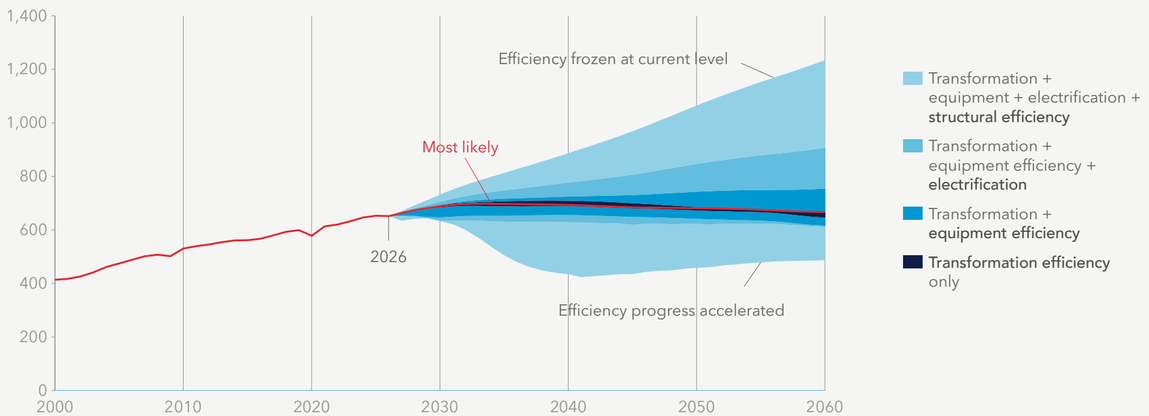


FIGURE 2.2

The cost of delay and the value of acceleration

The trajectories above and below the central forecast are two complementary sets of analytical bounds, not alternative forecasts (Figures 2.2 and 2.3).

The upper trajectories successively freeze each form of efficiency at its current level. They show how much primary energy would be required if transformation efficiency, equipment efficiency, electrification, and structural efficiency stopped progressing. These cases isolate the contribution of the changes already contained in our most likely forecast and demonstrate the energy consequences of implementing none of the further improvement included in our forecast.

The lower trajectories estimate how rapidly the primary energy needed could decline under accelerated change. Transformation, equipment, and electrification pathways retain physical constraints such as infrastructure development, new additions and the natural retirement of existing assets. For structural efficiency, we assume that the useful-energy intensity otherwise reached by 2100 is achieved over 15 years. These pathways are deliberately ambitious and exclude many economic, political, and institutional barriers. They indicate how quickly efficiency could progress when only selected physical constraints are retained, rather than presenting likely outcomes.

Electrification consequently produces much larger savings than incremental equipment improvements alone.

Transformation efficiency offers the smallest additional savings. Equipment improvements make a larger contribution, while electrification provides a substantially greater opportunity. Structural efficiency creates the widest range of outcomes, although it is also the most uncertain and depends on changes extending well beyond energy technology.

Distinguishing between these mechanisms matters because each requires a different response. Transformation efficiency depends mainly on energy infrastructure and the power mix; equipment efficiency on standards, innovation, and stock replacement; electrification on enabling infrastructure and investment in new technologies; and structural efficiency on wider decisions about buildings, transport, and material use. Identifying their relative contributions helps policymakers, planners, and investors target the largest opportunities and recognize where slow turnover or inadequate infrastructure will delay progress.

Useful energy is a bridge, not the destination

Useful energy provides a more meaningful measure of energy productivity than final energy, but it is still an intermediate output. People do not demand motion, heat, or cooling for their own sake. They seek mobility, thermal comfort, food, communication, materials, and other services that meet human needs.

This logic is embedded in our ETO modelling framework. Energy demand is derived from concrete activity measures that include passenger- and vehicle-kilometres, freight activity, building floor area, and industrial output. These service drivers determine useful-energy requirements, which are then translated into final and primary energy demand.

As conversion losses diminish, we must turn our attention beyond useful energy. The next frontier is to deliver the same prosperity and well-being with less need for motion, heat, and material production in the first place. AI may prove to be a powerful enabler of this next stage of energy productivity.

The range of potential outcomes from efficiency in 2060 is greater than today's total energy use
Global primary energy consumption in 2060 under various efficiency pathways, difference from most likely future (EJ/yr)

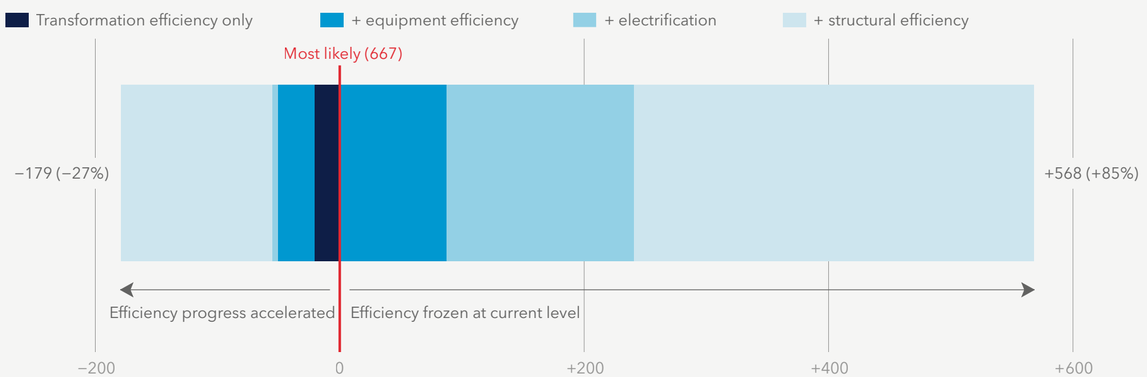


FIGURE 2.3 | 2060 zoom of Figure 2.2: range between efficiency frozen at current level and accelerated efficiency progress.

2.3 SYSTEMS THINKING

Why systems thinking matters for the energy transition

Energy has always been a systems challenge. Since humans first gathered firewood for winter, energy has involved interconnected decisions about supply, storage, transport, and consumption. Over the millennia, those systems have become larger and more complex; the modern oil supply chain, for example, is among the most sophisticated industrial systems ever created. Today, the interconnections and feedback loops within the energy system are deepening and accelerating. The system is becoming more dynamic.

Consumer decisions increasingly influence how energy is produced, stored, transported, and valued. In other words, consumers are becoming more consequential system actors. EVs can charge when renewable generation is abundant and, eventually, discharge when it is scarce. Heat pumps can shift demand, batteries can both consume and supply electricity, industrial loads can respond to hourly prices, and, most recently, data centres affect local networks.

Success in understanding, forecasting, planning, and ultimately, operating the emerging energy system depends on understanding not just its components, but the interactions between them. Accurately quantifying the flexibility needs in a renewables-dominated system, for example, is vital to avoid under- or over-dimensioning storage installations. There are similar complexities involved in quantifying links like the long-run dependencies between clean electricity and clean hydrogen to produce green shipping fuels. Yet system subtleties – which can be pivotal – invariably fall victim to dogma in polemical arguments about the energy transition.

Systems thinking, which is the foundational philosophy of our forecast, is superior to conventional linear approaches for capturing interconnectedness, constraints, and feedback loops. It seeks to understand not only how individual technologies evolve, but how they influence one another over time and collectively shape long-term outcomes.

Industry leaders are increasingly recognizing the need for such an approach. In our survey (DNV, 2025), 96% of power sector leaders call for urgent grid investment, while 86% believe that policymakers do not fully grasp the system-wide challenges.

Dynamics over static snapshots: capturing multifaceted system behaviour

Energy transitions do not unfold in straight lines or at constant rates. They are shaped by feedback loops, delays, and fierce competition between technologies. Learning effects, infrastructure bottlenecks, policy interventions, consumer trends, and investor responses all influence the pace of change.

These dynamics cannot be captured in static spreadsheets. A structured, dynamic model is necessary to reflect how different parts of the energy system interact to reinforce or offset one another. Without this systems-level view, even the most well-informed forecasts risk overlooking the interactions in the background that ultimately drive or stall the transition.

What is system dynamics?

System dynamics is a specialized method for simulating how complex systems behave over time. Built on the broader principles of *systems thinking*, it is uniquely designed to capture how feedback loops, time delays, and imperfect decision-making shape reality.

System dynamics is ideal for environments where the overall behaviour of a system is driven by internal interactions rather than external forces. In the global energy transition, this is critical. The long-term development trajectory is primarily driven by the interaction between the economics of supply and demand through markets. External stimuli like government policies and subsidies serve to kick-start or steer these underlying feedback processes.

How do we apply systems thinking?

The ETO Model is a feedback-driven system dynamics model built on the principle of bounded rationality. Instead of assuming perfect foresight, we model how decision-makers (investors, producers, consumers, and governments) actually behave – responding to changing signals under uncertainty – and how this totality effects energy systems change. We also model how technologies themselves (like ever cheaper solar PV) change the environment in which future investment decisions are made. These two examples provide a small flavour of the complexity involved, the first showing how we consider multiple factors affecting decisions and the second how dynamics interact through time:

- **Consumer choices:** the choice between electric and internal combustion engine (ICE) vehicles is not based purely on the lowest price. Our model weighs multiple real-world factors (upfront cost, operating expenses, charging density, and convenience) and applies different historical preferences for passenger versus commercial buyers.

- **Power station investments:** rather than calculating a theoretical 'optimum mix' that minimizes total system costs or emissions, our model determines the technology mix based on market competitiveness. We use revenue-adjusted levelized costs, factoring in both the capital required and the revenue generated from selling power, which mirrors how actual power markets operate.

Modelling real-world constraints: the case of grid bottlenecks

Nowhere is the application of systems thinking more vital than in addressing grid bottlenecks. The electrical grid is a shared, limiting resource that is facing an unprecedented surge in connection requests from high-demand users (data centres, industry, EV networks, hydrogen projects) and new generators (renewables).

While planning and permitting processes (which average five to ten years in advanced economies) are frequently cited as the primary roadblock, local capacity expansion is also constrained by severe shortages in both transformer and non-transformer equipment and specialized labour. Global supply chains are struggling to deliver critical infrastructure equipment like high-voltage direct current (HVDC) cables and specialized switchgear, with lead times stretching beyond three to five years (BCG, 2025). Furthermore, the global grid workforce is heavily constrained; the industry must add at least 1.5 million specialized workers by 2030 just to meet projected baseline demand (IEA, 2024).

In the ETO Model, we address this interplay of constraining factors dynamically. We estimate the desired connection requests from each sector based on their growth targets and the prevailing technology mix. We then compare this total connection demand to the 'regional capacity expansion capability' – a metric that dynamically limits how fast networks can add lines and substations based on these real-world labour, equipment, and administrative constraints.

Viewed through a systems lens, this dynamic represents three classic system archetypes, all of which we explicitly model in the ETO:

1. Limits to growth: the rapid expansion of electrification naturally slows down as it hits the hard ceiling of regional expansion capability. We model this ceiling to reflect the physical reality that grids cannot scale infinitely just because capital is available; they are bound by the availability of specialized labour, raw copper and aluminium for cables, and supply chain capacities.

2. Tragedy of the commons: the transmission network acts as a shared resource. Individual developers optimize for their own benefit by rushing to secure grid capacity. To hedge their bets, developers frequently submit duplicate requests in congested regions. In the ETO Model, we explicitly capture this behaviour by simulating a 'phantom connection queue', which in turn creates an additional paperwork burden that further drains administrative resources and extends delays for all projects.

3. Shifting the burden: Because the root structural issue – insufficient long-term transmission investment – is too difficult or slow to solve, developers rely on local workarounds. Instead of waiting years for a transmission connection, developers build behind-the-meter generation, sign local curtailment agreements, or deploy on-site gas turbines and batteries for facilities like data centres. We model this 'shifting the burden' behaviour in the ETO, allowing grid delays to actively alter the final technology mix and increase local emissions.

3 ENERGY DEMAND

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3.0 SUMMARY

Demand for energy services continues to grow
Demand for useful energy – the energy that directly delivers a service such as heating, lighting, or motion – will rise 23% from today to 2040 and another 10% to 2050. Buildings account for more than half of this additional demand.

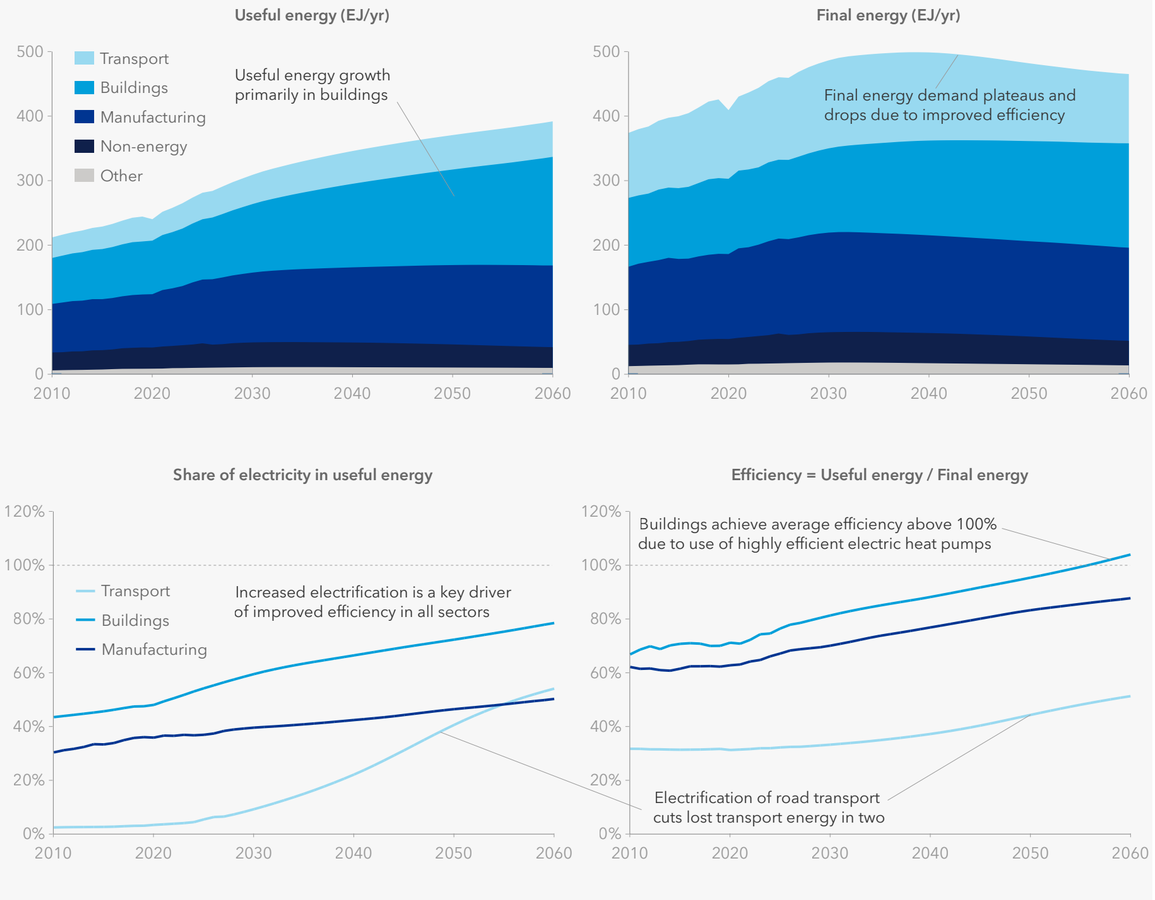
A variety of drivers explain this growth. Primarily, economic growth per capita leads to higher demand per person, particularly in emerging markets, and the global population will grow 20% by 2050. In addition, some energy demand categories are growing more than others, such as space cooling and AI data centres.

Final energy demand will plateau
Energy supplied to the final user (i.e. the energy supplied to consumers' houses via the grid or cars via a petrol pump) constitutes 'final energy demand', which will only grow 8% until it plateaus in the 2030s and starts to decline.

The fact that useful energy demand continues to grow, while the final energy demand levels off, is due to a gradually more efficient conversion of final energy to useful work. For instance, efficiencies increase 0.8 percentage points (pp)/yr in buildings and 0.6 pp/yr in transport and manufacturing (bottom right panel).

Electrification drives efficiency gains
Efficiency gains are strongly linked to increased electrification in all sectors (bottom left panel). Electricity demand is surging, growing 80% by 2050. Transport, AI and data centres, and green fuel production (which demand little electricity today) make up half of the added demand. Read more details on electrification in Chapter 4.

Demand for energy services will continue to grow strongly throughout our forecast period, but final energy demand largely stabilizes because electrification and efficiency improvements offset much of the growth.



Transport continues to electrify where possible

Transport services will expand as more people and goods travel more, yet the sector's final energy use will start to fall. By 2050, transport will waste only half as much energy as it does today, largely because EVs convert energy into motion far more efficiently than internal combustion engine vehicles. While aircraft and ships can only be electrified on shorter routes, they will improve efficiency and increasingly switch to low-carbon fuels.

Buildings, including AI and cooling, demanding ever more

By 2050, buildings stands alone as the only sector that will demand more final energy than today. Most of the growth is from new AI and data centres (38%) and from an increased need for space cooling in other buildings (35%). Final energy demand growth would be much larger without the efficiency gains introduced by heat pumps in both space heating and cooling. The resulting global average energy efficiency of buildings surpasses 100%.

Manufacturing will decarbonize gradually

While demand for manufacturing output will continue to grow, recycling of used plastics and scrap metal will limit the growth. Through electrification and other efficiency measures, final energy demand from manufacturing in 2050 is the same as today. Decarbonization will be gradual, held back by the slow turnover of long-lived industrial assets and the lack of a single dominant low-carbon solution. Therefore, carbon capture and storage (CCS) remains essential for reducing emissions beyond fuel switching.

3.1 TRANSPORT

50%
battery EV sales share by 2035

2035
peak transport energy demand

1 EV uptake is transforming transport energy demand

- We expect passenger vehicle market share to be over 50% EVs by the mid-2030s, driven by rapid adoption in China and Europe.
- Final energy demand for transport will peak in 2035, mainly due to the electrification of the road transport subsector, despite rising demand for transport services in all subsectors except shipping.
- As road transport electrifies, demand for biofuels shifts to maritime and aviation where there are higher barriers to electrification. By 2060, road transport accounts for just 11% of biofuel demand (1.2 EJ/yr), compared with 25% in maritime (2.8 EJ/yr) and 63% in aviation (6.9 EJ/yr) (Figure 3.2).

2 Policy support is critical for hard-to-electrify transport

- Price parity for decarbonization options in aviation, shipping, and long-haul trucking will only be reached in the long term. Policy support is therefore critical in the short to medium term.
- In shipping, efficiency measures followed by a steady shift to low- and zero-carbon fuels will see oil and gas covering less than a quarter of final energy demand by 2060. However, continued uncertainty on the IMO's *Maritime Net-Zero Framework* (NZF) has resulted in a two- to three-year delay in the decarbonization trajectory we forecast in 2025.
- Falling emissions from shipping also reflect a long-term decline in high-seas tonnage as global demand for hydrocarbons falls away. The opposite holds true for aviation, where demand for air travel will double from today to reach 1 billion passenger-trips by 2060.
- Aviation initially transitions through bio-based sustainable aviation fuels (SAF) in the 2030s; hydrogen and synthetic e-fuels will be used starting in the 2040s as production cost barriers come down.
- *ReFuelEU* SAF targets will likely require imports from regions with strong SAF production capacity expansion policies, such as Greater China.

3 Iran war oil shock will resolve in the short term, but is likely to have lingering effects on transport demand

- Global aviation and road transport will both experience a short-term oil demand reduction due to the oil shock, but we expect a rebound in 2027.
- EVs' share of new registrations spiked in Europe and Asia, further tipping the scales to BEV normalization and dominance in the coming decades.
- A diverse array of policies, from urgent to structural, have been enacted globally following the oil crunch. These have boosted electrification, increased biofuel blending, promoted public transport, introduced fuel subsidies, and implemented travel restrictions and bans. Most of these interventions are likely to prove temporary, but they offer a glimpse of how economies might reduce their exposure to future oil and gas disruptions.

Electrification high in road, rail; low in aviation, maritime
Share of electricity in final energy demand

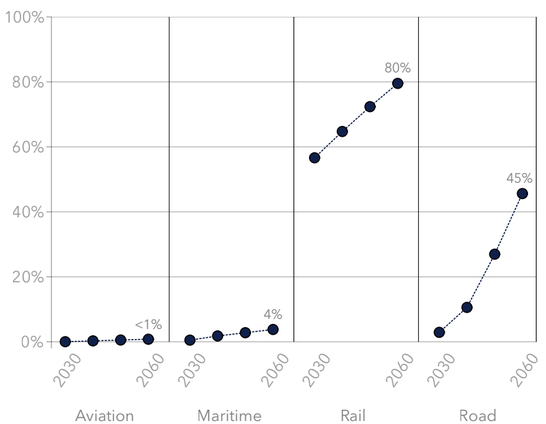


FIGURE 3.1 | Maritime excludes on-board nuclear power.

Bioenergy use shifts from road to maritime and aviation
Final bioenergy demand (EJ/yr)

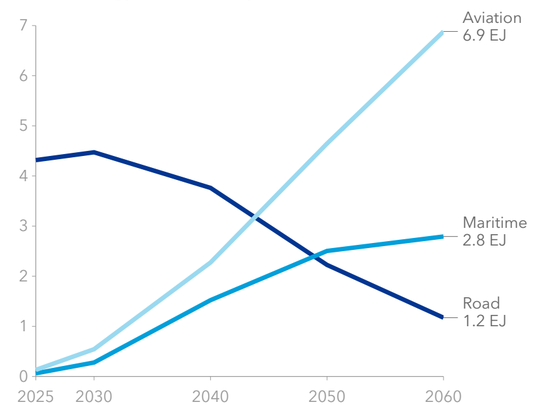


FIGURE 3.2 |

Road

Road transport demand grows from 1.8 billion vehicles in use globally in 2025 to 2.5 billion in 2060. However, despite all those extra wheels on the road, we forecast the total energy demand will reduce from 95 EJ/yr in 2025 to 65 EJ/yr in 2060 due to the electrification of the fleet globally reaching 73% by 2060, squeezing out oil.

Oil demand for road transport will peak in 2031, but it will remain the majority energy carrier as measured by share of final energy all the way through to the 2050s. The remaining ICE vehicles in the 2050s (less than 20% of the total) represent such a disproportionate share of final energy demand because final energy is useful energy plus waste. EVs convert over 90% of electricity into motion, three to four times better than combustion engines. Therefore, batteries and charging technologies are the key enabling technologies in the transition.

Oil supply shock strengthens the case for electrification. We expect the current disruption to oil supply resulting from US and Israeli strikes on Iran to only have a short-term impact on global road transport demand, but it illustrates the strategic importance of the transition away from oil. The sudden closure of the Strait of Hormuz in early 2026 disrupted 20% of global oil supply. Emergency consumer-protection measures have been deployed globally, including EV purchase incentives, fuel subsidies, public transportation promotion, and travel restrictions (IEA, 2026a). Asian markets felt the brunt of the supply crunch as more than 80% of oil from the Strait was exported to the region prior to the disruption (EIA, 2025). However, on a global scale, we forecast a temporary oil demand reduction of only 0.8% from total road transport (from 85.7 EJ/yr in 2025 to 84.8 EJ/yr in 2026) with a minor rebound effect in 2027 when oil demand increases to 87.1 EJ/yr. More significantly,

Oil demand to peak by 2031 as EVs take over
Road energy demand by carrier (EJ/yr)

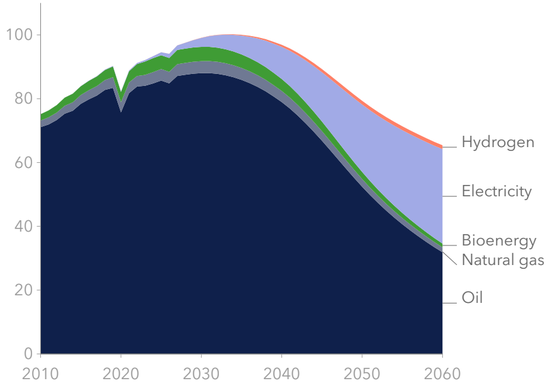


FIGURE 3.3 | Historical data source: IEA WEB (2026)

the Middle East supply shocks and price increases are drawing widespread consumer attention to the benefits of EVs. For example, global EV sales rose sharply (35%) in Q2 2026 compared with Q1, 2026 (IEA, 2026b). However, this is merely giving an additional impetus to a much stronger electrification trend already underway.

Passenger EVs entering the mass-market phase
EVs are entering the mass-market phase in the passenger vehicle segment, with battery EVs (BEVs) expected to dominate new car sales across major markets during the 2030s. Our analysis shows that already in 2026, nearly one in four new cars purchased will be a BEV. By 2035, we forecast EV sales to be one in two cars, and half of vehicles in use will be EVs by 2049 (Figure 3.4). Europe and China drive this momentum, both regions having now crossed a tipping point where EV uptake is now 'self-propelling' (Mercure et al., 2025). The trend of EV uptake has matured to a point where its dominance is now the likely path in these markets. China is leading adoption, with BEVs already accounting for 33% of passenger vehicle sales in 2025 and forecast to reach over 50% of sales by 2029. Europe follows closely and will reach majority BEV sales by 2031. Due to increasing electrification, passenger vehicle final energy demand will decline from 55 EJ/yr in 2025 to 36 EJ/yr in 2060, of which 46% will be supplied by electricity.

We forecast that electric two- and three-wheeler vehicles will exceed 50% global market share by 2031, two years ahead of passenger cars. The high utility of mopeds and three-wheel 'tuk-tuks' particularly in the Indian Subcontinent, South East Asia, and China is due to a confluence of dense urban traffic congestion in these regions and their affordability compared to cars. These regions make up 80% of the global two- and three-wheeler fleet, and we expect this share will remain as the global fleet grows from 1 billion in 2025 to 1.7 billion in 2060. We forecast that low- to middle-income regions such as Sub-Saharan Africa and Latin America will also increase electrification through their two- and three-wheeler fleet in the 2040s.

EVs will dominate sales by the 2030s and in-use by 2050
Global share of EVs in passenger vehicles

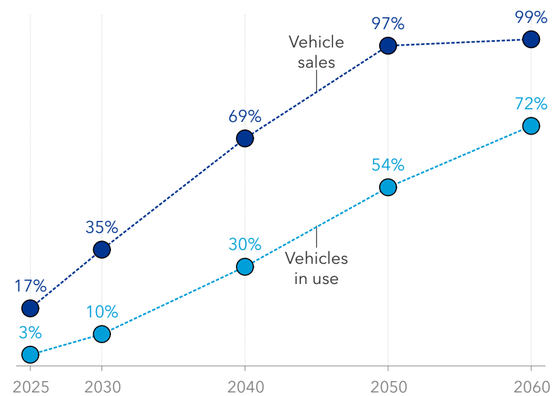


FIGURE 3.4 |

Policy, infrastructure, and technology are essential for uptake

Policy has been essential in driving rapid EV uptake. China has already exceeded its 2035 target of 50% sales share of new energy vehicles (NEVs), which include BEVs, plug-in hybrids, and fuel cell EVs (FCEVs). Countries in the OECD Pacific and Indian Subcontinent have national or regional EV sales targets in the 2030s. South Korea aims to have 4.5 million EVs by 2030 while Japan aims to have 100% electrified vehicle sales, including conventional hybrids, by 2035. Australia also has regional sales targets ranging from 50% to 100% zero-emission vehicles by 2030 to 2035.

Our analysis finds that access to public chargers and driving range are key factors influencing EV purchase decisions. A US study shows that although 86% of drivers have access to a home charger, 60% still rely on public chargers weekly (Auto Remarketing, 2024). In 2026, China and Europe are global leaders in EV charger deployment, at 4.7 million and 1.2 million public charging points respectively (IEA, 2026c). Battery technology improvements have bolstered driving ranges, with average battery capacities in new passenger EV models quadrupling to 76 kWh over the last 15 years (McCay, 2026). We forecast that all ETO regions will overcome these barriers at varying timelines, as growing EV ownership and infrastructure buildout further make purchasing EV an increasingly attractive choice.

Average battery capacities in new passenger EV models have quadrupled over the last 15 years.

Commercial road transport: still harder to abate

The commercial fleet will eventually follow the same electrification trend as the passenger vehicle fleet, but near-term barriers mean the transition will be slower. Annual EV sales share in this segment was only 4.5% in 2025 but will reach 95% by 2060. The size of the commercial EV fleet will also grow from 3 million vehicles to 423 million vehicles in use in this time.

To effectively operationalize commercial EVs, trucks need access to fast public chargers along highways. Long-haul truck logistics require DC megawatt-charging compared to passenger fast DC charging of 150 kW to 400 kW. The EU *Alternative Fuel Infrastructure Regulation* (AFIR) affirms this need by requiring that by 2030 the main corridors of the Trans-European Transport Network (TEN-T) must have megawatt charging available every 60 km (Bernard, 2023). China also has parallel ambitions, where 3,000 stations for fast recharging and battery swapping will be operational in the main highways by 2030 (Jiying, 2026). Greater China will be the first region to reach 50% market share of EVs in the commercial fleet, which we forecast will be achieved in 2030. Europe will soon follow in 2034. This trajectory broadly aligns with the EU's mandate for new heavy-duty vehicles – i.e. trucks, buses, and coaches – which requires 43% CO_2 reductions by 2030 and 90% by 2040 (EC, 2026).

Bidirectional charging and vehicle-to-grid (V2G)

The growing number of battery EVs creates a vast distributed energy source and sink through *bidirectional charging* that helps to balance electricity supply and demand and serves as an important tool for grid flexibility. Vehicle-to-grid (V2G) is a specific bidirectional charging application that enables EVs to provide power to the electrical grid. We forecast that by 2060, the total storage capacity potential of road vehicle batteries will total 200 TWh, but only 25 TWh will be utilized due to operational constraints (Figure 3.5).

Passenger EVs can be a valuable energy source if they are available to feed into the grid during demand peaks. During off-peak time, they can also support grid frequency and voltage regulation (IEA, 2026d). As an energy sink, V2G can support renewable energy integration by storing excess wind and solar generation during the day, reducing curtailment.

Widespread V2G adoption requires a confluence of multiple factors: technology, regulation, and participation. Every component of the V2G system – EVs, charging terminals, and grid networks – must be interoperable to allow for bidirectional flow of power. Regions and countries like Europe, the US, Japan, and China are advancing regulations to overcome barriers to V2G adoption.

Even with technology and regulations in place, vehicle owners must be willing to participate in the V2G scheme. One of the main concerns is that constant charging and discharging may degrade the battery faster. However, studies indicate that while V2G power cycling marginally increases battery degradation, up to 90% of capacity loss comes from ageing (Sagaria et al., 2025).

Expected operational V2G is a fraction of total capacity

EV fleet battery storage capacity (TWh)

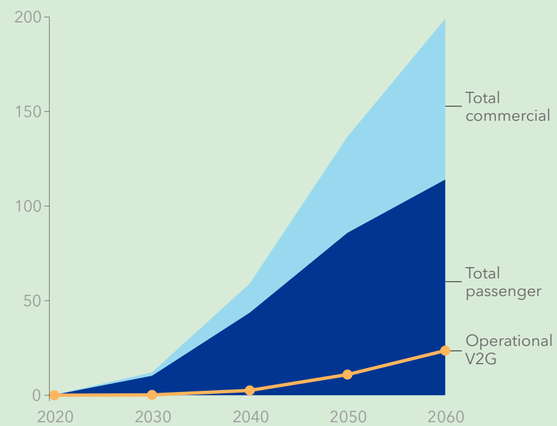


FIGURE 3.5 | Historical data source: IEA EV Outlook (2026d)

FCEVs will have a niche application in long-haul trucking, where there are operational demands of rapid refuelling and extended driving range. Overall, we forecast low commercial FCEV uptake due to high vehicle cost, high cost of production and distribution of hydrogen, and the slow rollout of refuelling infrastructure. We expect hydrogen to play a minor role at 2.5 million vehicles in use globally by 2060, just over 1% of final transport energy demand.

Role of bioenergy in future of road transport

With oil still supplying over half of road energy demand up until the late 2050s, biofuel blending will continue to be an important interim decarbonization pathway. Globally, we forecast biofuel will comprise just over 2% (2.1 EJ/yr) of road transport final energy demand by 2060, while ICE vehicles decline from 2.6 billion vehicles in 2025 to just 550 million by 2060. North America leads in bioenergy demand for road transport due to the US's volumetric blending mandates for both bioethanol and biodiesel (US EPA, 2026), which effectively standardized to 10% bioethanol blends (E10) nationally, with E15 available in many states. In terms of blending fractions, Brazil is a global leader with E27, owing to abundant availability of sugarcane. Europe's *Renewable Energy Directive (RED) III* targets a 29% share of renewable energy in transport by 2030, including conventional and advanced biofuels (EC, 2023). India has had rapid bioethanol uptake in the past decade, achieving an E20 bioethanol blending mandate in 2026 (PIB, 2026).

Maritime

Energy demand in maritime transport peaks at 13.7 EJ/yr in the early 2030s, before starting to decline 20% over the subsequent 30 years (Figure 3.6). The fall off in energy demand is due primarily to a plateauing demand for shipping cargo, but is also related to improved operations and energy efficiencies across the fleet.

Changing volumes

Total global shipping work will plateau at 77 trillion tonne-miles (Figure 3.7) in the 2030s, even as the world economy continues to expand. This decoupling from GDP (set to double by 2060) separates maritime from all other transport sectors where underlying demand for services continue to grow. However, while total tonne-miles level off in shipping, there is a clear shift in the work needed by different vessel types.

Maritime will need less energy due to efficiency gains
Final energy demand in maritime by energy carrier (EJ/yr)

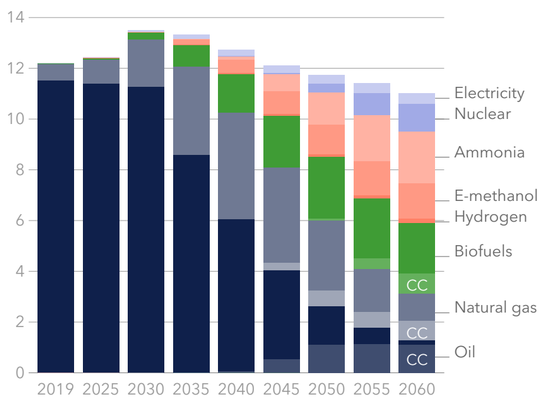


FIGURE 3.6 | CC stands for on-board carbon capture. Historical data source: IEA WEB (2026)

Shipping of fossil fuels will fall, cancelling other growth
World seaborne trade by segment (thousand Gt-nm/yr)

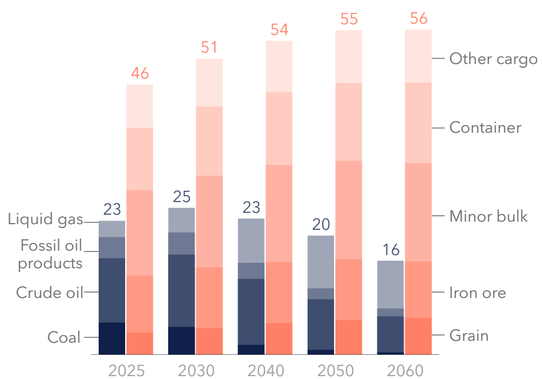


FIGURE 3.7 | Non-fossil oil products and alternative fuels are included in 'other cargo'. Historical data source: Clarksons (2026)

Fossil-energy carrier shipping, which currently makes up a third of total tonne-miles, will start to decrease in the mid-2030s as the world transitions away from fossil-energy carriers. Coal shipping will start to decrease by the end of this decade and oil carriers from 2030 onwards. The situation in the Strait of Hormuz has decreased crude oil shipping in 2026 and led to a reduction of available crude oil carriers. While the demand for this shipping segment will bounce back when the strait eventually opens and countries begin to restock their reserves, the situation is causing permanent oil demand destruction. Hence, the long-term demand for crude oil carriers is lower than before this event. Liquid gas shipping, on the other hand, will continue to grow to mid-century. This includes shipping of liquified natural gas (LNG) and liquid petroleum gas (LPG). Their extended relevance is partly due to gas releasing less CO_2 than coal and oil when burned. In addition, ongoing supply diversification and 'friend-shoring' efforts are likely to further boost the inter-regional shipping of gas.

Shipping of other commodities will continue to grow, almost offsetting falling levels of fossil fuel shipping. Nevertheless, diverse commodity shipping will also gradually decouple from GDP as the global economy becomes more centred on services than goods. Additionally, containerization, the trend of moving more cargo into containers, will continue to reduce the need for other vessels. Importantly, our forecast assumes that global chokepoints will not be closed permanently over the period.

Efficiencies

Energy efficiency has, and will continue to be, the primary lever of decarbonization both for shipping and more broadly. Compared with business-as-usual operations, 15–25% of energy/tonne-mile has been saved over the last 20 years, and a further 20–30% will likely be saved by 2050. These efficiency gains are driven largely by economics: fuel efficiencies lead to savings for charterers and more attractive assets for owners, in some instances justifying extensive retrofitting (DNV, 2026a, Section 3.2). Efficiency measures can be linked to operations (e.g. slow-steaming), maintenance (e.g. hull cleaning), and technical solutions (e.g. air-lubrication systems).

Emission reductions flow not just from efficiencies, but from switching to low- or zero-carbon fuels and onboard carbon capture. This transition is primarily driven by regulation, which is needed to create a level playing field in a market with small margins. Compared with 100% reliance on oil, emissions were already 2% lower in 2025; we estimate that they will be 5% lower in 2030, and 26% lower in 2040. This latest forecast differs from our 2025 forecast when we estimated that emissions would be 36% lower in 2040 relative to fully oil-based propulsion (DNV, 2025). Instead, this year we forecast that decarbonization will be delayed by two to three years thanks to regulatory uncertainty.

Regulatory uncertainty

The International Maritime Organization (IMO) did not adopt its *Net-Zero Framework* (NZF) in October 2025. We find it more likely than not that a version of the NZF will be adopted eventually. Nevertheless, the delay in agreement leads to weaker demand signals for low-carbon fuels in shipping, and consequently, a delay or even cancellation of low-carbon fuel production projects. In the meantime, regional regulations such as *EU ETS* and *FuelEU* are in effect. The impact of different regulatory scenarios, from immediate adoption of NZF to no binding global agreement ever, are discussed in greater detail in DNV's *Maritime Forecast to 2050* (DNV, 2026a).

Shipowners are increasingly hedging against risk by prioritizing flexibility in their fuel choices. LNG and different biofuels rank highest amongst maritime options that will be most viable in the next decade (ICS, 2026). While the share of alternative fuel capable ships in gross tonnage has grown to 5%, excluding LNG carriers, most new orders are on LNG (DNV, 2026b). Biofuels can be used as drop-in fuels in existing ships (e.g. as bio-diesel or bio-LNG) but uptake will be limited by biomass availability. Hydrogen and its derivatives (methanol and ammonia) are now regarded less favourably than previously and have thus been adjusted down in our forecast.

Energy efficiency is, and will remain, the primary lever for decarbonization in shipping.

Aviation

Decarbonizing a growing sector is challenging. Aviation energy demand grows to 26 EJ/yr (13 Mbpd) by 2060, 77% higher than today (Figure 3.8). Despite a rising use of low-carbon SAF, oil will provide as much energy in 2060 as today. The transition will vary by region: Europe focuses on demand mandates, North America on locally-produced biofuel, and Greater China on industrial-scale SAF production for export.

Growing demand

Demand for air travel will grow to 1 billion passenger-trips by 2060, a doubling from today, but with clear regional differences. Demand will almost quadruple in Sub-Saharan Africa, the Middle East and North Africa, and South East Asia due to growing populations and prosperity. The demand will double to triple in Latin America, North East Eurasia, and Greater China, even without a growing population. The mature markets in Europe, North America, and OECD Pacific will, in contrast, see a limited growth of 42% over the same period.

Energy intensity per passenger-trip will decrease during our forecast period, but not sufficiently to counter the growth in travel demand. Historically, each new generation of aircraft has been 10-20% more efficient, thereby justifying the expense of new aircraft. However, the rate of improvements has slowed for the traditional tube-and-wing design, with only 7% better efficiencies on average for the aircraft fleet over the last decade (ATAG, 2025). In the meantime, a post-pandemic supply chain squeeze has slowed the delivery of new planes. As a result, the average fleet age is at an all-time high of close to 15 years (IATA, 2025).

Amidst the conflict in Iran, the fuel market first responded by increased production (particularly in North America) and drawing from inventories, which saw the prices drop temporarily (EIA, 2026; Natal 2026). As the conflict has extended longer than many first thought, jet fuel prices are back up to USD 185/bbl as we enter October, twice the average of last year (IATA, 2026a). However, past conflicts in the Middle East suggest the situation is unlikely to have a long-term effect on aviation demand, which remains at an all-time high and is more affected by supply constraints (new planes) than fuel constraints (IATA, 2026b).

Fossil fuel dominates

SAF, primarily bio-based now and renewable fuels of non-biological origin (RFNBO, e-SAF) later, are starting to enter the energy mix (Figure 3.8). However, SAF will not start to displace fossil oil before the 2040s. The scale-up in production is currently slow due to lack of binding offtake agreements between SAF producers and airlines. Currently, bio-SAF is three times and e-SAF is 10 times more expensive than fossil jet fuel (EASA, 2026). Strong policies supporting production or enforcing demand are necessary to drive the uptake of SAF, which we expect to remain more expensive than fossil jet fuel.

Regional regulations will be necessary to drive SAF uptake as the current global policy framework is rather ineffectual. The International Civil Aviation Organization (ICAO) has a long-term aspirational goal of net-zero carbon emissions by 2050. However, the *Carbon Offsetting and Reduction Scheme for International Aviation* (CORSIA) targets through 2035 are relatively weak. Airlines can meet requirements largely by acquiring low-cost Eligible Emission Units rather than fuel switching, and the post-2035 regulatory framework remains uncertain. In the meantime, stronger regional regulations such as *ReFuelEU* are driving regional SAF uptake.

Aviation energy demand growth is met by low-GHG fuels
Final energy demand in aviation by carrier (Billion bbl/yr)

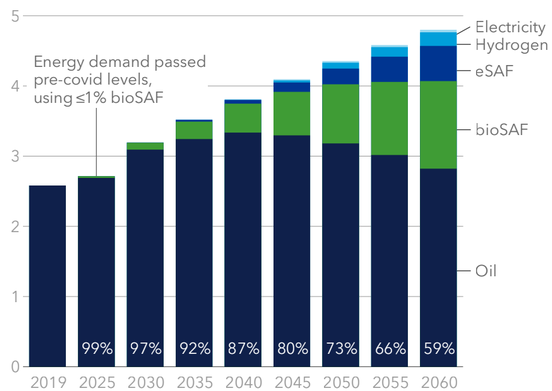


FIGURE 3.8 | Energy in units of barrels of Jet A-1 fuel, equal to 5.52 GJ. The percentages are fossil fuel shares. Historical data source: IEA WEB (2026)

Total SAF production is constrained by feedstock availability. The main SAF production pathway, hydro-processed esters and fatty acids (HEFA), is limited to feedstocks sufficient for 10% of aviation fuel demand. Many of the other approved pathways under *ASTM D7566* face similar limitations. Moreover, SAF is only one product of broader refinery output, meaning that production also depends on demand for these co-products. Even as more pathways are approved, there will not be enough sustainable biomass yield globally to meet all demand. That creates a long-term demand for the more expensive (but less feedstock-constrained) e-SAF.

Co-processing biogenic feedstocks in existing petroleum refineries may be the most viable pathway in the near-to-medium term to scale SAF. While typically limited to around 5% of refinery feedstock, the scale of existing refining infrastructure means that co-processing could substantially increase SAF volumes from today's very low levels. However, its long-term potential is inherently constrained. As demand for crude oil products declines during the energy transition, the available capacity for co-processing will also decrease, making it a transitional rather than permanent solution.

Regional diversity

Significant differences in aviation decarbonization strategies exist between the regions representing more than half of aviation's energy demand: Europe, North America, and China (Figure 3.9).

Europe has set the highest demand mandates for SAF, in part to become less dependent on import of fossil fuels. The 2% blending mandate for 2025 seems to have been met (Plucinska, 2026), but the 70% blending mandate for 2050, of which 35% should be e-SAF, is unlikely to be achieved. Notably, as other regions demand more SAF, the share of SAF available for export to Europe will decrease.

North America aims to primarily supply its own demand through bio-SAF by leveraging its large agricultural and forestry base (AFFO, 2026; C-SAF, 2023). Current production capacity matches 2% of regional aviation fuel demand. Their goals by 2030, of 11.4 billion litres in the US and a further 1 billion litres in Canada, equals 9% of North America's jet fuel demand. However, the recent downward adjustment of the *45Z Clean Fuel Production Credit* will limit the short-term scaling of SAF supply in the US.

Greater China has a multi-path build up for its SAF industry (Pei, 2026). Short-term production is primarily HEFA, based on used cooking oil, but there are plans to scale other bio- and e-SAF production pathways. Domestic demand is still voluntary, and much of the production is currently exported to Europe. Should a domestic blending mandate policy come into effect in Greater China, as we expect, local consumption could scale quickly by diverting a share of export volumes.

Large regional diversity in decarbonization of aviation

Share of oil in aviation energy demand by region (%)

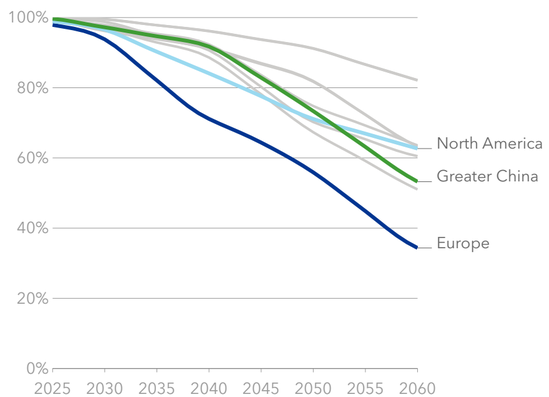


FIGURE 3.9 | Other regions shown in grey.

Rail

Rail will continue to make up 2% to 3% of transport final energy demand, growing slowly from 2.6 EJ/yr today to 3.1 EJ/yr in 2060. However, it carries a much larger share of transport activity, around 7% to 9%, because it is highly energy efficient (IEA, 2019).

Rail is the most electrified transport segment. 50% of rail energy demand is already electric and we expect this to reach 80% by 2060 (Figure 3.10). Much of this future electricity growth comes from new rail instead of electrifying existing diesel-driven railways in operation.

Shifting passenger and freight transport from road, air, and sea to rail can use less energy to transport the same volumes. Expanding urban rail can, in theory, replace passenger road vehicles and inner-city buses; freight trains can replace long-haul road transport; and high-speed rail (HSR) can replace short- and medium-haul flights. The key factors deciding whether these replacements happen are overall capacity and door-to-door travel time (ALG, 2024). With the current pace of building capacity, substantial modal shifts to HSR are mostly in Greater China.

The greatest expansion of rail capacity is currently happening in urban rail and HSR in Asia. In 2025, China opened 930 km of urban metro lines, 72% of global additions, bringing their total urban network length up to 12,500 km, five times as much as they had in 2013. It is also five times as much as the EU has combined (Freemark, 2026). China has also built out a 50,000 km HSR network since 2007 (Railway Pro, 2026). Other countries in Asia are also building out urban transport, including India, which will open 330 km of urban metro lines in 2026. Europe is similarly planning a significant upgrade to its current 12,000 km HSR network by 2040, both in speed and connectivity, in line with its *Trans-European Transport Network* (TEN-T) regulation (DGMT, 2025).

Rail electrification brings huge efficiency gains

Final energy demand (EJ/yr)

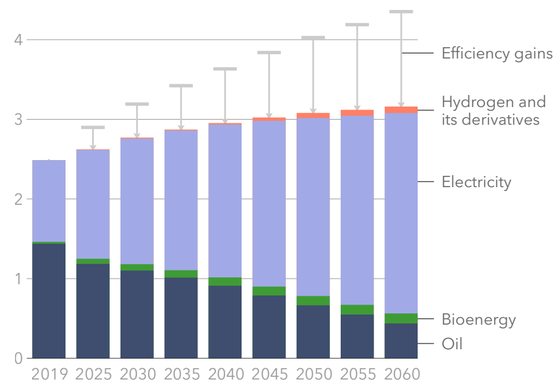


FIGURE 3.10 | Efficiency gains compared to the energy mix in 2019.

3.2 BUILDINGS

Final energy demand for buildings will rise by just over one-third between now and 2060. The main drivers are increasing population, expanding floor area, and rising GDP per capita. Efficiency gains from ongoing electrification will help moderate but not fully offset this growth which is also driven by new areas of demand, including space cooling needs due to climate change and data centre and AI development.

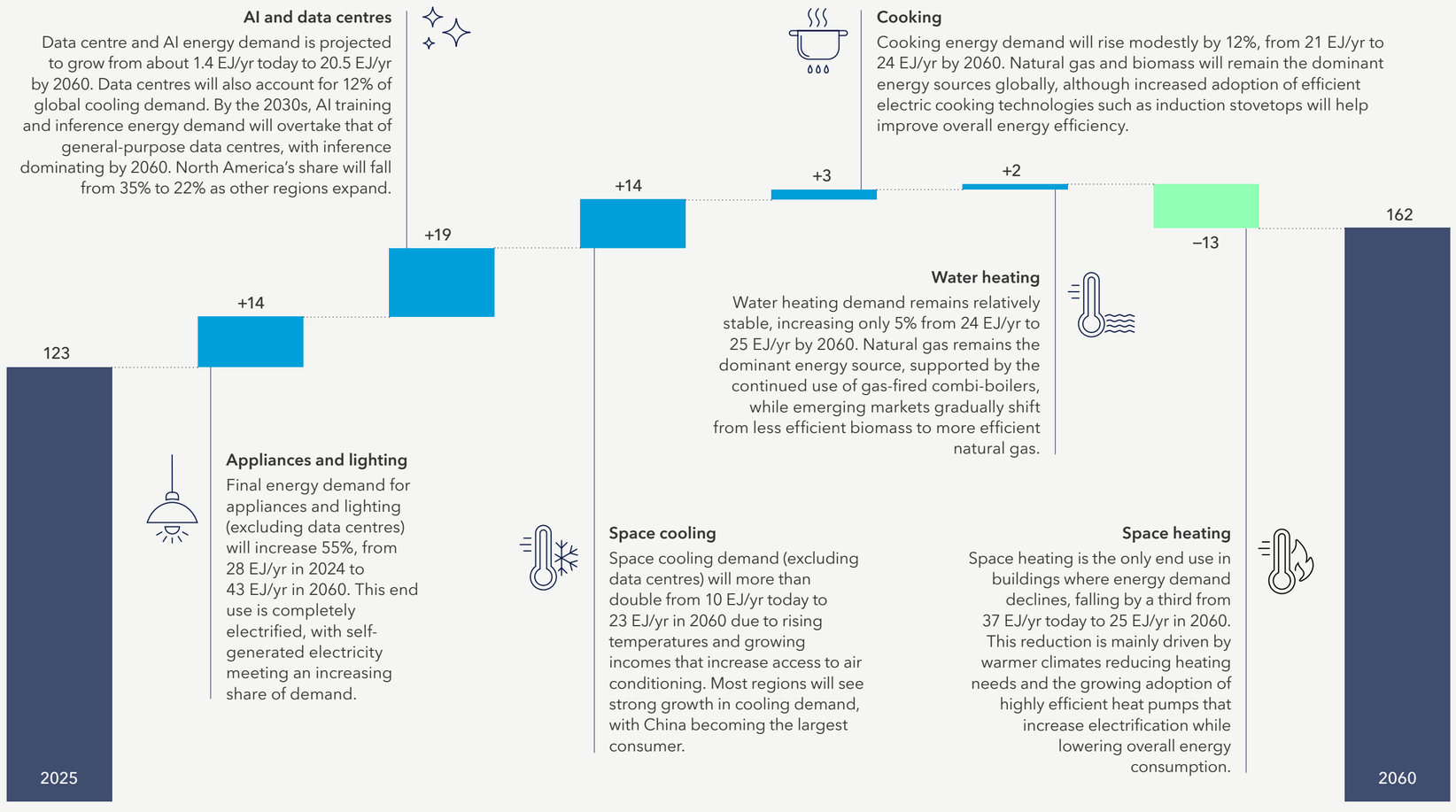
Supply side

- Electricity, including self-generated electricity, will grow to 62% of the buildings energy mix in 2060. That power will be much greener; by 2060, natural-gas-fired fired electricity will have reduced by 95%.

Demand side

- Final energy demand for space heating falls by 32% as fewer heating degree days, lower energy demand, and the uptake of heat pumps makes space heating significantly more efficient.
- However, space cooling demand (including data centres) will more than triple as rising per capita GDP in emerging economies makes cooling technology more affordable and cooling degree days increase with climate change. Demand is also driven by data centre cooling needs.

The drivers behind the increase of energy demand in buildings to 2060
Final energy demand by sector (EJ/yr)



Buildings energy demand

Of our five demand sectors, buildings stands alone as the only sector where final energy demand increases towards 2060. In fact, it grows by 32%, from around 123 EJ/yr today to 162 EJ/yr in 2060 to become our largest demand sector. It makes up 34% of the mix in 2060, just ahead of manufacturing. Considerable efficiencies brought by electrification will be insufficient to offset spiralling energy demand from space cooling and data centres. Natural gas and biomass will continue to play important but diminishing roles as energy carriers in the final energy demand mix, with their use dropping 19% and 21% respectively as electricity, both grid and self-generated, takes over.

A growing population, with rising GDP per capita and increased floor area needs, is driving an increase in many of this sector's end uses. This is, however, subject to regional variation: regions with high electrification and a stable population have stable or even slightly decreasing final energy demand. This is still the case even though high-income regions will host most of the energy demand growth from data centres, which rises from a little over 400 TWh/yr today to almost 6,000 TWh/yr in 2060. Beyond the cooling requirements of data centres, increasing cooling degree days and increasing wealth in emerging markets will lead to a much higher uptake of cooling technology and thus final energy demand.

Several factors will temper what would otherwise be runaway energy demand from data centres and cooling in emerging markets. Chief among these is increasing electrification, especially where it replaces much more inefficient biomass or natural gas. This is especially true in space heating, where heat pumps can be three times as efficient as other heating technologies, and switching to this technology brings significant efficiency gains. We also expect technologies in all building end uses to become incrementally more efficient towards 2060 due to the learning that comes with time and increased installations.

Today, grid-generated electricity makes up 37% of buildings final energy demand mix; by 2060 this will have risen to 53%. We see high growth in electricity in both developed and developing regions. Regions like North America, Greater China, and South East Asia will be powered by over 70% grid-generated electricity in 2060. Europe, OECD Pacific, Latin America, and the Indian Subcontinent are not far behind, at between 50% and 70%. Electricity generation is also greening in all sectors including buildings; the use of biomass to produce electricity is negligible and the role of gas declines 95% towards 2060.

Self-generated electricity will make up 9% of the global final energy demand mix in 2060. Combined with grid-generated electricity, this will push the total share up to 62% in 2060. Self-generated electricity is mostly used in appliances and lighting, and is most often from rooftop solar with or without storage. Regionally, its share will be highest in North America, at 16% of worldwide generation, with Latin America, the Indian Subcontinent, and Europe following behind at around 13% each. In regions that today do not have extensive grid build-out and high electricity uptake, like Sub-Saharan Africa, behind-the-meter generation plays an even more important role in the future, making up a third of the electricity generation in this region in 2060.

Several factors will temper what would otherwise be runaway energy demand from buildings, especially in emerging markets. Chief among these is increasing electrification, particularly where it replaces much more inefficient biomass or natural gas.

Buildings final energy demand is divided into residential buildings, commercial buildings, and data centres for five end uses.

Appliances and lighting

Final energy demand for appliances and lighting (excluding data centres) will increase 55%, from 28 EJ/yr in 2024 to 43 EJ/yr in 2060. This end use is completely electrified, with self-generated electricity making up over 20% of the power used by the end of our forecast period. Rising prosperity and increased electricity access worldwide will drive this increase, with more households investing in labour-saving appliances. There will also be some marginal efficiency gains from new technologies tempering the rise in demand.

Data centres

Final energy demand from data centres and AI will rise from around 400 TWh/yr today to 5,700 TWh/yr in 2060. There will also be an uptake in cooling demand from data centres, they will make up 12% of cooling demand worldwide in 2060. At present, general-purpose data centres dominate energy demand, but AI training and inference energy demand will surpass them by the 2030s, with inference going on to make up most of energy demand in 2060. Though North America claims around 35% of data centre energy demand today, this will have decreased to around 22% by 2060 as uptake in other regions becomes more significant. For more on the energy demand of AI and data centres, see the factbox at the end of this section.

Space heating

Space heating is the only buildings end use where final energy demand decreases towards 2060, falling 32% from the present 37 EJ/yr to 25 EJ/yr. Natural gas will still be the main energy carrier in 2060, making up 35% of the mix, but this will have decreased from today's 55% share. Direct heat will make up 30% of the mix, with electricity (including self-generated) having risen to 25%, up from 12% today. Although electricity represents just 25% of final energy demand, it will account for over half of the useful energy consumed in the space heating subsector.

There are two main growth factors. First, owing to climate change, there will be fewer heating degree days in the future, reducing the need for space heating in the first place. Secondly, much of the increase in electrification is from the installation of heat pumps, often replacing gas boilers. Heat pumps are the most efficient of all heating technologies, including traditional electric heaters, because they use electricity to transfer heat from one place to another rather than producing it directly. This means heat pumps today can have coefficients of performance (COP) of around 300%, and we expect that this COP will continue to rise towards 2060. Heat pumps will rise from 6% of the technology mix today to 18% in 2060, a significant increase which we see clearly reflected in declining energy demand in this end use.

Water heating

Final energy demand for water heating stays relatively stable, increasing only 5% towards 2060, from 24 EJ/yr today to 25 EJ/yr in 2060. This end use will still be dominated by natural gas, at 44% of the mix, with biomass the second largest energy carrier at around 24%.

Sub-Saharan Africa will have the largest energy demand for water heating, as it is mainly powered by less efficient biomass. We do see more-efficient natural gas replacing some biomass in some regions, such as the Indian Subcontinent and South East Asia. Combi-boilers, which provide both space and water

Heat pumps gradually replace gas boilers in most regions
Distribution of installed space heating technologies

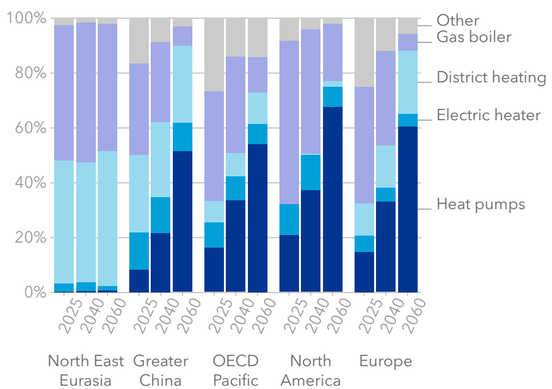


FIGURE 3.11 | Other includes e.g. biomass boilers and stoves, and oil boilers.

Increased efficiencies dampen the increase in energy demand
Buildings useful and final energy demand of various end uses (EJ/yr)

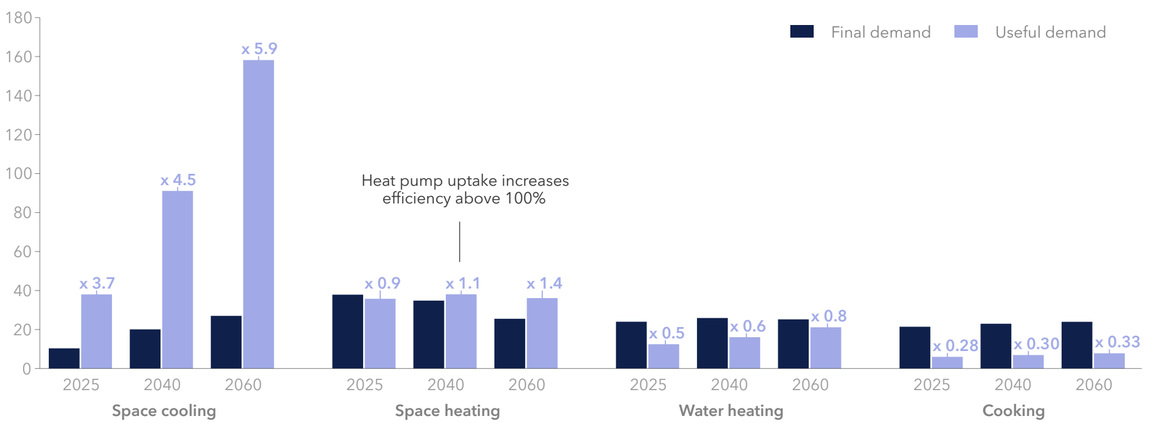


FIGURE 3.12 | Final demand is energy consumed; useful demand is energy service delivered. Their ratio reflects end-use efficiency.

heating, and are natural gas powered, will still be common in 2060 and are one of the reasons natural gas remains important in both the water heating and space heating energy mix. There will also be some water heating provided by district heating and air-to-water heat pumps.

Space cooling

Final energy demand for space cooling (excluding data centres) more than doubles from 10 EJ/yr today to 23 EJ/yr in 2060. This is due to many more cooling degree days – nearly doubling in regions like Europe and North East Eurasia – and increasing GDP per capita leading to increased investment in cooling technologies. North America and OECD Pacific, with high GDP per capita and an already large amount of cooling technology installed, will see relatively stable cooling energy demand. Notably, that is not because cooling needs remain stable but because replacement equipment, including heat pumps, will be more efficient, and building codes will drive better insulation, shading, and glazing. All other regions in the world will see steep increases in demand as their populations take advantage of more affordable and efficient air conditioning. China will demand the most energy for space cooling in 2060, at 28% of world demand.

Cooking

Cooking will grow 12%, from 21 EJ/yr today to 24 EJ/yr in 2060. Worldwide, the mix will still be dominated by natural gas and biomass, at 45% and 37% respectively, with electricity only making up 12%, though there are significant regional variations. Electric technologies, mainly induction stovetops, are much more efficient than gas or biomass cooking, so even a small uptick in electricity use implies a much larger uptake of electric appliances. This can have a significant impact on efficiency and final energy demand.

Electricity dominates the growth in demand
Buildings energy demand by carrier (EJ/yr)

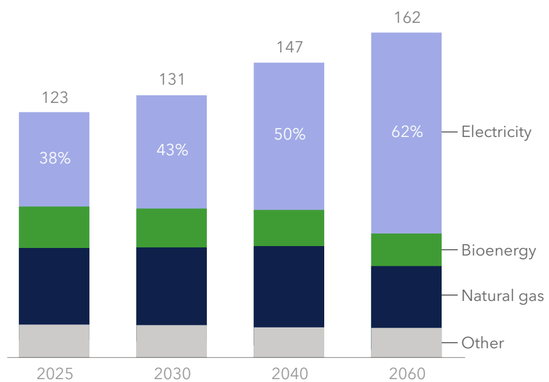


FIGURE 3.13 | Other includes oil, coal, and solar thermal.

Final energy demand from data centres and AI will rise from around 400 TWh/yr today to 5,700 TWh/yr in 2060.

The politics of cooling

Heatwaves are intensifying with global warming and making access to cooling an increasingly political issue around the world. Recent protests in both high-income and low-income countries have highlighted growing public frustration over the affordability and availability of cooling during periods of extreme heat (Economist, 2026a). International initiatives such as the *Global Cooling Pledge*, now signed by more than 70 countries, have helped elevate cooling on the global policy agenda. Yet a fundamental challenge remains: the cheapest AC units are invariably the least efficient and hundreds of millions of vulnerable households still lack access to affordable and efficient cooling. As temperatures rise, cooling is becoming not only a public health and energy issue, but also a question of social stability and resilience (UNDP, 2025).

Data centres and AI

Data centre and AI electricity demand grows fourteen-fold to 2060

We project global data centre electricity demand to increase to 1,100 TWh in 2030, 2,700 TWh in 2040, and 5,700 TWh in 2060, a fourteen-fold increase from today. As a result, data centres' share of global electricity consumption increases from 1.2% today to 2.7% in 2030, and 6.8% in 2060. Over the same period, we expect IT power capacity to rise from 82 GW to 860 GW. AI workloads exceed all conventional data centre use from 2031, with inference rather than training accounting for the majority of AI demand, while the regional distribution of demand shifts away from North America throughout the period. Growth is currently paced by access to firm power rather than by chip supply or capital availability.

AI and data centre electricity demand will grow 14-fold
Data centre energy demand (TWh/yr)

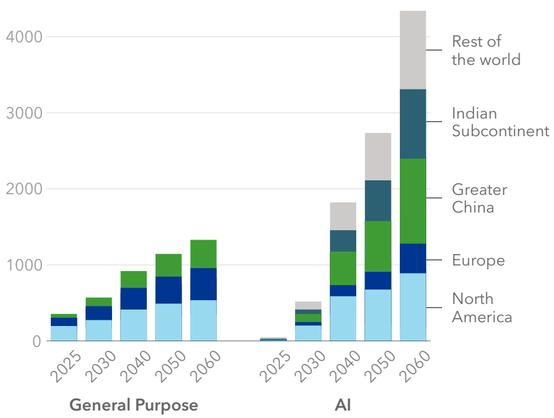


FIGURE 3.14 |

Data centres consumed approximately 400 TWh of electricity in 2025, or 1.2% of global electricity supply. Capital expenditure by the world's five largest technology companies exceeded USD 400bn in 2025 and is expected to rise by a further 75% in 2026, to approximately USD 700bn (IEA, 2026f). For comparison, global upstream oil and gas investment was USD 546bn in the same year (IEA, 2026g).

The investment focus is increasingly on physical assets, like semiconductors, buildings, cooling systems, and grid connections – and these are emerging as constraints on AI deployment, even as AI software advances at remarkable speed.

The AI energy demand forecast is among the most uncertain in our Outlook. Efficiency gains, rising use, and growing computation per task pull demand in opposing directions, and uncertainty grows over the forecast period. New computing architectures or algorithms could change how much energy AI needs to deliver a given outcome, a shift our assumption ranges cannot capture.

AI workloads exceed all conventional data centre use from 2031, with inference accounting for the majority of AI demand.

Demand more than doubles by 2030

We forecast data centre electricity demand growing 170% between 2025 and 2030 and a further 150% to 2040, equivalent to 22% annually over the first five years and easing to around 10% annually from 2030 to 2040. Capacity follows a comparable trajectory, from 82 GW of IT power in 2025 to 202 GW in 2030 and 440 GW in 2040. Two-thirds of compound growth occurs before 2040, and growth in the 2050s is distributed across a wider set of regions rather than accelerating in aggregate. Our estimate of installed capacity is broadly aligned with outlooks from McKinsey and JLL and lies toward the upper end of the power consumption estimates given by those organizations (Srivathsan et al., 2024; JLL, 2026).

Our forecast indicates a gradual rise from an estimated 42% load factor today towards 50% in a decade. While Goldman Sachs applies 70% explicitly in its US analysis (Wei et al., 2026) and IEA (2026f) make projections on the basis of a 50–70% load factor, the current fleet utilization is likely lower because newly energized facilities take several years to reach full IT utilization and therefore consume substantially less electricity than their nameplate capacities during the ramp-up period (IEA, 2026f). Grid Strategies finds that some utility load forecasts assume 95% to 100% load factors, which facilities cannot sustain once maintenance downtime is allowed for. On the same reasoning, it puts the 90 GW of additional US data centre load implicit in those forecasts at 65 GW (DiGangi, 2025). We foresee fleet-average load factors increasing over time as newly commissioned facilities complete their multi-year occupancy and equipment ramp-up. In addition, hyperscale and AI workloads account for a larger share of capacity with more sustained utilization, and tighter power availability and grid-connection constraints reduce excess capacity reservations and incentivize fuller use of connected infrastructure.

Inference is the driver and AI overtakes conventional workloads in 2031

We forecast AI workloads exceeding all other data centre use in 2031, from 12% of the sector in 2025. By 2040, AI accounts for 66% of data centre energy demand and 77% by 2060. Inference rather than training carries that growth, supplying half of AI energy demand in 2030 and 84% in 2060.

Inference demand continues to grow after training energy has peaked because more types of work become addressable by AI over time. In the near term, AI demand is driven mainly by cognitive work (such as analysis, synthesis, customer interaction and creative output) where deployed capability sits today. Later, demand will also come from embodied AI, where models see, decide, and act through robots and machines like industrial robots or autonomous vehicles. Embodied AI is likely to be much more token-intensive than transactional processing. As a result, embodied work accounts for around half of eventual token demand on its own in our model, while cognitive work in services accounts for slightly less than a third. Because embodied applications will take time to evolve, much of the associated demand enters the forecast late. This timing explains

How we forecast AI inference demand

Our inference forecast is built from the economic work AI can perform, not from chip supply or announced capacity. The chain has four steps (and a caveat).

The total addressable market. For each region we take value added by sub-sector, isolate the share attributable to human labour, and identify the portion of that labour AI can augment or replace. We term this the 'AI surface area'. It is a ceiling on the addressable market rather than a forecast.

The conversion. Addressable labour hours are allocated across eight work archetypes, classes of work that share a common token intensity. Tokens, the units of information an AI model processes, convert to compute through floating-point operations per token, compute to installed power capacity through chip efficiency, and capacity to energy through utilization and power usage effectiveness.

Rate of AI adoption. Uptake is set by model capability and by price rather than by the ceiling. Each archetype becomes addressable only when models can perform that class of work at acceptable quality and cost, after which diffusion follows an S-curve filtered by what users will pay, applied against a cumulative income distribution by region.

The feedback. Revenue from inference funds training investment, which can either improve model capability, which raises willingness to pay, or reduce the cost of the existing models through refinement. Falling cost per unit of computation feeds back into price with a five-year expectation lag. The result is a reinforcing loop on the demand side balanced by a construction and cost loop on the supply side.

Caveat. Our assumptions broadly assume the economic space for AI is driven by the value of work currently undertaken by humans. History shows that general-purpose technologies create substantial value beyond mere substitution through entirely new products, services and patterns of consumption. The labour-based AI surface area may thus be a useful estimate of substitution demand but is unlikely to represent the true upper bound on the long-run addressable market. We nevertheless use labour replacement and augmentation as a practical first order approximation of the size of the AI market, and the size of additional value creation from AI is linked to it.

Our inference forecast is built from the economic work AI can perform, not from chip supply or announced capacity.

why inference demand keeps increasing even after training energy has peaked.

Cognitive work is addressable now, but the evidence on how far it has been taken up is mixed. Frontier model capability is advancing quickly, with the length and complexity of tasks they can complete doubling approximately every seven months over the last six years (Kwa et al., 2025). Evidence on productivity so far remains mixed as well. In a randomized trial developers working on familiar codebases found that early-2025 AI coding tools increased task completion times despite participants expecting the opposite, highlighting the gap that can exist between technical capability and economically valuable deployment (Becker et al., 2025). Use also remains concentrated in a few information-intensive occupations, with computer and mathematical roles at approximately 30% of surveyed users against 4% of US employment (Massenkoff et al., 2026). In the near term, electricity demand reflects the expansion and utilization of computing infrastructure. Over the longer term, however, demand will depend on whether the economic value generated by AI exceeds the cost of its use.

Training and general-purpose computing still grow, but they become less important in relative terms. In our forecast, training energy peaks in the late 2030s as efficiency improves and frontier-scale training becomes more consolidated. Training loads do not disappear, but most of the training focus shifts to refinement of models through fine-tuning, distillation, domain adaptation, and iteration rather than frontier-moving training, which is further constrained by availability of novel and variable data. The largest uncertainty around training is its local peak-power requirement, not its contribution to total energy demand. General-purpose computing also grows, from 356 TWh in 2025 to 1,300 TWh in 2060, but its share of data-centre demand falls from 88% to 23% as AI workloads expand.

Growth in AI demand is limited by capability and price rather than by the size of the addressable market. We forecast that AI will address 0.5% of world GDP in 2025, 15% in 2040, and 25% in 2060, with implied service revenue rising from approximately USD 80bn to USD 1,200bn. The remaining upside comes from applications that consume substantially more compute per unit of value delivered, including always-on agents, video generation, and large-scale simulations. These become viable only as capability improves and cost per token falls, so they contribute less before 2040 and dominate the growth in energy demand thereafter across both commercial and consumer use.

Grid remains dominant, but hybrid supply grows where grid constraints bind
Share of new data-centre power capacity additions by supply architecture

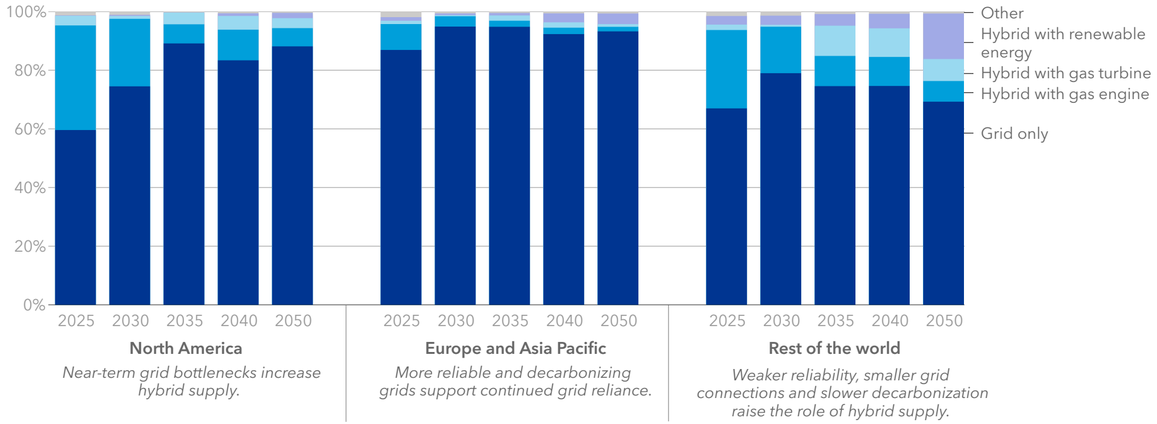


FIGURE 3.15 |

The largest uncertainty around training is its local peak power requirement, not its contribution to total energy demand.

There is a geographical drift from North America to the rest of the world

North America accounts for 35% of global data centre electricity demand in 2025. This falls to 31% by 2040 and 22% by 2060, with China becoming the largest single market. North America's IT capacity plateaus at approximately 160 GW between 2038 and 2050 before resuming growth to 190 GW in 2060, while China reaches 220 GW and the Indian Subcontinent 150 GW. China passes North America on energy demand in 2053 and on installed capacity in 2054. In absolute terms, North American demand still more than quintuples from 140 TWh to 1,200 TWh. Its share of incremental global capacity additions declines throughout the period.

The Indian Subcontinent represents the sharpest growth in the forecast, rising from 1.5% of global data centre demand in 2025 to 13% in 2040 and 18% in 2060 (1,000 TWh), second only to China. South East Asia reaches 7%, Latin America 5%, and the Middle East and North Africa 4%. Europe's share falls from 18% (580 TWh) in 2025 to a stable 10% (72 TWh) from 2040 onwards, on a substantially larger base. This shift is consistent with current planning locations; OECD Pacific capacity will likely grow from 32 GW to 57 GW by 2030, while North America holds approximately half of the global total (JLL, 2026). The redistribution reflects both changing demand geography and resource availability. AI capacity moves closer to users and industrial activities, particularly in Asia. Whether this capacity can be developed depends on locations offering suitable land, timely grid interconnection, and reliable power

supply. North America accounts for around 40% of global AI energy demand over the next decade, with China emerging as the second major growth centre. However, to 2040, cooling in buildings and EV charging contribute more to global electricity demand growth than data centres, including AI.

The regional impact differs markedly when measured against the size and growth of each power system. North America faces the greatest pressure, with data centres rising from 2.6% of electricity demand in 2025 to 11.4% in 2040 and absorbing around one-third of generation growth. This helps explain the emerging regulatory and political pressure and is broadly consistent with the IEA's estimate that data centres account for approximately half of US demand growth to 2030 (IEA, 2026e).

China presents the opposite pattern. The growth is almost as strong as in North America, but the sector remains below 5.5% of national electricity demand and accounts for only 8% of generation growth to 2040. Europe lies between North America and China, although national concentration can be much higher, as illustrated by Ireland, where data centres accounted for 23% of metered electricity consumption in 2025 (Curran, 2026).

Data centres remain primarily grid-powered, but power supply strategies will diversify

We forecast that grid-only supply remains the largest source of new data-centre power capacity, accounting for about 80% of global additions today and 85% by 2035. This is despite growing grid bottlenecks, connection delays, and local opposition in major data-centre markets because grid connection allows data-centre operators to purchase electricity as a service and avoid the cost and complexity of developing and operating dedicated power generation. This advantage is strongest for AI inference and general-purpose data centres where proximity to demand and high reliability remain difficult to compromise. The clearest near-term example of grid bottlenecks affecting the mix is North America: grid-only supply is relatively lower in 2025, at around 60% of additions, before rising again in 2030s as connection pressure eases.

The shift away from grid-only supply is not a simple move 'off grid'. It happens where one of three constraints becomes binding: grid access is too slow or contested, grid reliability is weak enough to make resilience costly, or companies need lower-carbon power faster than the grid can provide it. On-site assets have different roles in this context. UPS systems, batteries, and backup generators are resilience assets used to bridge interruptions, while behind-the-meter generation becomes part of the main supply only when it reduces grid-connection size, accelerates energization, improves reliability, or helps meet firm low-carbon power needs. Market evidence points in the same direction (Fischer, 2026). US grid interconnection timelines of three to seven years are common in many markets, compared with 18 to 24 months to build a data centre. This explains why behind-the-meter power is becoming a speed-to-power strategy rather than a universal replacement for grid supply.

Different data-centre workloads have distinct power requirements and flexibility potential. AI training is large and power-intensive, but it is also the workload with the greatest ability to move in time or space: jobs can often be queued, checkpointed, delayed, or shifted to locations where power is available. AI inference is user-facing, latency-sensitive, and reliability-critical, so it cannot respond to grid constraints mainly by moving far away or shifting large volumes of load over time. General-purpose data centres sit between these two cases. This means flexible training loads can make better use of remote grid connections, hybrid behind-the-meter supply, or flexibility agreements, while inference and many general-purpose loads still favour reliable grid-connected sites close to demand.

Alternative supply architectures become attractive when grid power cannot deliver speed, firmness, and carbon performance at the same time. Gas engines and turbines are the bluntest response to the power bottleneck; they can bring firm power to large AI campuses faster than waiting for grid reinforcement, but they shift fuel, emissions, permitting, and operating risk onto the developer. Notably, gas-fired generation invariably requires access to piped gas, which effectively moves the constraint from an electricity network to a gas network. Chevron's dedicated gas-power agreement for Microsoft in West Texas (Chevron, 2026) shows why this route is attractive when speed and firmness matter, while tightening turbine delivery slots show that gas generation is itself becoming a constrained supply chain. Renewables plus batteries offer a cleaner answer, but they still need land, overbuild, storage, and grid. Pantheon AI's planned Croatia campus (which combines on-site solar, large-scale batteries and grid infrastructure) shows how power supply is increasingly being designed into the data-centre site rather than procured separately (Lago, 2026). Nuclear-backed supply can provide firm, low-carbon electricity, but long development times make it a strategic procurement option, not a near-term workaround.

Societal pushback

Data centres, specifically those built in service of AI, are garnering more attention as buildout occurs rapidly. Residents and communities around the world, and not just the US, have raised their concerns about data centre impacts. These include rising electricity bills, noise, light, air pollution, and overwhelming use of local resources like energy, water, and land. Opposition has been especially strong in instances where data centres have built on-site gas-fired power generation, bypassing grid connection to avoid lengthy wait times. Polling in the US shows the majority opposes constructing AI data centres in their locality, with over half of respondents citing effect on resources and environmental impacts (including energy consumption) as a main reason for opposition (Jones, 2026).

Pushback against data centres is also heavily linked to the high-level uncertainty surrounding AI: the perceived distribution of burdens and risk (in this case, living with a data centre and all its impacts on your community) versus the benefits, which are viewed as increasingly inequitable. Job security has been a specific concern that has incentivized pushback; nearly one in five workers in the US expect AI or automation to eliminate their job in the coming years (Economist, 2026b). In Seattle, opposition to AI data centres has arisen from within the tech sector itself due to concerns over AI-driven job losses in the wake of Amazon and Microsoft laying off thousands of workers whilst spending hundreds of billions on AI investments (Willems, 2026).

Community opposition can result in project delays, cancellations, and construction bans. In July 2026, New York became the first US state to enact a data centre moratorium, that pauses state environmental permits for new data centres for up to one year to allow time for energy, environmental, and community impact assessments. Texas followed in August 2026, pausing approvals for new data centres until existing grid connection requests from data centres could be audited to understand electricity and water consumption patterns and make plans for mitigating impact on local communities. Several other state legislatures have considered moratoriums and over 100 towns and cities have implemented them, the largest being Seattle in June 2026. If the AI data centre expansion is to continue at pace, developers and local governments must consider how their buildout and governance will impact communities.

Does the AI energy demand accelerate or slow down the energy transition?

AI energy demand both slows and accelerates the energy transition, depending on the timeframe. In the near term, additional electricity demand from AI data centres increases reliance on fossil-fired generation, raises electricity prices, and slows emissions reductions. Over the longer term, however, it stimulates investment in low-carbon generation, accelerates cost reductions for emerging clean technologies, and supports the expansion of both renewables and nuclear power. Overall, we estimate that AI-related electricity consumption delays global power-sector decarbonization by approximately one year by 2060.

To quantify these effects, we compare a scenario with AI-driven electricity demand against a counterfactual scenario without it. This allows us to move beyond the direct impact of additional load and capture the broader feedbacks that influence generation, capacity investments, technology costs, and electricity prices. The analysis focuses exclusively on the direct effects of AI-related electricity demand and excludes any wider impacts of AI-enabled technologies on the planning, operation, or optimization of power systems (or oil and gas exploration and production systems for that matter).

The primary effect of AI is higher electricity consumption. Without data centres serving AI workloads, global electricity demand would be 4.0%

lower in 2040, 4.8% lower in 2050, and 7.1% lower in 2060. The effect is even more pronounced in North America, where electricity demand would be around 10% lower.

AI demand increases average electricity consumption more than peak demand. Global peak load would be 3.1% lower in 2040, 3.8% lower in 2050, and 5.7% lower in 2060 without AI-related data centres. This reflects the highly stable consumption profile of data centres, which operate continuously throughout the year. As a result, data centres reduce the ratio between peak and average demand and partially offset the growing seasonality of electricity consumption driven by electrified heating and increased air-conditioning use. Alongside industrial demand, data centres effectively act as a source of baseload consumption.

This demand profile has important implications for the generation mix. While both fossil and renewable generation would be lower in the absence of AI demand, the effects are not evenly distributed. By 2040, in the absence of AI demand the largest reduction occurs in fossil-fired generation because AI demand extends the utilization and economic viability of conventional thermal power plants. In other words, AI demand slows the pace of power sector decarbonization in the near term.

Over time, however, the effect shifts. From around 2050 onwards, the impact on renewable generation becomes larger than the impact on fossil generation, with wind power showing the strongest response. This suggests that higher electricity demand from AI creates stronger incentives to invest in renewable capacity, particularly offshore wind, onshore wind, and solar PV. Additional deployment also accelerates learning effects and cost reductions. Without AI-driven demand, floating offshore wind costs would be around 6% higher in 2060, while fixed-bottom offshore wind and onshore wind costs would each be approximately 2% higher.

We see the strongest long-term impact in nuclear power. By 2060, global nuclear capacity would be 12% lower without AI-related electricity demand, with the largest effects occurring in North America, Europe, the Indian Subcontinent, and South East Asia. This result reflects the convergence of two trends: declining costs of small modular reactor technologies during the 2050s and the strong alignment between nuclear generation and the stable electricity demand profile of data centres.

AI demand also impacts electricity prices, particularly in the near term. Rapid growth in data-centre electricity consumption creates supply-demand imbalances and requires additional investment in generation and network infrastructure. These costs are ultimately passed on to consumers. In North America, residential electricity prices in 2035 would be 8.4% lower in the absence of AI-related data-centre demand. Note that this result is based on a region-wide power market model without any explicit geographical representation of supply, demand, and the grid. The effect is smaller in Europe (2.1%) and Greater China (1.3%), where data centres account for a smaller share of demand growth and supply constraints are less pronounced.

The emissions impact follows a similar pattern. The largest difference occurs in the near term, with global power-sector emissions around 2% higher in 2040 due to increased fossil-fired generation. Cumulatively, AI-related electricity consumption adds 12.6 $GtCO_2$ between now and 2060, equivalent to roughly one year of global emissions in the late 2050s. As a result, AI-related energy demand delays global power-sector decarbonization by approximately one year.

Taken together, these findings suggest that AI demand does not fundamentally alter the direction of the energy transition, but it does mildly affect its pace. In the near term, AI increases emissions, prices, and dependence on fossil generation. In the longer term, it strengthens the economic case for investment in renewables and nuclear power and contributes to lower costs for key clean-energy technologies. The net effect is therefore best characterized as a modest delay in decarbonization rather than a reversal of the transition.

Global electricity supply by source with and without AI energy demand
Electricity supply (PWh/yr)

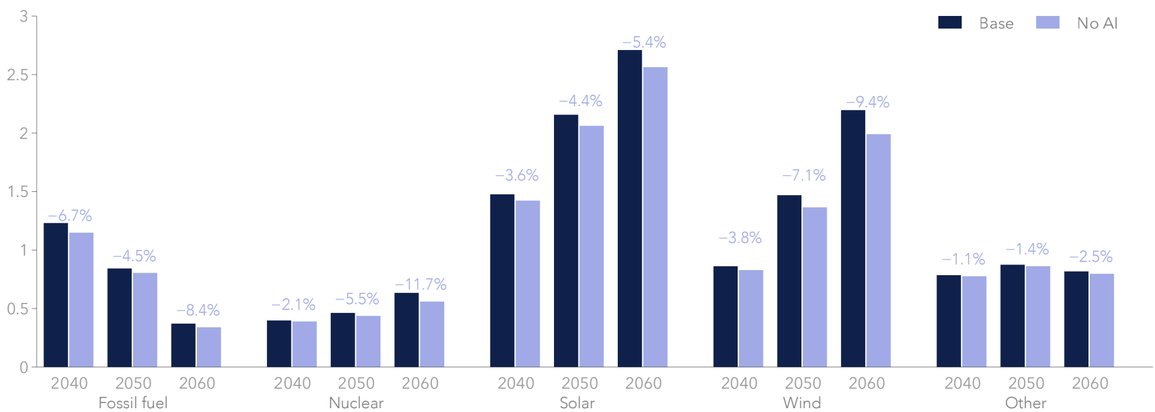


FIGURE 3.16 | 'Base' represents the ETO most-likely future.

By 2060, global nuclear capacity would be 12% lower without AI-related electricity demand.

3.3. MANUFACTURING

Manufacturing decarbonization is progressing gradually. Investment decisions are likely to determine the pace of change more than technology availability.

Existing assets slow the transition
We expect many industrial facilities operating today to remain in service for decades, making retrofits more common than replacement. Emissions reductions in the medium term therefore depend largely on efficiency improvements, electrification, fuel switching, and, to a lesser extent, carbon capture leading to a gradual decline in fossil fuel use.

Heavy industry has no preferred decarbonization pathway
Technology requirements differ by sector. Cement relies heavily on CCS to address process emissions, chemicals require changes to feedstocks as well as fuels, and steel combines recycling, electrification, hydrogen, and CCS. This dilutes investment and increases complexity and uncertainty. High costs, infrastructure needs, and policy inconsistency also slow decision processes.

Electricity is becoming decisive for competitiveness
As electrification expands, access to reliable, low-cost power increasingly influences investment and manufacturing location decisions. This favours regions with abundant affordable electricity.

Electricity will soon become the main energy carrier for the manufacturing sector, while coal will slowly lose ground

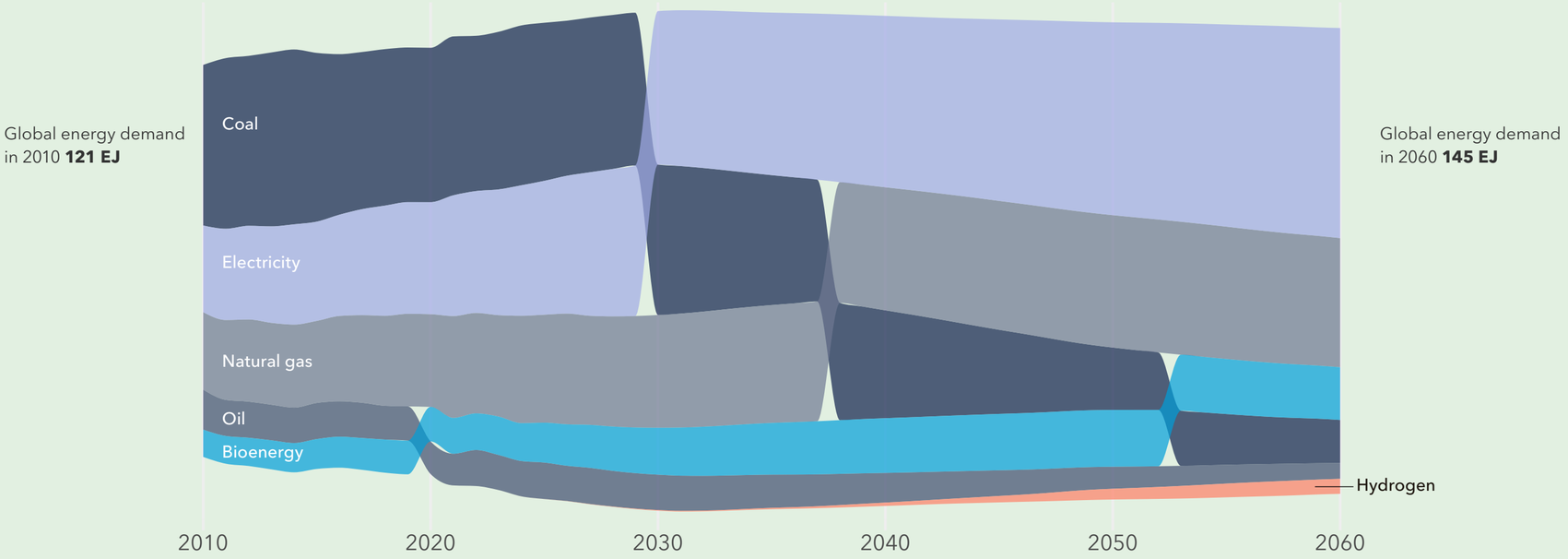


FIGURE 3.17 | Small amounts of geothermal and solar thermal not shown.

Coal declines only gradually because long-lived assets continue to shape investment decisions.

Electricity grows rapidly as electrification becomes the dominant near-term decarbonization pathway.

Natural gas provides a lower-carbon bridge between existing industrial assets and emerging technologies.

Oil steadily loses market share as industrial processes adopt more efficient and lower-carbon energy sources.

Hydrogen scales after the mid-2030s as costs fall and ageing facilities reliant on legacy energy sources reach replacement age.

Bioenergy complements electrification by providing renewable heat where direct electrification is difficult.

CCS remains essential for addressing process emissions that cannot be avoided through fuel switching alone.

Manufacturing energy demand

Manufacturing is on a slow decarbonization trajectory to 2060. Three structural constraints govern its pace: existing assets determine how quickly the sector can change, no single technology fits the full range of manufacturing processes, and access to cheap and low-carbon electricity is becoming a determinant of competitiveness.

Existing assets determine the pace of change

Most industrial capacity that will still be operating in 2040 has already been built. Heavy industrial plants are designed to operate for several decades, with capital costs recovered over long investment cycles. Owners rarely retire them early unless regulation, energy prices, or market conditions make continued operation unattractive.

Much of today's capacity dates from the rapid expansion of global manufacturing in the 2000s and 2010s. Around 40% of global steel capacity and global cement capacity is under 20 years old. These assets retain sufficient economic value to make continued operation preferable to replacement in most cases. Industrial decarbonization therefore depends as much on the economics of retrofitting or replacing legacy technologies as on the availability of cleaner technology.

Fossil fuels are more difficult to displace in industry than elsewhere because they often serve functions beyond energy supply. In steelmaking, coal acts as both fuel and reducing agent, while in cement most emissions arise from calcination, the chemical process that converts limestone into clinker. As coal demand declines more rapidly in other sectors, industry accounts for an increasing share of the remaining market, reaching nearly half of global coal demand by 2050.

Coal on the decline, electricity on the rise in manufacturing

Evolution of energy demand by carrier (EJ/yr)

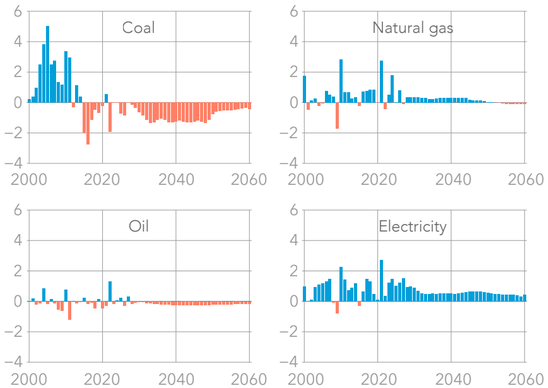


FIGURE 3.18 | Blue: new demand; Orange: decline in demand.

Retrofitting existing facilities is consequently the dominant route to emissions reductions over the near and medium terms. Efficiency improvements, electrification, fuel substitution, and carbon capture can reduce emissions, but retrofits rarely transform a production system as completely as replacement. Industrial decarbonization therefore proceeds gradually rather than through disruptive change.

Heavy industry lacks a preferred transition pathway

No single low-carbon technology serves heavy industry's full range of processes because those processes differ substantially. Different products require different combinations of electrification, hydrogen, bioenergy, recycling, material efficiency, and carbon capture. This diversity increases investment uncertainty because no single technology cost trajectory can anchor capital planning across the sector.

Cement and chemicals illustrate the challenge. Roughly two-thirds of cement-sector emissions arise from calcination rather than combustion, making carbon capture and supporting CO_2 transport and storage infrastructure important for deep decarbonization. In chemicals, fossil fuels serve as feedstock as well as fuel, requiring broader supply-chain and product-specification changes rather than a simple fuel switch. Cost, scale, and logistics constrain these alternatives, leaving multiple pathways progressing at different rates.

We also expect Hydrogen and CCS will slow the transition timeline. Manufacturing will likely account for more than 80% of clean hydrogen demand through 2040, but high green hydrogen costs will likely limit adoption until the mid-2030s. Hydrogen therefore plays a substantial role primarily in the longer term.

CCS is often presented as the most direct retrofit option because it can partly remove emissions at an existing plant without changing feedstock or process chemistry. Yet uptake has lagged this apparent fit. A CCS retrofit preserves the existing production asset, but only by committing it to an entirely new, long-lived CO_2 transport and storage network. Capture, compression, transport, and storage add cost without creating saleable output, limiting opportunities for cost reduction. Combined with insufficient carbon prices in most regions, this limits deployment. In our forecast, CCS captures less than 10% of remaining manufacturing CO_2 emissions (500 Mt/yr) by 2060.

Fossil fuels are more difficult to displace in industry than elsewhere because they often serve functions beyond energy supply.

We expect more than a third of CCS uptake to be in Europe, where high carbon and energy prices improve project economics. Regions with decarbonization goals but access to cheaper energy, such as Greater China, will likely decarbonize primarily through other routes.

The combined effect is prolonged investment uncertainty. Industrial companies must commit capital to assets that may operate for decades without clarity on future technology costs, carbon prices, or infrastructure availability. This uncertainty delays investment decisions and slows the pace of transition.

Electricity becomes decisive for competitiveness

Electrification lowers industrial emissions and improves energy efficiency. It also increases exposure to regional electricity markets, making electricity cost an increasingly direct determinant of industrial competitiveness. We project industrial electricity demand will rise by 25% over the next decade, equivalent to an additional 3,000 TWh.

Unlike internationally traded fuels such as oil, coal, or gas, electricity is supplied largely through regional markets shaped by local generation capacity, grid infrastructure, and permitting. Access to affordable, reliable low-carbon electricity is therefore becoming a more prominent factor in industrial investment decisions.

This is producing measurable regional divergence. Greater China and the Indian Subcontinent continue to benefit from relatively low-cost, state-supported electricity for industrial consumers. Together, we expect them to add four times more industrial electricity demand than Europe and North America combined over the next decade to reinforce their position as manufacturing centres.

North America faces a different challenge. Data centre electricity demand will likely rise by almost 700 TWh over the next decade, compared with approximately 15 TWh for manufacturing. Industrial users may increasingly compete with data centres for low-carbon electricity.

In Europe, industrial policy seeks to preserve domestic manufacturing while pursuing decarbonization. The EU *Carbon Border Adjustment Mechanism* aims to reduce carbon leakage and raise the cost of high-carbon imports. However, carbon policy cannot fully offset higher energy costs where access to competitive low-carbon electricity remains limited.

Electricity price differentials affect competitiveness even where electricity represents a modest share of total energy demand. In cement production, electricity accounts for around one-sixth of total energy demand, yet a regional power-price difference of USD 100/MWh can raise total production costs by approximately 20%.

Industrial decarbonization therefore depends not only on technologies deployed inside the factory gate, but also on the performance of regional energy systems, including electricity supply, grid expansion, hydrogen infrastructure, and CO_2 transport and storage networks.

Steel illustrates the transition challenge

Asset lifetimes, technology choice, and electricity competitiveness interact in steel production. We project global steel demand will rise from 1.9 billion tonnes today to just over 2 billion tonnes by 2060, while direct emissions fall from 2.8 to 1.1 $GtCO_2$/yr.

Every region will have a different fuel mix, depending on local resources and industrial footprint

Manufacturing energy demand in selected regions (EJ/yr)

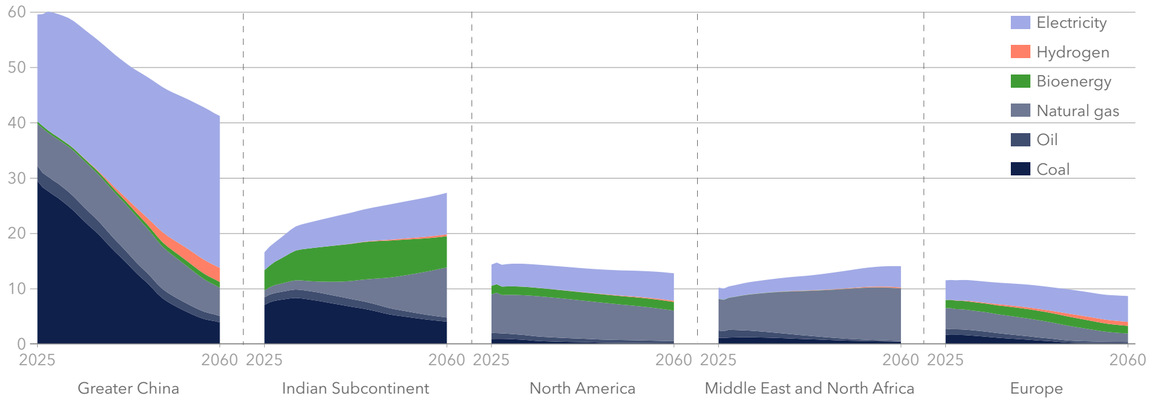


FIGURE 3.19 | Small amounts of geothermal and solar thermal not shown.

Manufacturing's varying electricity regional priorities

Share in additional electricity demand, 2025–2035

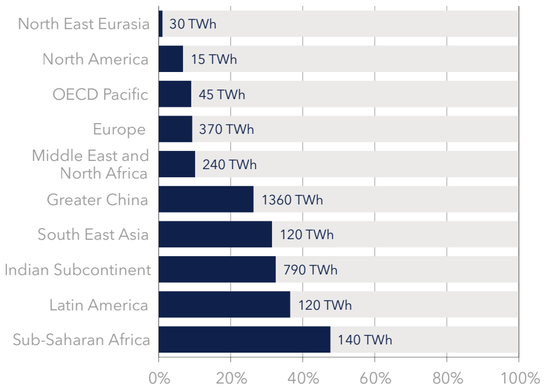


FIGURE 3.20 | Grid-connected electricity only.

This reduction depends primarily on changes in production methods rather than output volume.

The lowest-cost decarbonization route is greater use of scrap steel in electric arc furnaces (EAFs), which already offer low-emission production when supplied with low-carbon electricity. Their share of global steel production will rise from 23% today to 50% by 2060. However, scrap availability is constrained by historical steel consumption, product lifetimes, and collection systems, meaning primary steel production remains necessary.

Electricity ahead of hydrogen for green steel production
Global steel production by route (Mt/yr)

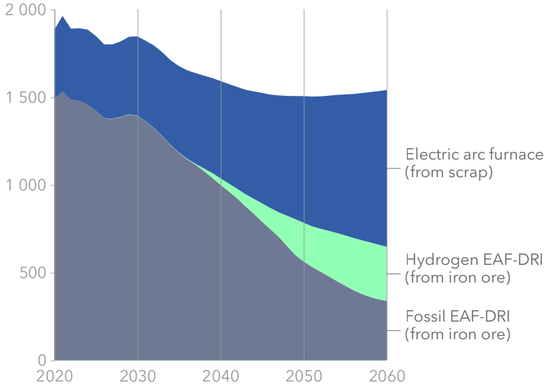


FIGURE 3.21 | EAF-DRI: Electric Arc Furnace with Direct Reduced Iron.

Around 70% of global steel production currently relies on coal-based blast furnace-basic oxygen furnace (BF-BOF) technology. Hydrogen-based direct reduced iron (DRI) offers a viable low-carbon alternative, but deployment is constrained by cost and a limited market. Existing steelmaking overcapacity and growing scrap-based production reduce the requirement for new primary capacity and limit space for nascent hydrogen-based routes. Continued investment in conventional BF-BOF capacity across India, South East Asia, and parts of China reinforces this dynamic.

When a blast furnace reaches the end of its operating cycle, operators can either replace the furnace lining and continue operating the asset, or invest in alternatives such as hydrogen-ready DRI. Under current conditions, relining typically remains the lower-cost and lower-risk option, reproducing the retrofit-over-replacement dynamic observed across the sector.

Policymakers are responding by tightening capacity-replacement requirements to encourage EAF and hydrogen-based routes rather than continued reliance on relining. Natural-gas-based DRI-EAF systems will likely serve as an intermediate pathway, particularly in gas-rich regions such as the Middle East. We expect more substantial scaling of hydrogen-based DRI after the mid-2030s as older blast furnaces retire, hydrogen costs decline, and policy increasingly favours replacement over life extension.

Unlocking faster decarbonization

The limiting factor in manufacturing is often not technology, but investment. Most low-carbon production routes become attractive only when existing assets reach replacement points or when policy and market conditions change the economics of replacement.

Faster decarbonization would therefore depend on bringing forward investment decisions. This could be achieved through stronger carbon prices, lower-cost low-carbon electricity, faster infrastructure deployment, or policies that favour replacement over life extension. Without such changes, industrial decarbonization is likely to continue at the pace set by asset turnover rather than by technological progress.

Feedstock demand

Feedstock demand remains an important source of oil demand growth, but we expect growth in virgin feedstock demand to peak in the 2030s as recycling, efficiency, and material substitution increasingly offset growth in demand for fossil-based materials.

Our forecast for a mid-2030s peak in feedstock demand reflects opposing market forces. Population and economic growth continue to increase demand for fossil-based materials such as plastics, asphalt, and fertilizers. Unlike in many energy applications, alternatives to oil and gas are not yet available at scale for producing these materials. At the same time, rising recycling rates, efficiency improvements, and substitution reduce the need for virgin fossil feedstocks. Together, these factors limit further growth and lead to a peak in feedstock demand during the 2030s.

Growing recycling limits virgin feedstock demand

Plastics production remains the primary driver of feedstock demand. We project that demand for plastics will nearly double by 2050. However, demand for plastics from virgin fossil fuel feedstock rises by only a third and peaks in the early 2040s with the remaining growth in plastics demand supplied by recycled material.

Mechanical recycling expands globally and meets nearly a third of demand by 2060. At the same time, chemical recycling progresses towards commercial readiness and supplies recycled feedstock directly to steam crackers. The increasing use of recycled material reduces the volume of crude oil and natural gas required for petrochemical production over the long term.

Geopolitics reshapes regional feedstock choices

Feedstock choices have historically been determined primarily by local availability and cost. However, supply security is becoming a more important factor, particularly for regions that rely on imported feedstocks. Oil and natural gas currently dominate the feedstock mix. North America relies largely on domestically available ethane, while Europe and Greater China have historically depended on imported naphtha.

The war in Iran has disrupted maritime trade routes and highlighted the vulnerability of some supply chains. In response, import-dependent regions are seeking greater feedstock diversification. On top of short-term expansion of alternative import channels, China is reinforcing its domestic coal-to-chemicals pathways, while Europe accelerates uptake of renewable feedstock (from biomass or electrolytic hydrogen).

Although these developments alter regional feedstock choices, they do not change the broader outlook for slowing growth in virgin feedstock demand.

Petrochemicals remain a limited growth opportunity for oil demand

As demand from road transport declines, oil and gas producers increasingly target petrochemicals as a source of future demand. We project that feedstock demand will represent a quarter of total oil demand by 2060.

However, petrochemicals do not provide an unlimited source of long-term oil demand growth. Demand for virgin feedstocks peaks after the mid-2030s, increasing competition for market share. As demand growth slows, the risk of global overcapacity and stranded assets increases. Recent supply disruptions in the Middle East and force majeure-related closures have contributed to margin pressure across the industry and illustrate the challenges that may emerge in a more competitive market.

Feedstock demand to plateau in the next few years

Feedstock demand (EJ/yr)

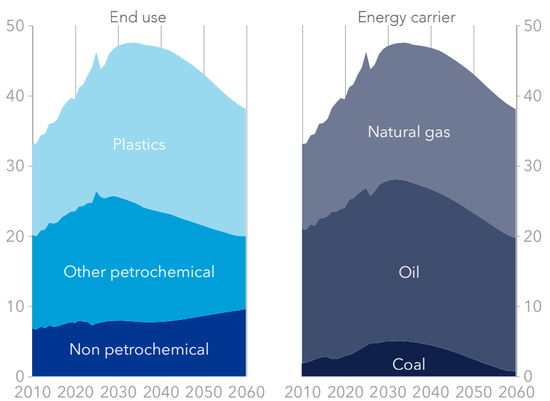


FIGURE 3.22 | Non petrochemical: asphalt, bitumen, lubricants, etc.

Gas dominates in production regions

Feedstock demand mix in selected regions

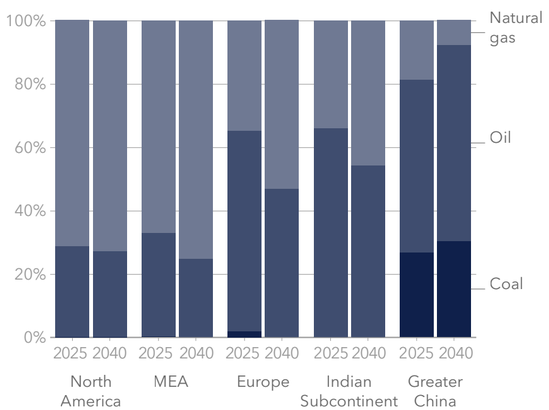
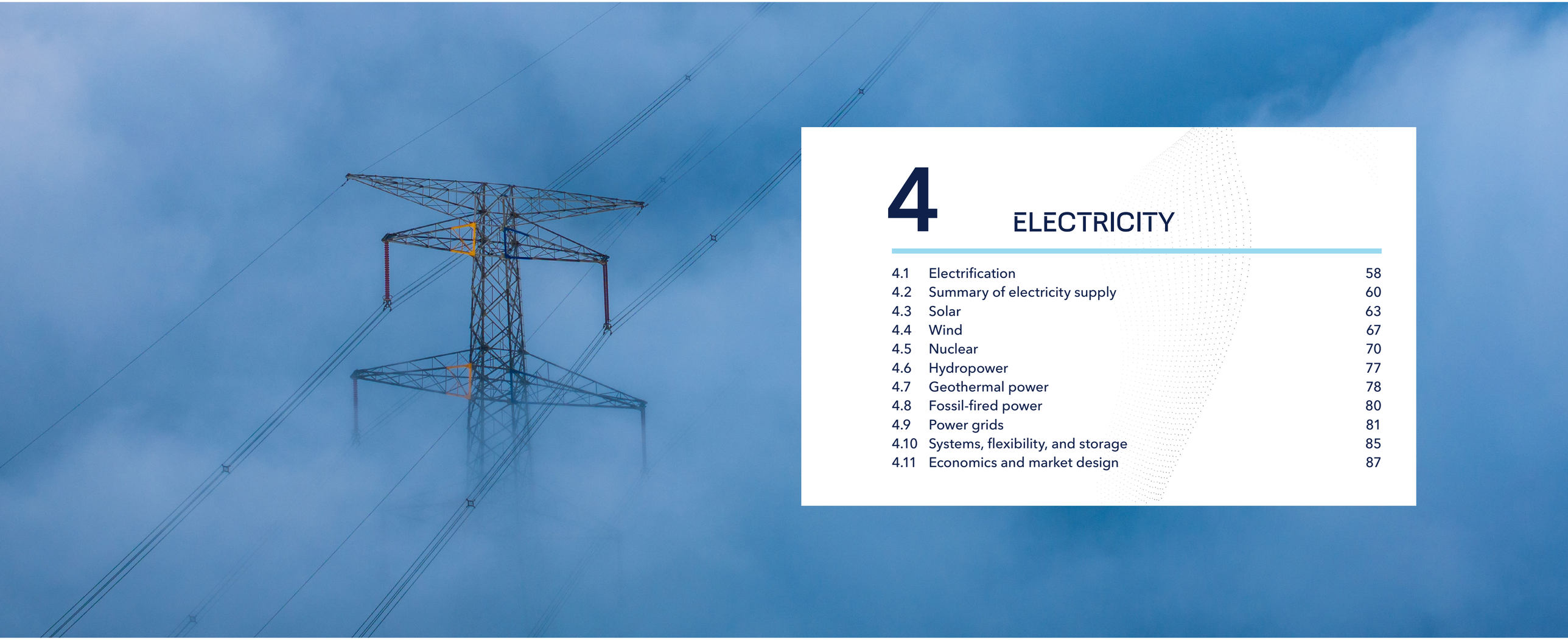


FIGURE 3.23 | MEA: Middle East and North Africa

Virgin fossil fuel plastics feedstock peaks in the 2040s with remaining growth coming from recycling.



4 ELECTRICITY

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4.1 ELECTRIFICATION

The energy transition is largely a story of electrification. As transport, buildings, and industry increasingly switch from direct fossil fuel use to electricity, global electricity demand doubles over our forecast period, growing much faster than overall energy demand. The shift is further reinforced by the emergence of new electricity-intensive applications, particularly data centres and hydrogen production. As a result, the energy transition becomes increasingly dependent on the ability to expand generation, grid infrastructure, storage, and system flexibility at unprecedented scale.

Buildings and transport are the main drivers of demand growth. Rising populations, urbanization, and higher living standards increase buildings electricity consumption by 75% by 2060, excluding data centres. In transport, electrification is increasing rapidly through EVs: the global vehicle fleet will be 84% electric by 2060.

Manufacturing is currently one of the largest sources of electricity demand globally and that demand will increase by 52% over the forecast period. However, because electricity demand grows even faster in transport, buildings, and data centres, manufacturing's share of total electricity demand declines from 42% today to 31% in 2060. Electrification is increasingly used to improve efficiency and reduce emissions, with electricity replacing coal and gas in a growing range of industrial processes. The strongest electricity demand growth occurs in industrial regions such as Greater China, the Indian Subcontinent, and Sub-Saharan Africa.

Data centres currently represent a small but growing share of global electricity use, consuming 1.2% of total demand, mostly for non-AI applications. While advances in chips, cooling systems, and data-centre design continue to improve energy efficiency, the rapid growth of AI workloads is driving the construction of more and ever-larger facilities and pushing overall electricity demand sharply upwards. By 2060, we expect AI to account for 77% of total data-centre electricity use, raising the sector's share of global electricity demand to 10%.

Uneven progress across regions

Electrification remains uneven across regions. At one-fifth of global final energy consumption, the share of electricity in regional final energy mixes ranges from 9% in Sub-Saharan Africa to 27% in Greater China. By 2060, only three regions (Greater China, Europe, and OECD Pacific) derive at least half of their final energy demand from electricity.

In low- and middle-income regions, electrification is driven by a sharp increase in air conditioning (which in turn is driven by rising temperatures as the planet warms) and by rising GDP per capita. Falling unit prices and elevated inventories across the supply chain are lowering the barrier to air conditioning adoption (IEA, 2026a), although poorer households typically lack the means to invest in the most energy-efficient equipment (Economist, 2026). As a result, global space cooling more than doubles by 2060, with the strongest growth in Greater China, the Indian Subcontinent, the Middle East and North Africa, and South East Asia.

In high-income regions, electricity already plays a large role in end-use sectors, and so further electrification is driven primarily by technology substitution where EVs replace ICE vehicles or fossil-fired equipment such as space heaters are converted to electric heat pumps. EVs are among the largest sources of demand growth in North America, Europe, and OECD Pacific. Data centres also emerge as a major contributor to electricity demand growth, ranking as the second largest growth driver in North America and OECD Pacific.

While electrification accelerates globally, the pace and drivers of electrification vary regionally
Electricity demand (PWh/yr)

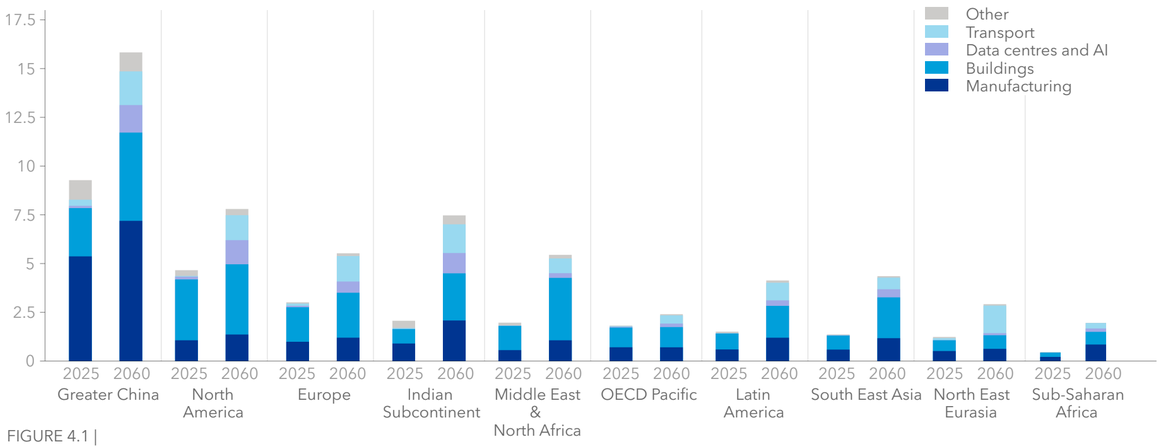


FIGURE 4.1 |

Demand becomes more flexible and complex
Electrification is changing not only how much electricity is consumed, but also how electricity systems operate. As sectors become increasingly electrified, new loads (such as EVs and heat pumps) can help provide flexibility by shifting demand in response to system needs. For example, buildings can be pre-heated or pre-cooled and use thermal storage to temporarily reduce electricity consumption during periods of system stress. This ability to adapt consumption becomes increasingly valuable in power systems with high shares of variable renewable generation.

The distinction between electricity suppliers and consumers is blurring. The growing deployment of rooftop solar PV, battery storage, and EVs is turning households and businesses into 'prosumers' that both consume and produce electricity. Through digital platforms and smart control systems, these distributed resources can be aggregated into virtual power plants that provide balancing services and support system operation.

These developments create new opportunities but also increase system complexity. Electricity flows are becoming increasingly bidirectional, demand patterns more dynamic, and power systems more dependent on digitalization and active consumer participation. At the same time, increasing electricity demand and growing shares of variable renewable generation are placing growing pressure on grid infrastructure.

As a result, the success of electrification will depend as much on grids, storage, price signals, market models, and system flexibility as on new generation capacity. In many regions, grid constraints are already slowing renewable integration and delaying connections for new electricity-intensive loads. Accelerating investment in transmission, distribution, storage, and digital infrastructure will therefore be essential to ensure that electricity systems can accommodate growing demand while maintaining reliability and affordability. Our forecast assumes substantial – unprecedented – investment in grid and related infrastructure in the next three decades, but we remind readers that our forecast is one that sees the world fall very far short of the goals of the *2015 Paris Agreement*.

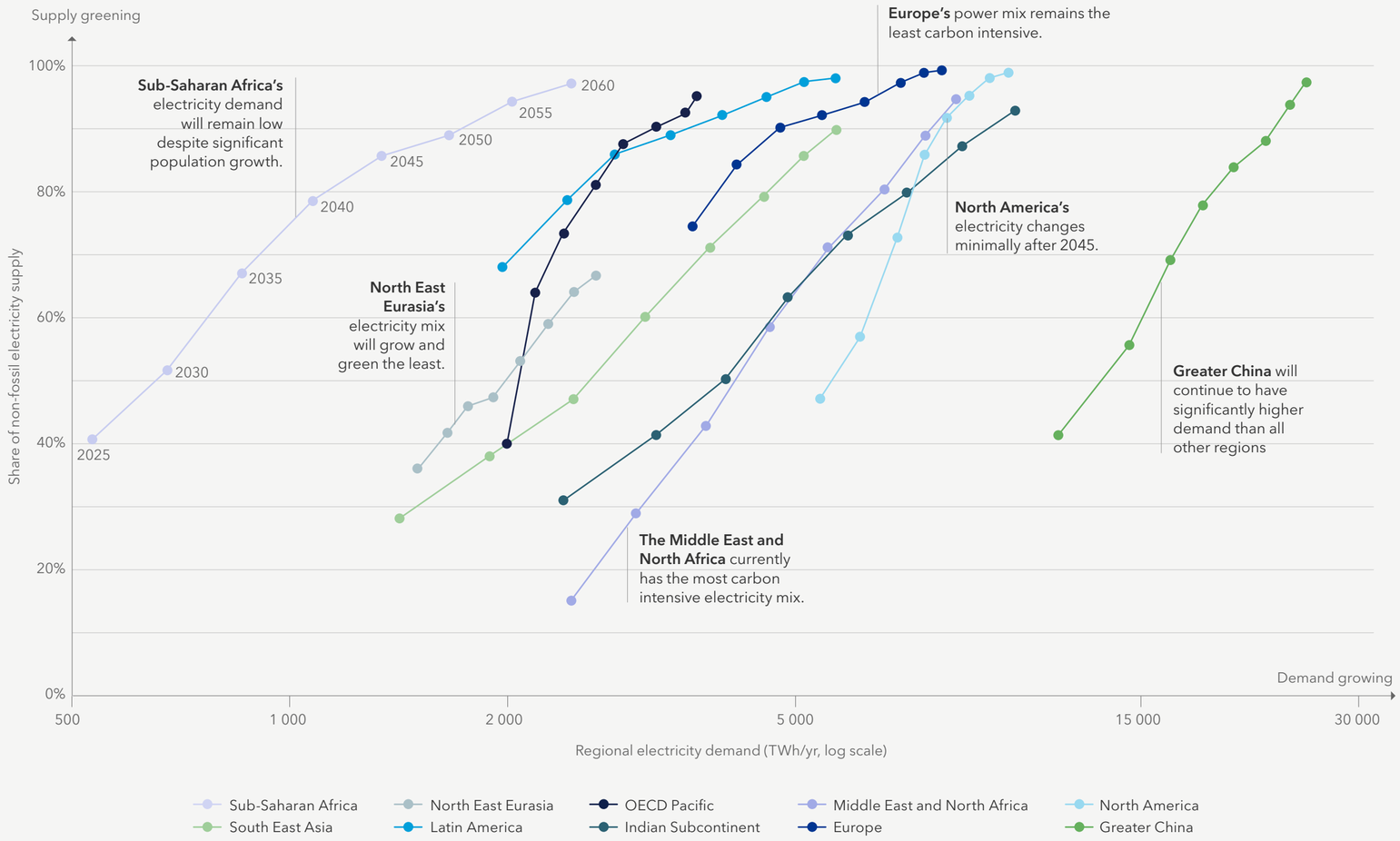
Electrification is changing not only how much electricity is consumed, but also how electricity systems operate.

4.2 SUMMARY OF ELECTRICITY SUPPLY

The growing demand for electricity is being supplied by an increasingly decarbonized power system:

- In 2025, electricity generation from renewables surpassed coal-powered generation for the first time (IEA, 2026b). In 2033, generation from wind and solar alone will be greater than from all fossil-fired plants.
- By 2060, solar and wind will supply 46% and 31% of electricity, respectively. By then, electricity demand will be 2.5 times greater than in 2025.
- All regions, except North East Eurasia, will have less than 10% fossil generation in their electricity mix in 2060. Europe is already the region with the greenest electricity supply, and it will be 99% decarbonized from the mid-2050s.
- Battery storage is now the fastest growing power capacity technology. Twice as much utility battery storage was installed in 2026 as in 2025.
- Overall global electricity generation rises from 33,000 TWh in 2025 to 84,000 TWh in 2060, an increase of 155%. This growth reflects both rapidly expanding electricity demand and the additional generation required to support storage, system flexibility, and transmission losses in a highly renewable power system.

Almost all regions' electricity systems will be more than 90% decarbonized by 2060, despite demand more than doubling



Wind and solar buildout are increasingly an energy security story

Stoked by ongoing geopolitical tensions, energy security is an increasingly prominent corner of the energy trilemma. Following the 2022 gas shock in Europe, the war in Iran is once again laying bare the risk of relying on imported fossil sources for electricity generation. In 2025, eight of our ten global ETO regions generated most of their electricity from fossil fuels. While all regions prioritize the use of domestic fossil fuels in power stations, Asian regions demand more fossil fuels than they produce. India, China, and Japan are among the countries with the heaviest reliance on LNG from the Middle East Gulf. Since March 2026, this supply has been subject to supply disruption and price hikes that impact lower-income countries the most.

In contrast, renewable generation does not require fuel imports and existing capacity is much more resistant to supply shocks. Most countries import the generation technology from only a handful of producer countries, but sudden changes only impact future installations, providing time to adjust supply chains. Many countries, particularly in Europe and North America, are increasing their focus on homeshoring wind, solar, storage, and nuclear technologies to reduce reliance on China's electro-tech (IEA, 2026d).

By 2030, five more regions will generate over 50% of their electricity from non-fossil sources. Of the remaining three, the Indian Subcontinent and the Middle East and North Africa will achieve this milestone in the mid-2030s, while North East Eurasia will trail all other regions and only surpass 50% by 2045.

In 2060, solar and wind will generate 46% and 31% of global electricity demand. These technologies will continue to be selected due to their low cost and ease of installation and management. 41% of solar generation and 16% of wind generation will be distributed away from utility power stations, either as rooftop solar on residential or commercial properties or as excess generation sold to the grid from dedicated renewable hydrogen electrolysis production sites.

Storage: growing fastest of all

Due to the inherent variability of wind and solar power, it is now imperative to firm the generation such that electricity can be used when it is needed, not just when it can be produced. Storage is no longer valued solely for its ability to enhance the economics of variable renewables; it has become a cornerstone of energy security, providing the flexibility and resilience needed to maintain reliable power systems amid growing geopolitical and climatic uncertainty.

In 2060, 7% of grid electricity supply will pass through storage before its final use, up from 0.7% in 2025. This represents a 20-fold growth in the same period in which overall electricity demand doubles.

Between 2020 and 2026, annual storage capacity additions have grown 25-fold; in the same period total solar and wind additions have grown five- and two-fold respectively. Storage, particularly battery storage, is currently the fastest-growing energy supply technology.

Battery storage installations are driven by the demand for flexibility and the falling costs of battery technologies. The levelized cost of storage (LCOS) of lithium-ion batteries has halved since 2018 and we expect it will halve again by 2028 and continue to fall before stabilizing in the mid-2040s. Much of the decline in costs has been driven by a rapid expansion in Chinese manufacturing leading to economies of scale and improved chemistries, mirroring the decline in costs in solar modules over the past 15 years (IEA, 2025c).

Currently, the primary challenge for most global power systems is providing short-term flexibility to counteract sub-daily fluctuations in renewable power generation. In subsequent decades, maintaining firm capacity during extended periods of low wind and solar output will become more important due to the retirement of gas- and coal-fired generators. Various types of storage can provide both requirements, and we expect to see their expansion continue.

Electricity supply greens and grows
World electricity generation (PWh/yr)

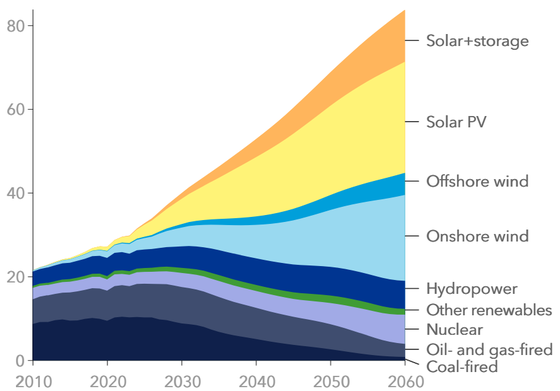


FIGURE 4.2 | Generation includes behind-the-meter and off-grid. Historical data source: IEA WEB (2026), GlobalData (2026), Rystad Energy (2026).

Storage is the fastest growing power capacity
Share of power capacity additions (%)

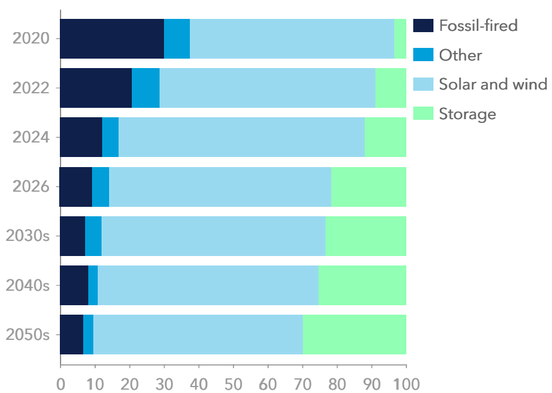


FIGURE 4.3 | Includes distributed additions. 2030s, 40s, and 50s are decade averages. Historical data source: Rystad Energy (2026), GlobalData (2026).

Levelized cost trajectories diverge

The cost of power generation remains a key decision factor when installing new capacity. Between 2015 and 2020, solar PV and onshore wind had comparable levelized costs of energy (LCOE) to those of coal- and gas-fired power plants. In 2025, the LCOEs of solar PV and onshore wind were around half those of coal and gas. We expect the costs of these renewables to continue to fall, but at a lower rate, throughout the forecast period.

Both solar co-located with storage and offshore wind have higher LCOEs than standalone solar and onshore wind, but offer other advantages that result in installations growing at a rate comparable to their cheaper alternatives. Solar+storage offers flexibility to the grid and higher income per unit of electricity sold because operators can shift generation to more lucrative hours. Offshore wind has higher capacity factors due to generally higher wind speeds offshore and benefits from more straightforward planning procedures as these are usually federally mandated rather than subject to municipal approval.

Floating offshore wind is much earlier in its cost trajectory than onshore and fixed offshore wind and is currently the most expensive of the power technologies included in our model. However, as more pilot projects come online in the next few years and standardization proceeds for full-scale projects, we expect the cost trajectory to fall similarly to other variable renewables. In 2060, the LCOE of floating offshore wind will be 20% higher than fixed offshore wind, but by then some regions will only have deeper ocean sites available that are only suitable for floating turbines.

The LCOE of nuclear has remained stable for decades, but we expect it to decline once small modular reactors (SMRs) emerge. The average LCOE of SMRs is forecast to halve between 2025 and 2040. The LCOE of hydropower will increase 20% by 2030 as most regions run out of available sites for large projects. It will then remain steady, as technological improvements for smaller projects are balanced by declining technical availability overall.

Fossil fuel LCOE trajectories trend dramatically upwards due to falling capacity factors and increasing carbon prices.

Geothermal energy's LCOE will increase sharply in the short term as more expensive deep geothermal, available in more places, becomes the standard rather than shallow geothermal which has few available unutilized sites remaining.

Bioenergy's LCOE also rises over time, driven in part by lower capacity factors as its role shifts towards dispatchable generation and the increasing deployment of BECCS. While CCS adds to generation costs, BECCS operators can offset some of these costs by monetizing the value of negative emissions.

Fossil fuel LCOE trajectories will trend dramatically upwards due to falling capacity factors and the adoption and increase of carbon prices. Both factors have already taken effect on coal power. The LCOE of coal-fired projects has increased by 50% in the past ten years and will double by 2033. We expect the LCOE of gas to fall until 2031, particularly in regions with natural gas resources. Near-term additions benefit from turbines procured before recent supply shortages drove up costs; the LCOE will rise as more expensive equipment is deployed. Carbon pricing and lower utilisation then drive a much sharper increase between 2045 and 2060.

LCOE is not the sole determinant of investment decisions. Technologies that provide electricity during high-value periods, contribute firm capacity, or enhance power system flexibility can remain attractive despite higher LCOEs.

LCOEs of power generation technologies diverge

Global weighted average levelized cost of electricity for new capacity installations (USD/MWh)

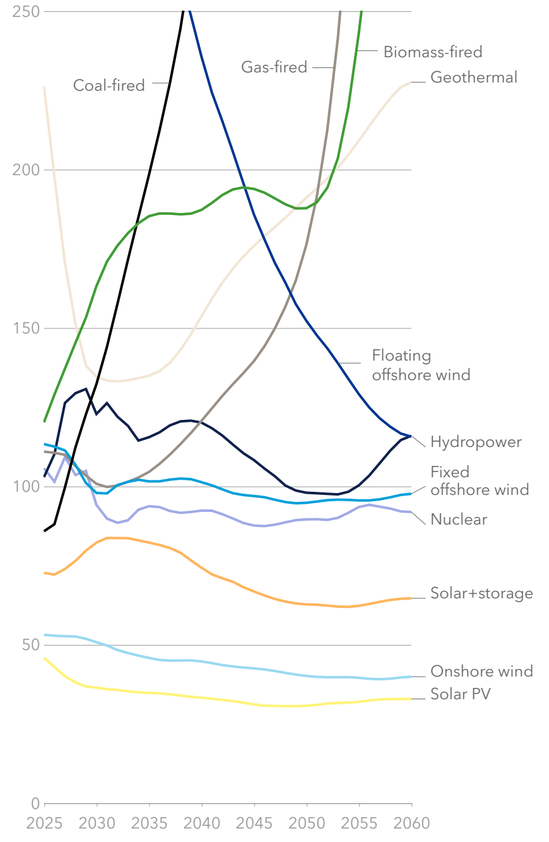


FIGURE 4.4 | LCOEs weighted by regionally installed capacities and averaged over three years. Regional cost differences and capacity additions shape the LCOE curve, so its decline does not directly reflect regional cost trends.

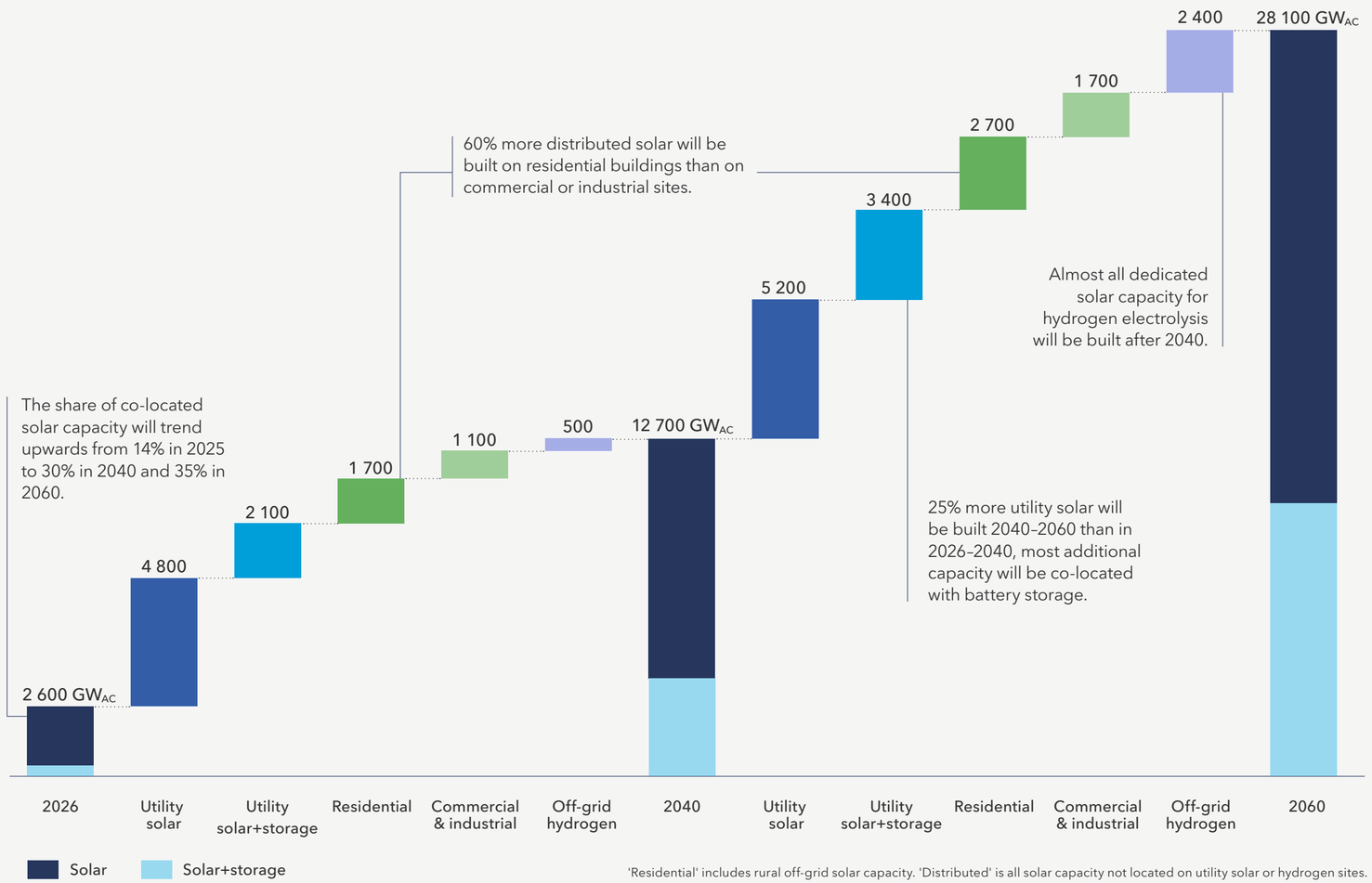
4.3 SOLAR

Solar power will be the largest contributor to the electricity mix from 2031.

- The total installed solar capacity in 2025 was 2,600 GW. This will double by 2030 and be ten times greater by the late 2050s. Solar power will generate more than 40% of electricity in all regions except North East Eurasia by the early 2050s.
- Over the past decade, solar capacity grew rapidly, by 26% annually. This rate will slow as the transition unfolds, reaching 2% annual growth in 2060.
- About 40% of installed capacity is currently located on residential and commercial rooftops and industrial sites, rather than at utility-scale solar power plants, and we expect this share to remain stable.
- Global module imports have increased in 2026 compared with last year as countries dependent on fossil fuel generation accelerate installations amid energy security concerns. Notably, South East Asia will install 25% more solar in 2026 than we expected in last year's report.

Solar power's cheap and modular design enables installed capacity to grow 10 times at all scales

Global installed solar capacity and additions (GW_{AC})



'Residential' includes rural off-grid solar capacity. 'Distributed' is all solar capacity not located on utility solar or hydrogen sites.

Drivers of solar power installations vary by region

After two decades of accelerating growth, we expect global annual solar capacity additions to peak in 2026 and then remain broadly stable for several years. This is because the two regions with the highest existing capacity, Europe and Greater China, are facing significant solar curtailment and grid constraints. This is prompting rising installations of solar plants co-located with battery storage and standalone storage in both regions. Between 2024 and 2035, the share of annual co-located installations will grow from 25% to 40% in Europe and from 22% to 40% in Greater China. The OECD Pacific is experiencing the same trend at a smaller scale.

Most other regions are earlier in their solar capacity build-out and are yet to experience significant negative pricing during the day due to high solar generation. These regions include the Indian Subcontinent, Sub-Saharan Africa, and South East Asia. There, solar power plants without comparatively expensive batteries are still the norm and they rely on fossil generation for flexibility. Since the war in Iran began in spring 2026, many countries in these regions have accelerated solar buildout in response to increased fossil prices and supply shortages (Stringer, 2026). We expect South East Asia to build 25% more solar capacity in 2026 than we forecast in our 2025 report, and that new distributed solar capacity additions across that region will be 50% greater than what we predicted last year.

China's role in the exponential growth of solar capacity over the past five years cannot be overstated. In 2025, Greater China installed slightly more solar than the rest of the world combined. However, 2025 installations were accelerated to come online before the switch from fixed feed-in-tariffs to market-driven pricing. The surge in spring 2025 installations has exacerbated grid constraints and curtailment, which has risen to 9.2% (Bloomberg, 2026). China's 2026 solar capacity factor is 30% lower than the global average, due mostly to grid curtailment. This contributed to rising emissions from China's power sector in the first quarter of 2026 (Myllyvirta, 2026), though they remain below the 2024 peak.

After many years of market over-saturation, China now manufactures 97% of solar wafers and 78% of modules (IEA, 2025d). However, this share may begin to decline as certain countries prioritize homeshoring solar equipment production including the US, India, and some European and South East Asian countries. Homeshoring policies include tariffs on imported equipment and tax incentives for producing or installing domestically produced equipment.

The increased CAPEX of home-shored modules (and therefore LCOE) is tightening the already close competition for the cheapest source of new power capacity between solar and onshore wind. This is particularly relevant for India which is now installing Chinese-produced wind turbines but increasingly locally manufactured solar modules. In the US and Europe, most wind turbines are currently from European suppliers; switching to European or American solar modules would bring solar's LCOE closer to that of onshore wind. When adjusted for energy-revenue, the average global LCOE of solar rises above that of onshore wind in the mid 2030s. This is caused by increasing solar capacity driving down solar capture prices because generation is concentrated during midday when demand is reasonably low.

Types of solar installed in the near future varies by region
Installed capacity by type, 2026 to 2040

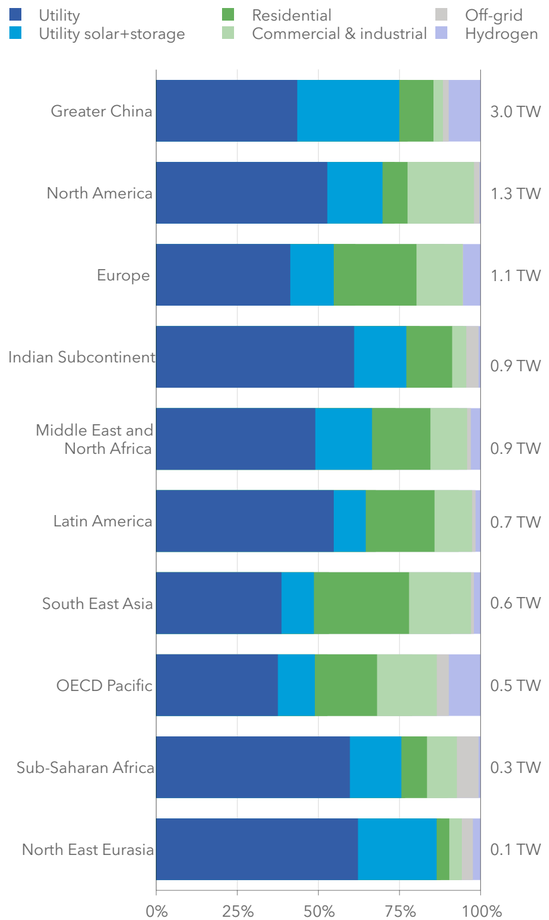


FIGURE 4.5 | Residential and Commercial & Industrial include systems with storage.

The long-term role of solar

Solar power has been proven to be low cost and straightforward to install. Its challenge lies in its inherent variability. Not only does the sun only shine half the day, but sunset and sunrise often coincide with peak power demand in the mornings and evenings. Solar's enduring role in the power system is reliant on our ability to shift its generation. The most straightforward way to do so is to install battery storage on the same site as solar modules, at any scale.

Co-locating storage with solar has many advantages, including sharing grid access points and enabling site operators to store their own power at times it is not needed, rather than buying from the grid and paying associated grid fees. 70% of battery storage is already located on solar power sites, a share which will trend upwards to reach 80% in the 2040s. The average duration of battery packs is also trending upwards. From two hours in 2020, we expect it to reach four hours in the early 2030s. There are already many examples of significantly longer battery duration enabling 6, 8, 12, or even round-the-clock baseload generation (Energy Connects, 2025).

Sufficient financial motivation to pair battery storage with both utility and distributed solar exists in established energy markets such as Europe and North America. In the autumn of 2025, the EU's day-ahead power market switched from hourly to 15-minute trading intervals to reflect short-term fluctuations in demand and supply in electricity prices and to allow trading within a day (Nord Pool, 2025). These changes improve the flexibility and reliability of the electricity system, but rely on sufficient inter-connectors, responsive grid control, and accurate weather forecasting.

Energy markets elsewhere will need to follow the trend of liberalization and flexibility to ensure full utilization of the affordability of variable renewables and battery storage. In China, the prevalence of yearly contracts provides insufficient market motivation not to use coal plants while electricity from solar is abundant. This is delaying the shift away from fossil generation (Civillini, 2026).

In the 2020s, solar generation provides 18% of average European hourly electricity demand during the midday peak. This increases to 41% in the 2030s and 51% in the 2050s. In the 2020s, solar already covers 7% of total daily demand, and we expect this to jump to 20% in the 2030s and then trend upwards to 29% in the 2050s.

Solar supplies a growing share of daily power demand

Average European solar generation and demand by decade and hour (GWh/h)

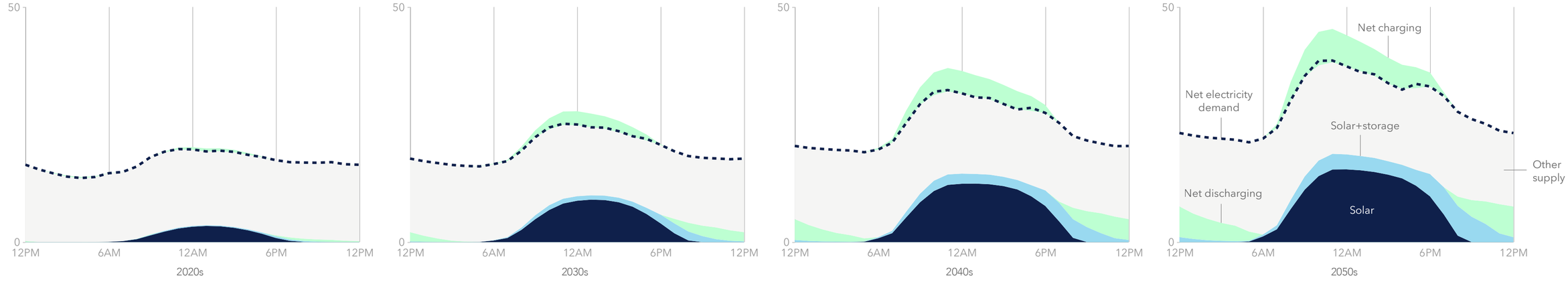


FIGURE 4.6 | Supply and demand data averaged over each decade for each hour.

Solar+storage plants have a slightly lower capacity factor than standalone solar due to technical limitations. However, by shifting output from low-price periods of abundant solar generation to higher-value hours, solar+storage plants achieve a higher capture price that outweighs the loss in generation. This advantage is particularly strong in countries with limited grid and storage capacity, where midday solar generation depresses prices and causes curtailment. The premium peaks globally in 2050, when the average capture price of solar+storage is more than twice that of standalone solar. By then, co-located systems account for a third of installed solar capacity. These price distortions are reduced as additional grid infrastructure and storage are deployed, narrowing the capture-price premium enjoyed by solar+storage plants.

Distributed solar modules on residential and commercial properties and off-grid systems are already globally widespread. Behind-the-meter (BTM) solar power paired with batteries can shield households and businesses from peak electricity prices and benefit the grid. The decision to invest in BTM solar is based on pay-back periods in wealthier regions while off-grid solar installations are driven by lack of grid access in lower income regions. In both cases, wealthier households are able to invest in a technology that will decrease their energy costs in the longer term, an opportunity that is usually not available to those with less disposable income. While distributed solar systems are, and will remain, more expensive than their utility counterparts, costs will trend downwards enabling access to more households.

Currently, there is 50% more solar installed on residential rooftops than on commercial & industrial (C&I) properties worldwide. There are some regional variations; North America and OECD Pacific have higher shares of C&I solar owing to high retail electricity prices, plentiful existing commercial roof space, and supportive policies. 10% of residential installations are currently co-located with storage compared to 7% of C&I systems. This gap will widen to 70% of residential and 35% of C&I solar by 2060. This is because batteries increase residential self-consumption enough to reduce the pay off period whereas the stronger alignment between commercial load profiles and solar time of generation minimizes the economic benefit of co-located storage.

Further into the future, demand for hydrogen produced through dedicated renewable electrolysis will increase, reaching 130 MtH_2/yr in 2060 from negligible amounts today. These electrolysis facilities will be able to participate in electricity market dynamics by producing hydrogen when grid prices are very low and selling excess electricity generation when it is beneficial to the grid. In 2060, there will be 2,800 GW of solar capacity located on electrolysis sites, 10% of total capacity. Half of this will be in China, with significant amounts in Europe and the OECD Pacific.

4.4 WIND

- Fixed offshore wind additions are regaining some momentum after several years of reduced optimism due to continued cost inflation, global supply chain disruptions resulting from geopolitical unrest, and slow permitting processes. China and Europe remain the top markets for buildout, while the North American market continues to be uncertain due to unfavourable federal policy in the US.
- The role of wind power in the energy system has shifted in recent years. Initially driven by climate concerns and decarbonization goals, wind power is now considered a key strategy for energy security, representing decentralized and affordable power that can be built at scale domestically.
- Chinese original equipment manufacturers (OEMs) continue to play an important role in the global wind buildout. Historically, these manufacturers have primarily met domestic demand, which is the biggest global market and growing. However, they are increasingly meeting global order intake from countries outside of Europe and North America, especially in emerging markets like India and parts of the Middle East and North Africa. Europe and North America remain core markets for European OEMs.

1.4 TW

Total grid-serving wind capacity installed today

+150%

Growth of grid-serving installed wind capacity to 2040

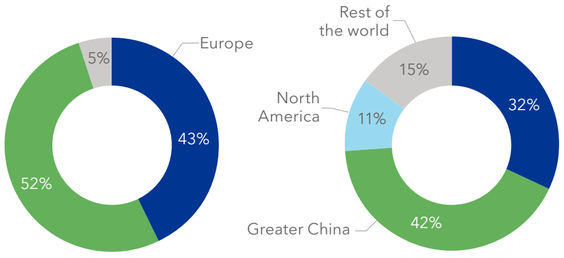
19%

Wind's share of global grid-serving electricity in 2040

2025 ⋯⋯→ 2040

Fixed offshore wind is dominated by two markets today

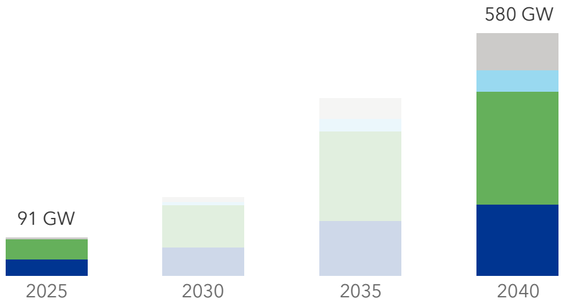
Greater China and Europe account for the large majority **share of fixed offshore wind installations** today



How will it look in 15 years?

These two markets are still dominant in 2040, but North America emerges as the third largest market, and we expect the rest of the world will buildout offshore capacity

Grid-serving power-first installed capacity today is around 91 GW and Greater China has around 10 GW more than Europe



We forecast grid-serving installed capacity to grow to 580 GW by 2040. Greater China will outpace Europe, with 270 GW compared to 170 GW. North America will have over 50 GW

Today, the global average **levelized cost of energy (LCOE)** for fixed offshore wind is around USD 113/MWh, over twice as much as onshore wind

The LCOE will likely fall to around USD 100/MWh by 2040. The plateauing in cost is due to regions with higher LCOEs starting to build out capacity, keeping the global average elevated

Wind is significant for electrification and security

Wind is key for renewable energy buildout, and thus the global energy transition, and is second only to solar in global installed renewable capacity. Wind uptake remains strong and will continue to grow rapidly. Grid-serving wind capacity today is almost 1.4 TW and is set to grow two and a half times to 3.5 TW in 2040, with even higher growth expected when including wind power used for hydrogen production. Grid bottlenecks, global supply chain disruptions, lengthy permitting processes, community opposition, and turbulent policy landscapes have posed challenges but have not significantly set the buildout back thanks to decades of cost reductions.

With energy security top of mind in many countries after several years of energy market turbulence, wind is increasingly central in strategies for securing supply. Wind power is decentralized, strengthens energy autonomy by reducing energy import dependence, and is appealing in large part due to its low cost.

Levelized cost of wind energy

The LCOE of wind is a critical factor in the quantity of wind capacity coming online around the world. LCOE is impacted by two main factors: the annualized CAPEX and OPEX over the wind power plant's lifetime, and the total amount of electricity it generates, determined by the turbine size and the wind conditions of the site.

The global LCOE for wind will continue to fall across all wind technologies (Figure 4.7). Today, the average global LCOE for onshore wind is less than USD 55/MWh, only marginally more expensive than solar PV, making it one of the cheapest available technologies. The LCOE for fixed offshore wind is about twice as much as onshore wind (USD 113/MWh) and floating is three to four times as much as fixed (USD 350–450/MWh). The LCOE for offshore wind will fall as more projects are realized in the coming decades. We expect that the global average LCOE for fixed will decrease by 14% from today to 2040, and floating will halve from 2030 to 2050.

Global LCOE falling slowly for onshore and fixed offshore
Global average levelized cost of wind energy (USD/MWh)

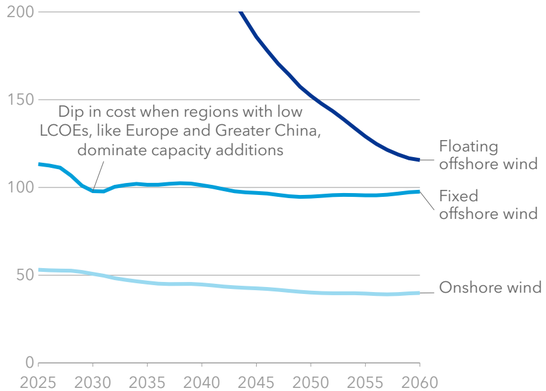


FIGURE 4.7 | Global LCOEs are running three-year averages.

Indian Subcontinent to have second largest onshore share
Global installed onshore wind capacity (GW)

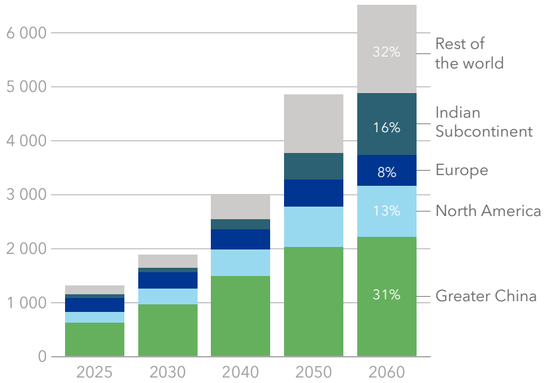


FIGURE 4.8 | Grid-connected capacity only.

Onshore wind dominated by three regions for now

Onshore wind will continue to make up the vast majority of installed wind capacity globally throughout our forecast period. At a regional level, we expect that Greater China will continue to dominate installed capacity (Figure 4.8). Today, nearly half of all onshore wind installations are in Greater China. The remaining capacity today is split almost evenly between Europe, North America, and the rest of the world.

After 2040, we expect that regions beyond these three leading markets will account for an increasingly significant share of onshore wind capacity as their markets mature. Most notably, we forecast that the Indian Subcontinent will have around 180 GW of installed capacity in 2040, which will increase more than two and a half times to 2050 and more than double again to 2060. This growth in wind power is driven by rapidly increasing electricity demand from the region's growing population – which is also becoming richer, with more people gaining access to electricity – as well as a desire to reduce reliance on fossil fuels as outlined in India's nationally determined contributions.

Turbine costs in Greater China are very low, which is a significant contributor to a region's LCOE. Today the LCOE for onshore wind in Europe is twice that of Greater China, and 67% higher in North America. Producing turbines in Greater China has several advantages that ensure turbine costs remain low: economies of scale, lower labour and material costs, and a mature domestic supply chain.

Offshore wind auctions gaining traction

After several years of waning optimism due to continued disrupted supply chains, cost inflations, and sluggish permitting, offshore wind is regaining some momentum. Despite some failed auctions in Europe in previous years, activity has picked up in the past year globally: the UK's largest ever auction (Allocation Round 7) saw over 8.4 GW awarded; France opened the largest offshore wind auction ever recorded; the Philippines has emerged as a market after offering 3.3 GW at its most recent auction; South Korea awarded 1.8 GW (including some floating capacity); and Turkey, Canada, and Australia are moving towards their first offshore wind tenders.

Installed fixed offshore wind capacity in the short and medium term is slightly higher in this year's forecast

than in previous years. This is mostly due to Greater China's offshore wind market growing faster than we had anticipated. At the same time, Europe and North America's capacities have been scaled back – reflecting the pragmatic ambition of today's offshore market.

Floating offshore wind faces significantly greater logistical challenges than fixed offshore wind. We expect that sizable floating offshore wind capacity will only come online in the 2040s, as consolidation of floating foundation and turbine technologies accelerates standardization across both technology and operations, reducing the LCOE. We forecast global floating offshore wind capacity will reach about 30 GW in 2040 and grow to over 120 GW by 2050.

Regional differences in wind electricity share

Wind's share of electricity generation varies greatly between regions today: from less than 2% in fossil fuel dependent regions like North East Eurasia and South East Asia, up to 18% in Europe, which has historically been a forerunner of supportive wind policies. Europe's wind market is mature, with strong technical capabilities as demonstrated by its leading position in offshore wind as a share of electricity today and to 2060.

In the medium term, we project that wind will make up around 33% of Europe's electricity mix in 2040, the highest share of the ETO regions. North America and Greater China are not far behind. Wind energy buildout to 2040 will largely meet new electricity demand rather than replace fossil fuels, especially in middle-income regions where overall demand is growing rapidly. Although wind power has begun to replace coal and gas in the power mix in some European countries, we do not expect it to replace fossil fuels in any substantial way in the rest of the world until the 2040s.

Policy is a decisive factor

Wind deployment depends not only on technology and economics, but also on public and political support. Anti-wind narratives, often rooted in dis- and misinformation, can undermine both. As public opposition grows, political support may weaken, leading to longer permitting processes, increased project investment risk, and the introduction of anti-wind legislation. Motvind Norge, an anti-wind group in Norway, has been identified as one of Europe's leading disseminators of wind disinformation and has contributed to conflicts and delays in Norwegian wind projects (WindEurope, 2026).

The current hostile policy stance toward wind in the US has led to the reduction and removal of federal investment and production tax credits. The US Department of Defense has effectively halted national security reviews for hundreds of onshore projects, creating significant uncertainty for developers and investors regarding project feasibility and bankability.

Conversely, supportive policy can fast-track uptake of wind. Policies like subsidies, tax incentives, and feed-in-tariffs played an important role in the early development of onshore wind by reducing investment risk, encouraging technical learning, and supporting market growth. Strategic policy interventions are therefore necessary to support uptake of offshore wind.

Several governments have announced or enacted policies to help support offshore wind in its early phases. The European Commission approved a 25-year, USD 71bn (EUR 63bn) state aid scheme in France to support offshore wind; Japan has announced plans to revise offshore wind auction rules to improve bankability; and India continues to pursue offshore wind auction plans backed by viability gap funding. China, which has a more developed offshore wind sector due to early technical and political support, has now shifted from fixed tariffs to market-based pricing.

Regional dynamics have changed for fixed offshore wind
Installed fixed offshore wind capacity (GW)

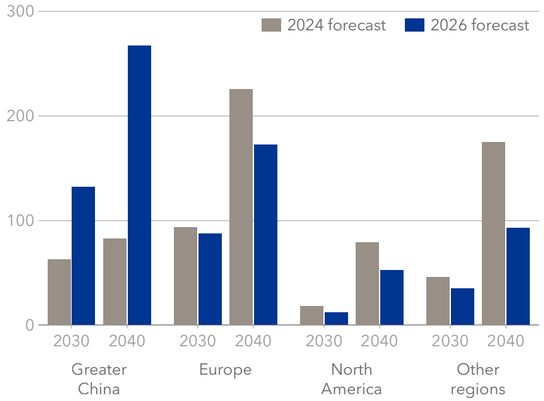


FIGURE 4.9 | Includes grid-connected and off-grid capacity.

Global wind share of electricity grows to 34% in 2060, where wind is more than a quarter share in the majority of regions
Share of wind in utility-scale electricity generation by region

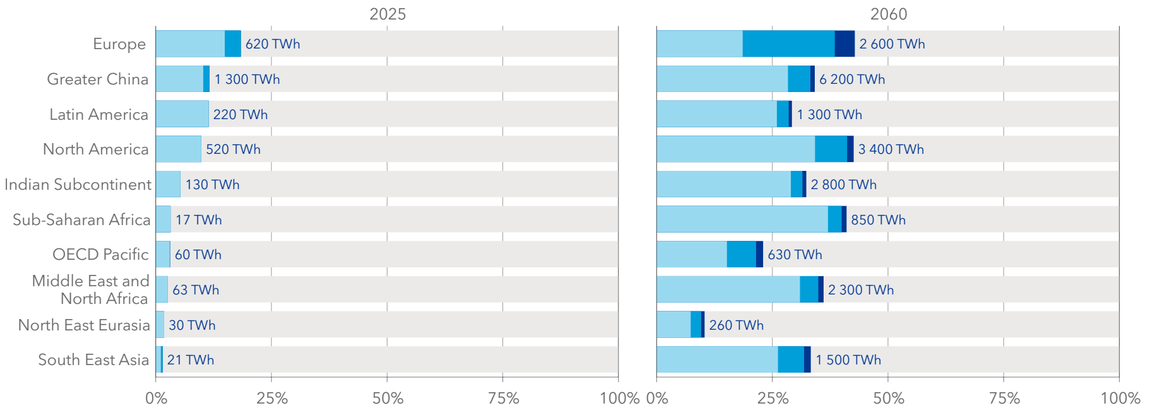
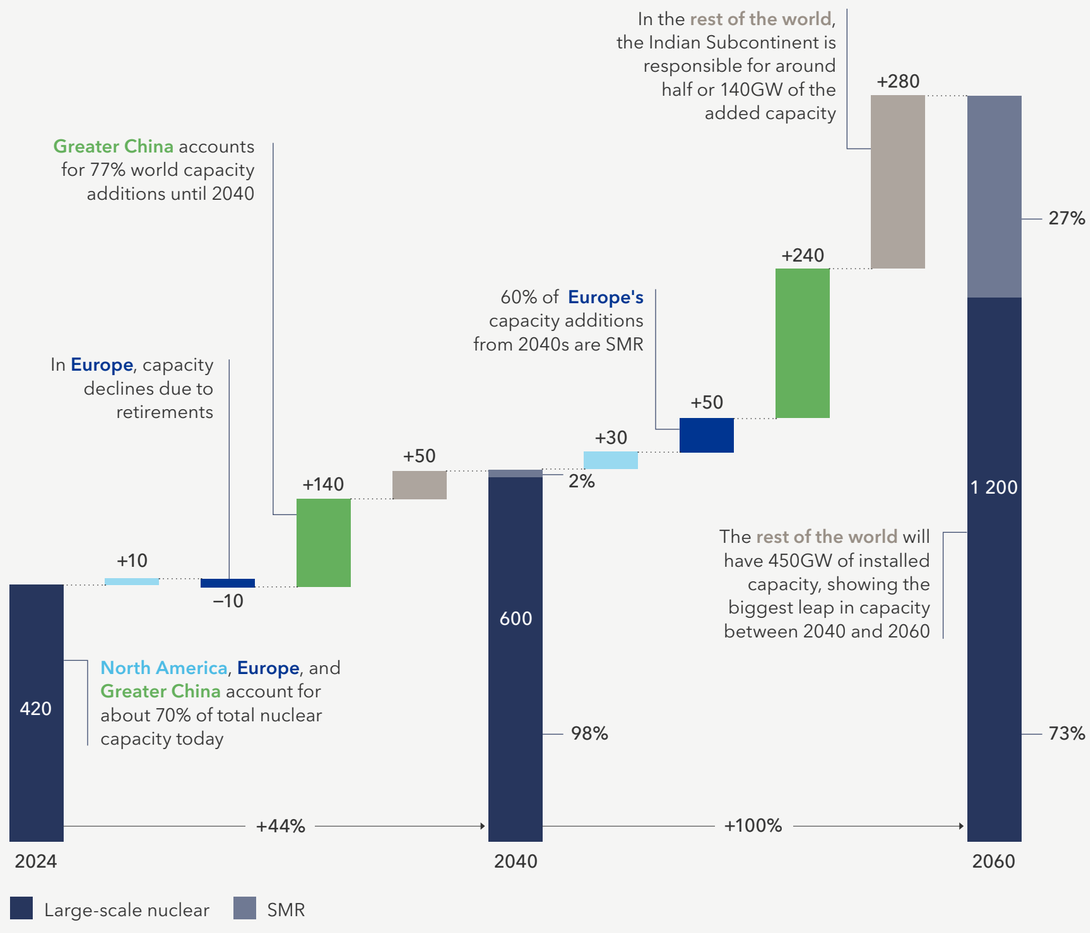


FIGURE 4.10 | Grid-connected capacity only.

4.5 NUCLEAR

- **Nuclear capacity grows almost three-fold, from 415 GW in 2025 to 1,200 GW in 2060, while generation reaches 7,000 TWh.** Our forecast falls short of the COP28 pledge to triple 2020 capacity to 1,220 GW by 2050, crossing that level in 2061. Nuclear's share of global electricity falls from almost 9% in 2025 to 7% in 2050 before recovering to 8% in 2060.
- **China accounts for 77% of net capacity additions to 2040, but its share falls to 40% thereafter as growth broadens geographically.** China adds 140 GW of the 180 GW global additions between 2025 and 2040. Between 2040 and 2060, the Indian Subcontinent rises from 31 GW to 170 GW, while the Middle East and North Africa reaches 90 GW, and South East Asia reaches 33 GW.
- **Small modular reactors (SMRs) rise from 6% of capacity commissioned in the 2030s to 46% in the 2050s.** Installed SMR capacity reaches 13 GW in 2040 and 310 GW in 2060, supplying 54% of fleet growth in the final decade and lifting nuclear generation growth above total electricity generation growth.
- **We forecast that North American SMR capital costs fall by almost 60% due to learning and standardization.** North American SMR capital cost falls from USD 12,100/kW to USD 5,100/kW, a rate of decline that has not been demonstrated in a Western market and a key assumption in our forecast. Modular units undercut large reactors from 2038 in Europe and 2039 in North America. In Greater China, the Indian Subcontinent, and the Middle East and North Africa, they do not reach parity at any point to 2060 and deployment there depends on state support.
- **Hyperscaler nuclear agreements total up to 13 GW, but only approximately 7 GW is covered by power purchase agreements deliverable by the early 2030s.** The contracted tier primarily supports restarts and existing plants, while most proposed new-build and SMR capacity remains at the development-agreement or option stage.
- **Industrial delivery capacity, SMR cost reduction, and advanced-reactor fuel supply are more immediate constraints than broad national public support in Western markets.** Public support has strengthened, but local acceptance of greenfield projects remains a risk. Heavy-forging capacity is concentrated in a small number of facilities, skilled workforces are ageing, and enriched-fuel capacity constrains some advanced reactor designs.

SMR vs Conventional nuclear by region
Units: GW



Capacity rises nearly three-fold to 2060, but share of global electricity falls

We forecast global nuclear capacity rising from 415 GW in 2025 to 600 GW in 2040 and 1,200 GW in 2060, a 2.9-fold expansion of the current fleet. Generation grows from 2,900 TWh to 7,000 TWh over the same period. Nuclear's share of global electricity nevertheless falls from 8.7% in 2025 to 7.6% in 2040 and a low of 7.0% in the mid-2040s, before recovering to 8.3% by 2060. In other words, total electricity generation grows faster than nuclear generation for most of the period. The expansion does not meet the COP28 tripling pledge by 2050 (tripling the 2020 fleet requires 1,220 GW), and the forecast crosses that level just after 2060.

Three factors are key to the trajectory we forecast:

- **Concentration.** Greater China supplies 140 GW of the 180 GW added worldwide to 2040. Thus, the near-term path is largely one region's construction programme.
- **Replacement.** 260 GW of the 1,000 GW commissioned to 2060 compensates for retiring reactors rather than adding to the fleet, and almost all that replacement sits in North America and Europe.
- **Cost.** Growth after 2040 comes mainly from SMRs at a forecast volume potentially sufficient to see their capital cost falling by 60% in North America. That rate of decline in nuclear costs has not been demonstrated in any Western market and thus our forecast rests on the industry successfully managing to reduce costs in line with our estimates.

The starting conditions are strong. Global nuclear generation reached a record in 2025 and installed capacity stood at approximately 415 GW at the end of the year. A further 78 GW was under construction across 15 countries, which the IEA describes as among the highest construction levels since 1990. There were 10 construction starts in 2025, nine of them in China, adding 12.2 GW (IEA, 2026b). Annual investment runs at approximately USD 65bn, and the IEA projects USD 70bn to 120bn by 2030, though this relies on a different assumed policy delivery (IEA, 2025a).

Policy support in many OECD countries has moved from statements of intent into licensing law and public budgets. The US has capped decisions by the Nuclear Regulatory Commission (NRC) on new reactor applications at 18 months (The White House, 2025). Finland is replacing its 1987 Nuclear Energy Act, and Sweden received its first new-build application in 50 years (WNN, 2026a; Regeringskansliet, 2026).

Capacity and generation diverge

Absolute output and share move in opposite directions because overall power generation grows faster. Total electricity generation rises from 33,000 TWh in 2025 to 84,200 TWh in 2060, an increase of 155%, against 144% for nuclear generation. Today's record generation also reflects investment decisions taken up to two decades earlier, so it describes the fleet built in the 1980s and 1990s rather than the one being ordered now. Nuclear supplied 17.5% of global electricity at its peak in 1996 (Thunder Said Energy, 2025).

Capacity and generation also diverge because the fleet runs fewer hours as the period progresses. The implied load factor of the global fleet falls from 0.79 in 2025 to 0.67 in 2060, and to 0.50 in North America and 0.47 in Europe. The mechanism is due to merit order power generation. Solar and wind generation at near-zero marginal cost displaces nuclear output in a rising number of hours. Where that happens, nuclear runs at part load rather than continuously. The commercial response differs by reactor type. Large units in most markets cost more to shut down and restart than to keep running at reduced output, so they stay on the system at a lower load factor. Modular units are better placed to follow demand, but through unit size rather than through faster ramping. A multi-module site can vary output by taking whole modules offline, which a single large unit cannot do (NEA, 2021). This is why the fleet expands 2.9-fold while generation expands 2.4-fold.

Nuclear's share of electricity falls fastest in regions where it is currently largest

The regions with the highest nuclear shares today record the largest declines. Europe generates 20% of its electricity from nuclear in 2025, 10% in 2040, and 8% in 2060. North America falls from 17% to 10% in 2040 and 7% in 2060. Both regions commission substantial new capacity over our forecast period, but end the period below the world average of 8.3% because retirements and faster growth in total generation outweigh the contribution of new units.

The regions gaining share start from a low base. China rises from 4% to 11% and the Indian Subcontinent from 3% to 10%. OECD Pacific is the only OECD region where nuclear gains share, rising from 14% in 2025 to a peak of 20% in 2030 on Japanese restarts and then easing to 16% in 2060. North East Eurasia has the highest nuclear share of any region in 2030 and 2035, peaking at 20%, before OECD Pacific overtakes it from 2038.

China, North America, and Europe on different pathways

China is the dominant driver of global nuclear growth before 2040, contributing 77% of the 180 GW of net global additions between 2025 and 2040. China' s current capacity is 60 GW and we forecast

that will rise to 200 GW in 2040 and 440 GW in 2060. All other regions contribute to the growth in global nuclear capacity in single or low double digits, with the Middle East and North Africa at 17 GW, the Indian Subcontinent at 17 GW, North America at 11 GW, and OECD Pacific at 8 GW. Europe declines from 110 GW in 2025 to 96 GW in 2040 as retirements exceed new units; from that point there is a net gain from newbuild capacity such that Europe exceeds its 2025 level by 2050.

Gross commissioning and net capacity growth diverge substantially. The world commissions 260 GW between 2026 and 2040 and retires 74 GW over the same period. To 2060, 1,000 GW are commissioned globally, with the world ending the period at 790 GW above its 2025 capacity (i.e. 26% of global commissioning replaces retiring capacity). Almost all of that replacement occurs in the OECD. Between 2026 and 2060, North America retires 100 GW, Europe 79 GW, China 13 GW, and the Indian Subcontinent 7 GW.

North America carries the widest range of uncertainty in our Outlook. Electricity demand is rising from data centres and electrification more broadly, but conversion of the announced fleet is slow. We forecast commissioning of approximately 1 GW a year through the 2030s, rising to approximately 5.6 GW a year in the 2040s. Capacity is flat at 115 GW to 2030, reaches 126 GW in 2040, dips to 121 GW as the first retirement wave runs, and ends at 150 GW in 2060. The 2040 figure assumes significant utility and hyperscaler investment decisions taken between 2028 and 2030.

Nuclear power generation grows in Greater China and Indian Subcontinent while stable in OECD

Nuclear electricity generation (TWh/yr)

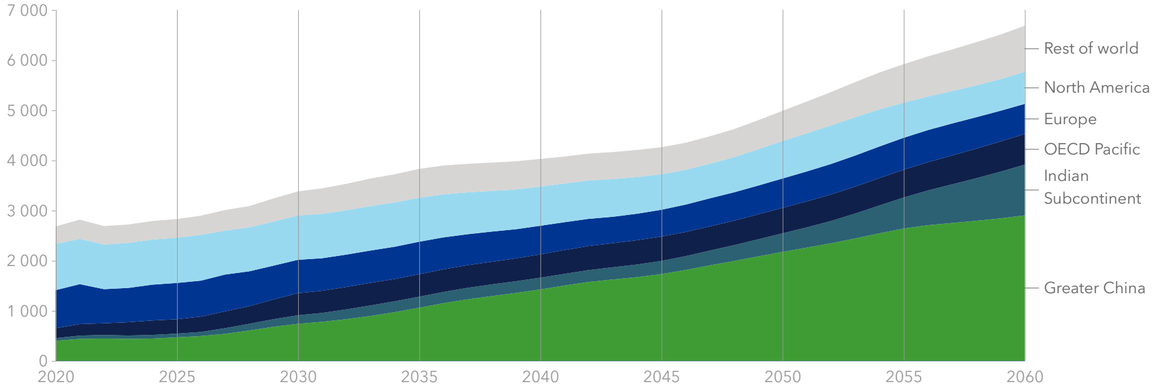


FIGURE 4.11 | Historical data source: IEA WEB (2026).

Three further sets of decisions are important for our forecast to materialize. The first is France's EPR2 programme, a standardized evolution of the EPR design, which has a final investment decision due at the end of 2026 and a USD 84bn (EUR 72.8bn) cost estimate from 2020 for six units. European Commission approval of state aid is still pending before the final investment decision can proceed (EDF, 2025). The second is the rate of Chinese approvals, where China's state plan treats nuclear power as a strategic technology to strengthen energy security, industrial competitiveness, and global influence (CSIS, 2026). The third uncertainty is whether India's *SHANTI Act* of December 2025, which replaced the single statutory cap on operator liability with a graded framework and opened limited private participation, will lead to a final commitment to develop either the Jaitapur or Kovvada nuclear project before 2030 (PIB, 2025b).

China dominates capacity additions through 2040, but after that the buildout becomes more geographically diverse. Although China still accounts for 40% of the 600 GW capacity added between 2040 and 2060, the Indian Subcontinent becomes the second largest source of additions, rising from a total capacity of 31 GW in 2040 to 170 GW in 2060. The Middle East and North Africa reaches 90 GW and South East Asia 33 GW, both from close to zero today.

SMR capacity reaches 13 GW in 2040 and 310 GW in 2060

Small modular reactors (SMRs) contribute little before 2040 but carry most of the growth thereafter. We forecast 13 GW of SMR capacity in 2040, rising to 88 GW in 2050 and 310 GW in 2060. Measured against capacity commissioned rather than installed, SMRs account for 6% of additions in the 2030s, 27% in the 2040s, 46% in the 2050s, and 33% of all nuclear capacity commissioned between 2031 and 2060.

That deployment profile is what restores nuclear's share of global electricity. Modular units supply 220 GW, or 54% of the net increase in nuclear capacity in the 2050s – a decade in which nuclear generation overall grows 38% while total electricity generation grows 22%, reversing the pattern of the 2040s.

Modular units reach cost parity with large reactors in the late 2030s in North America and Europe, but first deployments in both regions precede those dates and will require policy support. North America is likely to commission modular capacity from 2030 and Europe from 2036. In Greater China, the Indian Subcontinent, and the Middle East and North Africa, modular units do not reach parity at any point in our forecast period. We assume that the gap between deployment and unsupported economics is closed by state support, through direct funding, licensing priority, public offtake, and state ownership of the developer, which is the pattern of every modular programme announced to date. Modular deployment at the volumes we forecast is therefore a policy outcome before it is a market outcome.

Four regions account for 230 GW of the global 310 GW SMR capacity in 2060: North America reaches 72 GW, Europe 57 GW, China 58 GW, and the Indian Subcontinent 47 GW. Modular uptake is highest in regions with large retirement needs and high large-reactor costs. By 2060 SMRs account for 67% of additions in North America and 69% in Europe, compared with 48% in China and 43% in the Indian Subcontinent. China remains one of the least modular markets because it can deliver large reactors on schedule at approximately half the capital cost of modular.

Most SMR capacity is added after 2050

Nuclear installed capacity by reactor type (GW)

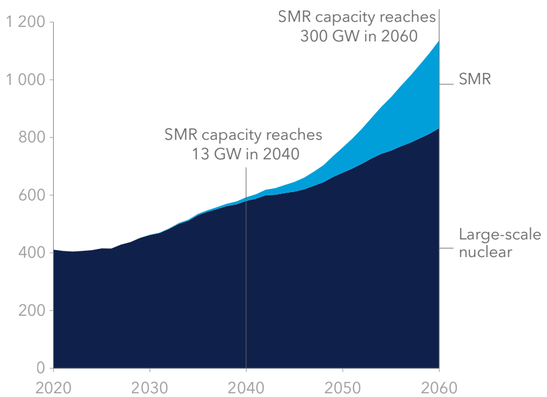


FIGURE 4.12 | SMR: Small modular reactor.
Historical data source: GlobalData (2026).

Theoretical capital cost decline (outside of China)

Our forecast for SMR uptake hinges on cost/kW declining relative to cumulative capacity deployed – for example reaching USD 5,100/kW from a starting point of USD 12,100/kW. That kind of cost compression has so far not been achieved in the nuclear industry outside of China. We detail here the reasons why we believe that a falling cost trajectory for SMRs globally is likely.

The first reason has to do with first-of-a-kind capital costs. At present, SMR capital costs are set at 48% and 71% above large-scale nuclear in Europe and North America respectively (USD 12,100/kW in North America and USD 9,500/kW in Europe). The corresponding large-reactor costs are USD 7,100/kW and USD 6,400/kW. These values reflect projected average costs for future projects and not historical costs for new builds, which have been substantially higher. The SMR premium is much larger in regions where conventional reactor costs are already comparatively low, reaching 106% in Greater China and 188% in the Indian Subcontinent.

The second consideration is delivery time and asset life. Large reactors have a technical lifetime of 60 years compared with 40 for modular units, and both can be extended by some 15 years. However, large reactors also take a lot longer to build: eight years in North America and Europe, falling to six by 2050, compared with six years for modular units, falling to three. Shorter construction times lower the amount of capital tied up before revenues are earned, reducing financing costs, which are a large component of nuclear generation costs. This is one reason why modular units narrow the gap on levelized cost, the lifetime cost per unit of output that determines which technology gets built.

A third factor is learning. Cost reduction with cumulative deployment is confined to each technology geopolitical bloc, because OECD countries do not adopt Russian or Chinese designs and vice versa. The regional cost paths therefore separate rather than converge. North American SMR capital cost falls to USD 6,500/kW by 2040 and USD 5,100/kW by 2060 (a 60% reduction), while Chinese SMR capital cost reaches USD 2,900/kW. Large-reactor capital cost is close to flat throughout, declining by around 1% over the period. The pace of capacity growth is insufficient for learning effects to materially reduce costs within the forecast horizon. The consequence is that modular units undercut large reactors at different dates in different places, with the timing driven more by the cost of large-reactor projects than by reductions in modular-reactor costs. The crossover occurs in the late 2030s in Europe and North America, and in the mid-2040s in South East Asia and Latin America, regions where large-reactor costs are high or delivery is unreliable. It occurs late or not at all where large reactors are already cheap, hence around 2050 in North East Eurasia and OECD Pacific. Greater China, the Indian Subcontinent, and the Middle East and North Africa do not cross within the period.

There is clearly uncertainty attached to these assumptions. Empirical evidence supports near-term capital cost estimates for SMRs but is too thin to

provide a basis for projecting future cost developments. OPG's Darlington programme, using GE Hitachi's 300 MW BWRX-300 boiling water design, is the first Western SMR with a published budget. Its cost estimate is USD 15.1bn (CAD 20.9bn) for four units totalling 1.2 GW, with the first unit at USD 4.4bn (CAD 6.1bn). That is approximately USD 12,400/kW at programme level and USD 14,500/kW for unit 1 (WNN, 2025). Our North American first-of-a-kind assumption of USD 12,100/kW sits just below the programme figure.

However, the rate of decline has not yet been demonstrated. Darlington Unit 1, due in 2030, provides the first test. The difference between the final completion cost of Unit 1 and Units 2 through 4 will be the first learning-rate observation available for the technology in the West. If that difference is smaller than we assume, the 2050s deployment curve flattens and nuclear's share of global electricity remains close to 7%.

We note that unlike capital cost compression, the 15% premium for SMR fixed operating and maintenance cost relative to large reactors persists throughout our forecast period in every region. Modularity is therefore assumed to improve capital cost and build schedule, and thereby reduce project risk, but not running cost. That OPEX disadvantage should also be weighed against the assumption that SMR can operate more flexibly than large reactors and may therefore enjoy a higher capture price for electricity delivered.

The rate of cost decline for small modular reactors has not been demonstrated yet.

Hyperscalers create a new source of long-term nuclear demand

Hyperscaler agreements are establishing a new class of nuclear offtaker, primarily in North America. The four largest hyperscalers – Microsoft, Amazon, Google, and Meta – all hold nuclear agreements as of mid-2026.

Carnegie's July 2026 assessment identifies up to 13 GW of announced agreements between technology companies and nuclear developers, although other trackers place the total closer to 10 GW (Carnegie, 2026; smr.intel, 2026). Approximately 6.9 GW is comprised of power purchase agreements and other commercial contracts deliverable by the early 2030s, while the remainder consists primarily of development agreements and options for future SMR projects.

The contracted capacity is concentrated in existing plants and reactor restarts. Microsoft's 835 MW, 20-year power purchase agreement supports the restart of Three Mile Island Unit 1 from 2027, while Amazon's 1.9 GW agreement with Talen secures output from the Susquehanna nuclear plant through 2042 (Carnegie, 2026). Google's 25-year agreement with NextEra underwrites the 615 MW restart of Duane Arnold, expected online in 2029 (WNN, 2026b).

Most of the remaining announced capacity relates to prospective SMR deployments. These include Meta's agreement with TerraPower covering eight Natrium units, Oklo and Switch's agreement for up to 12 GW, Amazon and X-energy's plans for up to twelve Xe-100 units, and Google's agreement with Kairos Power for up to 500 MW. These arrangements represent development partnerships and deployment options rather than final investment decisions (smr.intel, 2026). For the long term, we forecast a growing volume of nuclear capacity sited behind the meter, and most of it serves data centres. Behind-the-meter nuclear reaches 1 GW in 2040, 13 GW in 2050, and 40 GW in 2060, against a total fleet of 1,200 GW. The largest volumes are in China at 10 GW, the Indian Subcontinent at 6 GW, and North America at 5 GW in 2060. Almost all of it arrives after 2045, which is later than the announced hyperscaler tier implies. North America holds less than 0.1 GW of behind-the-meter nuclear in 2040 against up to 13 GW of announcements, and 5 GW by 2060.

Hyperscalers contract as offtakers rather than as equity owners, which limits their exposure to construction cost and schedule risk. That risk sits with the developer. If AI demand growth falls short of current projections the announced ambitions will be reduced, as the 2008 investment cycle contracted when US gas prices fell. Data centres and AI factbox on page 46 covers data centre power sourcing in more detail.

Public support rising but key constraints remain

Public acceptance of building new nuclear has improved in the US (Gallup, 2025; Pew, 2025) and Europe (NucNet, 2026). However, it is not clear that acceptance covers greenfield operations as opposed to existing sites (Good Energy Collective, 2022). We also note that India has a history of siting protests at Kudankulam and Jaitapur, and Japanese restarts proceed at the pace of host-community consent.

Beyond public acceptance, constraints remain, including enrichment and fuel fabrication. The US ban on Russian low-enriched uranium (LEU) has been in force since August 2024 (NRC, 2024). In response, the US Department of Energy (DOE) selected four enrichment contractors in January 2026 under a USD 2.7bn programme, with Centrus awarded approximately USD 900m (Shale Magazine, 2026). At full capacity, the first plant would cover approximately 30% of projected 2030 demand for high-assay fuel.

Nuclear waste management remains a challenge. Finland's Onkalo repository, the first deep geological disposal facility, is expected to begin receiving spent fuel in 2027 (Euronews, 2026). The US has no permanent disposal route, and utilities identify waste management among the leading barriers to advanced nuclear adoption (Woods, 2025).

Ramping up build capacity in Europe and North America depends on whether the industrial base rebuilds delivery capacity ahead of firm orders. Heavy forgings for pressure vessels, steam generators, and pressurizers come from a small number

of facilities worldwide, many of them in East Asia, with lead times of several years (Nuclear Scaling, 2026). The workforce that built the 1970s and 1980s programmes is retiring (nucleareurope, 2020). Rebuilding requires suppliers to invest before the orders exist.

There are some signs that commitments have been made. The UK targets 40,000 new nuclear roles by 2030 and a doubling of apprenticeships, with USD 89m (GBP 65.6m) of doctoral funding awarded beginning 2026 (Nuclear Skills Plan, 2026). Framatome is expanding its Montbard plant and workforce for the EPR2 programme (Framatome, 2026). We expect to see the results of these recent efforts in the 2030s. China faces neither constraint because they are building continuously enough to keep forging capacity and site labour employed between projects.

Alongside delivery capacity, cost remains a critical dependency in our forecast. Achieving the projected deployment requires North American SMR capital costs to fall by 58% from their 2025 level. The regional distribution of deployment reflects differences in delivery capability and cost more than differences in policy ambition. Thus, North America commissions only 14 GW between 2026 and 2040, compared with 140 GW in China, despite comparable policy commitment and higher announced demand.

Nuclear's strategic value extends beyond carbon-free electricity

Decarbonization is not the only force supporting nuclear development. Energy security, rapid electricity-demand growth, constraints on land and grids, and new approaches to power procurement are increasing the strategic value of firm low-carbon generation. These factors strengthen the case for nuclear, but do not remove the cost, schedule, supply-chain, waste-management, and delivery constraints.

Energy security has returned as an important policy driver. Geopolitical shocks, including Russia's invasion of Ukraine and instability in the Middle East, have increased the value placed on domestic and diversified energy supply. Governments are repositioning nuclear as a component of energy security alongside decarbonization and industrial competitiveness (IEA, 2022; NEA, 2026). The physical basis is energy density. A 1 GW plant requires approximately 30 tonnes of fabricated fuel a year, compared to over 3 million tonnes of coal for comparable output. Plants typically hold up to two years of fuel on site, and strategic inventories are correspondingly inexpensive to establish (WNA, 2024). European utilities held 39,900 tonnes of natural uranium equivalent at the end of 2024, more than three reloads on average, sourced 34% from Canada, 24% from Kazakhstan, 16% from Russia, and 11% from Australia (ESA, 2025).

The nuclear supply chain is nevertheless becoming increasingly segmented. OECD countries, Russia, and China largely rely on their own reactor technologies, suppliers, and fuel-cycle infrastructure, while countries without domestic capabilities draw selectively on competing technology blocs. Of the 52 reactors that started construction worldwide between 2017 and 2024, 25 are of Chinese design and 23 of Russian design (IEA, 2025a). Concentration is sharper upstream. Russia holds approximately 46% of global enrichment capacity, 20% of conversion capacity, and 10% of fuel fabrication capacity. One reactor in four worldwide operates on Russian technology, was built by Russia, or is under construction using Russian designs (Haneklaus et al., 2025). India illustrates the mixed approach. It has historically relied on Russian technology at Kudankulam but is also pursuing cooperation with French suppliers at Jaitapur and US suppliers at Kovvada as it works towards a target of 100 GW by 2047 (PIB, 2025a; WNA, 2026).

Security of supply should be distinguished from the physical security of nuclear facilities. Nuclear provides firm and predictable generation, but plants and associated infrastructure must also be protected against natural hazards, military conflict, cyber threats, and malicious attacks. Ukraine is the first case of operating commercial reactors inside an active conflict zone. Commercial power reactors are neither designed nor regulated for that condition, and a reactor shut down for safety still requires cooling systems that a conflict can disable (Squassoni, 2024). The IAEA has monitored the Zaporizhzhia plant against seven indispensable pillars of nuclear safety and security since March 2022, and against five concrete principles set out at the UN Security Council in May 2023 (IAEA, 2023). The agency's core conclusion is that the nuclear safety and security situation at the plant remains extremely fragile and unstable. Exposure is not confined to Europe. More than half of the countries expressing interest in new nuclear programmes rank at 'Warning' or higher on the Fragile States Index (Squassoni, 2024).

Nuclear offers land-use and grid-system advantages. Its compact physical footprint and high load factor can be valuable in regions where renewable development is constrained by competing land uses, biodiversity commitments, public acceptance, or limited interconnection capacity. Nuclear has the lowest land-use intensity of the generation technologies assessed in the peer-reviewed literature. Its median is 7.1 hectares per TWh a year, against 58,000 for dedicated biomass at the other end of a wide range (Lovering et al., 2022). Nuclear does not eliminate the need for transmission investment, and large units can themselves require substantial grid connections and system reserves. Building at existing nuclear sites and at retired thermal power sites nevertheless reuses established grid infrastructure. A US Department of Energy screening of 394 operating and retired coal plant sites found 80% of them suitable for advanced reactors below gigawatt scale. It estimated that reusing the existing infrastructure reduces construction cost by 15% to 35% (US DOE, 2022).

Nuclear provides firm generation, however its infrastructure must be protected against natural hazards and malicious attacks.

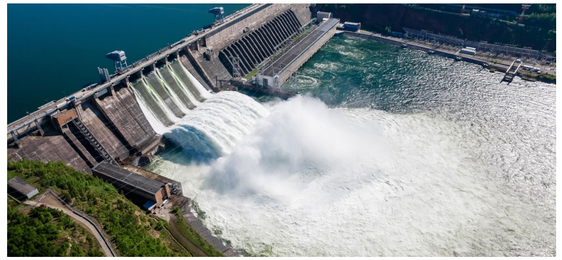
New contracting and support mechanisms could improve project bankability. Long-term power purchase agreements, contracts for difference, regulated-asset-base models, and direct public financing can provide greater revenue certainty and reduce exposure to volatile wholesale electricity prices. The IEA describes these instruments as indispensable in markets with volatile electricity prices, because lenders extend credit against expected future cash flow rather than against expected cost (IEA, 2025a). The two dominant structures differ in when consumers begin paying and in who carries construction risk. A contract for difference supports revenue only once the plant generates electricity, so overruns stay with the developer while consumers carry the gap between the strike price and the market price for the contract term. A regulated asset base brings consumer payment forward into construction, which shortens the period before equity earns a return and lowers the cost of capital, at the price of moving part of the overrun risk onto consumers. The UK has now used both, a contract for difference at Hinkley Point C and a regulated asset base at Sizewell C. The choice sets the cost of capital by deciding who is exposed to overruns (NAO, 2017; NAO, 2026).

These mechanisms redistribute risk rather than remove it. Hyperscaler power purchase agreements allocate risk in the same way as a contract for difference, paying only for delivered electricity, so construction cost and schedule risks remain with the developer. Most of the announced hyperscaler volume depends on projects that have not reached a final investment decision.

Conclusion

Our forecast excludes low-probability, high-impact events, including the deliberate targeting of nuclear facilities and associated infrastructure during armed conflict. However, in an era of growing geopolitical fragmentation, such risks cannot be ignored altogether. If concerns about the physical security of civilian infrastructure begin to influence public acceptance, financing conditions, or government policy towards nuclear power, we would need to recalibrate our Outlook.

Security considerations aside, the developments described in this section broaden the rationale for nuclear beyond climate mitigation. Nuclear's relevance rests on its combined contribution to decarbonization, energy security, system adequacy, spatial efficiency, and reliable power for the digital economy. These attributes strengthen the demand case, but do not eliminate the appeal of cheaper renewables twinned with batteries. Moreover, the supply response remains conditional on financing, industrial delivery capacity, fuel availability, licensing, and demonstrated cost reduction. In our forecast, those constraints explain why nuclear growth accelerates primarily after 2040 rather than providing a rapid response to near-term electricity-demand growth.



4.6 HYDROPOWER

Global primary hydropower electricity generation will increase by 50% between today and 2045. Two-thirds of this growth will occur in the next decade. From 2045 onwards, most of the economically viable hydropower potential will have been built and generation levels out. Hydropower continues to play an important role in the electricity mix as an emission-free power source and as a flexibility provider to the grid, contributing almost 36% of hourly flexibility needs in 2040 (Figure 4.13). However, environmental and geographical constraints on new hydro and rapidly declining battery costs make batteries the preferred option for most new storage investments after 2040.

Regional differences in hydropower growth

China generates around one-third of global hydropower today, but this will dip below 30% in 2060. Generation in China will grow 64% by 2035 and then level off. Hydropower is growing fastest in India, with generation doubling before 2035 and tripling by 2050. South East Asia is another region with strong hydropower growth, with generation doubling by the early 2040s. North America and Europe have already utilized most of their hydropower potential, and primary generation will not grow in these regions in our forecast period.

The role of hydropower in a more variable power system

Pumped hydropower can help to integrate growing shares of renewable generation into the grid and reduce curtailment by storing surplus energy and releasing it during periods of high demand. Surplus generation from solar, wind, and conventional hydropower is used to pump water to a higher altitude reservoir, effectively storing the energy for later use. In our modelling, the electricity used for pumping is counted as the primary energy input. As both electricity demand and variable renewable generation grow, power systems require increasing flexibility to balance fluctuations in supply and demand. We measure flexibility demand as the extent to which dispatchable sources must change their output hour by hour within a day to maintain system balance. Daily flexibility demand will grow 180% over our forecast period, with most of this growth occurring up to 2045. We find that hydropower (both conventional and pumped) will play an increasingly important role in this period (to 2045), with its share of flexibility provision rising from 24% to 36% of total flexibility demand – which itself is growing rapidly.

Higher flexibility demand creates new revenue opportunities for storage operators. Greater differences between periods of low and high electricity prices increase the value of storing electricity and shifting generation across time. Existing hydropower and pumped hydropower facilities are well positioned to benefit from these arbitrage opportunities. However, declining battery costs mean batteries become the preferred option for most new storage investments after 2040.

India illustrates this trend. National policy measures, including pumped hydro auctions and other policy incentives (IHA, 2026), are driving rapid deployment leading to a doubling of pumped hydro capacity in India over the next five years. China is following a similar path. Pumped hydro also fits with the ambitions in China's *15th Five-Year Plan*, which emphasizes grid flexibility and renewable integration. Capacity will double in China in the next five years and will grow by 240% by the early 2040s. However, in both regions, the cost per pumped hydro capacity provision rises as the more optimal locations are used up at the same time the cost of utility-scale battery storage continues to fall.

Hydropower delivers flexibility

Hourly flexibility supplied by technology within a day (GW)

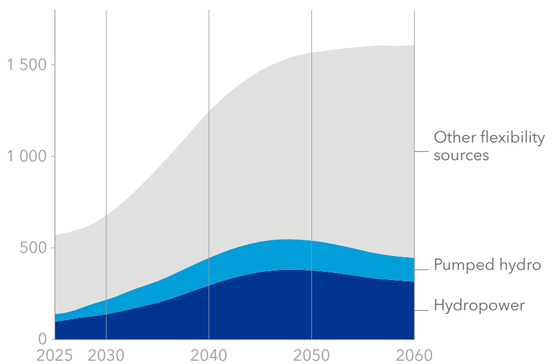


FIGURE 4.13 | Other sources include Li-ion batteries (incl. EV V2G), long duration storage, thermal power, nuclear, and power-to-hydrogen.

Climate risks and environmental constraints

Other factors also constrain hydropower uptake. Climate-driven shifts in weather patterns, including prolonged droughts in southern Africa, have increased uncertainty over hydropower availability and future deployment. This vulnerability directly threatens energy security in countries that depend heavily on hydropower. Zambia, for example, suffered a severe energy crisis after severe drought in 2024 and is now rapidly investing in solar to diversify its power mix (Down to Earth, 2024)).

Changing weather patterns also alter the investment case for hydro as investors face greater uncertainty regarding future project performance. Mounting environmental and social considerations add a further constraint. New hydropower projects increasingly face opposition related to ecosystem impacts, biodiversity, and water management. Projects may achieve a better profile when they are framed as broader water infrastructure projects that demonstrate they can support irrigation and flood control. However, environmental and social trade-offs, such as biodiversity disruption and community displacement, remain potential showstoppers for traditional newbuild dam projects.

4.7 GEOTHERMAL POWER

From approximately 0.3% of global electricity generation in 2026, we forecast geothermal power will increase to around 0.4% by 2060. While this remains a relatively small share of global electricity supply, it represents a three-fold increase in absolute generation over the period, in contrast to a roughly 2.5-fold increase in total electricity generation.

This expansion is driven by a fundamental technological transition within the geothermal sector. Historically, geothermal power deployment has been almost entirely limited to locations with favourable geological conditions. Over the coming decades, we expect enhanced geothermal systems (EGS) broaden the accessible resource base significantly. Declining costs, operational learning, and advances in horizontal drilling technologies further support deployment.

Geothermal power therefore evolves from a geographically constrained resource to a more scalable source of firm, low-emissions electricity.

Beneath the surface

Geothermal power converts underground thermal energy into electricity. However, geothermal technologies differ substantially in their dependence on local geological conditions and their level of technical maturity.

Three distinct geothermal technologies are currently emerging.

Hydrothermal geothermal exploits naturally occurring geothermal reservoirs where permeable hot rock and groundwater coexist. Such resources are geographically limited and largely concentrated in tectonically active regions. Significant hydrothermal developments exist in the US, Kenya and Ethiopia along the East African Rift System, and Indonesia's volcanic geothermal belt (Star Energy Geothermal, 2026).

Enhanced geothermal systems (EGS) exploit hot rock formations where sufficient heat exists but natural permeability is inadequate. Reservoir permeability is increased through engineered stimulation, to increase fluid circulation and extract geothermal heat. Commercial deployment remains at an early stage, although several projects are under development in North America, including Fervo Energy's Project Red (Fervo Energy, 2026).

Advanced geothermal systems (AGS) are the least mature category. These systems utilize closed-loop configurations that circulate fluids through engineered underground infrastructure and are largely independent of natural reservoir permeability.

The evolution from hydrothermal systems toward EGS and, eventually, AGS substantially expands the technically accessible geothermal resource base.

Drivers of future growth

We foresee a larger role for geothermal power for several reasons:

- Its strategic value as a firm, dispatchable, and low-emission source of electricity and heat.
- It provides continuous energy independent of weather conditions.
- It complements variable renewable resources such as wind and solar.
- Rising electricity demand in regions where geothermal resources already exist, particularly North America.
- Strong policy support in key markets where geothermal power is increasingly viewed as a secure domestic energy resource.
- Growing demand for 24/7, carbon-free electricity from corporate and industrial power purchasers.
- Significant technological overlap with the oil and gas sector that allows geothermal developers to leverage existing expertise, infrastructure, and skilled labour.

Collectively, these factors support both accelerated deployment and continued cost reductions across geothermal technologies.

Geothermal growth differs sharply by region and technology
Installed capacity of geothermal power (GW)

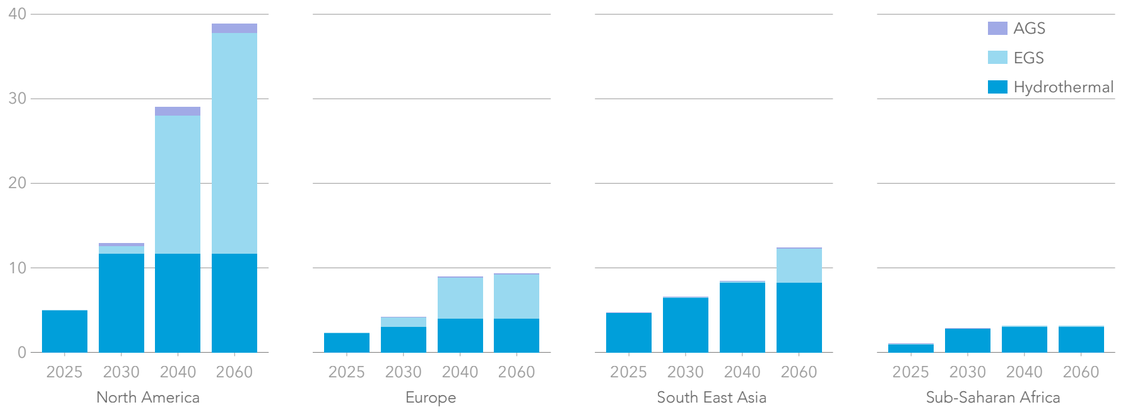


FIGURE 4.14 | AGS – Advanced Geothermal Systems; EGS – Enhanced Geothermal Systems. Historical data source: GlobalData (2026).

Capacity expansion

Global geothermal power capacity increases from 23 GW in 2026 to 95 GW by 2060. More than half of this capacity is concentrated in North America, Europe, Sub-Saharan Africa, and South East Asia.

In 2026, almost all installed geothermal capacity is hydrothermal and therefore dependent on highly favourable geological conditions. By the 2040s, EGS begins to gain market share as technological improvements reduce costs and expand the range of economically developable resources.

Hydrothermal geothermal continues to dominate in regions with exceptional natural resource endowments, particularly Sub-Saharan Africa and South East Asia. In contrast, EGS becomes the primary source of incremental geothermal capacity growth in North America and Europe where the availability of undeveloped hydrothermal resources is more limited.

Cost dynamics drive the technological transition

Cost reductions are the primary catalyst for the transition from hydrothermal geothermal to EGS. EGS is currently the highest-cost commercially available geothermal technology, but costs will decline rapidly, driven by learning effects, technology maturation, and the application of advanced drilling techniques developed within the oil and gas industry. By the early-to-mid 2040s, EGS becomes the lowest-cost geothermal option in both North America and South East Asia.

The transition occurs earlier and more decisively in North America. Hydrothermal costs rise over time as the highest-quality resources are progressively developed, increasing the attractiveness of EGS deployment. In South East Asia, abundant volcanic hydrothermal resources maintain a relatively low hydrothermal LCOE throughout the period, delaying but not preventing the emergence of EGS as the most competitive geothermal technology.

Advanced geothermal systems also experience cost reductions, but they remain substantially more expensive than both hydrothermal geothermal and EGS through 2060.

EGS costs decline rapidly compared to hydrothermal power
Levelized cost of electricity (US¢/kWh)

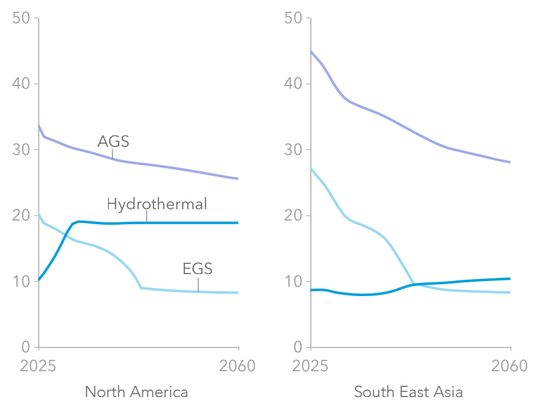


FIGURE 4.15 | AGS - Advanced Geothermal Systems; EGS - Enhanced Geothermal Systems.

4.8 FOSSIL-FIRED POWER

Fossil-fired power is central to today's energy system, supplying 56% of global electricity generation. However, as renewables, grids, and storage expand, fossil sources shift from providing bulk electricity to delivering flexibility and backup capacity. Constrained by economics and climate policy, their share of global electricity generation declines to just 5% by 2060.

From bulk generation to system flexibility

Fossil-fired power remains an important part of today's power system, but its role is changing from supplying baseload generation to providing flexibility and system reliability. This shift is driven by the expansion of renewables and battery storage, which become increasingly cost-competitive. In addition to supplying low-cost, clean power, renewables reduce exposure to fuel price volatility and strengthen the security of supply, particularly in import-dependent regions. Gas-fired power plants remain valuable due to their ability to ramp up quickly, making them well suited to balance intermittent renewables.

Over the coming decades, fossil-fired generation declines much faster than installed capacity, meaning fewer operating hours and falling utilization rates. This increases the cost per unit of electricity. By 2060, fossil-fired generation declines by 79%, while installed capacity increases by 28%, with gas-fired plants accounting for most new fossil capacity additions.

Fossil-fired power shifts from the backbone of electricity supply to a marginal source in most regions

Share of fossil-fired power in electricity generation by region

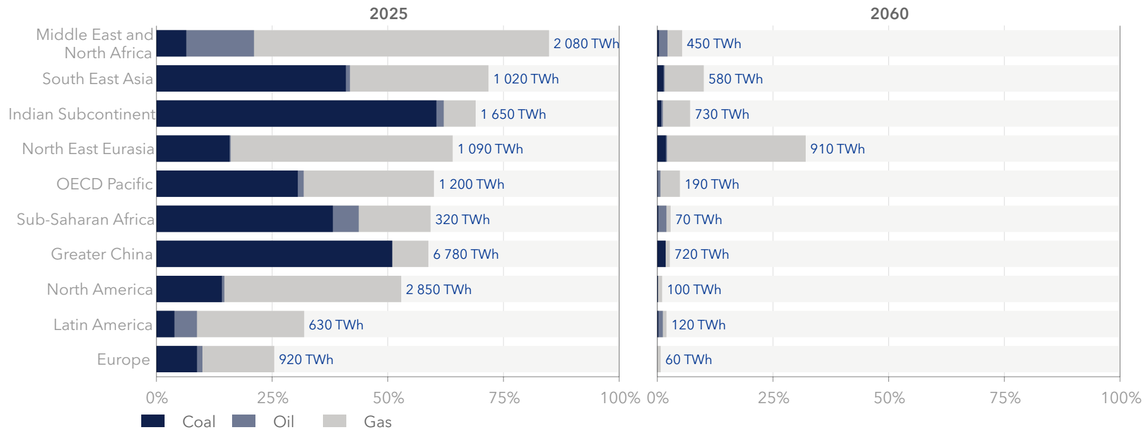


FIGURE 4.16 | Generation includes behind-the-meter and off-grid.

Reliability concerns sustain demand for dispatchable power

Despite declining fossil-fired generation globally, concerns about reliability continue to support investments in dispatchable technologies. Geopolitical instability, extreme weather events, and the rapid expansion of AI data centres are increasing the value of reliable, firm backup power. As a result, both diesel generators and gas-fired generation continue to attract investment even as renewables gain market share.

Diesel generators remain widely used in regions with weak or unreliable grids and are increasingly deployed as backup power for critical infrastructure, including data centres. Their appeal lies in their ability to provide electricity on demand. However, they are expensive to operate, inefficient, and highly polluting. Falling solar and battery storage costs, combined with higher fuel prices, are making cleaner alternatives increasingly attractive, particularly for off-grid and backup applications. Nevertheless, reliable supply is often prioritized over cost and environmental considerations and diesel generators continue to expand in some markets, especially as backup power for data centres (Harder, 2026).

Similar dynamics are supporting investment in gas-fired generation. In North America, rapidly growing electricity demand from data centres has renewed interest in gas-fired power projects because they can provide firm and flexible capacity. However, strong global demand for gas turbines is creating manufacturing bottlenecks, leading to longer lead times and spiralling equipment costs.

OEMs are prioritizing limited slots to high-margin, lower-risk buyers in developed markets, especially the US, where capacity is urgently needed (Infrastructure Investor, 2025). As a result, developers in emerging economies face growing challenges in securing gas turbines and financing new projects. Rising costs and supply constraints are encouraging developers to accelerate investment in alternative technologies, strengthening the business case for utility-scale solar and batteries.

Regional differences

Dependence on fossil-fired power persists longest in emerging economies, making the global transition uneven across regions. Advanced economies have largely passed peak fossil generation, with ambitious coal phase-out targets and rapid renewable expansion driving the decline. By 2060, only 5% of global electricity will be supplied by fossil fuels and fossil-fired power becomes a marginal source of electricity in most advanced economies.

Emerging economies remain more dependent on fossil generation due to rapid demand growth, domestic resource availability, and slower deployment of renewables in some regions. Greater China alone accounts for 57% of global coal-generation today, while we expect India's coal use to peak in the early 2030s. The Middle East and North Africa, North East Eurasia, the Indian Subcontinent, and South East Asia remain heavily reliant on fossil-fired power in the near to medium term.

4.9 POWER GRIDS

Grid Bottlenecks: Beyond Permitting

Electricity grids are assets no longer being taken for granted as mature infrastructure. They are becoming one of the fastest-growing and most complex aspects of the energy system. Countries are on the path of expanding grids at a rate unmatched in recent decades, driven by the need to connect new generation, accommodate fast-growing demand, strengthen system resilience, and increase cross-border power flows. This expansion is occurring simultaneously across many regions. As demand for cables, transformers, substations, power electronics and specialized labour surges, pressure is mounting across the entire value chain. Permitting remains a major obstacle, but approvals alone will not deliver faster grid expansion. Once projects enter construction, they increasingly encounter a second layer of constraints: limited manufacturing capacity, long equipment lead times, shortages of specialized workers, and fragmented procurement.

The supply chain is running at capacity

The most severe equipment constraints concern large power transformers and high-voltage cables. Large transformers can now take up to four years to procure outside China, while standard transmission cables typically require two to three years (IEA, 2025a). For specialized HVDC cables used in long-distance and cross-border links, waiting times can exceed five years. Since 2019, cable prices have nearly doubled in real terms and transformer prices have risen by around 75% (IEA, 2025e).

These are not interchangeable, mass-produced components. Transformers are frequently designed for specific voltage, capacity, and network requirements. HVDC cables and converter equipment depend on a small group of qualified manufacturers. Production also relies on constrained inputs, including grain-oriented electrical steel, copper, aluminium, and specialized insulation materials. New factories have been announced, but expanding output requires substantial capital, trained workers, qualification processes, and a credible long-term order book.

Labour shortages are specialized, not general

The labour constraint is not simply a shortage of construction workers or electricians. Grid projects require power-system planners, protection and control engineers, relay technicians, high-voltage cable jointers, substation commissioning specialists, tower and line crews, and workers capable of manufacturing and testing large transformers.

Around 8 million people currently construct, operate, and maintain grids worldwide. At least 1.5 million additional workers will be needed by 2030, an increase of almost 19% (IEA, 2025e). This constraint is difficult to resolve quickly because many roles require several years of technical training and experience on live power systems. In a 2025 European industry consultation, 96% of participants said skilled-workforce shortages were already slowing progress towards energy-transition targets (ENTSO-E et al., 2025).

The bottleneck differs by region

The exposure of ten regions is unlikely to be uniform. Europe and North America face particularly strong pressure from ageing assets, simultaneous network programmes, specialist-labour shortages, and dependence on long-lead equipment. Greater China has deeper domestic manufacturing capacity and extensive experience with high-voltage network construction, although the scale of expansion creates continuing pressure on materials and factories. The Indian Subcontinent and South East Asia benefit from expanding domestic supply chains, but remain exposed to shortages of advanced transmission equipment and engineering capacity. In Sub-Saharan Africa, financing, utility capability, procurement scale, and access to equipment can reinforce one another

The grid-equipment waiting list

- **Transmission cables:** 2–3 years
- **Large power transformers:** up to 4 years
- **HVDC cables:** more than 5 years
- **Cable prices:** nearly +100% since 2019
- **Transformer prices:** approximately +75% since 2019

Where labour becomes binding

- **Protection and power-system engineers:** long training and experience requirements
- **Cable jointers and HV commissioning staff:** small, mobile specialist workforce
- **Lineworkers and substation technicians:** shortages rise when construction accelerates
- **Transformer manufacturing workers:** production cannot expand through capital alone
- **General construction labour:** important, but usually not the decisive constraint

to make apparently small projects difficult to deliver. These regional ratings should be presented as structural exposure, rather than as measured project-delay statistics.

The emerging problem is therefore one of delivery capacity. Securing a permit does not create a factory slot, a protection engineer, or a cable-laying crew. Governments and grid operators will need to coordinate project pipelines, standardize equipment where possible, make longer-term procurement commitments, expand technical training, and develop regional supply chains. Otherwise, faster permitting may move projects quickly through the first gate only to leave them waiting at the next one.

Physical infrastructure

Despite short-term constraints, particularly in Europe and North America, where permitting and supply chain delays continue to slow project delivery, we project sustained and widespread grid expansion through to 2060 (Figure 4.17). Transmission line length is set to nearly triple, driven by rising electrification, changing spatial patterns of supply and demand, and the operational needs of variable renewables. While the bulk of new build occurs in Asia, high-income regions also see renewed growth after a period of stagnation. The growth in high-income regions is linked to structural changes in energy use (i.e. electrification) rather than growth in overall demand.

Distribution networks – moving power from the high-voltage transmission grid to consumers – will follow a similar trajectory, doubling in length by 2050. In low- and middle-income regions, this growth is driven by new connections and spatial expansion. In high-income regions, it reflects asset renewal and the expanding role of distribution in system flexibility and balancing. As investment needs grow, attention is turning to how costs are recovered. This places network charges at the centre of the debate on financing the energy transition and fuelling innovation and discussion on new ways to finance grid expansion beyond the rather crude mechanism of utility charges.

Historically, power grid capacity has grown in line with peak electricity load; peak demand has generally served as a reasonable proxy for grid capacity need. But this relationship is changing. The location of solar and wind power sites, which hold a growing share of capacity additions, are dictated by natural resources which are usually farther away from the demand centres than the conventional power stations. Moreover, the average size of a solar or wind project is smaller than most nuclear, hydropower, gas, or coal projects and requires more cables per unit capacity, including equipment capable of handling bi-directional flows. Recent grid expansion has also been driven by the broader need to strengthen grids – both within countries and

Region	Equipment	Skilled labour	Planning and delivery capability
North America	Material constraint	Potentially binding	Material constraint
Latin America	Material constraint	Material constraint	Material constraint
Europe	Potentially binding	Potentially binding	Material constraint
Sub-Saharan Africa	Potentially binding	Potentially binding	Potentially binding
Middle East and North Africa	Material constraint	Material constraint	Material constraint
North East Eurasia	Material constraint	Material constraint	Material constraint
Greater China	Emerging constraint	Emerging constraint	Generally manageable
Indian Subcontinent	Material constraint	Material constraint	Material constraint
South East Asia	Material constraint	Material constraint	Potentially binding
OECD Pacific	Material constraint	Potentially binding	Emerging constraint

Generally manageable · Emerging constraint · Material constraint · Potentially binding

Power grids are set to grow by 110% in length by 2060, led by growth in Asia
Transmission and distribution power line length by region (Million circuit-km)

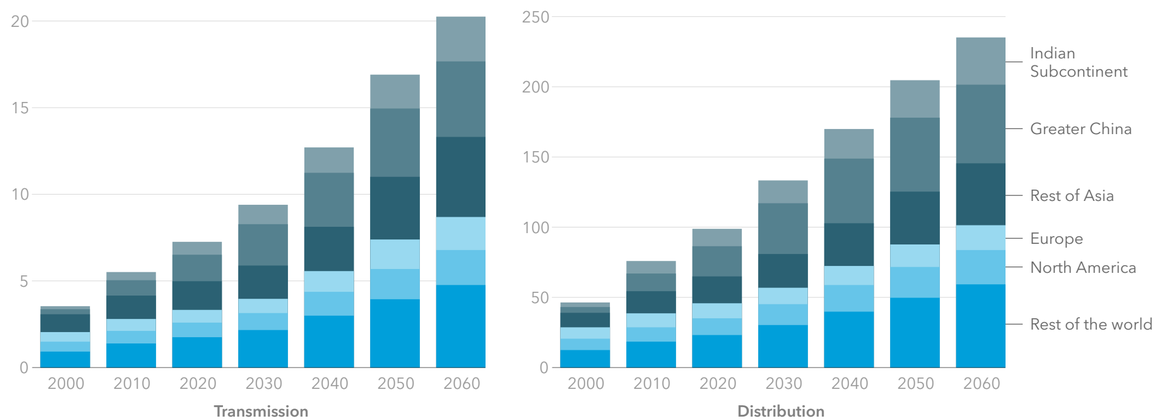


FIGURE 4.17 | Historical data source: GlobalData (2025)

across borders – to ensure system stability under greater variability from new sources of concentrated demand like EV charging, data centres, and heat pumps. Furthermore, in regions with older grids, grid hardening against natural calamities, such as wildfires, is an urgent requirement (Think Power Solutions, 2025).

Countries are addressing grid-strengthening needs through a combination of domestic transmission expansion, cross-border interconnections, and regulatory coordination. Germany is investing heavily in the SuedLink and SuedOstLink HVDC corridors to transport renewable electricity from the north to industrial demand centres in the south (Bundesnetzagentur, 2024). The UK is expanding offshore transmission infrastructure through the Electricity System Operator's strategic network plans and interconnectors with Norway (North Sea Link) and continental Europe (NESO, 2025). France, Spain, and Portugal have increased interconnection capacity across the Pyrenees to improve market integration and system resilience (European Commission, 2023). The Nordic countries have long relied on extensive cross-border transmission through Nord Pool and projects such as NordLink between Norway and Germany (ENTSO-E, 2024). In North America, the US regional transmission organizations are advancing multi-state transmission plans, while Canada and the US continue to expand cross-border electricity trade (DOE, 2026). China has built a nationwide ultra-high-voltage (UHV) network to transfer power over long distances from resource-rich western regions to coastal demand centres (IEA, 2023).

Expenditures and network charges

As of 2026, we expect global investment in electricity grids to exceed USD 500bn/yr (IEA 2026c), covering transmission, distribution, digital infrastructure, and grid support functions. With growing network length and complexity, total expenditures are rising. We project combined investments and operating costs to exceed USD 1trn/yr in the 2030s, and close to USD 1.4trn/yr by the 2050s (Figure 4.18). Operating expenditure (OPEX) is set to double from USD 300bn to USD 600bn annually. Transmission accounts for 30–35% of total investment.

These costs are ultimately passed on to end users. In addition to investment recovery and OPEX, system operators increasingly charge for a broader set of services:

- Ancillary services such as frequency regulation, spinning reserves, inertia support, and black start capability
- Capacity payments to ensure firm supply in systems with falling price duration curves
- Loss compensation, congestion management, and curtailment costs
- Grid access and connection charges, particularly for distributed energy resources

As these components grow in cost and coverage, unit grid charges are rising faster than electricity demand. We project the global average grid charge will increase from USD 45/MWh in 2025 to USD 76/MWh by 2060 (Figure 4.19). Three quarters of this increase is driven by expanding capacity payments that are needed to ensure stable revenue streams in systems increasingly dominated by zero-marginal-cost generation. As solar and wind set wholesale prices for longer durations, market revenues alone will no longer suffice to cover total system costs. This

Grid expenditures almost double by 2050s

Average annual grid expenditures by decade (Trillion USD/yr)

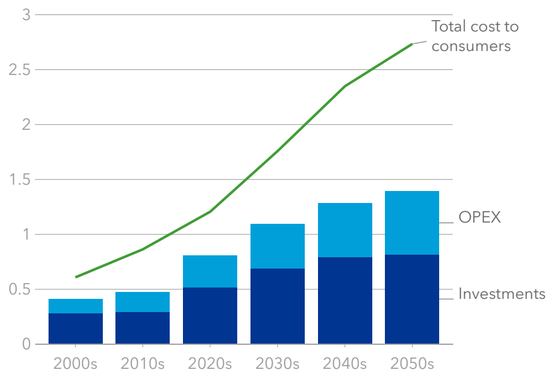


FIGURE 4.18 | Each bar shows average yearly spending for each decade.
Historical data source: DNV estimates based on IEA, GlobalData

Rising grid charges will absorb much of the benefit from falling wholesale prices

Average grid charge globally and by region (USD/MWh)

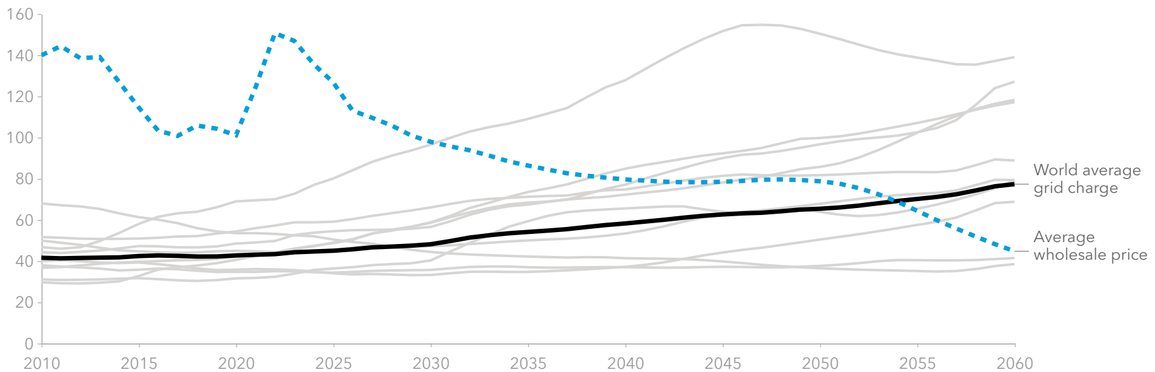


FIGURE 4.19 | Each grey line shows the average grid charge by region, averaged across residential and industrial users.
Historical data source: DNV estimates based on data from ENTSO-E (2025), EIA (2025), IEA (2024) and local sources

will require reform and redistribution, much of which will be reflected in end-user tariffs.

While grid charges rise by USD 30/MWh to 2060, wholesale electricity prices fall by USD 75/MWh over the same period. This means that globally, nearly half of the cost benefit of switching to a renewable-based power system will be offset by rising grid-related charges. As we highlight in our sidebar discussion on financing a 50-year asset, how those costs and benefits are distributed is an increasingly important aspect of energy policy.

The probable outcome is a hybrid model: longer asset lives, continued consumer charges, rising adequacy payments, more capacity-based tariffs, selective generator contributions and public support for investments with wider societal benefits. This will spread the cost more widely and defer part of it, but will not reduce the underlying bill. The operating and fuel costs of the largely combustion-based energy system that electrification is replacing must be weighed against these costs. Over time, the shift from an OPEX-dominated to a CAPEX-dominated system can make higher upfront investment costs economically acceptable, even before accounting for the benefits of avoided climate damages.

Nearly half the savings from renewables are offset by rising grid charges.

Grid financing: paying for a 50-year asset

A new grid asset may operate for 50 to 70 years, but somebody must finance its construction now. Recovering the cost over 20 to 30 years limits interest costs, but makes current consumers pay heavily for infrastructure that future consumers will use. Extending recovery towards 40 to 50 years lowers near-term bills, but keeps capital outstanding longer. The UK, for example, has moved towards a 45-year regulatory asset life for new network investment. The likely response to sustained investment will be longer repayment periods (Ofgem, 2024). This is politically easier today, but liable to leave future consumers paying returns on an ever-larger asset base.

Who pays will also change. Consumers currently bear most costs through network charges, although grids deliver wider benefits such as cleaner air, energy security, and industrial development. Public budgets and state guarantees will therefore finance more strategic projects, particularly infrastructure built ahead of identifiable demand. This will likely be selective rather than a wholesale shift from consumers to taxpayers.

Generators may increasingly be required to contribute to network costs. ACER (2025) has recommended that regulators assess generator charges rather than recovering costs exclusively from consumers, while Germany is explicitly considering feed-in charges in its redesign of network tariffs (BNetzA, 2026). Charging generators according to capacity, or reinforcement they trigger, could improve siting decisions and shift some of the costs from end users. However, generators may respond through higher required power prices, relocation, or reduced investment. Generator charging changes incentives and the initial incidence of costs, and hence does not guarantee a lower final consumer bill.

Future grid charges will become less dependent on electricity consumed. Grid investment has been driven chiefly by capacity at peak times, so regulators are moving towards charging capacity, peak demand and time-of-use, rather than pure consumption (ECRB, 2026). Bills will include larger capacity or adequacy charges, paying generators, storage, and flexible demand to be available when variable generation is scarce. These charges do not finance wires, but they will become an increasingly visible part of the overall cost of maintaining a reliable electricity system. They also carry the risk of paying for too much backup if they are designed poorly.

Network tariffs typically bundle together investment, maintenance, losses, system operation, and resilience. Separating these components would make cost increases more transparent; losses are affected by electricity prices and network use, while redundancy is essentially an insurance cost. Most major grids in the world follow the N–1 principle, meaning they should withstand the failure of one important component. There is no universal redundancy percentage because the spare capacity required varies by network. However, a blanket N–1 engineering will face greater scrutiny in locations where storage, flexible demand, or curtailment can actually provide resilience more cheaply. N–1 orthodoxy is also likely to come under pressure as digitalization and AI improve grid visibility, forecasting, and system management.

Zonal pricing could go further by giving different areas different wholesale prices. It may reduce congestion and steer generation and large consumers towards better locations, lowering future grid investment. Its disadvantages are also real: regional winners and losers, more volatile revenues, and higher investment risk. Governments are therefore more likely to strengthen locational network charges than embrace radical zonal market reform.

4.10 SYSTEMS, FLEXIBILITY, AND STORAGE

- **Total grid-connected storage expands from 2.7 TWh to 45 TWh by 2060.**
- As power systems integrate growing shares of variable renewables and shift away from large central thermal power plants, storage becomes increasingly important to balancing supply and demand across daily, seasonal, and weather-based fluctuations.
- **Li-ion batteries surpass pumped hydro to become the most dominant form of storage; from 16% of energy capacity in 2025, to a little more than half by 2030, and reaching almost 85% by 2060. At the same time, short-duration storage becomes more prominent than longer-duration storage, like pumped hydro.**
- Li-ion batteries proliferate in many configurations: as stand-alone and grid-connected; co-located with solar and grid-connected; distributed and stationary in the form of behind-the-meter batteries; and distributed and mobile, e.g. as vehicle-to-grid (V2G).
- This mass-scale adoption of Li-ion batteries is made possible by a continuous decline in manufacturing cost. Globally, battery CAPEX costs are down 20% from 2020 to 2026, and we expect their costs to decrease by about 30% from 2026 to 2040.
- **Distributed storage, in the form of behind-the-meter (BTM) storage and V2G grows, and with it the attractiveness of virtual power plants.**
- BTM batteries co-located with solar systems, both in residential and commercial buildings/premises grow from 120 GWh in 2026 to 1.8 TWh by 2040, and 5.4 TWh by 2060. In 2060, BTM batteries have almost 12% of the energy storage capacity as total grid-connected storage.
- V2G batteries grow from minuscule amounts in 2026 to about 2.5 TWh by 2040 and 24 TWh by 2060, a little more than half of total grid-connected energy storage capacity globally.
- This distributed storage growth is aided by two key factors: heavy digitalization, including virtual power plants (software-controlled networks of distributed energy resources), and better consumer-facing pricing/market models which enable consumers to become active participants in the electricity market.
- **As the power system shifts from centrally planned thermal-heavy power systems to renewable-based systems with hourly variability, flexible demand becomes as important as or more important than supply.**
- As the need for flexibility grows, supply starts changing globally from thermal power plants supplemented by hydro and pumped hydro to battery storage capacity, both short-duration (Li-ion batteries) and long-duration storage.

Li-ion batteries surpass pumped hydro by 2028
Share of storage technology (%)

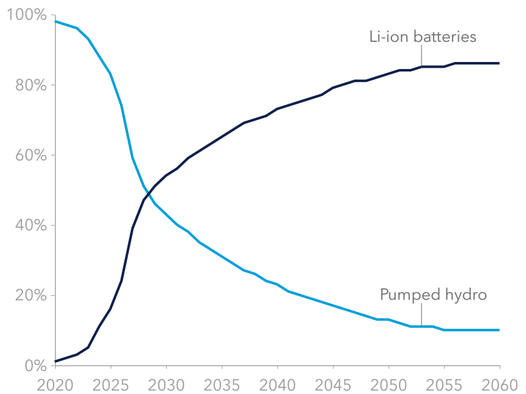


FIGURE 4.20 | Historical data source: GlobalData (2026)

As more solar and wind capacity is added, more storage is needed in the grid
Installed capacity (GW); Share of storage relative to variable renewable energy (%)

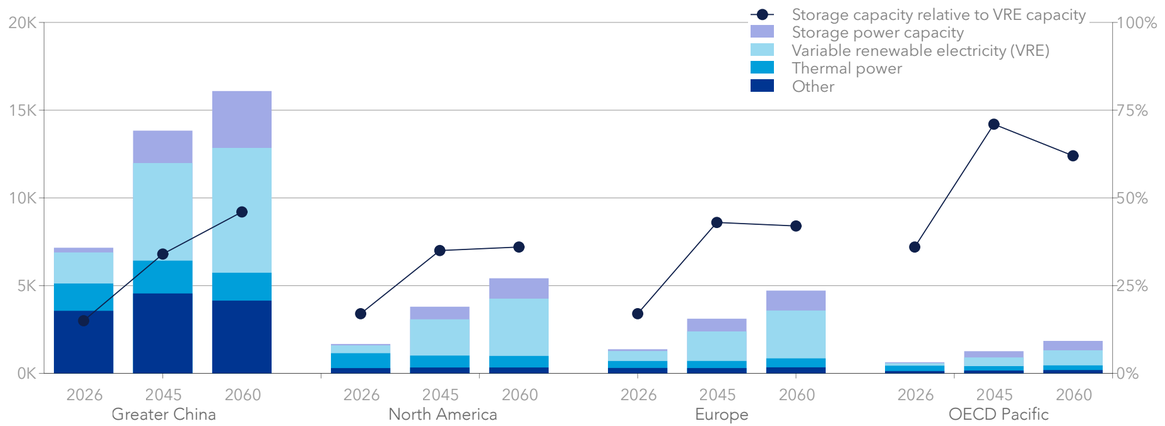


FIGURE 4.21 | Other includes hydropower, nuclear, and geothermal. Historical data source: GlobalData (2026)

Energy storage: from nature-constrained systems to fast-acting batteries

Pumped hydro has long dominated utility-scale storage, accounting for nine-tenths of global capacity in 2025. However, its expansion is increasingly constrained by geography and environmental considerations. Its share of utility-scale storage declines to one-half by 2030 and one-tenth by 2060 as the economics of Li-ion batteries continue to improve.

Li-ion batteries are quickly becoming the leading source of short-duration flexibility. They provide rapid response, firm capacity (typically up to four hours), frequency regulation, and reserve replacement. Batteries already represent almost four-fifths of global storage additions in 2026. By 2060, global Li-ion capacity reaches 58 TWh, with a substantial share co-located with solar to shift midday generation into evening demand peaks.

Falling costs reinforce this transition. Average global CAPEX of Li-ion batteries, including battery energy storage system (BESS) equipment, declines from some USD 160/kWh in 2026 to USD 100/kWh in the early 2040s (Figure 4.22).

Near-term deployment in certain regions may be slowed by geopolitical risks, lack of access to critical minerals, and supply chain volatility (IEA, 2024). A recent example is China's introduction of export controls on cathode technology that expose the vulnerability of battery supply chains in Europe and North America (Laha, 2025).

However, North America, especially the US, has significantly expanded manufacturing capacity, from cell-level manufacturing to pack-level assembly (Spector, 2026). Although much of this capacity is still ramping up, we expect it to relieve the pressure of sourcing batteries needed to accommodate expanding electricity demand. However, we estimate this domestic manufacturing strategy carries a premium of around 12% relative to comparable regions, such as OECD Pacific.

While alternative chemistries such as sodium-ion are emerging, we expect Li-ion batteries to remain dominant due to their established manufacturing ecosystem, scale advantages, and market maturity. Given the similar applications of Li-ion and sodium-ion technologies, current ETO battery projections should be interpreted as representing both.

Distributed and mobile batteries

Battery flexibility extends beyond utility-scale systems. Distributed battery systems adopted behind-the-meter by residential and commercial customers are an increasingly important storage resource.

These solutions are often attractive because they are economically beneficial to the customer or because consumers want a measure of control over their energy production and consumption patterns.

The levelized cost of residential prosumer electricity in three key regions (Figure 4.23) shows that residential battery-coupled solar systems are becoming more cost-effective over time, particularly in regions with high residential electricity prices and time-of-use tariffs.

The value proposition for prosumers depends on electricity tariffs, export compensation, incentives, financing, and battery and solar costs. Beyond cost, adoption also depends on customer willingness to invest, access to capital, and opportunities to participate in prosumer and flexibility markets and the infrastructure to support them.

Another distributed storage technology is V2G technology, where EV batteries act as mobile storage, returning electricity to the grid during weather-related scarcity and demand peaks. We forecast V2G capacity to exceed 10% of global storage energy capacity by 2029 and to surpass pumped hydro by 2040. By 2050, V2G reaches one-fifth of global storage capacity and maintains that share through 2060.

V2G and BTM are part of a broader portfolio of prosumer resources that includes technologies capable of demand response. When aggregated as virtual power plants, distributed solar, batteries, and, on the demand side, smart appliances can be coordinated through advanced software to operate collectively and respond to grid needs in real time.

Li-ion battery costs have reduced in the last three years
CAPEX of Li-ion battery systems (USD/kWh)

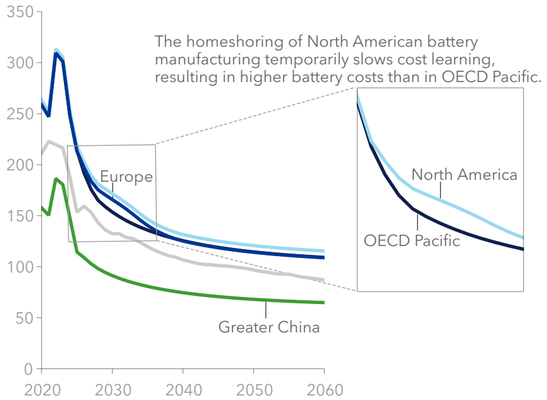


FIGURE 4.22 | CAPEX includes full stack costs, and support and tariffs.

Residential battery-coupled solar systems are becoming more cost-effective over time
Levelized cost of electricity (USD/MWh)

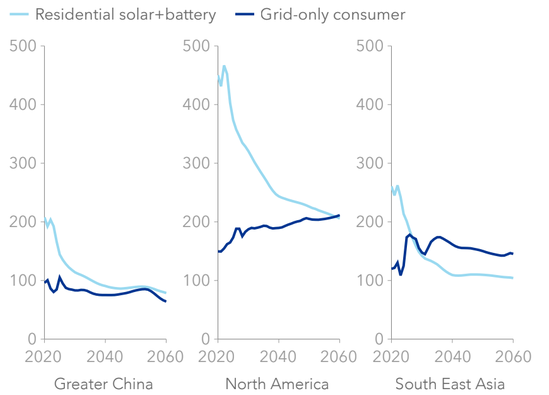


FIGURE 4.23 |

4.11 ECONOMICS AND MARKET DESIGN

The central economic challenge of the energy transition is shifting. Solar and wind are becoming increasingly competitive sources of new electricity, but their growing share is changing when electricity has value and where system costs arise.

As solar and wind energy penetration increases, market revenues become less closely aligned with generation costs and more dependent on the availability of flexibility that can absorb or shift output to periods of higher value. At the same time, reliability depends increasingly on storage, firm capacity, ancillary services, and stronger networks. The economics of electricity are therefore moving from the cost of producing a kilowatt-hour towards the cost of integrating, balancing, and delivering electricity when and where it is needed.

In this section, we examine the widening gap between LCOE and capture price and the principal responses available to developers and policymakers. We then show how these responses contribute to a broader shift in electricity expenditure from wholesale energy towards adequacy, flexibility and infrastructure.

Widening gap between LCOE and capture price

The dramatic decline in solar PV and onshore wind costs has made them among the cheapest sources of new electricity globally. However, while LCOE remains an important metric, it is no longer sufficient to assess project viability or system value. This is because LCOE does not account for when electricity is produced or the price it receives in the market, defined as 'capture price'. As variable renewable energy penetration increases, solar and wind generators often earn lower capture prices than the average wholesale market price. This 'cannibalization effect' occurs because large volumes of generation may arrive simultaneously, depressing prices during periods of high output. In some cases, it may take years for demand to adapt to new price patterns and take advantage of periods with low prices. Australia is providing an early example of this with its 'Solar Sharer' scheme launched in July 2026 to induce householders to use electricity in the middle of the day when solar power is abundant (Galvin, 2026).

The gap between LCOE and capture prices widens as solar and wind penetration rises. Consequently, the lowest-cost project is not always the highest-value project. For example, the upfront costs of an offshore floating wind project might seem prohibitive, yet its economics may rest on a wind regime that generates strongly during periods of high demand and limited solar output that allow it to capture disproportionately high market value.

Solar and wind see declining capture prices and stagnating LCOEs

Levelized cost of electricity and capture price (US¢/kWh); Share in utility-scale generation (%)

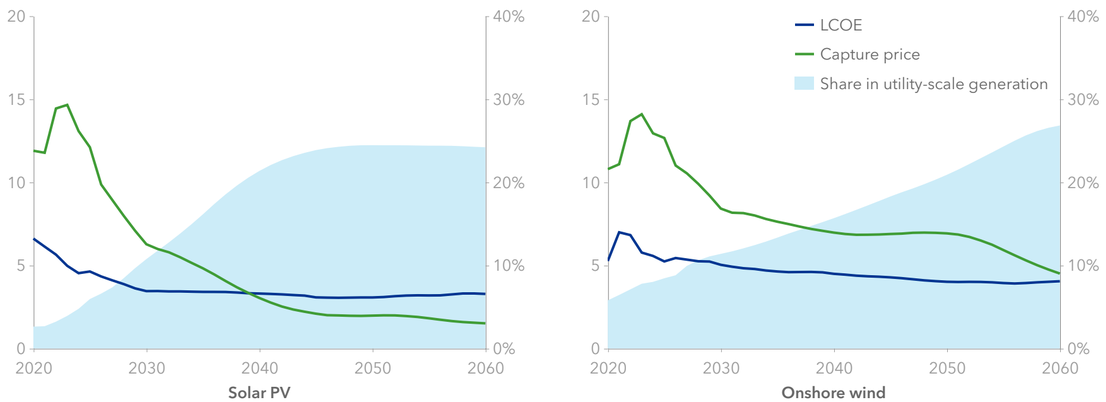


FIGURE 4.24 | Historical data source: IEA WEB (2026).

Solar and wind LCOEs continue to decline over the outlook period because of ongoing technology improvements and cost reductions. However, as the share of variable renewable generation increases, capture prices fall more rapidly than costs due to growing cannibalization effects. For solar, where its share of global utility-scale generation rises to around one-quarter by mid-century, the concentration of output during daylight hours increasingly depresses market prices and causes capture prices to fall below LCOE by the late 2030s.

Our forecast explicitly accounts for value erosion as solar PV penetration increases. Consequently, investment decisions are not driven by LCOE alone. Instead, technologies compete based on the relationship between their levelized costs and the capture prices they can earn, with investment flowing towards options that deliver the strongest value-adjusted economics. As solar PV deployment rises, its capture price gradually declines relative to average capture prices, reducing the attractiveness of additional capacity. As a result, solar PV's share of global utility-scale electricity generation plateaus at about 25% from the 2040s.

Wind experiences similar value erosion as deployment expands, although the decline is more gradual because wind generation is more temporally diverse. These trends highlight that project competitiveness is increasingly determined not only by cost reductions, but also by the ability to deliver electricity when and where it is most valuable to the system.

Declining BESS partially offsets this effect by enabling increasing volumes of low-cost solar and wind generation to be shifted into higher-value hours. However, widespread storage deployment also reduces arbitrage opportunities over time as the shape of electricity prices adjusts to a new market equilibrium.

Three responses to capture-price erosion

Developers and policymakers have three broad responses to declining capture prices: shift electricity supply to higher-value hours; monetize capacity, flexibility, and other system services; and replace part of *energy-only price* exposure with contracted revenues. Each response improves revenue resilience, but none removes the underlying system cost of integrating variable renewables.

1. Shift electricity supply to higher-value hours

The most common example of shifting supply hours of zero-marginal cost electricity generation, such as solar and wind, to higher value hours is through BESS, whether it is through stand-alone grid-connected systems or co-located with solar power.

The most common example – a BESS co-located with solar – can be observed in many regions, especially North America, Greater China, and OECD Pacific. However, solar+storage power plants also have higher LCOEs and if a region has higher penetration of solar+storage power plants, their time arbitrage value also erodes (Figure 4.25).

2. Monetize adequacy, flexibility, and ancillary services

Storage and other flexible resources can earn revenues beyond the energy-only market by providing capacity, reserves, frequency response, and related grid services. These value streams can materially improve project economics, although they also decline as participation increases and scarcity is reduced.

In many jurisdictions, these revenue streams are supported by explicit resource-adequacy mechanisms and flexibility markets that were introduced to maintain reliability as traditional energy-only market signals became insufficient.

Capacity and grid service revenues improve the economics of solar+storage projects

LCOE and capture price (US¢/kWh)

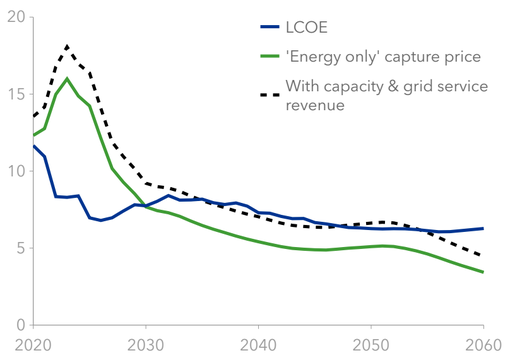


FIGURE 4.25 | Capacity and grid service revenue increases from 10% in to 20% of 'energy only' capture price.

While the energy-only capture price falls below LCOE from around 2030 as growing solar penetration depresses market prices, additional revenues from capacity markets and grid services keep total project revenues above LCOE until the mid-2050s. Over time, however, increasing competition among battery assets and declining scarcity value decrease these revenue streams, causing total revenues to gradually converge towards and eventually fall below LCOE.

3. Stabilize revenue through long-term contracts

PPAs and Contracts for Difference (CfDs) reduce exposure to capture-price erosion by replacing part of merchant revenue with a contracted price.

While PPAs are typically commercial agreements between private counterparties, CfDs are usually policy-backed instruments that combine revenue stabilization with an explicit or implicit public-support mechanism.

Corporate PPAs are becoming particularly important where electricity demand from AI and data centres is growing faster than utilities can provide new connections and capacity. For this reason, many data centres are being required to bring their own additional clean power. Therefore, for data-centre operators, PPAs improve visibility over future electricity costs, support clean-energy procurement, and help secure supply in constrained markets. For renewable developers, they provide more predictable revenues and improve project bankability.

The value of such contracts increases as renewable penetration grows and merchant revenues become more volatile. They allocate price risk between generators, buyers and, in the case of policy-backed contracts, consumers or governments.

Figure 4.26 compares the annual revenues of a simple and hypothetical solar plant generating 1,000 MWh per year under two commercial arrangements. In the fully merchant case, all generation is sold into the energy-only market at the annual capture price. In the hybrid case, 50% is sold at the capture price and 50% through a long-term PPA priced at the project's LCOE plus a 10% margin.

Solar project with a PPA earns around 10% higher lifetime revenue than one without

Project revenue per year (USD/yr)

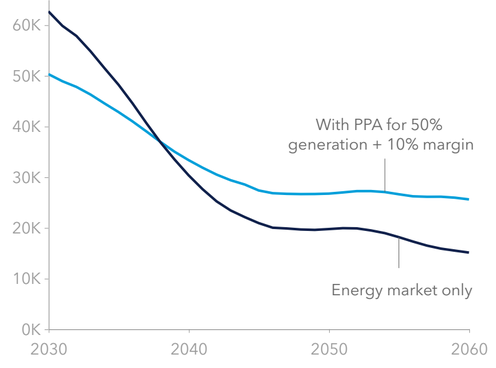


FIGURE 4.26 |

The plant enters operation in 2030 and remains operational through 2060. As capture prices decline, the contracted share provides an increasingly valuable hedge against market-value erosion and stabilizes revenues relative to full merchant exposure.

In our analysis, a solar power plant that enters into a PPA for 50% of its annual generation earns around 10% higher lifetime revenues than a project exposed entirely to energy-market revenues. Long-term contracts therefore provide a partial hedge against capture-price erosion, helping maintain project profitability even as renewable penetration increases.

Conversely, CfDs are the public-policy counterpart to corporate PPAs. They are designed to maintain investment in low-carbon generation when capture-price erosion would otherwise weaken the business case for new projects. In most cases, they involve a support or subsidy element as well.

A CfD provides a renewable power developer with a strike price settled against a market reference price, for which the capture price is a proxy. When the reference price falls below the strike price, the developer receives a top-up payment; when it exceeds the strike price, the developer returns the difference. The mechanism therefore provides revenue certainty, reduces financing risk and enables investment that may not proceed on merchant terms alone.

Were one to compare PPAs and CfDs as market mechanisms to handle the divergence between LCOE and capture price, PPAs between two corporate entities suffer from a lack of transparency, and because of that, have limited contribution to price recovery (Berkeley Lab, 2025).

Electricity bills increasingly recover expenditure on networks, adequacy and system integration, even as the wholesale cost of energy falls.

From energy cost to infrastructure cost

The strategies above address capture-price erosion from the perspective of projects, but they do not eliminate the wider costs of a high-renewables power system. Instead, part of the value lost in the energy-only market reappears through payments for adequacy, flexibility, networks, and long-term revenue support. This is driving a structural shift in both the level and composition of electricity costs.

Historically, a large share of electricity expenditure reflected fuel costs. This exposes consumers to commodity-price volatility and geopolitical risk. As solar, wind, and other low-marginal-cost technologies expand, the wholesale energy component declines, particularly during periods of abundant renewable output. However, lower wholesale prices do not translate automatically into equivalent reductions in total system costs (Figure 4.27).

While the wholesale energy component declines, adequacy-related costs rise in many regions. Capacity mechanisms compensate for resources that remain available during periods of system stress, while ancillary-service markets procure frequency regulation, reserves, voltage support, inertia, and other services previously supplied implicitly by synchronous thermal generation. Batteries, demand response, grid-forming inverters, and other flexible resources are therefore becoming essential, but they require dedicated remuneration streams.

Network costs are also increasing (Section 4.9). Renewable resources are often located far from demand centres, requiring transmission expansion, while electrification of transport, buildings, and industry increases the demands placed on distribution networks. These investments are capital-intensive, long-lived, and generally recovered through regulated network charges rather than wholesale energy prices.

The result is a shift from variable energy costs towards fixed and socialized infrastructure costs. Electricity bills increasingly recover expenditure on networks, adequacy, flexibility, and system integration, even as the wholesale cost of energy falls. In this emerging paradigm, the economics of electricity are determined less by the cost of producing an additional kilowatt-hour and more by the cost of ensuring that reliable power is available whenever and wherever it is needed.

Rising grid and adequacy charges offset falling wholesale prices

Residential electricity price components (USD/MWh)

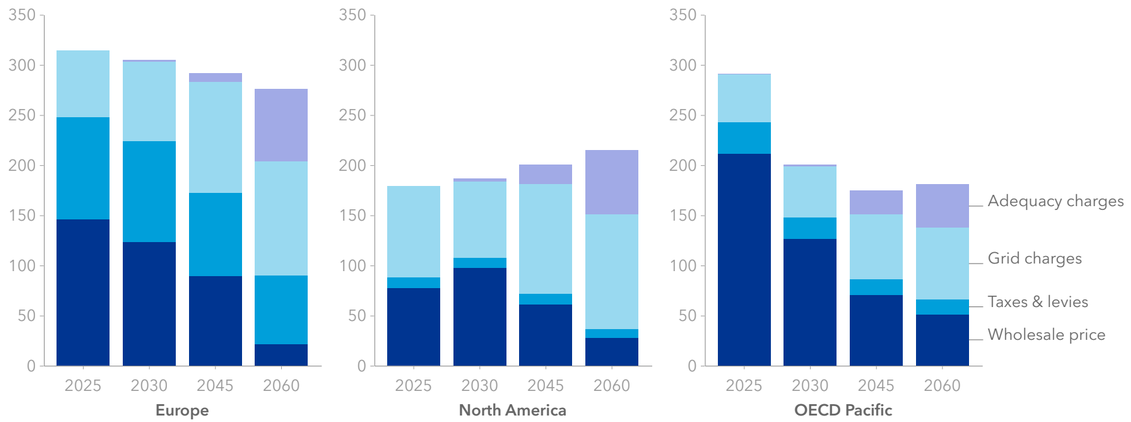
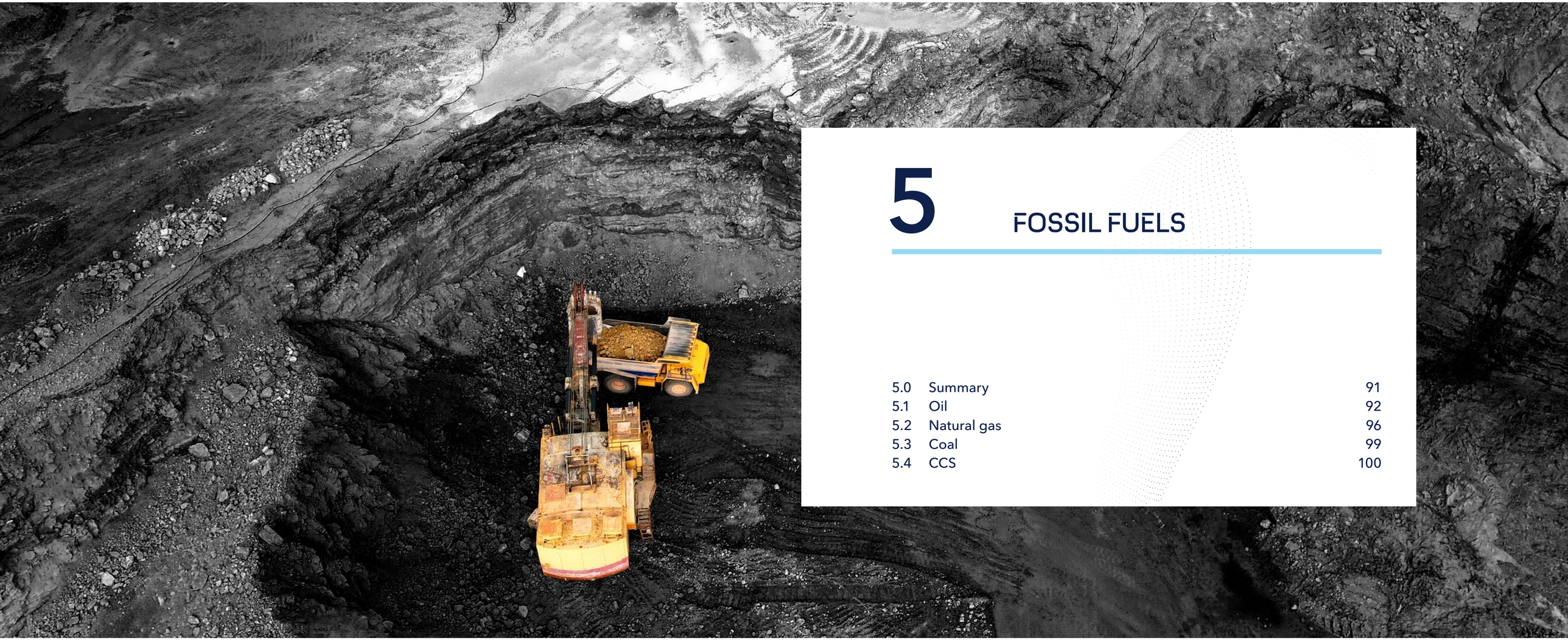


FIGURE 4.27 |



5 FOSSIL FUELS

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5.2	Natural gas	96
5.3	Coal	99
5.4	CCS	100

5.0 SUMMARY

A new fragility

2026 deeply challenged assumptions about the security of fossil fuel supply. Strikes by the US and Israel on Iran, and their wider aftermath, have removed not only barrels of oil and cubic metres of gas from the market, but also confidence in the reliability of Gulf energy supplies. While the disruption is among the most severe in history, its impact has been moderated by strategic reserves, particularly in China. Unlike the oil shocks of 1973 and 1979, consumers and nations, especially fossil-fuel importers, now have viable and permanent alternatives to fossil dependence, not least through transport electrification. The longer the disruption persists, the greater the likelihood that some demand will be destroyed permanently rather than merely deferred. This is already reflected to some extent in our forecast. However, the world is unlikely to respond with a dramatic acceleration of the energy transition.

Transport and power drive fossil fuel reduction
Energy demand by sector (EJ/yr)

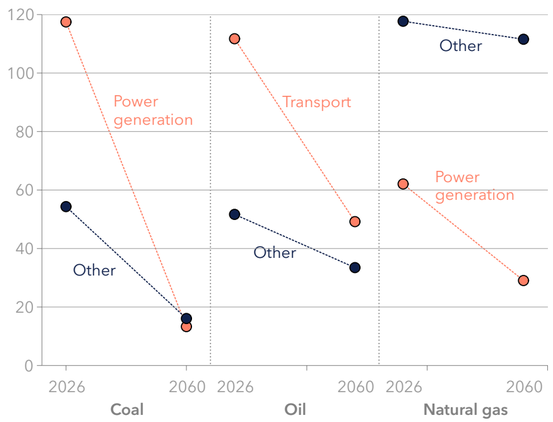


FIGURE 5.1 |

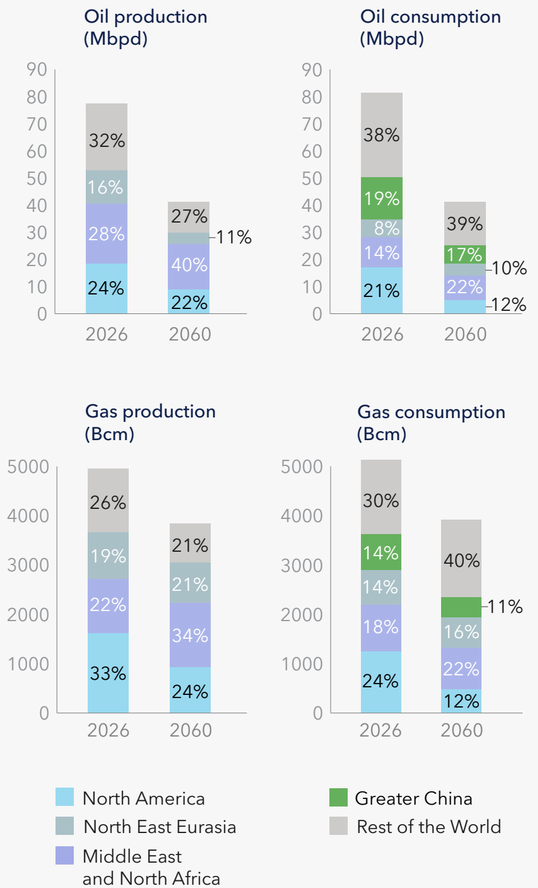
Fossil fuels have no fast exit

Our long-term prognosis for fossil fuels is largely unchanged from ETO 2025. Fossil energy sources are currently around 80% of primary energy and are on track to fall to around 50% in 2050 and 38% by 2060. This megatrend is largely unaffected by short-term perturbations like Asian utilities burning more coal to offset the scarcity of gas. Additional moves to electrify end-uses like transport, heating, and cooling are likely to have a more durable impact, as will renewed commitments to invest in nuclear power. However, natural gas, despite being hardest hit on the supply side in the near term, goes on to retain an even larger role in the mid-century energy mix than we have previously forecast owing partly to a smaller role for clean hydrogen.

Gas outlasts coal and oil

The oil demand shock of 2026 reverberates through our forecast: road transport and feedstock demand, the two largest uses of oil, both fall below our ETO 2025 projections as high prices push EV adoption and squeeze petrochemical cracker runs. However, oil demand recovers to around 170 EJ by 2028 from 163 EJ in 2026, then falls to 83 EJ by 2060. Gas demand peaks around 2035 at 190 EJ, five years earlier than last year's Outlook, as importers diversify away from vulnerable supply routes. However, the longer-term decline of gas is slower than previously forecast, as hydrogen scale-up proves costlier and gas continues to displace coal in manufacturing. Gas still accounts for over 20% of primary energy in 2060. The steady decline of coal is arrested this year as Asian utilities burn an estimated 70–90 million extra tonnes to cover for scarce gas. Its decline resumes in 2027, falling 82% to 30 EJ by 2060, when it will represent less than 5% of primary energy.

Oil and gas production and consumption overview (2026 vs 2060)



Production

In 2026, the Middle East and North Africa (MEA) absorbs most of the shock, with oil and gas production down 21% and 10%, respectively. However, the region recovers over the next decade and still supplies around 40% of global oil and 34% of gas in 2060, remaining by far the world's largest producer of both fuels.

Trade and investment

Energy trade patterns shifted markedly in 2026. Oil flows adapted through an expanded Red Sea pipeline capacity, Iraq's revived link to Turkey, and rising Latin American supply. Gas had fewer alternatives, with significant Gulf LNG capacity disrupted. North America overtook MEA as the largest gas exporter; MEA only reclaims the lead after 2030.

5.1 OIL

The long-term challenge for oil is a demand outlook that sees global consumption falling by almost two-thirds by 2060. The immediate challenge is the Hormuz crisis, which has produced one of the largest oil-supply disruptions in decades. At the time of writing, the Strait of Hormuz remains highly disrupted, and indications are that global production will fall more than 5% (about 4.5 Mbpd) in 2026.

Crude oil demand appears to have fallen more modestly, dropping some 2% (around 2 Mbpd), as high prices, fuel shortages, and a slowing Chinese economy curbed consumption almost everywhere. With demand running slightly ahead of supply, the shortfall has been covered by strategic and commercial inventories. We expect the market to re-balance by 2027, with both production and demand recovering by then.

The longer-lasting effect is on the demand trajectory. The supply shock has reinforced trends already under way, including the end of the post-pandemic travel rebound, weaker industrial activity and faster EV adoption. The more consequential legacy of 2026 is therefore not the immediate loss of barrels, but a lasting shift in demand, particularly in two major sectors: road transport and non-energy.

In 2026, Middle East and North Africa oil production dropped by 21% compared to 2025.

The drop in demand in 2026 gave impetus to a trend already under way: the end of the post-pandemic travelling rebound, weakening industrial activity, and growing EV adoption. In the resulting transformation of global sectoral oil demand, transport, the largest category, still peaks right before 2030 at roughly 117 EJ (a 70% share) before declining to 104 EJ by 2040 and 50 EJ by 2060 (60% of a much smaller total) (Figure 5.2). Non-energy (petrochemical) demand fell even more sharply in 2026 than transport, with lasting effects on its own recovery path.

Replenishing the strategic reserve is working in the other direction. Once the conflict settles, we expect production to run above consumption for a period as countries rebuild stockpiles that have been drawn down since March 2026 when the IEA coordinated the largest release in its history. China is the notable exception, having largely sat out the release and drawn only modestly on its own reserves.

The more significant finding is what happens after the shock fades. MEA production recovers to 30 Mbpd by 2030, with its share of global output returning to around 36% by 2030 and reaching 40% by 2060. That share is below our ETO 2025 projection, as the smaller demand pool matters more than the physical damage. North America is the main beneficiary: its 2060 production of 9 Mbpd is higher than the 7.5 Mbpd we projected last year, while its share of the shrinking global market rises to around 22%, from about 19% previously (Figure 5.3).

Transport is the biggest contributor to oil decline
World oil demand by sector (EJ/yr)

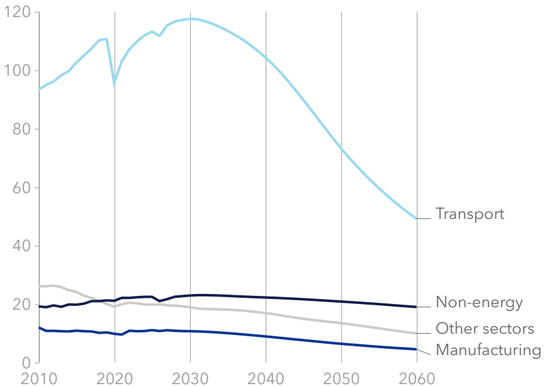


FIGURE 5.2 | Historical data source: IEA WEB (2026)

Middle East and North Africa production peaks around 2030
Oil production and consumption (Mbpd)

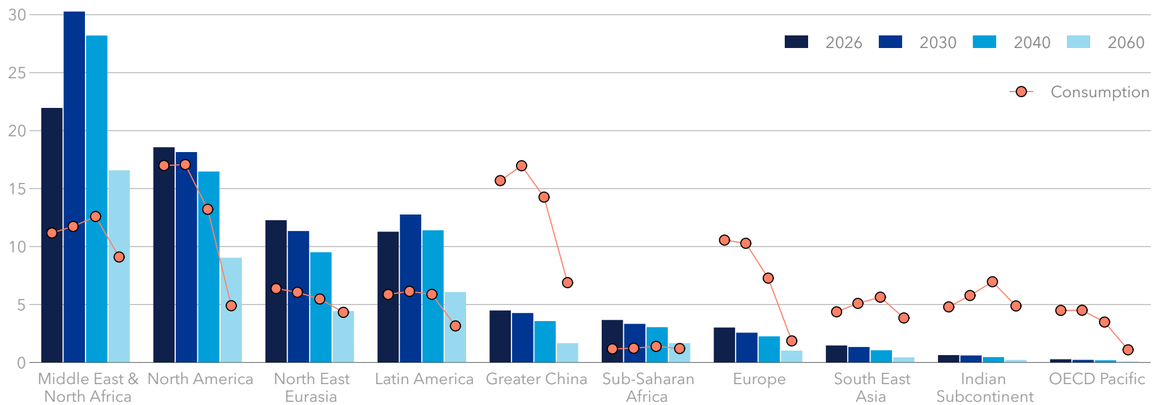


FIGURE 5.3 |

Sectoral demand

Transport sector expansion, driven by population and economic growth, has been the main historical driver of oil demand. EV sales are gaining momentum and will dominate new passenger and commercial vehicle sales around 2035 and 2037 respectively. Oil road transport demand fell roughly 1% in 2026 to 85 EJ, but it rebounds to 87 EJ in 2027 and plateaus by 2030. It then falls to 32 EJ by 2060 (Figure 5.4). This is a lasting, not temporary, downgrade.

Oil use in aviation, shipping, and rail transport ('other transport' in Figure 5.4) also fell in 2026, down 2% to 27 EJ from 27.5 EJ in 2025, as Gulf-region shipping disruptions pushed up jet fuel costs – the IEA specifically flagged aviation as one of the hardest-hit demand categories in the early months of the conflict. Growth resumes once the disruption passes, before the longer-term shift toward biofuel, green ammonia, e-kerosene, and other low-carbon fuels takes hold. 'Other transport' oil demand falls from a 2030 peak of 29 EJ to 17 EJ by 2060.

Oil's other large sectoral demand is as feedstock for non-energy use, mostly plastics production and non-petrochemical industries. This was the single hardest-hit category in 2026; non-energy demand fell 7%, from 22.5 EJ in 2025 to 21 EJ in 2026, as high feedstock costs and reduced Middle Eastern cracker runs curtailed output. This was consistent with the IEA's reporting of a 13% year-on-year drop in global naphtha demand during the worst of the crisis. Demand recovers to 23.0 EJ by 2030 (still 11% below the 26 EJ/yr we projected for 2030 in ETO 2025), holds close to that level through 2040 (22.2 EJ), then declines further to 19.0 EJ by 2060 (a 27% share).

Manufacturing, power generation, buildings, and others collectively halve from 31 EJ in 2026 to 14 EJ in 2060, mostly because of fuel switching to gas and electricity (generated by renewables) and increased efficiency.

Regional demand

Peak oil demand and production continue to vary significantly by region (Figure 5.5). High-income regions peak early, while low-income regions peak later or outside our forecast period. North America led global oil demand for decades, but we expect Greater China to overtake it after 2026. Beyond 2050, MEA overtakes China to become the world's largest oil-consuming region.

Although North America's absolute oil demand is falling faster than expected due to EV growth, its demand per capita remains the highest in the world through around 2040 (Figure 5.5). By 2060, North East Eurasia has the highest oil demand per capita (4.7 bbl/person-yr), with North America then in second place (4.2 bbl/person-yr), narrowly ahead of

EV growth is the main reason for oil decline

Oil demand by sector (EJ/yr)

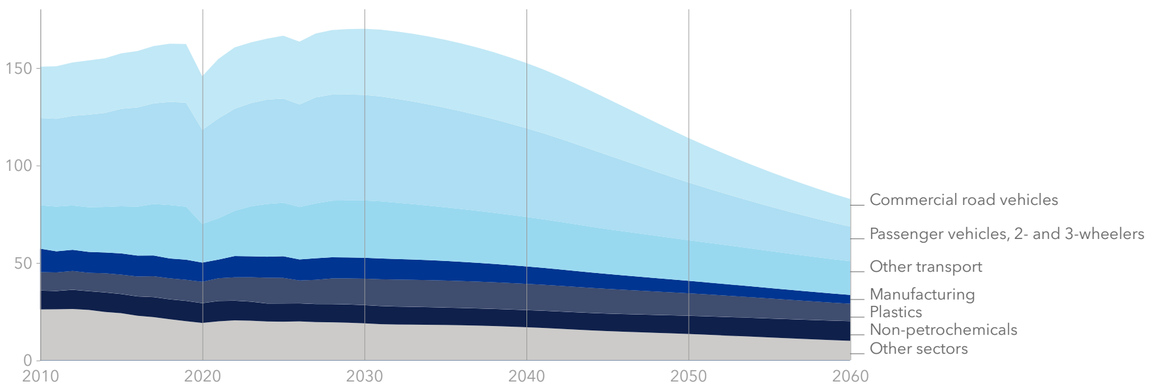


FIGURE 5.4 | Historical data source: IEA WEB (2026)

North America stays the biggest consumer per capita for the next two decades

Crude oil demand per capita by region (bbl/person-yr)

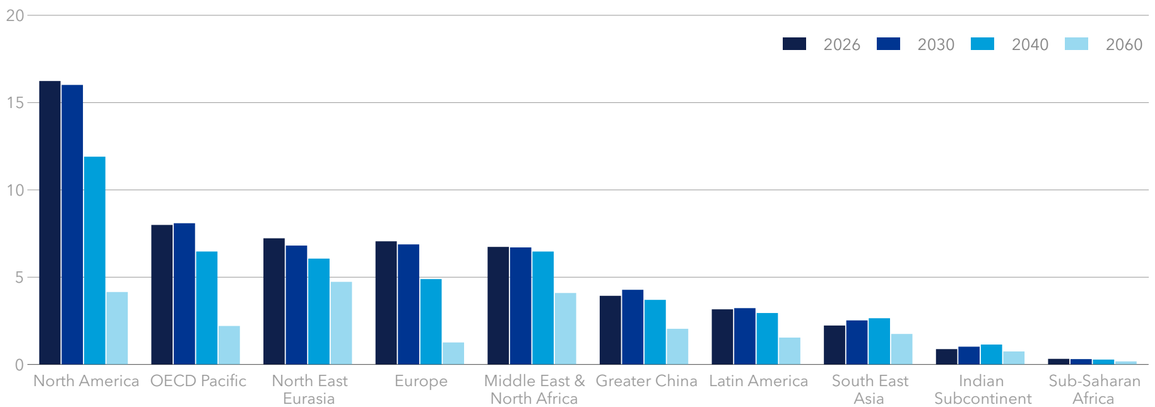


FIGURE 5.5 |

MEA (4.1 bbl/person-yr). Sub-Saharan Africa maintains a far lower oil demand per capita than all other regions throughout the forecast period.

Production and trade

There is a clear distinction between net oil exporters (Middle East and North Africa, North East Eurasia, North America, and Latin America) and net oil-importing regions (Greater China, Europe, the Indian Subcontinent, OECD Pacific, and South East Asia) – a pattern that persists to 2060, albeit at ever-lower volumes (Figure 5.6).

MEA 2026 production is on track to fall 21% year-on-year (from 28.0 to 22.0 Mbpd) as Iraq, Saudi Arabia, the UAE, Kuwait, and Iran itself all had volumes disrupted. Every other producing region grew or held broadly flat, as non-OPEC+ producers raced to fill the gap. The IEA's own balance shows what it calls the "Americas Quintet" (the US, Canada, Brazil, Guyana, and Argentina) lifting output by about 1 Mbpd in 2026, enough to more than offset the loss of Gulf oil supply (IEA, 2026b). In our model, Latin America rises 11%, North America grows a further 2%, and even Sub-Saharan Africa (+1%), Europe (+3%, off a small base), and Greater China (+2%, reflecting Beijing's push for self-reliant supply) post gains. Part of this shift persists beyond the crisis. Latin America becomes the main source of non-OPEC supply growth through 2030, and we

Middle East and North Africa increases export share
Oil export by region (Mbpd)

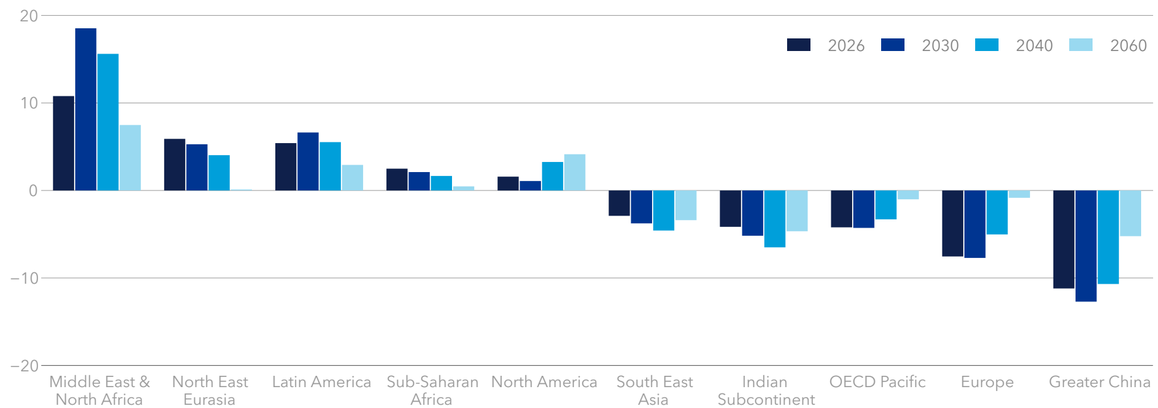


FIGURE 5.6 |

have revised the medium-term production paths for both Latin America and North America upward as importers diversify supply.

MEA remains the largest net exporter throughout the outlook and production and export rebound toward 30 Mbpd and 19 Mbpd respectively around 2030 as shut-in capacity returns and projects sanctioned before the conflict are completed. Bypass pipelines through Saudi Arabia, the UAE, and Turkey softened the immediate loss of Hormuz access, but could replace only a fraction of the roughly 20 Mbpd normally moving through the strait. Saudi Arabia pushed its East-West Petroline pipeline toward its expanded 7 Mbpd ceiling, the UAE ran its Habshan-Fujairah line above design capacity, and Iraq restarted the Kirkuk-Ceyhan link.

That bypass strategy is now itself at risk. The Red Sea route carrying Saudi Arabia's rerouted crude runs through Bab el-Mandeb, which remains exposed to Houthi attack. In June and July 2026, oil flow through the strait reached around 7.4 Mbpd, of which 4 to 5 Mbpd was Saudi crude specifically diverted via Yanbu to avoid Hormuz (Bloomberg, 2026). Renewed attacks on shipping constrained western-terminal loadings and support our expectation that MEA production will remain below its pre-conflict level until early 2027.

MEA's share of global production – which had held around 35% for three decades – has been hit in 2026, and will fall to 28%. It will recover to around 36% by 2030 and 40% by 2060 (Figure 5.4). Low production costs preserve the region's competitiveness, although war damage, higher investment risk, and importer diversification reduce its long-term advantage.

North America is the main relative beneficiary; its 2060 production of 9 Mbpd is higher than the 7.5 Mbpd we projected last year, with its share of a shrinking global market rising to around 22%, up from about 19% previously. It moves from a modest exporter in the near term (2 Mbpd in 2026) to a growing one by 2040–2060 (rising to about 4 Mbpd), as domestic demand falls faster than domestic production. North East Eurasia moves the other way: its surplus erodes steadily and turns to almost zero by 2060, as Kazakh and Russian output declines while regional demand holds up. Europe's import dependence falls away fastest of all, from 7.5 Mbpd in 2026 to just 1 Mbpd by 2060. That is something of a red flag for the US because Europe is currently almost 50% of total US crude exports.

Strategic stocks and investment

When the conflict genuinely ends, we expect production to run above consumption for a period simply to rebuild the stockpiles drawn down since the conflict began – a call on supply that exists independently of long-term demand developments.

The IEA coordinated a release of 400 million barrels from member countries' strategic reserves in March 2026, the largest in the agency's history. The US was the largest single contributor, supplying 172 Mb (from 413 Mb strategic reserves). Japan supplied 80 Mb (from 263 Mb, a mix of national and privately held reserves), with smaller volumes coming from South Korea (22.5 Mb), Germany (19.5 Mb), France (14.5 Mb), Spain (11.5 Mb), and the UK (3.5 Mb). Countries outside the IEA have not announced meaningful releases of their own. However, In May and June 2026, China also tapped 41–43 million barrels from reserves (Russell, 2026).

In 2025, lower oil prices and demand expectations saw a 6% fall in upstream oil investment in 2025, to just under USD 570bn (IEA, 2025). The war deepened that retrenchment further; the IEA's puts global upstream oil investment below USD 500bn for 2026, a third consecutive annual decline (IEA, 2026a). Gas has moved the opposite way, rising more than 10% to a decade-high USD 330bn (Hernandez, 2026), as buyers diversify away from disrupted Qatari and Iranian supply.

The regional allocation has also shifted in 2026. While upstream investment in Africa and Latin America increase by more than 10%, MEA oil and gas investment fell by around 1% (IEA, 2026a). In MEA, an estimated USD 34–58bn in war-damaged energy infrastructure will require reconstruction that diverts already constrained capital and competes with new developments for rigs, equipment, and investment (Martinsen, 2026).

The conflict has also reshaped OPEC+ itself. In 2026, the UAE left OPEC, exposing a widening split in Gulf strategy, with Saudi Arabia prioritizing price stability and the UAE focusing on market share.

Outlook uncertainty

The near-term numbers in this chapter remain conditional on a war that remains unresolved at the time of writing. Uncertainty over the regional source of future oil supply is unusually high. In our ETO Model, oil production follows demand and high-cost regions will not develop their resources if cheaper oil can be supplied elsewhere. However, regional distribution may be skewed by non-market factors like subsidies, trade barriers, and geopolitical risk, and hence may not follow the disciplined pattern our model assumes.

5.2 NATURAL GAS

Natural gas has staying power in the global energy mix throughout our forecast period – with demand falling by only one fifth by 2060.

The general pattern in the last few years has been one of rising demand. We expect this to continue to the mid-2030s. However, the 2026 geopolitical disruptions have seen oil production flatten after growing by 2% between 2024 and 2025. The fall in production in the Middle East has been largely offset by production in North America, Latin America, North East Eurasia, and elsewhere, with the result that global demand remains almost unchanged from 2025.

Because of the conflict, we now expect global gas demand to peak earlier and at a lower level than projected in ETO 2025. It will likely reach around 190 EJ in 2035 rather than 197 EJ in 2040 as supply-security concerns reinforce the shift towards renewables and nuclear power in many importing regions. Even so, gas remains a resilient part of the global energy mix. Having accounted for around 27% of global primary energy supply over the past decade, its share rises slightly to around 28% by the mid-2030s before gradually declining to around 25% by mid-century and just over 20% by 2060.

Importantly, the earlier peak does not translate into lower long-term demand. In fact, demand beyond 2050 is higher than in ETO 2025, reaching around 175 EJ in 2050 and 140 EJ in 2060, compared with 164 EJ and 130 EJ respectively in last year's Outlook. This reflects gas displacing more oil and coal than previously expected, while slower progress on hydrogen means less switching away from gas than we assumed a year ago. Our hydrogen forecast has been revised down accordingly, reflecting persistently high production costs, limited policy support, and continued inefficiencies in hydrogen production and trade (Chapter 6). Gas demand rises across most sectors from 2026 to 2035, then declines across almost all (Figure 5.7). This reflects fuel switching (electricity and hydrogen) and improved energy efficiency – except for manufacturing and hydrogen production, both of which continue to grow.

Global gas production is led by North America and MEA throughout this Outlook, but the order has been reshuffled this year (Figure 5.8). In 2026, North America is comfortably the world's largest gas producer at about 1,615 Bcm, while MEA output has dropped down to about 1,100 Bcm. MEA reclaims the production lead only around 2035, five years later than we projected last year.

Gas demand starts declining only after 2035
Natural gas demand change in different sectors (EJ/yr)

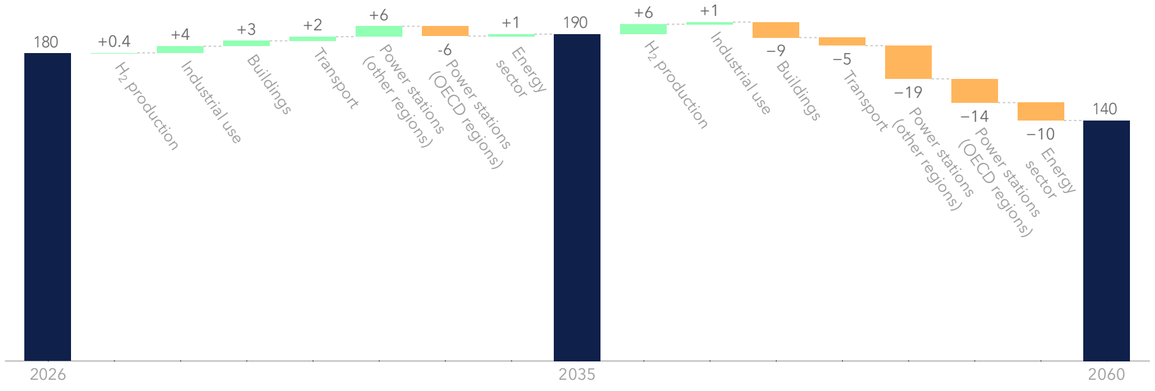


FIGURE 5.7 | Industrial use combines manufacturing and non-energy demand. OECD regions: North America, Europe, and OECD Pacific.

Middle East and North Africa overtakes North America in natural gas production
Natural gas production and consumption (Bcm/yr)

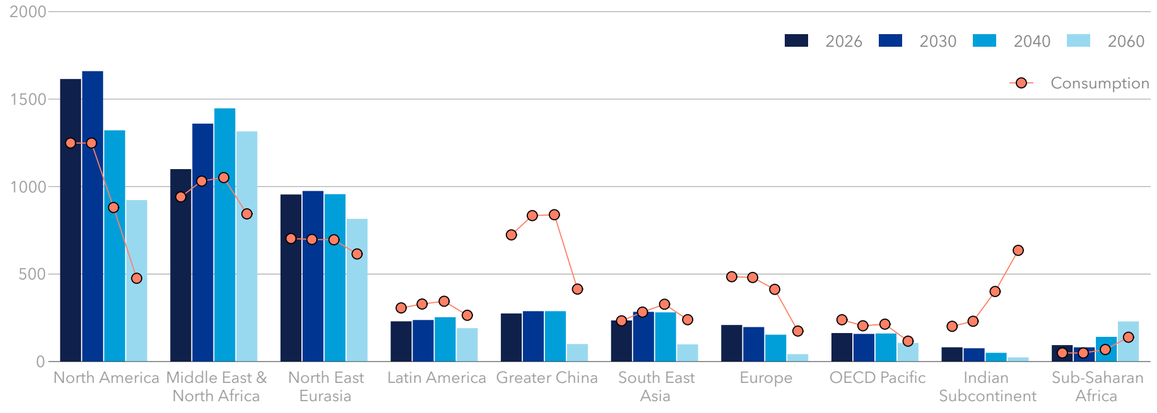


FIGURE 5.8 |

Sectoral demand

Natural gas demand has increased over the past two decades, and although the war has brought forward the peak in demand, the sectoral distribution has remained relatively stable (Figure 5.9). In 2026, total natural gas demand declined by around 1% from 2025, reaching approximately 180 EJ.

Power generation remains the largest source of gas demand throughout our forecast, accounting for around 35% of total consumption into the mid-2030s. Gas use in power declines thereafter, but more slowly than we projected in ETO 2025. By 2060, gas still accounts for around 32 EJ of power demand, compared with 23 EJ in last year's outlook. The difference is largely explained by the divergence between OECD and non-OECD markets: OECD gas-fired generation is largely displaced by renewables, storage, and nuclear (from today's 23 EJ to under 3 EJ in 2060), while gas retains a more substantial role in emerging Asia and the Middle East (from 39 EJ to 29 EJ in the same period).

Buildings and manufacturing provide the other major sources of resilience. Buildings demand increases from around 37 EJ in 2026 to a peak of 41 EJ around 2040 as gas replaces coal and biomass in emerging markets, before falling to 30 EJ by 2060 as efficiency gains and electrification (e.g. heat pumps) take hold. Manufacturing follows a different trajectory, growing from around 32 EJ in 2026 to 37 EJ by 2050 and plateauing thereafter for the rest of our forecasting period as the continued replacement of coal more than offsets losses to electricity and hydrogen in many industrial processes.

Gas demand peaks around 2035

Natural gas demand by sector (EJ/yr)

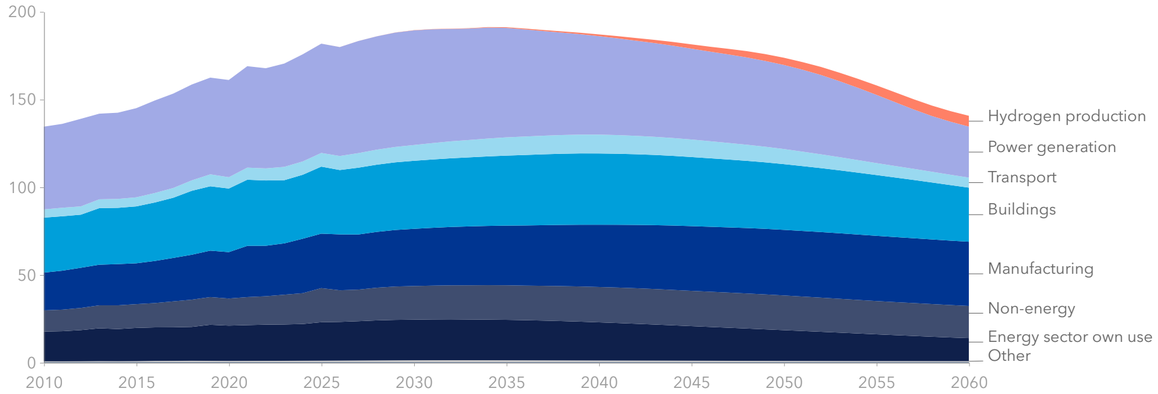


FIGURE 5.9 | Historical data source: IEA WEB (2026)

Demand in transport and non-energy use remains comparatively stable. As momentum builds to replace gas with hydrogen derivatives such as ammonia and e-fuels in shipping, gas demand from transport rises from 8 EJ in 2026 to about 11 EJ by 2040, before declining to around 6 EJ by 2060. Non-energy demand, mainly petrochemical feedstock, remains broadly unchanged at about 18-20 EJ throughout the outlook.

Own use within the energy sector also remains broadly stable into the mid-2030s before declining as production facilities become more efficient, increasingly electrified, and less reliant on flaring. Gas demand for hydrogen and ammonia production grows from a negligible base to around 6 EJ by 2060, although this is substantially below the level projected in ETO 2025 because of a weaker hydrogen outlook driven by persistently high production costs and limited policy support.

Regional demand

On the demand side, most regions still have room to grow before peaking sometime between the early 2030s and 2040, then declining (Figure 5.8). The Indian Subcontinent is the exception: its demand keeps climbing well beyond the point where the rest of the world peaks as gas continues to replace coal in power generation and manufacturing. Sub-Saharan Africa follows the same continuous-growth path from a much smaller base.

How that demand reaches these growing markets is itself being redrawn. Unlike oil, there is no pipeline bypass for Gulf LNG, so what shifts is which cargoes go where. Asian buyers have pulled Atlantic Basin cargoes away from Europe since the conflict began, putting the two regions into direct competition for the same shrunken pool of flexible supply. MEA's net exports fall by a third in 2026, briefly handing the world's largest-exporter title to North America – the first time in the modern LNG era – before MEA reclaims it in the early 2030s.

The disruption is also strengthening the case for reducing gas dependence in major importing regions. In Europe, the energy-security response increasingly centres on faster deployment of renewables, electrification, and nuclear. India is similarly expanding nuclear and renewable capacity. These shifts will accelerate the longer-term transition away from gas, although grid constraints and other infrastructure bottlenecks may limit how quickly alternatives can replace it. Ironically, the geopolitical crisis that underpins the US LNG production leadership over the coming decade has also accelerated European efforts to reduce gas dependence, potentially weakening the very market that justified much of the new export capacity.

Production and trade

Gas remains a much more regional commodity than oil, so production tends to follow regional demand more closely than oil does. North America is the clearest exception, with its own gas demand already declining even as its LNG export capacity keeps expanding. This effort is reshaping the region's production profile independently of anything happening in the Middle East.

North America leads gas production through the mid-2030s before MEA reclaims the top spot; by 2040 MEA is back in front (1,450 Bcm vs North America's 1,320 Bcm), and by 2060 it produces around 1,315 Bcm, still comfortably the largest single producing region even as its traditional export markets shrink. Its share of global production expands from around 22% in 2026 to roughly 34% by 2060, a smaller endpoint than the 40% we projected last year, since North America's own expansion is proving more durable than we had expected.

A few other regional production patterns stand out. North East Eurasia holds broadly stable output, roughly 950 to 1,050 Bcm across the forecast period, with Russian pipeline gas increasingly redirected toward China as European markets stay locked out by the EU's phase-out of Russian gas. Greater China's own production keeps climbing through 2040 (275 to 300 Bcm) on the back of the country's accelerated push for gas self-sufficiency, before falling back to around 100 Bcm by 2060 as domestic reserves mature. As legacy fields mature, Sub-Saharan Africa dips slightly through 2030 (95 to 85 Bcm) before new LNG capacity – including the restart of Mozambique LNG – lifts output sharply to 230 Bcm by 2060 to make SSA the one region where production is still growing at the end of our forecast period.

On the export side, MEA, North America, and North East Eurasia remain the top three exporting regions throughout our Outlook, but the order between them has been reshuffled (Figure 5.10).

North America's own trajectory has flipped entirely from what we projected last year: rather than the steady decline we previously forecast, war-driven acceleration of US LNG capacity – as new capacity at Plaquemines, Corpus Christi, and Golden Pass come online (Young, 2026) and buyers scramble for non-Gulf supply – means exports keep growing to around 450 Bcm by 2040, before easing only modestly to 430 Bcm by 2060. North East Eurasia's exports stay broadly stable in the 260–280 Bcm range.

On the import side, OECD Pacific's net import position narrows from 76 Bcm in 2026 to just 10 Bcm by 2060 as Australian production expands (Barossa, Scarborough Train 2) and demand drops. Europe's net imports grow slightly through 2030 before roughly halving by 2060 as gas demand decreases sharply and the EU's push to eliminate Russian gas entirely reinforces the broader transition away from gas. Greater China's imports keep rising until around 2030, then stabilize until 2040 before declining. The Indian Subcontinent – already a substantial net importer in 2026 – sees that position widen further and faster than we projected last year; rather than a threefold surge, we now expect their import demand to grow about fivefold by 2060.

Diverging regional gas demand trends are also reshaping LNG trade. As Asian buyers compete more aggressively for flexible cargoes, Europe and Asia are increasingly bidding for the same supplies. Even if the immediate disruption subsides, the crisis may encourage buyers and sellers alike to diversify supply chains and reduce dependence on single transit routes.

The disruption has also triggered a renewed wave of gas investment. We forecast global gas spending will rise by more than 10% in 2026, driven by LNG export projects and rising power demand (e.g. from data centres). Yet much of this investment appears aimed at sustaining existing supply and improving resilience rather than opening entirely new hydrocarbon frontiers.

The Middle East and North Africa remains the biggest exporter during our forecast period
Natural gas export by region (Bcm/yr)

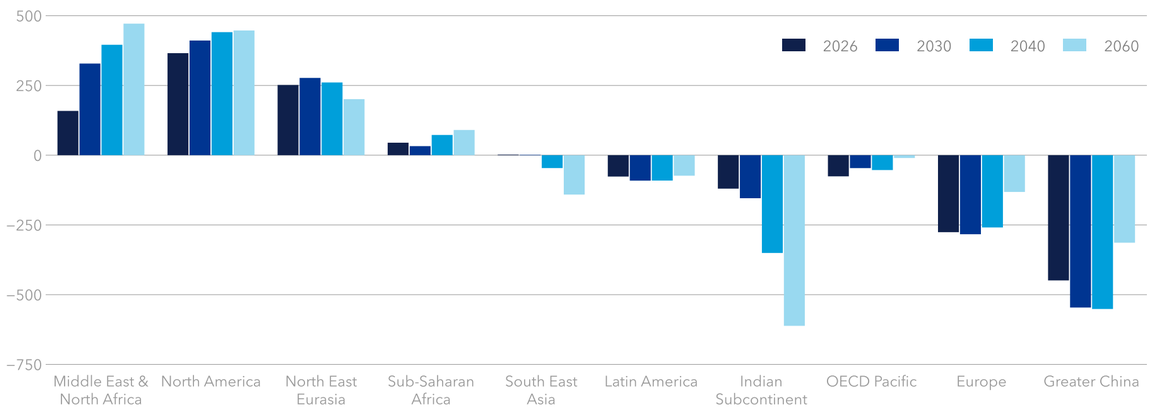


FIGURE 5.10 |

5.3 COAL

Global coal demand in 2026 was similar to 2025, around 8.8 Gt. This is not the modest decline we expected before the Iran-related conflict took roughly a fifth of Asia's LNG supply off the market for months. Utilities across China, India, Japan, and South Korea responded by burning more coal at levels sufficient to maintain demand (Zeitlin, 2026; Talukdar, 2026). This aberration is unlikely to persist. From 2026, our model has coal entering a long decline as electrification and renewables continue to expand. Global production falls from some 9 Gt in 2026 to 7 Gt in 2030, and 1.3 Gt by 2060.

Sectoral and regional demand

Over time, power generation gives way to manufacturing as the largest consumer of coal. Power took 69% of coal demand in 2026 and manufacturing accounted for 25%. By 2060, power's share falls to 40% of a much smaller market for coal. Manufacturing becomes relatively more important because steel-making and cement are hard to electrify (Figure 5.11).

- China remains by far the largest coal-consuming region, at 100 EJ in 2026 (58% of the global total), but also posts the sharpest decline: demand nearly halves by 2040 (55 EJ) and falls to 12 EJ – an 86% reduction – by 2060, as renewables continue displacing coal in power at the pace seen in recent years. Even so, in 2060 China still accounts for 40% of global coal demand.
- India is the one region where demand grows before it falls. Coal demand rises from 24 EJ in 2026 to about 30 EJ in 2032, driven by continued expansion in both manufacturing and power, before falling back to 21 EJ by 2040 and 5 EJ by 2060. Almost all of that later decline is concentrated in power, which falls 88% between 2040 and 2060 alone, while manufacturing demand declines far more gently.
- South East Asia's demand stays almost flat to the mid-2030s, then reduces to one-third (3 EJ) by 2060. However, its share of the global total more than doubles, from 6% to 12%, as demand elsewhere falls even faster.
- The rest of the world sees demand fall from 37 EJ in 2026 to 18 EJ by 2040 and 9 EJ by 2060, led by continued power-sector coal retirement outside Asia.

Production and trade

Because only around 10% of coal is traded internationally, compared with much higher shares for oil and gas, coal production largely follows regional demand. Production was 9 Gt in 2026 (8 Gt hard coal, 1 Gt brown coal), and we expect both Greater China and India to maintain domestic output close to current levels for longer than pre-war plans assumed (Agnolucci, 2026). Over the long term, coal production falls sharply in all regions, with China's output dropping fastest in absolute terms while India accounts for a larger share of a shrinking market (Figure 5.12).

The coal trade was sharply impacted by the conflict in Iran. Higher gas prices boosted demand for hard coal in parts of Asia, temporarily lifting coal prices (Samonte, 2026). However, we expect these effects to subside because of the small share of coal that is traded. Globally traded coal volumes will still likely contract in 2026 onward, as major exporters prioritize domestic supply security over shipments abroad. Most of that contraction lands early; we expect total trade volumes to fall by around 70% between 2026 and 2035 alone as demand cools across every region, and by roughly 90% by 2060.

Coal demand in power drops by 90% from today to 2060.

Global coal demand drops by 40% by 2040

Coal regional and sectoral demand (EJ/yr)

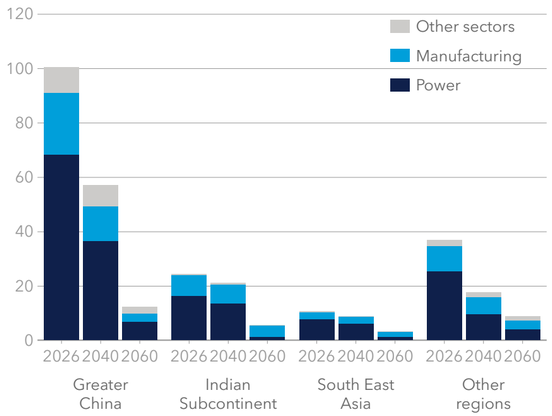


FIGURE 5.11 |

Brown coal almost disappears in 2040

Coal regional production (Gt/yr)

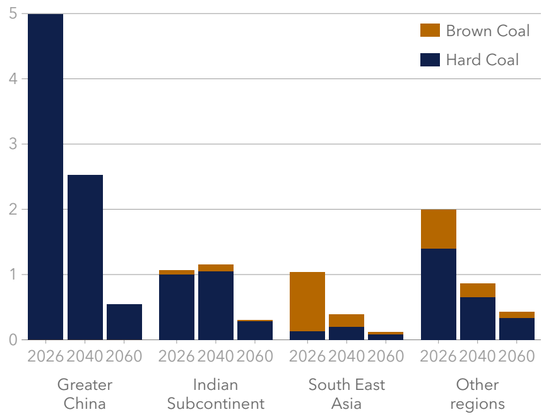


FIGURE 5.12 |

5.4 CCS

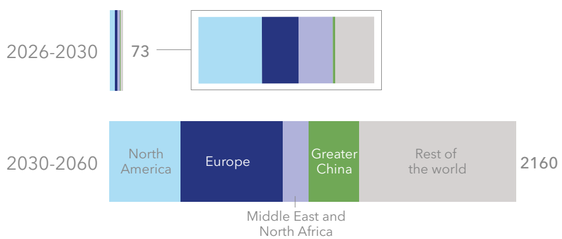
- **We forecast CCS will reach 120 $MtCO_2$/yr by 2030, representing an approximately 2.5-fold increase from CO_2 captured and stored currently.**
- **After 2040, CCS deployment in manufacturing rises to 860 $MtCO_2$/yr and makes up about 38% of all captured emissions by 2060.**
- **CCS will grow to capture nearly 14% of global CO_2 emissions in 2060, making it a critical, though insufficient on its own, enabler of decarbonization.**
- **Carbon dioxide removal (CDR) will capture 1,000 $MtCO_2$ in 2060, 43% of total captured emissions.**

For further information, please refer to our *Energy Transition Outlook: CCS to 2050*. While short- and long-term forecasts have been updated, many of the details on technology, costs, and policy and financing mechanisms are still relevant.

The CCS project pipeline remains strong, although the pace of near-term progress reflects the complexities inherent in CCS project delivery.

- Projects in development, particularly those without clearly defined financing and transport and storage infrastructure, are often experiencing delays or materialize at lower capacity. We have therefore revised downwards our assessment of the de-risked CCS project pipeline by 2030 compared with our *CCS to 2050 Outlook* (DNV, 2025)
- In the short term, North America will remain the leader in CCS, accounting for 40% of global CO_2 captured and stored in 2030. Another 40% will come from the Middle East and North Africa, Europe, and the OECD Pacific combined.
- We forecast 2,600 $MtCO_2$/yr will be captured and stored globally by 2060. This includes volumes from point sources and direct air capture (DAC). The leading regions in CCS will be Europe (27% of global total) and North America (19%), followed by Greater China and OECD Pacific (13% each).

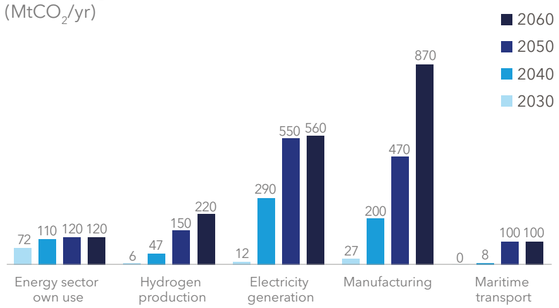
Additional emissions captured with CCS after 2025
($MtCO_2$/yr)



CCS deployment will be led by the power sector through 2050. Manufacturing gains momentum from 2040 and holds the largest share by 2060, at 38% of CCS applications.

- CCS in power will see substantial growth from 2030 to 2050, reaching close to 35% of total CO_2 captured. A large share of CCS in power (46% in 2060) will come from biomass in Europe and North America. Capturing CO_2 from fossil fuels, predominantly coal, will be most prominent in China.
- CCS is essential to address hard-to-decarbonize emissions (e.g. steel production and maritime transport). We expect onboard capture to scale from the 2040s in parts of the global shipping fleet. Manufacturing sectors will be the biggest users of CCS in Europe and to a lesser extent in North America.
- CCS in energy supply will be driven by production of hydrogen and ammonia in Greater China, the Middle East and North Africa, Europe and North America. However, this will represent less than 10% of total emissions captured globally owing to a strong preference for renewable-based electrolytic hydrogen limiting blue hydrogen to narrower applications (such as synthetic fuels that require biogenic CO_2 to meet lifecycle emissions criteria).

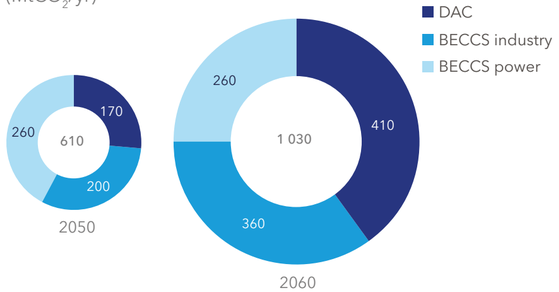
CCS deployment across sectors
($MtCO_2$/yr)



Carbon dioxide removal (CDR) expands after 2050, reaching 1,000 $MtCO_2$/yr by 2060.

- CDR becomes important in the second half of the century to reduce the large carbon budget overshoot. By 2060, CDR will be capturing 43% of total CO_2 emissions captured and stored.
- Bioenergy with CCS (BECCS) is generally the cheaper CDR option and by 2060, 60% of CDR will come from BECCS. Power will be the leading BECCS application until the mid-2050s, while manufacturing will see steady growth and eventually narrow the gap.
- DAC costs remain relatively high, at around USD 300/tCO_2 even in 2060. While voluntary markets continue to support early uptake, the long-term growth of DAC is expected to be driven primarily by compliance carbon markets, enabling deployment to reach 410 $MtCO_2$/yr by the end of the forecast period.

Carbon dioxide removal in 2050 and 2060 by sector
($MtCO_2$/yr)



Recent CCS developments and near-term project pipeline

While the long-term outlook for CCS remains positive, deployment over the last year has been progressing more slowly than anticipated. Final investment decisions (FIDs) have not materialized at the pace we assumed in earlier outlooks, and several projects have experienced delays. We have therefore revised our *CCS to 2050* estimate (DNV, 2025) of the de-risked CCS project pipeline in 2030 from 210 $MtCO_2$/yr to 120 $MtCO_2$/yr captured and stored.

However, the near-term pipeline continues to support significant growth. According to the Global CCS Institute, more than 60 $MtCO_2$/yr of CCS capacity is currently operating globally, with a further 40 $MtCO_2$/yr under construction (GCCSI, 2026). Together, this represents over 100 $MtCO_2$/yr of relatively low-risk capacity expected to come online before 2030. While the broader project pipeline continues to expand, the near-term conversion of announced projects into operating facilities remains uncertain.

While the broader project pipeline continues to expand, the near-term conversion of announced projects into operating facilities remains uncertain.

The industrial and energy supply sectors remain the primary drivers of CCS deployment, accounting for 22% and 59% of captured and stored CO_2 in our 2030 forecast, respectively. We expect large-scale developments such as Saudi Aramco's Jubail CCS hub to strengthen growth in the energy sector. CCS for managing emissions from fuel production and processing continues to be important.

Another notable development is the growing link between CCS deployment and rising electricity demand from AI and data centres. Technology companies are increasingly supporting gas-fired power generation equipped with CCS as a source of reliable low-carbon electricity. Recent announcements from Google, Microsoft, and Chevron suggest that data centre growth could become an important catalyst for new CCS-enabled power projects (Patel, 2025; Chevron, 2026). As a result, we forecast the power sector will account for around 10% of global captured and stored CO_2 by 2030.

Hydrogen plays a smaller role in our near-term outlook, accounting for only around 5% of captured and stored CO_2 in 2030. However, it remains an important source of near-term demand supported by both operating projects and facilities currently under construction in North America and Europe, including major blue hydrogen developments and associated CO_2 transport and storage infrastructure.

Greenergy Data Centers (GDC), Estland.

CCS outlook to 2060: regional and sectoral perspectives

The regional uptake of CCS will reflect both different policy priorities and sector-specific decarbonization needs.

Europe will overtake North America in the mid-2030s and become the leading CCS region by 2060, accounting for slightly more than one-quarter of global CO_2 captured and stored. European growth will be driven primarily by manufacturing, which will represent almost 70% of regional CCS deployment by 2060. Continued policy support for industrial decarbonization and a stronger carbon price, alongside more widespread availability of CO_2 transport and storage infrastructure, will help scale CCS across hard-to-abate sectors. In contrast, the role of CCS in the power sector will gradually decline after 2050 as other low-carbon technologies further decarbonize the electricity system.

North America leads CCS deployment today and will remain the largest CCS market through 2030 before falling to the second-largest, accounting for around one-fifth of global deployment in 2060. Anticipated ongoing support mechanisms in both the US and Canada will sustain investment across a broad range of sectors. The North American sector mix will feature a more prominent role for the energy sector's own use (15% versus 7% globally), reflecting the continued importance of oil, gas, and fuel-processing industries.

Greater China will emerge as one of the world's largest CCS markets, accounting for around 13% of global captured and stored CO_2 by 2060. Strong government direction and financial backing will help accelerate project deployment and establish CCS as an important component of China's carbon-neutrality pathway. Growth will be concentrated in the power sector which will account for around half of regional CCS deployment, largely driven by coal-fired power generation. CCS supports both low-carbon power and fuel production as hydrogen production will provide a further 23%.

The OECD Pacific will contribute a similar share of global CCS deployment, reaching around 13% by 2060. In Japan, South Korea, and Australia, CCS will remain an important part of long-term energy and climate strategies. Deployment will be heavily concentrated in power generation, which will account for 58% of regional CCS volumes, particularly in biomass, waste-to-energy, and natural-gas applications.

The Middle East and North Africa will also provide sizable contributions – around 6% of global CCS volumes. The growth will be closely linked to low-carbon hydrogen production, with the sector representing 45% of regional deployment.

Supported by emerging policy initiatives and growing public funding, CCS deployment in India will be dominated by manufacturing, which will account for nearly two-thirds of captured and stored CO_2 as industry seeks to reduce emissions while meeting growing demand for industrial output.

We project direct air capture (DAC) deployment to reach 410 $MtCO_2$/yr by 2060, with Europe, North America, and OECD Pacific accounting for around 40% of these volumes. This increase occurs despite DAC costs remaining around USD 300/tCO_2. The key enabler is growing demand for carbon dioxide removal (CDR) to address residual emissions from hard-to-abate sectors. While voluntary markets help scale the industry in the near term, compliance markets are expected to drive most deployment over the longer term. The UK is moving to integrate greenhouse gas removals into its Emissions Trading Scheme (ETS), and the EU has introduced certification rules for permanent removals while exploring their integration into the EU ETS. These developments provide a pathway to the durable demand needed for larger-scale DAC deployment.

CCS in manufacturing will see stronger growth after 2040, mostly in Europe, North America and the OECD Pacific

CO_2 captured by sector ($MtCO_2$/yr)

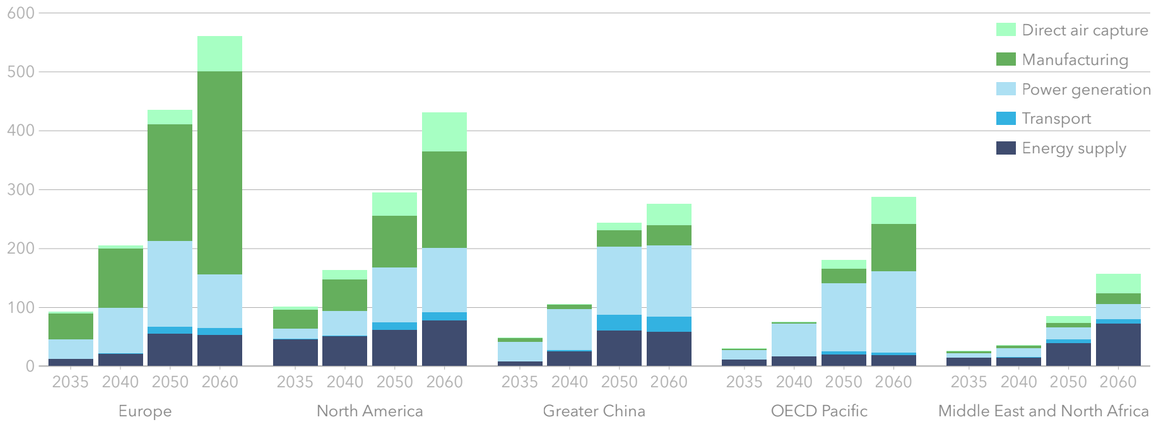


FIGURE 5.13 | Selected regions represent 75% of total capture in 2060.

European growth will be driven primarily by manufacturing, which will represent almost 70% of regional CCS deployment by 2060.

6 CLEAN MOLECULES & THERMAL SUPPLY

6.1 Hydrogen 104
6.2 Bioenergy 108
6.3 Direct heat 110

6.1 HYDROGEN

60 Mt of unabated hydrogen use will be replaced by renewable and low-carbon (clean) hydrogen

40% of new hydrogen production and use will be in China

1 Clean hydrogen will grow close to 100-fold* from today and mitigate over 2 Gt* of emissions from hard-to-electrify sectors annually by 2060

- The global hydrogen industry is currently a USD 240bn market by value, producing 100 million tonnes of fossil-based hydrogen per year.
- Overall hydrogen volumes will grow by 130%* and will see cumulative investments of USD 3.2trn to 2060*.

*We have updated these estimates since publishing our *Hydrogen Outlook* in May 2026, following updates to our ETO Model that reflect changes in a larger energy system. Specifically, we now expect a 30 MtH_2/yr reduction in 2060 relative to our *Hydrogen Outlook* forecast: 8 MtH_2/yr are from reduced estimate of demand for hydrogen and derivatives in aviation and maritime, 7 MtH_2/yr are in manufacturing (mostly due to reduced global 2060 energy demand for iron and steel), and 10 MtH_2/yr are in power.

2 Energy security will boost hydrogen production

- Renewable hydrogen and derivatives can increase security of supply and energy storage, for nations and companies reliant on global fossil supply chains to fuel hard-to-electrify assets.
- The current geopolitical situation will accelerate final investment decisions (FIDs). On top of the installed 1.5 Mt/yr of renewable electrolysis-based production in 2025, we forecast about 10 Mt/yr to be added by 2030 – with half of that in Europe and China.

3 Regulation and policy have not sustained early optimism, but are crucial for scaling

- Stalled and reversed hydrogen policies are the main reason our mid-century forecast for clean hydrogen is reduced by almost half* since our 2022 edition.
- High-level policies need to be translated into practical legislation, standards, and guidelines. Lessons from real-world projects and joint industry collaboration are key. Trust needs to be built through derisking, e.g. through technology qualification, verification and documentation, and certification.

4 To grow, hydrogen must close the safety confidence gap and document emissions reductions

- Operational learning from small-to-medium pilots is starting to inform inherently safe designs and procedures for large scale. However, it is not a copy-and-paste exercise to move from pilot to scale for both cost and safety assumptions.
- Hydrogen safety and performance require an understanding of the whole plant with subsystems. Best practices and increased standardization are still needed to unlock substantial investment capital.

Growth of hydrogen production, 2025–2035

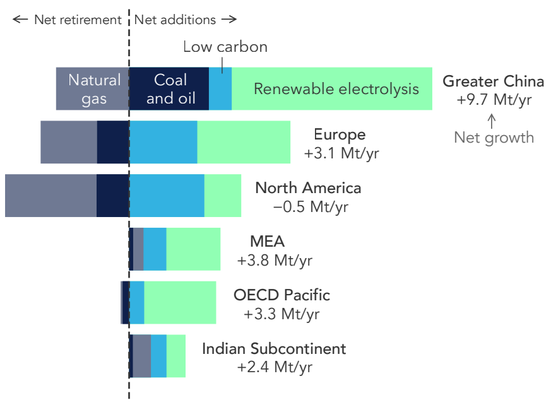


FIGURE 6.1 | Evolution of all hydrogen supply in selected regions, including derivatives. 'Oil' refers to process hydrogen in refineries.

Cost gap between unabated fossil and renewable hydrogen (USD/kg)

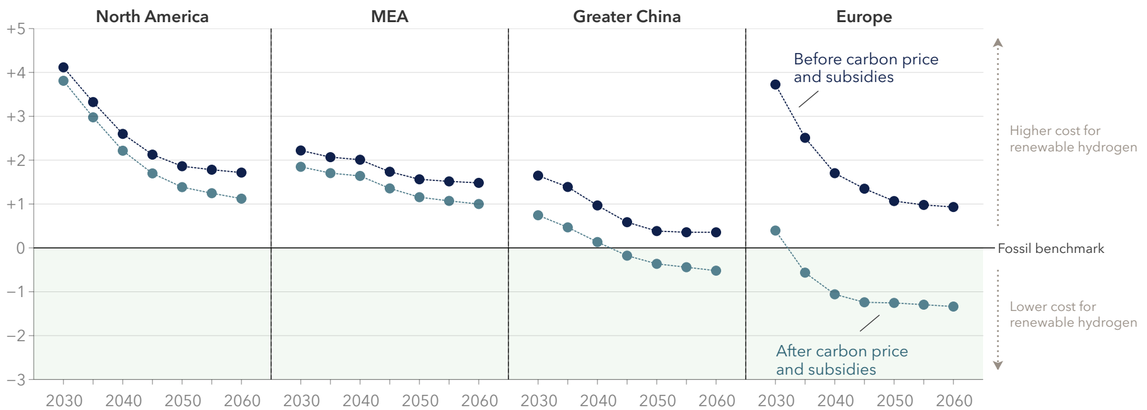


FIGURE 6.2 | Cost difference between the cheapest unabated fossil and renewable hydrogen. Production costs differ between regions.

Hydrogen demand

The market for clean hydrogen will attract over USD 3trn in investment over the next three decades. However, the role of hydrogen in the energy transition is narrower than we anticipated in our original hydrogen forecast (DNV, 2022).

Global hydrogen demand

(Mt/yr)

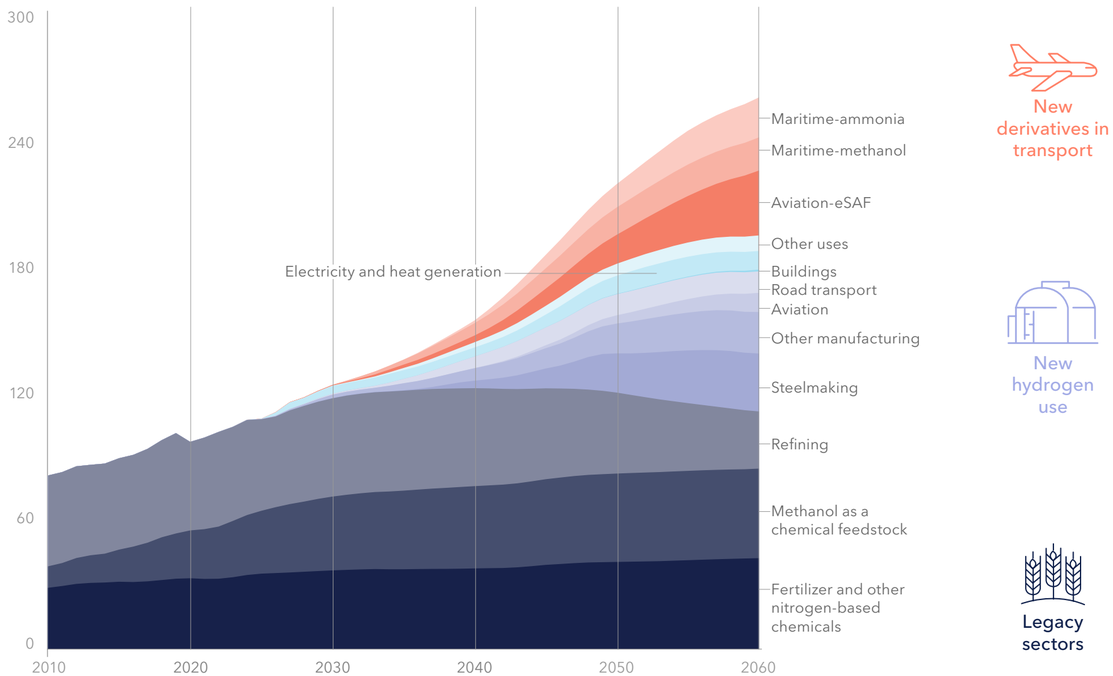


FIGURE 6.3 | Historical data source: IEA (2026). Other uses: transport and energy.

We now see hydrogen demand concentrating in three distinct use cases: transport fuels made from hydrogen combined with nitrogen or carbon, a limited set of new pure hydrogen uses, and a large base of legacy industrial demand that must be decarbonized and cannot be electrified. Across all three, policy and regulation remain the decisive drivers of uptake.

Legacy hydrogen users

Existing industrial users represent the most certain and stable source of hydrogen demand. Ammonia and nitrogen-based chemicals are likely to remain the largest market for clean (renewable and low-carbon) hydrogen, due to their critical role in fertilizer production and food security. Decarbonizing these industries offers a dual benefit: renewable hydrogen can both reduce the sector's substantial emissions and strengthen supply security by utilizing domestic renewable resources. However, a significant portion of feedstock ammonia will still come from fossil sources through to 2060, increasingly combined with carbon capture and storage (CCS). We expect China will lead green ammonia deployment, while the Middle East and North Africa (MEA) emerges as major export hub.

Methanol, which accounts for around 30% of hydrogen demand, will continue growing alongside economic development. However, decarbonization is challenging because methanol requires carbon as a feedstock, making fossil-based production dominant for decades to come. We expect China to lead renewable methanol deployment, while gas-rich regions continue relying on natural gas-based production.

Oil refining is the largest hydrogen consumer today, slightly ahead of fertilizer production. We expect demand to stay relatively resilient in the near term before gradually declining after 2030 as oil demand falls. Most refinery hydrogen production will continue to come from natural gas-based steam methane reforming, increasingly combined with CCS.

Hydrogen derivatives in transport

Transport represents the main source of new hydrogen demand, primarily through hydrogen-derived fuels. In shipping, batteries and pure hydrogen are generally unsuitable for long-distance vessels. Consequently, maritime hydrogen demand will arise largely through fuels where hydrogen is embedded in higher-density molecules that are easier to store, transport, and bunker at scale. E-fuels (including e-methanol) will gain market share first, benefiting from easier adoption, while ammonia will see broader deployment after the mid-2030s as infrastructure and technology develop.

In aviation, direct hydrogen use faces major technical and infrastructure barriers, particularly for long-haul flights. Hydrogen is therefore likely to enter mainly through synthetic sustainable aviation fuels (eSAF) rather than hydrogen-powered aircraft. Policy support, blending mandates, and carbon pricing will be essential for adoption. We expect bio-SAF to dominate initially, while eSAF scales up from the late 2030s as renewable hydrogen becomes more available.

Limited role as pure hydrogen uses

We have significantly revised our expectations for widespread new uses of pure hydrogen downward. Steel production will likely dominate long-term demand growth, with hydrogen-based direct reduced iron (DRI) emerging as the main pathway for low-carbon primary steelmaking where scrap recycling is insufficient. However, uptake will be gradual due to the long operational lives of existing blast furnace assets. Early hydrogen-DRI projects are likely to rely on unabated fossil-based hydrogen due to high clean hydrogen costs.

In road transport, we expect pure hydrogen will remain limited to niche heavy-duty trucking applications where there may be demanding range and refuelling speed requirements that cannot be fulfilled by batteries alone. Hydrogen use for heat in our forecast is restricted almost entirely to manufacturing (i.e. not used in building heating) where the use of pure hydrogen will be confined to a limited number of industrial processes where electrification is impractical and CCS is unavailable or undesirable. Pure hydrogen may also be deployed for selective applications within gas networks.

For further information, please refer to our recent Energy Transition Outlook: Hydrogen to 2060

Hydrogen supply

The next decade will mark a structural turning point for hydrogen. While volumes continue to grow broadly in line with historical trends, the main shift will be in how hydrogen is produced, with clean hydrogen accounting for most net additions.

Global hydrogen and derivatives supply by production route

(MtH_2eq/yr)

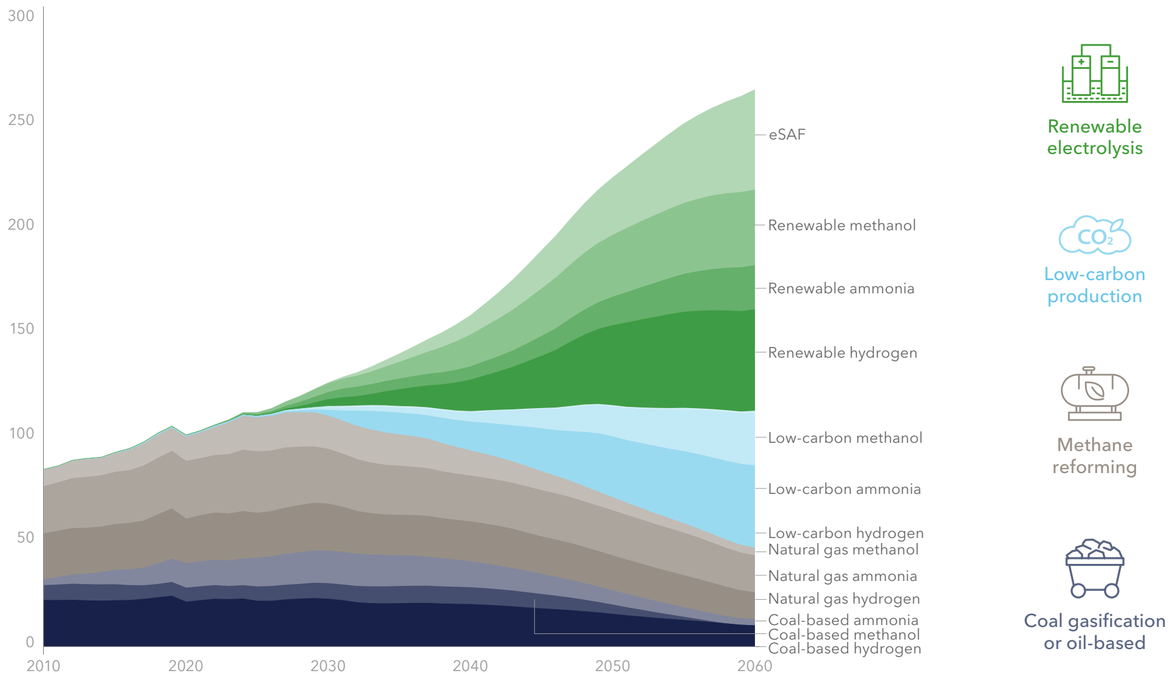


FIGURE 6.4 | Historical data source: IEA (2026). Low-carbon production includes coal gasification with CCS.

Global hydrogen and derivatives supply grows from 100 MtH_2/yr in 2025 to about 130 MtH_2/yr by 2035. Our ten-year outlook is derived from a risk-adjusted selection of projects from the DNV Alternative Fuels Insight database. Projects without identifiable demand are excluded. This means we only expect some 20% of announced projects to be realized. Clean hydrogen accounts for most net additions, while unabated production begins to decline in absolute terms (−8 Mt/yr). By 2035, a quarter of clean hydrogen supply will come from methane reforming with CCS; the remainder is electrolytic, the preferred route in regions like Europe, the OECD Pacific, Greater China, and the Indian Subcontinent.

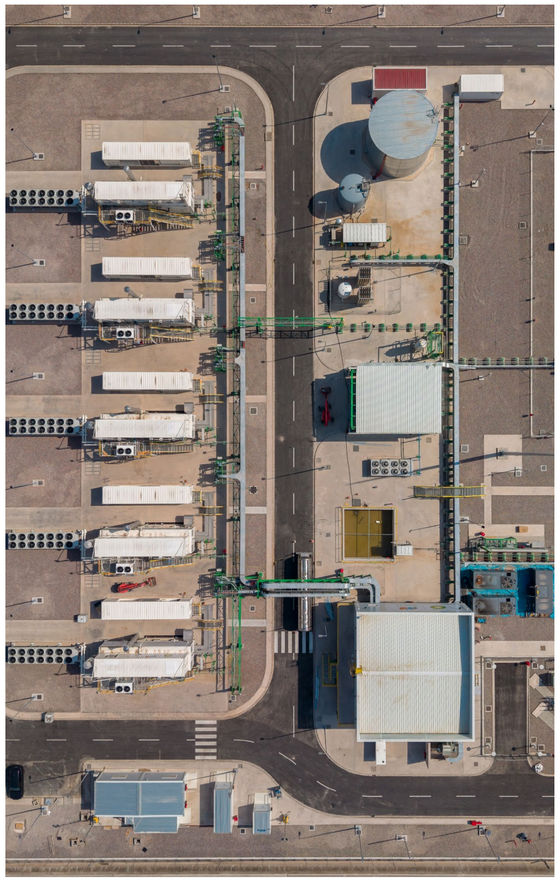
Beyond 2035, we expect global hydrogen and derivative production to accelerate rapidly, more than doubling to around 250 MtH_2/yr by 2060. This growth will be driven by three main demand sources, as outlined in the previous section. Continued cost reductions, maturing supply chains, and strong policy support in regions including Europe, Greater China, the Indian Subcontinent, and the OECD Pacific will further support growth.

Over time, we expect renewable and low-carbon hydrogen to develop in parallel rather than compete directly, sharing infrastructure and supporting the growth of integrated hydrogen markets.

By 2060, close to 85% of hydrogen production will be low-carbon or renewable, supplying roughly 210 MtH_2/year. About two-thirds of this clean hydrogen will serve new energy applications, while the remaining third will decarbonize legacy uses such as ammonia production and oil refining. Electrolysis becomes the dominant production route, accounting for close to 60% of total hydrogen and derivative supply, while low-carbon hydrogen from natural gas with CCS contributes another 23%. Unabated fossil-based production falls to about 17% of global supply.

Both low-carbon hydrogen and renewable electrolysis play important but regionally different roles. Natural gas reforming with CCS remains among the lowest-cost production pathways in many regions, particularly North America, which benefits from abundant low-cost gas and we expect to account for roughly one-quarter of global low-carbon hydrogen production by 2060. Europe, MEA, and China also develop significant low-carbon hydrogen industries as carbon pricing and emissions policies strengthen. However, we only expect a few regions (notably North America) to rely primarily on low-carbon hydrogen, as most governments favour renewable pathways for long-term decarbonization.

At the same time, renewable electrolysis becomes increasingly competitive. In Europe, China, and MEA, renewable hydrogen reaches cost parity with fossil-based or low-carbon alternatives between the early 2030s and 2040s. The preferred production model varies by region: grid-connected electrolysis supported by power purchase agreements dominates in regions like MEA and North America, while off-grid systems paired with dedicated renewable generation are particularly attractive in regions with abundant solar and wind resources.



A noteworthy insight from our forecast concerns hydrogen derivatives, which we expect will account for around half of clean hydrogen demand from the mid-2030s onward. Their increasing role is closely linked to strong demand patterns, particularly in sectors where direct electrification is difficult. Despite higher production costs, derivatives offer major advantages over pure hydrogen: they are easier to store, transport, and trade internationally, and they can often use existing fuel and logistics infrastructure or be blended at specific ratios. These characteristics reduce market risk for producers, enable larger-scale projects, and help establish supply chains before pure hydrogen markets become fully developed.

Among derivatives, ammonia maintains a central role throughout the outlook that reflects its importance both as a fertilizer feedstock and as an energy carrier. Falling production costs are expected to make renewable ammonia increasingly competitive, particularly in regions with low-cost renewables such as MEA and China. Carbon-based e-fuels, including e-methanol and eSAF, also grow steadily as demand from maritime transport and aviation expands. Their growth is somewhat constrained by the availability and cost of sustainable CO_2, but they remain essential for decarbonizing transport sectors where direct electrification is not practical. Consequently, we expect renewable hydrogen to supply virtually all long-term e-fuel production. This aligns with emerging policy requirements and lifecycle emissions standards.

6.2 BIOENERGY

While biomass has traditionally been used to provide heat through direct combustion for cooking and space heating, its future role increasingly lies in supplying low-emission fuels and gases to sectors where electrification is difficult and generating electricity with net negative emissions.

Biomass can be converted into liquid fuels, gaseous fuels, and electricity with low lifecycle emissions. When paired with carbon capture and storage, it can even deliver net-negative emissions. As a result, biomass is a versatile renewable resource that can supply both renewable electricity and low-emission fuels for hard-to-abate sectors. Stricter environmental regulations, particularly fuel mandates, and growing energy security concerns are accelerating the shift towards higher-value bioenergy applications.

Bioenergy will see steady growth in primary energy supply from 50 EJ/yr today up to a maximum of 72 EJ/yr in 2050. It will remain the largest renewable source of primary energy until 2040 when it is surpassed by solar power. Despite no lack of demand for bioenergy, it peaks in the primary energy mix by mid-century because while sources like solar and wind are primarily capital-constrained, bioenergy is also resource constrained.

Bioenergy demand by type, region and sector

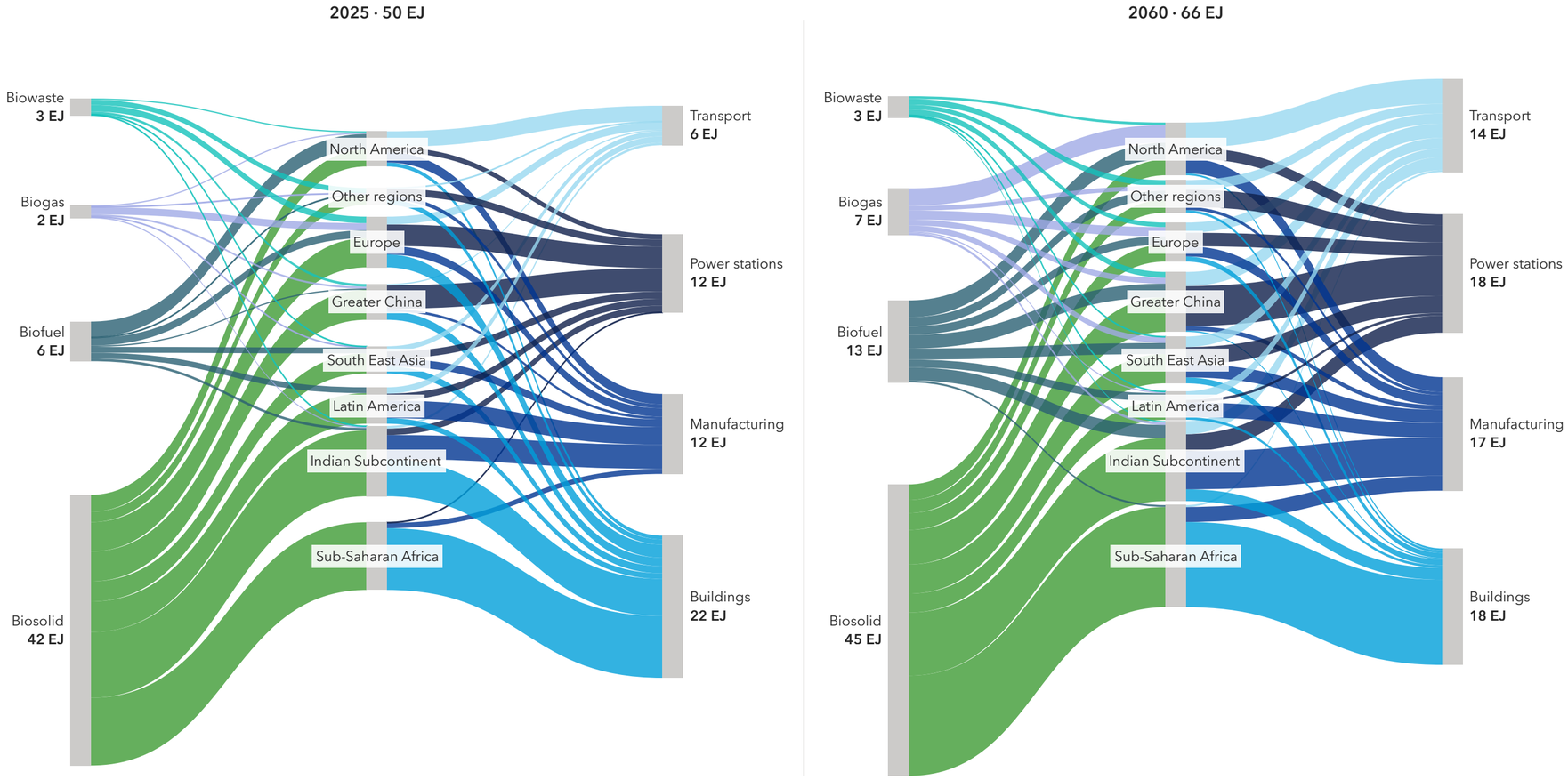


FIGURE 6.5 | Flow of bioenergy categories in 2025 and 2060 – from energy carrier to demand sectors (rounded numbers) and intermediate split for regions.

The trend towards higher value bioenergy

While bioenergy grows only modestly overall, future demand shifts away from crude combustion and toward biofuels and biogases are significant. Driven by new sectors – primarily manufacturing, aviation, and maritime transport – biofuel and biogas together will increase their combined share of total bioenergy consumption from 14% to 28% already in 2040. Growth is especially strong in North America where biogas accounts for 40% of bioenergy consumption by 2040. This reflects a favourable combination of demand-side drivers (mandates and voluntary carbon credit schemes) and supply (where bioenergy creates an attractive revenue opportunity for organic waste, especially from agriculture). Biogas consumption will more than double in Europe, partly owing to increased injection of biomethane into gas grids. South East Asia and Greater China will have exponential growth from late 2030s and reach global shares of 12-14% by 2060.

Biogases can meet the high-temperature needs of manufacturing where low-emission alternatives are scarce. Due to this, biogases are increasingly replacing natural gas in hard-to-decarbonize industry such as cement and metal production.

The regulatory regime in aviation, particularly in North America, Greater China, and Europe, is helping to expand the market for SAF. By 2060, biomass-based SAF reaches 7 EJ/yr. Although aviation oil demand peaks in 2024 and subsequently declines, conventional oil remains the dominant aviation fuel in 2060 at 16 EJ/yr, accounting for 59% of the sector's total energy use and more than twice the volume of SAF.

Biofuel is already used in maritime transport by blending biofuel qualities with fuels of fossil origin. It will grow from a very low volume to reach only a 2% share of maritime energy demand by 2030. From then on growth will be faster due to increased mandate requirements and improved and cheaper production technologies. By 2060, bioenergy will be the biggest non-fossil energy source in the sector, supplying 25% of total maritime energy demand.

Power and negative emissions

Power from biomass is an established technology but is poised for growth (by some 80% by 2050). Bioenergy to power with CCS (BECCS) adds an attractive financial component and spurs this growth, but by 2060 electricity from biomass will provide less than 2% of the global total electricity demand.

Thermal supply from bioenergy to be substituted in most of the world

Across much of the world, new technologies like heat pumps and induction stovetops will replace bioenergy demand for cooking and heating via direct combustion. The resulting efficiencies will see bioenergy for these end uses shrink by 20% by 2060. The exception is Sub-Saharan Africa where demand for traditional biomass is growing, from 9 EJ/yr today to 12 EJ/yr by 2060. This reflects a combination of rapid population growth, limited access to reliable electricity sources, and the high upfront cost of new cooking and heating equipment. As a result, bioenergy demand in SSA for heating and cooking will be more than of all other regions combined.

The supply constraint

The key limitation on bioenergy is supply, not demand. Bioenergy depends on sustainably sourced biomass, a resource with competing uses and finite availability. This limits the extent to which bioenergy can scale. Aviation provides a useful illustration: despite strong policy support and relatively high willingness to pay for low-emission fuels, bio-based SAF reaches only 7 EJ/yr by 2060, while total aviation energy demand remains more than three times larger. Biomass will not be available in sufficient quantities to fully decarbonize even the sectors with the strongest claim on it. Its scarcity reinforces bioenergy's role as a strategic resource for hard-to-abate applications rather than a universal substitute for fossil fuels.

Given the finite quantities of sustainably sourced biomass available, policymakers must be aware of the most feasible feedstocks and end-uses in specific locales. This is best achieved through the kind of data-led approach the IEA is pioneering through its Biogases Geospatial Resource Assessment Model (BioGRAM). BioGRAM shows that sufficient policy support will sustainably lead to global waste-derived production of biogases reaching nearly 6 EJ by 2035 in line with the COP30 pledge to quadruple sustainable fuels (IEA, 2025a). Our 'most likely' future forecast is for biogases to reach this volume before 2040.

North America supplies half of global biogas in 2040

Biogas supply by region (EJ/yr)

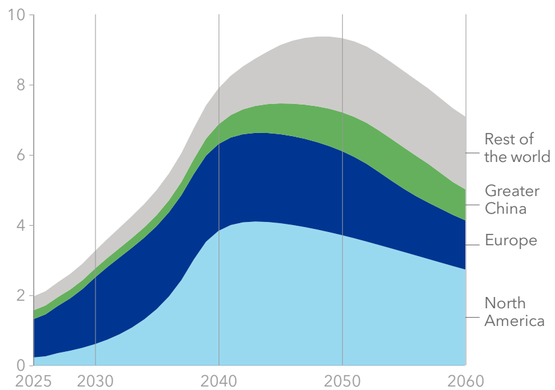


FIGURE 6.6 | Global supply of biogas.

Biofuel supply is evenly distributed

Biofuel supply by region (EJ/yr)

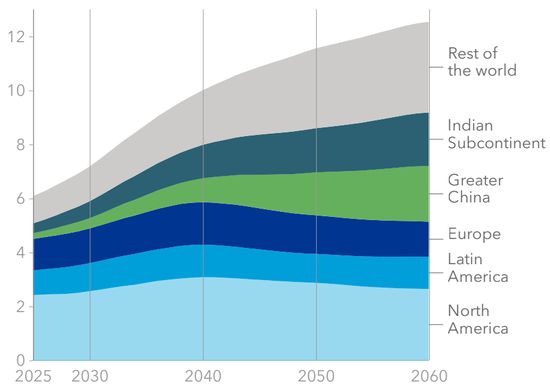


FIGURE 6.7 | Global supply of liquid biofuel.

6.3 DIRECT HEAT

Direct heat refers to the thermal energy from power stations and industrial processes, delivered as hot water or steam. This energy is either supplied to district heating networks serving residential and commercial buildings or used within industrial facilities for process heat. Direct heat demand remains concentrated in buildings and manufacturing, which account for 97% of the total heat demand today. Demand for direct heat in buildings grows from 8 EJ/yr today to 11 EJ/yr by 2060, while manufacturing remains unchanged at 9 EJ/yr.

While district heating grows steadily globally, its regional distribution remains largely unchanged. District heating potential is larger in regions with dense urban areas and high heating demand, such as Europe and North East Eurasia. India and Sub-Saharan Africa have smaller potential uptake due to lower urban density and lower heating demand. However, technical potential alone does not determine deployment. Uptake depends on policy support, municipal heat planning, and availability of resources. Consequently, North America and OECD Pacific show substantial potential, but uptake has historically been low.

North America's use of district heat is set to grow steadily, reaching 7% of global demand by 2050. However, uptake remains below its potential because gas prices are more affordable in North America and infrastructure there is typically built and dimensioned for heating at household rather than district level. Most growth occurs in the commercial sector, where consistent heat demand makes district heating viable. Emerging US state initiatives promoting clean heating infrastructure, thermal networks, and building decarbonization support this expansion (New York State Senate, 2022; Massachusetts Department of Public Utilities, 2024). There are similar developments happening in Canada, like the Zibi District Thermal System recovering industrial waste heat replacing conventional gas heating (NRCan, 2025). In contrast to North America's limited uptake, Germany is expanding district heating through its *Heat Planning Act*, that requires municipalities to develop local heat plans with a focus on decarbonization (BBSR, 2024).

Technical potential for district heating does not determine deployment alone. Uptake depends on policy support, municipal heat planning, and availability of resources.

District heating grows steadily on a global scale

District heat supply by region (EJ/yr)

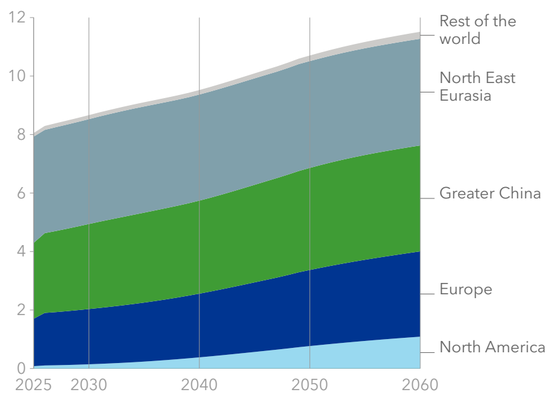


FIGURE 6.8 |

Decarbonization and localization

Coal, natural gas, and oil supply 80% of the global demand for district heating today, but this peaks in 2030. Natural gas is largest, and we forecast a sharp decline in use of coal towards 2040. Much of the reduction is driven by declining coal use in China, where consumption falls from 1.8 EJ in 2025 to 0.45 EJ in 2060.

As fossil fuel use declines, district heating decarbonizes and adapts to local resource availability. Most of the growth in district heating is covered by heat pumps and other heat only technologies, as renewable electricity reduces the value of electricity produced alongside heat in combined heat and power plants (CHP). Heat pumps increasingly provide baseload heat, but supplementary technologies remain necessary during peak demand when heat pump efficiency declines. Historically minor sources like bioenergy, municipal waste, and geothermal expand. Bioenergy and waste uptake remains largest in Europe, supplying 34% of district heat in 2040. Solar heating expands in Europe, increasing five-fold over the next 15 years. Geothermal district heating grows mainly in Greater China, supported by rich geothermal resources (Guo et al., 2023). A more diversified supply of direct heat adapted to local conditions reduces dependence on imports and contributes to increased energy security and more resilient heat supply systems (IEA, 2026).

Heat pumps will supply an increasing share of district heating globally. Air-source heat pumps remain the dominant technology due to low capital cost and broad applicability. Europe experiences the fastest growth, with heat pumps supplying 28% of regional district heating demand in 2040, driven by well-established and increasingly ambitious EU decarbonization policies. RED III sets targets for renewable heating, the *Energy Performance of Buildings Directive* (EPBD) pushes buildings towards zero-emission heating solutions, and the *Energy Efficiency Directive* (EED) sets requirements for efficient district heating and waste heat utilization (JASPERS, 2024).

Data centres

Heat pump efficiency increases when using excess heat rather than outside air. This makes low-temperature waste heat from data centres an attractive source for district heating – subject to site-specific design constraints. For example, if located close enough to urban areas, data centres can support heat pump expansion for district heating (IEA, 2025b). Europe is introducing policies to encourage this. Germany has introduced energy efficiency standards, requiring data centres to utilize waste heat (Bundesregierung, 2023), while Norway requires approved cost-benefit analysis from developers before construction (Regjeringen, 2025).

Heat pumps will cover most of new district heat demand

District heat supply by source (EJ/yr)

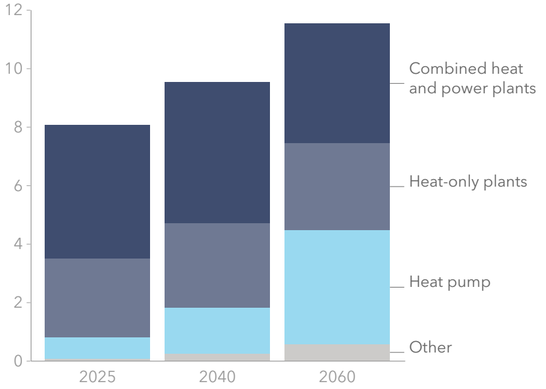


FIGURE 6.9 |

Fuel mix for district heating becomes more regionally diversified

Heat supply by source (EJ/yr)

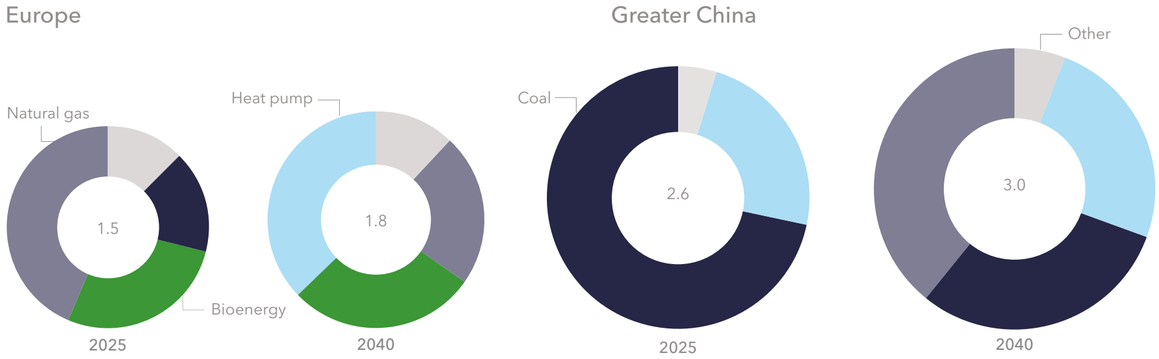


FIGURE 6.10 | Other includes: solar heater, heat pump, hydrogen, bioenergy, oil, geothermal, and natural gas.

7 POLICY

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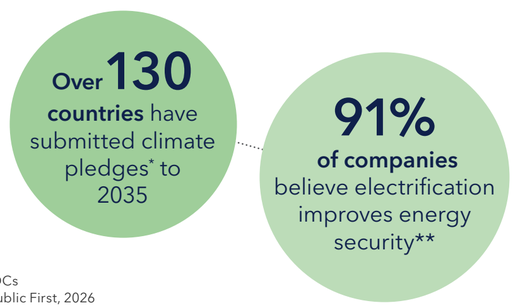
7.0 SUMMARY

The shift from climate-first to multi-objective policymaking is now firmly established: governments increasingly support technologies and infrastructure that simultaneously advance affordability, competitiveness, energy security, and decarbonization.

Policies that endure politically are those that strengthen sovereignty and domestic industry, reduce strategic dependencies, lower energy costs, and safeguard against climate and energy supply risks. As a result, resilience has become a guiding principle, replacing the debate between domestic cleantech production and lowest-cost imports with more balanced strategies.

Countries seek balance through supplier diversification and investment in domestic manufacturing of strategic technologies that align with national transition pathways. Renewable energy is central to energy security strategies because, as observers have noted (McKibben, 2026; Krugman, 2026), the sun and wind do not transit the Strait of Hormuz. However, policymakers are wary of replacing dependence on imported fossil fuels with reliance on imported cleantech equipment from concentrated supply chains, while also seeking domestic economic returns on public spending. As a result, local content rules, localization requirements, and trade measures are becoming more prominent, elevating strategic and political considerations alongside cost. This raises short-term costs in exchange for greater resilience, supply-chain security, and domestic value creation.

These multiple objectives contribute to a more fragmented global policy landscape. For energy companies, this means more complex projects, greater policy uncertainty, and a wider range of policy interventions driven by geopolitical and industrial priorities. In the Energy Transition Outlook (ETO) Model, technology selection is based on various factors, including cost, revenue, availability, utility, and policy (Chapter 11). We incorporate geopolitical dynamics by adjusting key assumptions (Chapter 1) that include GDP growth, manufacturing footprints, regional technology costs, supply chain development, and the energy technologies and resources that regions are likely to prioritize.



* NDCs
** Public First, 2026

Key energy policy trends

- **Climate policy for benefits beyond climate.** While momentum on climate targets has slowed, climate action drivers remain strong and increasingly diverse. The IEA's assessment of 2035 targets submitted through the 2025 nationally determined contributions (NDC) cycle found limited strengthening of near-term emissions reductions (IEA, 2026a). Climate concerns have slipped behind immediate economic and geopolitical priorities, and global climate consensus has weakened. Decarbonization remains a strategic priority, but is increasingly tied to energy security, cleaner air, industrial competitiveness, and climate resilience. The EU's carbon border adjustment mechanism (CBAM) is also driving climate policy and carbon pricing reforms among trading partners.

- **Deteriorating confidence in fossil fuel reliability strengthens the case for renewables.** Successive energy shocks and disruptions at fossil fuel chokepoints have highlighted the risks of fossil fuel dependence, including price volatility and supply disruption. Europe's exposure to Russian gas and Asia's dependence on Middle Eastern energy imports have reinforced energy security concerns. As a result, policymakers increasingly view renewable energy as a means of delivering affordability, security, and sustainability, while improving domestic control over energy supply and reducing exposure to external price shocks.

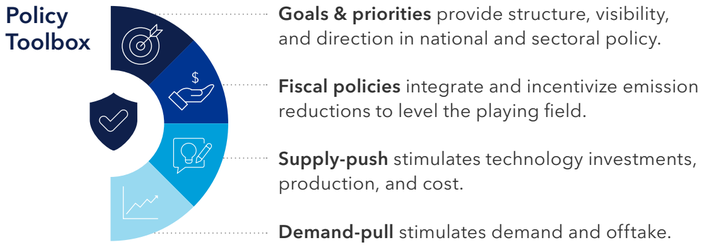
- **Electrification policy gaining momentum, but delivery lags.** Advances in electro-technologies and cheaper renewable electricity underpin electrification policy. The *EU Electrification Plan* (EC, 2026a), for example, sets an indicative target of 46% electrification (final energy demand) by 2040, up from 23% today, citing renewable deployment, efficiency, and electrification as drivers of strategic autonomy, lower energy costs, and competitiveness. Business support is also strong: across 18 countries, 91% of companies say electrification improves energy security, 79% say geopolitical instability increases its urgency, and 90% aim to electrify operations by 2035 (Public First, 2026). However, underinvestment in grids, permitting delays, and limited support for upfront equipment replacement constrain progress.

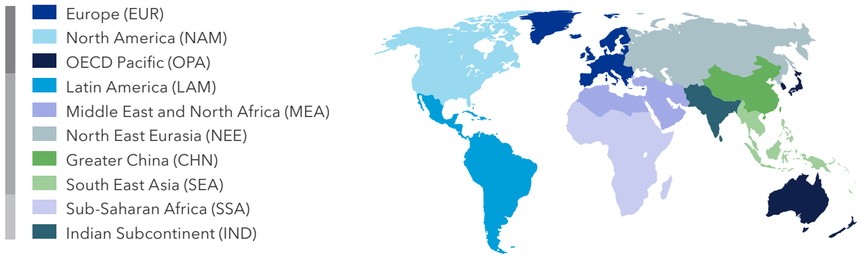
- **Affordability measures risk undermining transition alternatives.** With energy prices at around 25% above pre-Iran-war levels, governments are reintroducing subsidies and price controls after scaling them back since 2023 (OECD, 2025). Rising inflation risks and weaker growth linked to the war are affecting economies unevenly; fossil exporters generally benefit, while importers face economic headwinds. In response, many countries are expanding fiscal relief measures. While these can shield consumers and businesses, they distort price signals, strain public finances, and delay efficiency improvements and structural reforms unless coupled with clear sunset provisions (IMF, 2026).

- **Transition policy diverges across regions.** Policies are shaped by income levels, resource endowments, and economic priorities. Most high-income regions and China prioritize systemic decarbonization including in hard-to-abate and trade-exposed sectors through targeted support for emerging low-carbon alternatives and measures to reduce carbon leakage. By contrast, fossil fuel exporters, including the US, continue to support fossil fuel infrastructure and undermine climate action, recently seen in the delay of the *Net-Zero Framework* for shipping. Middle- and low-income regions prioritize affordable energy access, economic development, and infrastructure expansion, balancing decarbonization with broader socioeconomic needs. The favourable economics of solar and wind are accelerating renewable deployment in these regions and creating opportunities to leapfrog carbon-intensive energy additions.

7.1 THE POLICY TOOLBOX AT WORK IN ETO REGIONS

Below, we summarize the state of play of the 'policy toolbox' (as illustrated to the right) across the ETO regions.





High income

- **Goals:** Regions have divergent strategic priorities. Major economies in OPA and EUR provide long-term certainty on pursuit of energy independence and decarbonization (mid-century goals) while protecting industrial competitiveness. In NAM, US federal policy pursues fossil-based energy and AI dominance, while Canada maintains decarbonization goals amid fossil fuel development.
- **Fiscal policy:** Economy-wide carbon pricing frameworks are maturing and expanding. Japan launches its GX-ETS carbon market in 2026, while the EU is broadening CP coverage, implementing CBAM, and advancing electricity tax reforms (GBFR, 2026) to support electrification across transport, industry, and heating.
- **Supply-push measures:** Alongside renewable power, and nuclear power expansion in countries with established nuclear capabilities, governments promote hydrogen, carbon capture, and low-carbon fuels through R&D, funding, and tenders. Measures support CAPEX and revenue streams to bridge cost gaps and scale early supply. In NAM, support remains strong in Canada, while US federal policy is being rolled back.
- **Demand-pull measures:** Comprehensive policy mixes in OPA and EUR include offtake mandates, procurement, subsidies, or incentives to cover additional costs and strengthen market creation. These are deepening electrification and deployment of decarbonization solutions.

Middle income: China

- **Goals:** Rigorous policy for self-reliance, competitiveness, and carbon goals (peak before 2030 and carbon neutrality before 2060). Long-term planning permeates the entire energy system, shaping its development and providing state guidance on overseas investments to strengthen market access and supply chains.
- **Fiscal policy:** China uses a state-led investment model, providing low-cost capital to prioritized industries, aligning state-owned enterprise (SOE) with national plans, sustaining long-term policy support, and revising fiscal frameworks to support low-carbon development. The national carbon market is expanding to industry to encourage best practice and align with international trade-related carbon mechanisms.
- **Supply-push measures:** Five-Year Plans set priorities and targets for R&D and technology development. Supply-side initiatives are accompanied by innovation spending and long-term government support to strategically prioritized industry areas as outlined in the governments *Guiding Catalogue for Industrial Structure Adjustment*.
- **Demand-pull measures:** China combines efficiency standards, emissions limits, capacity and EV targets, and coal-to-gas and coal-to-electricity switching clean heating programmes. Its industrial upgrading strategy promotes electrification and hydrogen with demand stimulated through industrial clusters, demonstration zones, SOE-led offtake, and emerging hydrogen use mandates.

Middle income: Other regions

- **Goals:** Countries in MEA, LAM, SEA, and NEE regions generally prioritize economic growth and export competitiveness. Decarbonization targets are set for 2050-2065, and several regions balance entrenched fossil fuel interests with decarbonization.
- **Fiscal policy:** Carbon pricing is evolving in response to international trade-related carbon mechanisms but remains limited. Fossil fuel subsidies are prevalent.
- **Supply-push measures:** Policy is production focused with investments in both fossil and renewable energy. MEA in particular has large-scale renewable power, hydrogen and CCS projects benefitting from strategic state investments and favourable access to finance in GCC countries. Export-positioning is key to oil and gas exporter regions like NEE, MEA, and LAM; the latter also leverages advantageous renewable energy resources in power expansions.
- **Demand-pull measures:** Policy to advance economy-wide decarbonization and the transition outside power sectors are progressing somewhat slowly. Singapore, in SEA, is a notable exception with its cross-sectoral *Singapore Green Plan 2030*. In MEA, e-mobility policies are taking root, as are buildings efficiency standards on energy and water consumption. Industrial decarbonization focus is emerging to maintain access to low-carbon international markets.

Low income

- **Goals:** IND and SSA countries prioritize energy access and economic development while pursuing longer-term decarbonization. Net-zero targets are typically set for 2060-2070 and often depend on international support.
- **Fiscal policy:** Carbon pricing remains limited. India's national carbon market for industry launches in 2026, while both IND and SSA countries seek opportunities in international carbon markets under the *Paris Agreement*, Article 6. Fossil fuel subsidies remain widespread to support energy affordability.
- **Supply-push measures:** India stands out for its mission-driven industrial policy that combines national and state support for domestic manufacturing, renewable deployment, hydrogen production, and CCS. In SSA, policy focuses mainly on renewable power development, with some countries pursuing hydrogen export opportunities, though investments largely depend on external finance.
- **Demand-pull measures:** Limited overall, but India supports deployment through renewable purchase obligations, e-mobility incentives, emerging industrial decarbonization initiatives, and hydrogen procurement tenders. SSA policies remain concentrated on low-carbon power supply, with less policy progress in transport, buildings, and industry.

7.2 POLICY PROGRESSION IN SELECT SECTORS

About **60%** of global energy spending is electricity-related (IEA, 2026b)

124 **countries** have RES-E targets

Policy for renewables-led power systems: system integration and adequacy

Electricity-related investments are growing due to converging policy trends: the expansion of increasingly cost-competitive renewable power and pursuit of electrification of energy end uses for greater efficiency when shifting from fuel combustion to electro-technology.

Policy levers: power

Targets for electricity from renewable energy sources (RES-E) are set by 124 countries (REN21, 2026) driven by favourable economics, environmental benefits, domestic resource availability, and energy security concerns that reduce vulnerability to fossil fuel volatility risks. As renewable technologies have become increasingly cost competitive, private investments now drive much of deployment and government spending is decoupling from capacity growth (IEA, 2026a). Policy support is shifting toward market-based mechanisms, with contracts for difference, tariffs, and premiums from reverse auctions providing revenue certainty while avoiding excessive remuneration. Combined with RES-E targets, these mechanisms help sustain long-term investment pipelines. Renewables receive varying levels of support in different regions through multiple mechanisms that have changed over time, and will continue to change (Figure 7.1). As renewable penetration rises, costs come down, and markets mature, the need for generation support reduces. Attention then shifts from deployment support towards system integration and market design (Chapter 4). This includes technical requirements and remuneration frameworks that value adequacy, firm capacity, and short-term flexibility to maintain system reliability.

Grids are central to electricity policy worldwide. Governments are expanding investments to modernize and extend grids and enable greater renewable electricity deployment. India's National Transmission Plan has allocated USD 110bn to integrate more than 600 GW renewable capacity by 2032 (MOF, 2024), while China plans to invest around USD 700bn in grid infrastructure by 2030 (Xin, 2026). In Europe, the 2023 *EU Action Plan for Grids* (EC, 2023) identified EUR 584bn of investment needs over the next decade, followed by the *2025 Grids Package* (EC, 2025) including measures to accelerate permitting and increase funding through the *Connecting Europe Facility*'s energy budget five-fold from approximately USD 6.6bn to 34bn (EUR 5.8bn to EUR 29.9bn in the proposed *Multiannual Financial Framework, 2028-2034*). Globally, the COP29 *Global Energy Storage and Grids Pledge*, supported by 58 countries, highlights the critical role of grid modernization in achieving the COP28 target of tripling renewables capacity by 2030.

From intermittent to 'round-the-clock' renewables. Policies increasingly pair renewable generation with storage to reduce curtailment and meet evening peak demand. Solar+storage auctions are driving investment in utility-scale projects that combine solar PV with battery energy storage systems (BESS). This is helping manage grid constraints in countries including Australia, Brazil, India, Germany, Saudi Arabia, and the UAE. India has introduced dedicated tenders for standalone BESS, solar+storage, and firm dispatchable renewable energy, while the UAE's Masdar round-the-clock project combines 5.2 GW of solar PV with 19 GWh of BESS to provide continuous

renewable power. Major technology companies – including Amazon, Google, and Microsoft – are also driving demand through 24/7 carbon-free power procurement, reinforcing demand for storage-backed renewables. At the residential level, Australia expanded its *Cheaper Home Batteries Programme* in 2025 from approximately USD 1.6bn to 5bn (AUD 2.3bn to AUD 7.2bn), targeting 40 GWh of storage by 2030.

- **Adequacy policies extend the role of thermal power, particularly natural gas.** Capacity markets for dispatchable capacity are designed to ensure security of supply. Reliability is the key criterion and policy has generally not yet incentivized emission-free alternatives. Analysis of European capacity markets shows that, of the EUR 90bn allocated between 2014 and 2024, more than two-thirds went to thermal generators, with gas-fired assets receiving the largest share (Aurora Energy Research, 2025). ACER highlights a ten-fold difference in capacity auction prices across the EU, attributing much of this inefficiency to limited cross-border coordination in capacity planning (ACER, 2025). Overall, policy frameworks across many ETO regions continue to support natural gas as a peak demand and balancing resource. While many countries, particularly in South East Asia, have viewed gas as a transition fuel from coal to renewables, it is increasingly valued for the flexibility and firm capacity it provides alongside renewables, storage, and interconnections.

- **Coal power lacks exit policy.** While several European and other OECD countries have coal phase-out policies, many within the 2030s, only 12% of global coal power operating capacity (265 GW) has planned phaseout commitments in line with *Paris Agreement* goals (GEM, 2025). Moreover, substantial coal capacity continues to be added, particularly in India and several South East Asian countries. These governments prioritize coal to guarantee security of supply by relying on domestic coal resources, ensure firm dispatchable generation capacity for grid stability and to buffer renewable variability, and as insurance against gas supply disruption. These coal-reliant regions do not have phase-out policies and because of rising electricity demand are unlikely to retire plants prematurely (i.e. before natural plant lifecycles and/or before renewables have been sufficiently established for coal capacity factors to go down) which in our modelling happens in the 2040s.

- **China** is responsible for about 50% of global installed coal capacity, but does not have a conventional coal phase-out policy. Instead, it is gradually reducing coal's role through power system transformation and state-led industrial reform while retaining coal capacity for energy security and flexibility. Renewables, storage, grid expansion, and demand-side flexibility will play larger roles with coal utilization expected to decline even as installed capacity increases. China's *15th Five-Year Plan* identifies industrial transformation as a key decarbonization priority, and stricter regulation and industrial modernization will likely reduce coal demand and constrain emissions. Reflecting this shift, a three-year action plan from 2026 requires energy efficiency and carbon-reduction upgrades across nine high-emission industries, including power generation. Facilities failing to meet the standards by 2028 will be phased out (CGTN, 2026). This signals continuation of China's long-standing supply-side reform approach and state intervention which combine energy efficiency, decarbonization, and overcapacity reduction to constrain coal use and retire outdated production capacity.

Overall, policy frameworks across many ETO regions continue to support natural gas as a peak demand and balancing resource.

Support for variable renewables follow a rise-and-decline trajectory

Expected support for variable renewables (US¢/kWh)

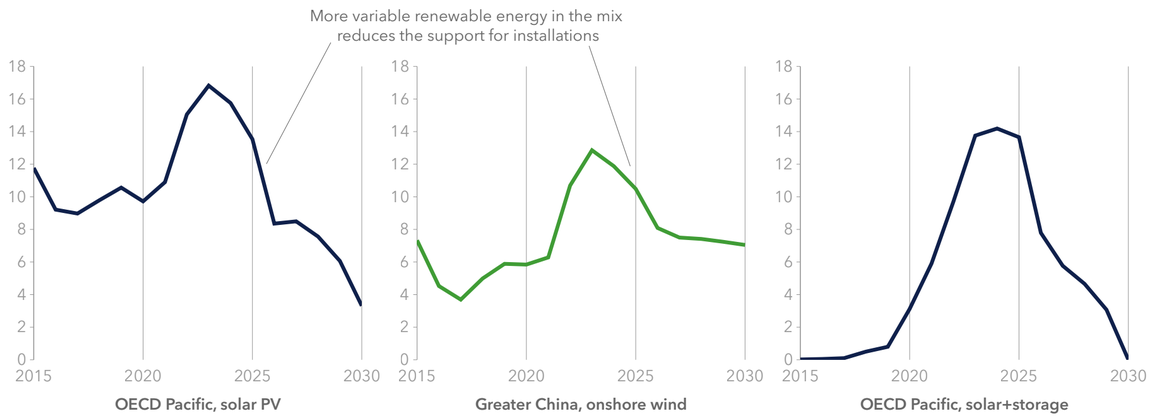


FIGURE 7.1 | Support also includes some revenue.

Buildings and road transport policy: mature solutions but difficult implementation

As the clean energy transition advances in power systems, the frontier of emissions reduction is shifting to end-use sectors. Buildings and transport account for around 30% of global energy-related emissions and are projected to reach 36% by 2050. Both sectors continue to rely heavily on direct fuel use.

Over 130 countries have at least one energy efficiency standard

50% of new floor area in emerging economies is under building energy codes

Decarbonization options are available for road transport and buildings, notably electrification technologies (such as solar systems, heat pumps, and EVs) as well as biofuels for transport. Many of these technologies are substantially more energy efficient than fossil-fuel-based alternatives, making their deployment critical for accelerating efficiency gains. However, despite supportive policies such as efficiency standards, incentives, and carbon pricing, progress towards the COP28 target of doubling annual efficiency improvements from 2% to 4% remains off track, with gains averaging only around 1.3% per year since 2019 (IEA, 2025a).

Decarbonization in both segments depends heavily on household investment decisions, making progress sensitive to affordability and public acceptance concerns that slow policy implementation and create inertia. Governments often shield consumers from energy price increases through subsidies or compensation measures (as seen in Europe, South Korea, and India) to cushion the impact of the war in Iran (Asmelash, 2026; Stott, 2026). Similar concerns have shaped the EU's ETS-2 design for buildings and transport, including price stability mechanisms, the Social Climate Fund, and a delayed start date of 2028.

Despite long-term benefits such as reduced exposure to fossil-fuel price volatility, high upfront costs and split incentives in building retrofits have led to underinvestment and remain major barriers.

Policy levers: buildings

Most countries lack integrated building transition strategies. Europe leads with a binding framework combining efficiency-first policies, 2030 energy savings and renewables targets, carbon pricing, zero-emission new-build requirements, solar mandates, fossil-boiler subsidy phase-outs from 2025, and dedicated funding. China also has comprehensive policies, including heat-pump deployment, electrification, district heating, ultralow-energy buildings standards, and renewable integration.

- **Mandatory building energy efficiency codes are adopted unevenly.** While more than 130 countries have adopted at least one energy-efficiency standard (IEA, 2026a), the 2025 *International Efficiency Scorecard* finds only six countries (France, Germany, Japan, Saudi Arabia, South Africa, and the UK) with mandatory national codes for buildings requiring minimum standards for insulation, windows, lighting, heating/cooling systems, and water heating (ACEEE, 2025).
- **Emerging economies lack minimum energy performance standards (MEPS).** Only half of new floor area in emerging economies is under building energy codes (IEA, 2025a). Adding building stock without robust MEPS results in the adoption of inefficient technologies, which can be seen in Sub-Saharan Africa (UNIDO and EELA, 2025). However, MEPS are emerging. Ghana's buildings efficiency standard came into force in 2022 and Kenya's first building energy code was effective from 2025.
- **Loosening of fossil fuel prohibitions.** Several revisions to standards scheduled to take effect through to 2035 reflect a setback in buildings efficiency policies, in particular scaling back or delaying fossil-fuel boiler bans (IEA, 2026a). The UK, parts of Australia and Canada, and Germany are all softening restrictions on gas heating to allow fossil-fuel boiler installation to continue. Renewable energy targets for buildings are scarce, existing in merely 28 countries (REN21, 2026). In 2026, Germany rolled back the *Buildings Energy Act*'s 65% renewable energy requirement for new heating systems. Globally, the share of building emissions covered by explicit carbon pricing was around 13% in 2025 (WB, 2025).
- **Insufficiently supported retrofit policies.** Globally, renovation rates sit at around 1% annually. The EU's *Renovation Wave* aims to double retrofit rates by 2030 and Canada's *Green Building* strategy targets 3% by 2050. However, only four countries (France, Germany, Italy, and the UK) are assessed to have robust policies for whole-building retrofits and financial incentives (ACEEE, 2025). The UK, for example, launched the *Warm Homes Plan* in 2026 with around USD 20bn funding (a mix of grants and loans) aiming to upgrade 5 million homes.

Policy levers: road transport

Transport improvements are driven by parallel policy measures that both incentivize cleaner alternatives and discourage conventional fossil fuels through restrictions, efficiency requirements, and price signals. In the NDCs submitted for 2035, the deployment of zero-emission vehicles (ZEVs) is among the most frequently cited mitigation action type (Changing Transport, 2025) with a strong focus on electrification. Global energy spending in 2025 mirrors this policy direction with a reported USD 893bn spent on vehicles and charging infrastructure (BloombergNEF, 2026). This reflects the converging trends of falling prices, improved technologies and supportive policies that trigger broader consumer uptake.

- **Electric mobility targets are increasingly common,** supporting both decarbonization and domestic manufacturing objectives. By 2024, 68 countries and 26 subnational jurisdictions had EV targets, including 49 countries with 100% EV sales goals or internal combustion engine (ICE) phase-out commitments (SLOCAT, 2025). Some recalibration emerged in 2026, with Canada and the EU replacing 2035 ZEV sales mandates with stricter vehicle GHG emission standards that continue to incentivize ZEV production and uptake while offering greater regulatory flexibility.

- **ZEV CAPEX support and other incentives** are present in all ETO regions, except North East Eurasia. Incentives for EVs vary, spanning purchase subsidies, tax incentives, and import duty reductions. While EV adoption remains uneven, most regions have an upward sales trend driven by a mix of incentives, infrastructure expansion, and motivations to reduce fuel import costs. The OECD Pacific region, driven by incentives in Japan and South Korea, have the highest support levels for fuel cell electric vehicles (FCEVs) among ETO regions.

- **Norway, an early pioneer, used tax incentives and operational benefits** to raise EVs to over 90% of passenger vehicle sales. Other leaders include the other Nordic countries, the Netherlands, and China – the latter dominating global EV, electric bus, and truck markets. Developing economies, such as Ethiopia, Nepal, and Sri Lanka, have also seen rapid growth driven by affordable Chinese imports (Jaeger, 2025) rather than robust incentives. Ethiopia's transition accelerated following its 2024 ban on ICE vehicle imports, which aimed to reduce fuel import costs.

- **Fuel efficiency regulations are widespread** covering over 80% of global passenger vehicle sales, with North East Eurasia and Sub-Saharan Africa the main exceptions (IEA, 2026a). Policy frameworks in Canada and the EU increasingly state vehicle performance standards in terms of emissions (gCO_2/km) to enable linkage to climate goals and compatibility with electrification policies like the EU targeting 90% reduction in CO_2 emissions in new passenger vehicles and vans by 2035 and heavy-duty vehicles by 2040. Several major markets, such as China and Japan, continue to tighten fuel-economy requirements (km/L), and India focuses on both efficiency and CO_2 output in proposed CAFE III norms (2027-2032). The US has moved in the opposite direction by relaxing CAFE standards through 2031, ending EV tax credits, and weakening the regulatory basis for vehicle emission controls (EPA Endangerment Finding Repeal).

- **Biofuels mandates** drive biofuel use across markets as set in policy mandates and blending requirements. Biofuels are the leading renewable energy source in transport in South East Asia. In June 2026, India approved a regulatory framework for the use of 100% ethanol (E100) as transportation fuel, paving the way for commercial roll out and marking a significant step beyond its national E20 ethanol-blending programme. Brazil similarly focuses on ethanol-based biofuels, supported by existing refuelling infrastructure, and increased blending to 30% in August 2025.

- **Fossil fuel support** exceeded USD 916bn globally in 2024 (OECD, 2025), though down 11% from 2023. Many countries still have measures supporting production and consumption of fossil fuels in place, for example in North East Eurasia, South East Asia, and the Middle East and North Africa, hampering efforts to reduce fossil fuel use by sustaining low fuel prices.

Maritime and aviation regulation

International shipping faces regulatory uncertainty after the *Net-Zero Framework* (NZF), expected for adoption in October 2025, was postponed until late 2026. The delay leaves GHG reduction requirements and implementation timelines unclear. While we still expect an amended NZF, the setback complicates progress toward the IMO's 2050 net-zero target, weakens incentives for decarbonization investment, reduces confidence in future demand for zero or low-carbon fuels, and limits the offtake prospects needed to support expanded fuel production and supply. Further discussion is provided in the *Maritime Forecast to 2050* (2026).

International aviation operates under the *Carbon Offsetting and Reduction Scheme for International Aviation* (CORSIA) under the International Civil Aviation Organization (ICAO). Starting in 2021, airlines on international routes aim to hold net CO_2 emissions at 2019 levels. While CORSIA helps limit net emission growth, it does not require use of cleaner fuels, and airlines can comply entirely by buying offsets. We refer the reader to the *ETO 2026 Hydrogen to 2060* for further discussion of global regulation.

Beyond global frameworks, both the maritime and aviation sectors are influenced by regional and national policies, such as the EU's *ReFuelEU Aviation* and *FuelEU Maritime* regulations; the *Renewable Energy Directive* (RED III) sets binding minimum levels for RFNBOs for industry and transport for 2030; and the EU ETS covers both sectors. European mandates, the EU, and the UK are driving clean jet fuel uptake.

Policy for hydrogen and CCS: early stage and commercially immature

Clean hydrogen and carbon capture and storage (CCS) remain highly policy-dependent technologies. Deployment accelerates where policy and regulatory certainty help bridge cost gaps with conventional alternatives or create strong incentives to reduce emissions.

Continued support is justified by their potential role in hard-to-electrify segments such as aviation, maritime transport, and emission-intensive trade-exposed industries including cement, steel, and chemicals. However, demand remains sensitive to costs and policy design.

Policymakers are often reluctant to mandate rapid uptake where competitiveness may be affected, making sustained support measures and, in some cases, trade or fiscal protections critical to scaling deployment.

Policy levers: CCS

CCS is a critical component of net-zero pathways. It addresses three distinct mitigation challenges: industrial process emissions that cannot be eliminated through fuel switching; emissions from hard-to-decarbonize sectors where direct electrification is difficult; and carbon dioxide removal (CDR) needed to offset residual emissions and achieve net-negative emissions.

Recognition of CDR is growing in long-term climate planning: by January 2026, 27 of the 80 long-term strategies submitted to the UNFCCC referenced CDR technologies, most commonly bioenergy with carbon capture and storage (BECCS). However, dedicated public support for CDR remains limited to a small group of countries, including Australia, Canada, Japan, and the UK (System Change Lab, 2026).

Our *Energy Transition Outlook: CCS to 2050* forecast accounts for regional differences in policy ambition, regulatory readiness, and financial support. To the right, we include a high-level overview of CCS policies across ETO regions that have dedicated policy and highlight the frameworks and incentive mechanisms supporting deployment. For more detailed analysis, explore the *Energy Transition Outlook: CCS to 2050* published in 2025.

- **Europe** drives CCS through continuous support across the value chain. Investment is driven by legally binding net-zero targets, carbon pricing (CP), and EU storage targets. Funding combines EU and national CAPEX support on a co-investment basis (50-60% public), while OPEX support includes CCfD-type mechanisms linked to CP.
- **OECD Pacific:** CCS is a priority in Australia, Japan, and South Korea. 60% of Japan's shortlisted projects rely on cross-border CO_2 export, encouraging partnerships. Robust policy frameworks are backed by substantial value-chain CAPEX support, including co-financing grants, but dedicated OPEX mechanisms are limited, except Australia's ACCU-based carbon credit system.
- **North America:** Canada and the US continue to support CCS through tax credits. Canada extended its highest credit to 2035.
- **Greater China:** CCS is central to China's carbon-neutrality strategy. State planning, public finance, carbon market integration, and SOE-led projects are driving a shift from pilots to large-scale deployment. CAPEX support is substantial – often reaching 20% and fully funding some SOE projects – while OPEX support remains limited.
- **Middle East and North Africa:** GCC countries are advancing CCS through state-led investment, storage targets, and value-chain development to decarbonize industry and hydrocarbon production while establishing low-carbon hydrogen and ammonia exports to remain competitive amid global carbon rules. However, limited explicit CP constrains commercial deployment. See our *Energy Transition Outlook 2026 – Oil and Gas Decarbonization in the Gulf Region* for a more in-depth discussion.
- **Indian Subcontinent:** India is the CCS frontrunner in this region. While policy support remains nascent, a planned CCS Mission – USD 2.2bn in federal funding between 2026 and 2031 – and a national CCUS R&D Roadmap (December 2025) signal growing momentum. A national carbon market launching late 2026 will extend CP to emission-intensive industries.

50-60%
CAPEX support to CCS in Europe

65
countries have hydrogen strategies

Policy levers: hydrogen

Policy motivations for clean hydrogen vary by region. Increasingly, countries prioritize it to decarbonize hard-to-electrify sectors, replace fossil-based hydrogen in existing industrial uses, and strengthen energy and food security. As of October 2025, 65 countries had adopted hydrogen strategies (IEA, 2025b).

Early optimism has given way to a more pragmatic assessment of hydrogen's systemic complexity, the scale-up required across the value chain, and the need for sustained policy and regulatory support. As with other emerging energy technologies, public support remains essential to mobilize investment, create demand, and narrow the cost gap with fossil alternatives.

Support measures have hitherto overwhelmingly concentrated on the supply side, while the key determinant of bankability is robust, long-term offtake (Rushton and Patonia, 2026).

An effective policy mix supports both supply and demand, including measures that:

- Combine subsidies to derisk and accelerate early phase projects across production infrastructure and end-use alongside carbon pricing to narrow the structural cost gap to unabated, fossil-based hydrogen.
- Create predictability and stronger offtake certainty to further investor confidence through public procurement of low-emission material, mandated end-use quotas, and regulatory obligations in hard-to-electrify sectors.
- Align regulations on emissions criteria, certification, and methodologies for calculating GHG emissions to improve market traceability and ensure hydrogen delivers a climate benefit.

Policy plays a role in ensuring hydrogen reaches cost parity with unabated fossil hydrogen across ETO regions (Figure 7.2 and Table 7.1). We encourage the reader to explore the *Energy Transition Outlook: Hydrogen to 2060* forecast on clean hydrogen and derivatives published in May 2026, which has more details on the regional policy and regulatory developments, as well as maritime and aviation developments.

> The key determinant of bankability is robust, long-term offtake.

Renewable hydrogen will only reach cost parity with unabated fossil hydrogen through strong policies

Cost gap between fossil and renewable hydrogen (USD/kg)

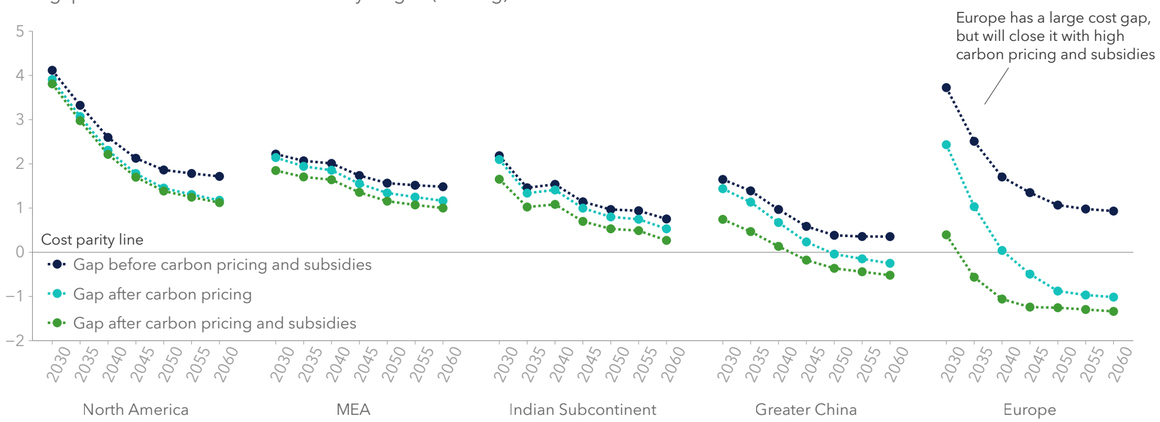


FIGURE 7.2 | MEA: Middle East and North America.

Region	Goals & priorities	Fiscal policies	Supply push	Demand pull
Europe	●	●	●	○
Greater China	●	○	●	○
OECD Pacific	●	○	●	○
Indian Sub-continent	●	●	○	○
North America	○	●	○	●
Latin America	○	●	○	○
Middle East and North Africa	○	●	○	●
South East Asia	○	●	○	●

● Well-defined
○ Defined
● Early stage
● Insufficient

TABLE 7.1 | Summarizes how comprehensive the policy mix is across ETO regions.

7.3 CARBON PRICING

Carbon pricing (CP) is the main market-based instrument that incentivizes CO_2 emissions reduction financially. CP policies, like taxes and emissions trading systems (ETS), have lost some momentum as political priorities shift in a turbulent world. In response, existing schemes have been adjusted and those under development, especially in middle-income countries, seek to learn from mature schemes.

Highlights on the current state of CP from global expert reports, such as those from the World Bank (2026) and the International Carbon Action Partnership (2026), include:

- 87 policies worldwide cover 29% of global emissions.
- Countries with implemented CP account for around two-thirds of global GDP and around half of the world's population.
- CP mechanisms raised over USD 100bn in 2025, over half of which was in Europe. Most funds raised were directed to climate mitigation and energy investments.

For our analysis, we derive a best estimate of the average regional CP trajectories to 2035 based on research into the policy trends and future CP levels in existing and expected schemes. From 2035 onwards, CP in the ETO Model is determined endogenously, based on regional emission intensity trends in the manufacturing and power sectors (Figure 7.3).

CBAM

The massive difference in regional CP underscores why Europe – the leader in CP – needs its carbon border adjustment mechanism (CBAM) to level the economic playing field and preserve competitiveness. Coming into effect from January 2026, the EU CBAM has been a major change in the CP landscape. It aims to combat 'carbon leakage' by equalizing carbon costs between domestic and foreign producers through a carbon cost on specific imported goods. CBAM will be phased in gradually, initially with free emission allowances which originally were set to be fully removed by 2034.

CBAM will overwhelmingly impact a concentrated group of trading partners; we predict Russia, China, Turkey, India, and Ukraine will account for more than half of total CBAM certificate demand by 2030. CBAM appears to be working as intended; several of these and other impacted countries have been expanding existing CP mechanisms or exploring pilots or new measures. Although CBAM applies to less than 1% of global emissions, it is a motivator for climate policy development.

Europe's mature ETS means carbon prices remain highest
Carbon price in select regions (USD/tCO_2)

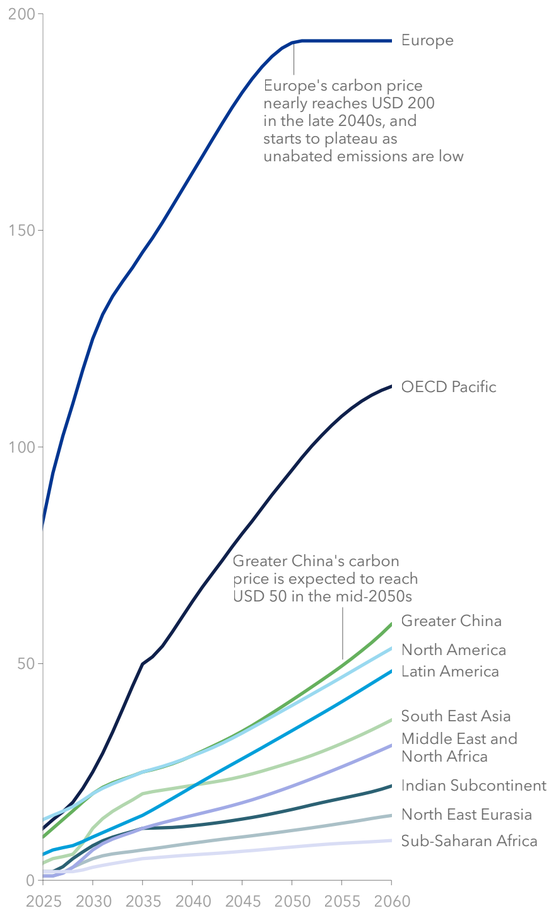
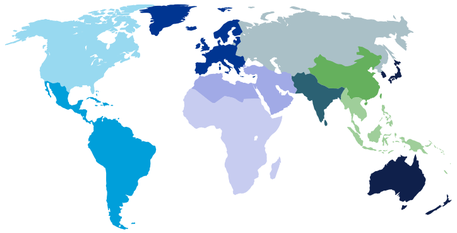


FIGURE 7.3 | Model calculation takes over estimates from 2035.

Regional carbon price highlights



Regional summary ordered by descending 2060 carbon price level

Europe
- Europe remains the clear leader in CP policy, though proposed ETS-1 reforms are moderating stringency (EC, 2026b) to balance climate and competitiveness objectives. Proposed changes include: slower cap reductions after 2030 with allowances remaining into the 2040s; integration of domestic carbon removals and limited international credits; reduced Market Stability Reserve withdrawals; continued conditional free allocation beyond 2030; and earmarking 50% of ETS revenues for decarbonization.
- We expect these supply-side changes will increase allowance availability and put downward pressure on ETS-1 prices from early 2030s.
- ETS-2, covering road transport and buildings, will likely become fully operational in 2028.

OECD Pacific
- CP mechanisms active in all countries, but several factors limit impact, rendering the effective price low for now.
- Japan's ETS transitioned from voluntary to mandatory in 2026, covering 50-60% of national emissions. South Korea's ETS continues to struggle with overallocation and market volatility.
- Australian emissions benchmarks remain high, meaning few installations pay. New Zealand's ETS is the only CP instrument in the world covering forestry and land use.

Greater China
- Already covering over 60% of national emissions, China's ETS aims to cover all main industrial polluters by 2027, ahead of the original 2030 schedule.
- Free allowances continue until auctions commence in early 2030s, meaning abundance from banked allowances dampens the price. Absolute cap controls could be introduced before the planned 2030 schedule.
- CBAM is a motivator for CP development, but current exposure is less than 2% of China's exports.

North America
- Canada has extended the long-term trajectory of its CP but also reduced prices. Washington, California, and Québec are linking their carbon markets.
- In the US, CP remains state-led; existing schemes, including California, Washington, and RGGI states, continue – often under 'cap-and-invest' branding – but limited climate policy appetite makes further expansion unlikely.

Latin America
- We expect Mexico's ETS to be fully operational in 2026, and a new carbon standard and registry is under development. Brazil established an Extraordinary Secretariat to implement the ETS in the coming years, which is expected to have broad sectoral coverage.
- Smaller economies like Colombia and Chile continue to move towards ETS implementation. Bolivia's ETS is under legislative consideration.

South East Asia
- Established policies include Indonesia's ETS and Singapore's carbon tax. Vietnam's ETS joins this list after major advancements to the pilot including operationalizing the design, clarifying allowance and compliance rules, and introducing offset credit use.
- ETS development continues, with functional systems expected in Malaysia and Thailand in the next few years, and the Philippines thereafter.
- CBAM is an incentive to pursue CP mechanisms in the region.

Middle East and North Africa
- Turkey's pilot phase ETS (covering power and manufacturing) is expected later in 2026, motivated by CBAM considerations.
- Other large economies like Saudi Arabia and the UAE are continuing their involvement in voluntary carbon markets, which we expect to evolve into effective CP in the long term.
- Fossil fuel subsidies remain common across the region.

Indian Subcontinent
- India's carbon credit trading scheme will cover nine hard-to-abate sectors, and power is excluded for now. The scheme will account for 15-20% of national emissions, with trading slated to start in late 2026. Unlimited banking of credits will likely result in an oversupply in early phases, dampening prices.
- Sri Lanka has a carbon crediting scheme, whilst Pakistan's carbon market framework is under discussion.

North East Eurasia
- Kazakhstan's ETS price remains low, as all allowances have been distributed for free.
- Ukraine's ETS planning is in the early stages and its carbon tax remains low.
- Outside of these two schemes, CP adoption is slow across the region.

Sub-Saharan Africa
- The region focuses heavily on voluntary and international carbon market participation, but ETS schemes are under consideration in Senegal, Cote d'Ivoire, Kenya, and Botswana, and Nigeria has approved its national carbon market framework.
- South Africa has the region's most developed carbon tax, with annual step-ups.

8 FINANCE OUTLOOK

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8.0 SUMMARY

17%
uptick in annual global energy expenditures by 2060

96%
rise in global GDP by 2060

Capital is becoming a key constraint of the energy transition

The energy transition is entering its most capital-intensive phase. Over the coming decades, the world must maintain and manage down the existing fossil energy system while building a new electricity-based system centred on renewables, grids, and storage. As investment shifts towards electrification, access to affordable capital will increasingly shape the pace of deployment across technologies and regions.

A temporary surge in expenditures … and delayed rewards

The 2020s and early 2030s represent the most expense-intensive stage of the transition. Society must finance an ageing fossil energy system and the rapid expansion of low-carbon infrastructure before the benefits of lower operating costs are fully realized. As a result, overall energy expenditure (including OPEX) rises during the build-out phase. Over time, however, lower operating costs offset higher upfront investment needs, and energy production expenditure becomes a smaller share of economic output as global GDP continues to grow.

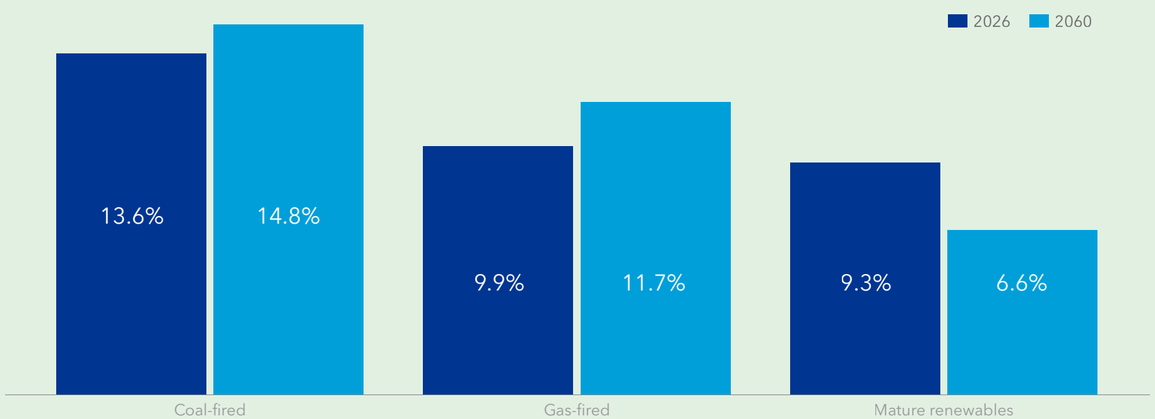
Electrification attracts the majority of future capital

Investment in grid, storage, and renewable power will grow three-fold to 2040. CAPEX is shifting heavily from fuel supply towards electricity infrastructure. Solar becomes the largest destination for energy-sector investment, while spending on grids and storage grows steadily as electricity expands into transport, buildings, and industry.

Financing conditions determine transition speed

Technology performance alone will not determine the outcome of the transition. Financing costs – themselves a reflection of both technical and policy risk – influence technology competitiveness, deployment rates, and emissions reductions. Mature renewable technologies benefit from relatively low costs of capital, while emerging technologies such as hydrogen and CCS continue to face financing premiums. Regional differences remain significant, with higher financing costs slowing deployment in many lower-income economies. Reducing the cost of capital therefore remains one of the most effective levers for accelerating decarbonization.

Global average cost of capital for new investments in selected power generation technologies



8.1 ENERGY INFRASTRUCTURE EXPENDITURES AND INVESTMENTS

We are now paying for two energy systems at once
The period from 2020 to 2035 is the most painful part of the energy transition from the perspective of total energy expenditures.

Total energy expenditures are the costs related to the production of energy and its transport to the user, including operational expenditure (OPEX) and capital expenditure (CAPEX).

Global energy expenditures grew more slowly than global GDP from 1980 to 2020. That downward trend is being temporarily reversed by the energy transition, with a window encompassing both fossil production and renewable CAPEX. The phase from 2020 to 2035 requires a higher share of GDP spent on energy production, storage, and transport, both from (initially) higher fossil CAPEX and maintaining fossil OPEX while introducing renewable CAPEX.

We forecast that, starting in 2030, the average annual reduction in expenditures in the fossil-fuel system will be USD 36bn. This number will be matched by a slightly higher amount of USD 39bn spent on the renewable-based energy system, excluding grid investments.

Total energy expenditure will plateau in the 2030s
From the late-2030s, overall expenditure on energy production, storage, and transport will start to plateau, even as global GDP steadily rises. The average annual expenditure on energy around 2060 will be USD 6,000bn (compared with USD 5,100bn in 2026, a 17% uptick) while GDP will have grown by 96%.

That energy expenditure becomes an ever-smaller proportion of global GDP during our forecast period is testament to the very low OPEX of the dominant technologies in the mid-century energy mix and the vast efficiencies enabled by electrification.

Solar and grid are the financial growth sectors of the future
Electrifying energy consumption will require large and increasing investments into grid capacity (IEA, 2023) and power generation (IEA, 2025a), with solar as the clear winner, followed by wind. We forecast solar energy production will continue to improve its competitive advantage compared with other energy sources. We expect the market for solar investments to match the combined CAPEX needs for oil and gas from the 2030s and continue to grow after that.

We forecast solar and wind energy will cover most of the increase in power demand (see Chapter 3 for details on the demand development). This is the case for every geographical region except North East Eurasia (dominated by Russia), where most of the electrification will be covered by nuclear power.

The demand for oil and gas will remain strong into the 2040s. However, the era of mega projects that require massive investments is coming to an end for the oil and gas industry. Instead, they will increasingly focus on maintaining production through smaller upgrades, tie-ins, and add-on projects. The overall direction for the fossil-based industry is from CAPEX to OPEX, reducing overall expenditures gradually while maintaining output.

Total energy expenditure will plateau in 2030s
World energy expenditure (Billion USD/yr)

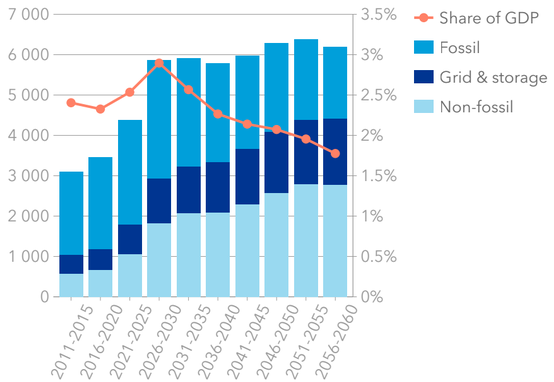


FIGURE 8.1 |

Fast electrification requires earlier grid investments
Regional grid investments (Billion USD/year)

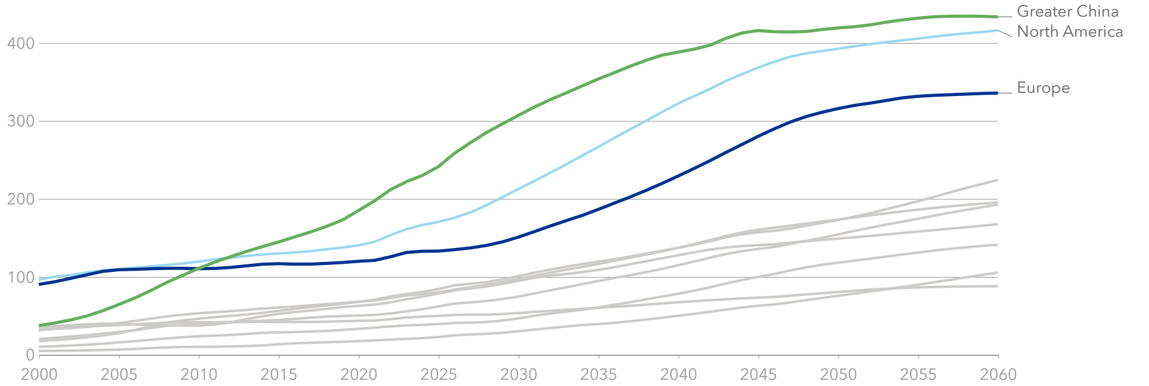


FIGURE 8.2 | All other regions are grey.

Electrification will move at different speeds in different regions. While Europe, North America, China, South East Asia, and OECD Pacific are on a fast track towards electrification, Latin America, the Middle East and North Africa, and India are moving slower, and North East Eurasia and Sub-Saharan Africa are on a very slow path. In most of the regions, oil is the main energy carrier being switched to electricity. However, electricity replaces both coal and oil in China, and electrification gradually reduces the need for biomass for energy production in Sub-Saharan Africa.

The largest investments into grid capacity will come in China and North America, which are also where the electrification is happening the fastest in terms of absolute volume of added capacity. Europe is next on the list. China is already on a rapid growth in total costs for grids and will likely spend 61% more on grids in 2045 than in 2026. North America will see annual grid costs rising a little later, by 73% from 2030 to 2045, and rising more moderately after that. Europe's growth will come even later than that, with moderate growth until the mid-2030s. From 2035 to 2050, Europe's spending on grids will increase by 69% before reverting to more moderate levels.

Global energy investments shifts to wind, solar, and grids
Global energy investments (Billion USD/yr)

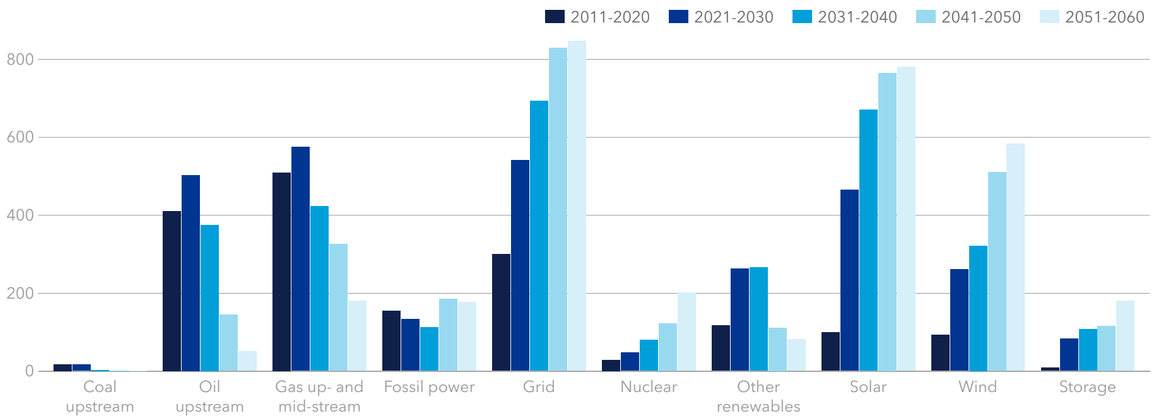


FIGURE 8.3 |

Variable and firm power generation

Annual investments in solar (including combined solar and storage) rose from USD 56bn in 2010 to USD 560bn in 2025. By the end of the next 15-year period to 2040, we forecast annual investments to reach USD 740bn and the average annual investment level to be USD 710bn from 2026 to 2060. Wind power saw higher annual investments than solar 15 years ago, reaching USD 67bn in 2010. The growth till 2025 has also been formidable, albeit slower than for solar, with wind investments reaching USD 280bn in 2025. Annual investment in wind will reach USD 390bn in 2040 and will continue to climb after that, such that we forecast average annual investment in wind will be USD 450bn from 2026 to 2060.

For the time periods when variable power plus storage proves insufficient, firm power generation (including other renewables, nuclear, and fossil power) will meet demand. Firm power generation will continue to attract USD 470bn on average in investment annually in between 2026 and 2060. The biggest contributors to firm power generation will be hydropower with USD 120bn and nuclear with USD 130bn in average annual investments in the same time period.

Storage

Investments in energy storage, such as batteries (not as a combined solar and storage plant) and pumped hydro storage, were very low until 2020 when they started to grow. From USD 13bn in 2020, annual investments have reached USD 66bn in 2025, and we expect them to double to USD 110bn on average between 2026 and 2050. From 2050 there will be a steady growth to USD 180bn invested per year by 2060.

8.2 COST OF CAPITAL

The transition is highly sensitive to financing costs

We forecast a capital-intensive energy transition that requires large new investments in new power generation, energy carriers, and end use. This makes the forecast sensitive to financing conditions. These conditions are set by perceived risk, and those perceptions vary widely between countries and technologies. Our forecast accounts for these differences by setting regional- and technology-specific cost of capital assumptions. By adding this level of detail to our model, we reflect the time value of money in a specific region when comparing new generation alternatives.

Long construction times raise financing costs and hurt competitiveness

Cost of capital is not the only driver of financing costs per technology. Construction time also impacts financing costs through the time value of money. If large amounts of capital are locked in during a period without earnings (e.g. in large-scale nuclear power or hydropower projects with construction periods of 10 years or more), the importance of financing costs increases compared to, e.g. solar and onshore wind projects with one to three year construction periods. It is therefore the combination of the cost of capital and the construction period that sets the relative financing costs of power generation alternatives.

Our cost of capital predictions

The cost of capital reflects the minimum return a company needs to earn on its investment to justify the cost of financing the investment. It is a composite of three elements:

- The cost of debt, which is the return required set by the risk-free rate and the risk premium (often called margin)
- The cost of equity, which is the return required by equity investors when they invest their capital
- The ratio between debt and equity

Our regional assumptions are validated by regional experts, within and outside DNV, who observe firsthand how governments, banks, and investors perceive risk and thereby set the cost of financing for various technologies across the 10 global regions in the ETO.

All regions see mature renewables as low risk

We predict that mature renewable energy will have structurally lower CoC than fossil sources in most regions, except for the Middle East and North Africa (MEA) and North East Eurasia (NEE) regions. Regardless of regional differences in funding sources, most regions consider mature renewable power to be low risk with the average spread between renewables and oil and gas increasing from around 1% in 2026 to 5% in 2060 (except for the oil exporting regions of MEA and NEE). Between renewables and coal, the average spread in CoC increases from around 4% in 2026 to 8% in 2060. This benefits the pace of decarbonizing power generation, especially because the cost of renewables is very sensitive to the cost of capital on account of being capital-intensive and having no fuel costs.

However, risk-free borrowing costs in a region can still elevate financing costs, irrespective of technology, and thereby slow the transition. We account for increases in interest rates in most regions due to persistent inflationary pressure, most recently revived by the war with Iran.

Hydrogen and CCS cost learning curves will bring down financing costs

Investors perceive new technologies (like floating offshore wind, hydrogen, and CCS) that also require commercial maturing through market designs, regulatory frameworks, and offtake structures as high risk and therefore require high investment returns over a longer period. Our forecast assumes high WACC for these technologies initially, in the range of 11-16%, that will lower to 5-8% once the installed capacities increase globally. The WACC over time is not firmly set but depends on the forecast deployment of the technology. This feedback loop in our model allows technologies to scale and accelerate once risks (and therefore costs, including financing costs) come down. Higher capital costs create significant short-term hurdles for projects like clean hydrogen and floating offshore wind – which are policy-dependent – but the powerful underlying dynamic of technology cost learning curves allows these technologies to scale over the longer term.

Petrostate financing of new oil and gas investments drives high emissions towards 2060

The spread between fossil fuel technologies and renewable technologies impacts the competitiveness between such technologies and is therefore also an important assumption in our forecast. We see that this spread differs per region, for different reasons.

Regions with abundant oil and gas resources and exports continue investing with favourable financing structures. On the other hand, regions with high dependence on fossil fuel imports, dramatically impacted by the recent Iran war disrupting oil and gas deliveries, see a strong strategic rationale for renewable build-out. This pushes the risk of such investments, and therefore financing costs, down. At the same time, such regions are also willing to invest in new fossil fuel projects nationally if this can lower their dependence on fossil-fuel imports (assuming those resources are available). This complexity is reflected in our cost of capital predictions: the main differences are explained by regions having large oil and gas resources.

To illustrate the sensitivity to the spread between fossil and renewables, we tested an increase in the spread between fossil and renewables of 2% in our

De-risking energy investments in high-risk economies is a key lever for speeding up the transition.

model (in other words, we make new fossil generation less competitive by increasing financing costs with an additional 2%). This would reduce global CO_2 emissions by an additional 11% in 2060 compared with our central forecast, while energy expenditure as part of GDP would be 3% higher. This shows that attractive financing terms for new oil and gas investments within petrostates remains a key driver of persistently high CO_2 emissions in 2060.

Location matters – power generation costs more in low-income countries

By accounting for region-specific parameters like country risk, policy risk, tax rates, and access to capital, our WACC assumptions for renewables range from 6% in Europe or China to 11% in Sub-Saharan Africa or the Indian Subcontinent. In other words, financing costs for the same technology can double based on where the technology is built. This increases the cost for new electricity generation in higher risk regions, which coincidentally happens to be the regions where we forecast the highest need for new power. Access to affordable capital is therefore a real constraint on meeting this demand effectively.

The importance of financing costs differs dramatically between regions (IEA, 2025b). Financing the costs of a solar park in Europe or China might account for only 20% of the levelized cost of energy (LCOE), while this can be 70% in other regions like Sub-Saharan Africa or the Indian Subcontinent. To illustrate this effect, we ran a sensitivity where all regions in the world enjoy the same financing assumptions. When all regions have the same capital costs as Europe, global CO_2 emissions would reduce by another 8% compared with our forecast. This shows that de-risking energy investments in high-risk economies is a key lever for speeding up the transition (IEA, 2025a).

Financing sources differ per region

Europe is a high-income transition leader, with effective allocation of capital through commercial financing mechanisms. North America is fairly similar, although abundant fossil fuel resources set the financing agenda to a greater extent and a lower risk threshold for oil and gas investments (BloombergNEF, 2026). The recent surge in new electricity demand to power data centres shows a technology-agnostic approach where the emphasis for financing is primarily 'time to power', with less emphasis on decarbonization.

Resource-rich regions like North East Eurasia and the Middle East and North Africa finance their energy investments by the immense cashflows from their oil and gas operations and are therefore less dependent on bank financing. Here, oil and gas prices are the primary drivers of capital access for new fossil energy investments.

In China, most coal-fired power is state-owned and funded with equity from public entities and debt from state-backed banks offering favourable terms. These investments serve national priorities: ensuring baseload power, enhancing energy security, and reducing the influence of geopolitical factors.

For the Indian Subcontinent, the strategy is equally complex: while locally mined coal continues to receive favourable financing, a serious solar PV expansion effort is underway as a means of weaning India off the expensive and geopolitically sensitive fraction of coal use reliant on imports. In this context, renewables are not just a parallel path to coal, but a direct tool to enhance energy security by reducing exposure to international markets.

For many low-income regions, particularly in Sub-Saharan Africa, the primary challenge is a large funding gap to meet investment needs in new electricity generation. The investment environment is defined by high-risk premiums driven by a combination of economic, political, and institutional factors. These consist of political and security risks, sovereign credit risks, currency risks, and energy system risks, all of which deter private capital. Available funding comes from a patchwork of commercial and public finance, with less than 20% of total energy investment financed through development finance institutions. The financing structures vary widely. In South Africa, most renewable energy projects are predominantly financed through debt in local currency, whereas concessional finance and equity play a more significant role in Senegal, Malawi, and Botswana. Overall, however, debt financing across the subcontinent remains high risk and thus costly. This can only be partially mitigated by measures like legally binding power purchase agreements and sovereign guarantees (Koefoed and Selvakkumaran, 2025).

Renewable power generation enjoys better cost of capital than coal- or gas-fired generation

Cost of capital for power generation for selected technologies and regions

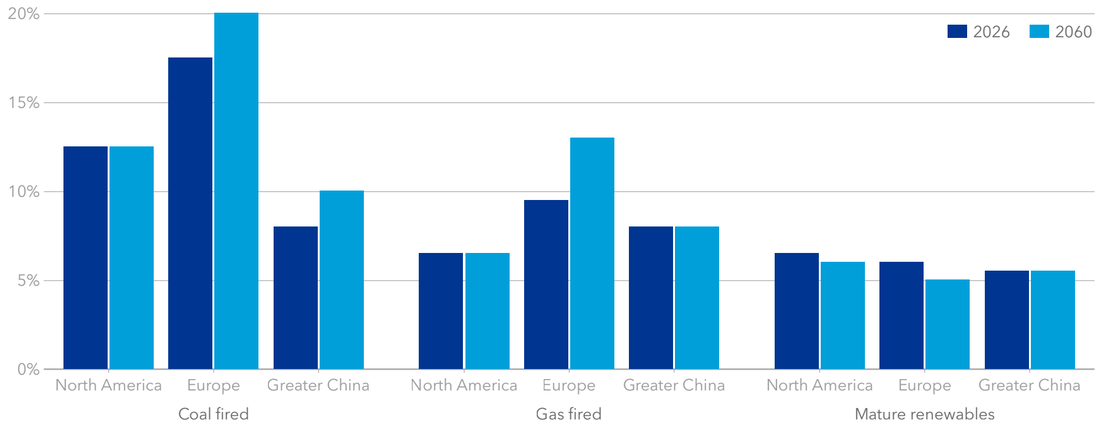


FIGURE 8.4 |

8.3 PRIVATE HOUSEHOLDS FINANCING THE ENERGY TRANSITION

Households are essential to the energy transition: They finance many of the required end-use investments, with the expectation of lower energy bills over time. Policy support broadens participation.

While the overall effect of transitioning is generally positive for household energy bills, the benefits tend to be unevenly distributed because lower-income households are often least able to invest in the EVs, heat pumps, and insulation required to capture money-saving efficiencies, or rooftop or balcony solar to reduce consumption and (potentially) generate revenue in selling power back to the grid.

While energy producers and governments can prepare for the transition by investing in power system capacity and through conducive legislation, the momentum of the transition is also dependent on households investing in new technologies.

Households are central to end-use energy demand

We estimate 28% of the world's energy demand is for transport and 28% goes to buildings (IEA, 2025c). 75% of transport energy is for road transport, and 60% of road transport is for passenger vehicles. 70% of the energy needed for buildings is for residential buildings. Based on these two sectors alone, households directly account for about one third of the world's final energy demand.

The need for upfront investments

Because consumers are directly responsible for a significant share of energy demand, the energy transition is strongly influenced by households investing in technologies that reduce the energy demand – notably, EVs, heat pumps, and building efficiency (EC, 2026).

Globally, the price of new EVs is still generally higher than that of ICE vehicles, but this will change rapidly, and has already changed for certain car segments and certain regions. China will be the first region to reach parity in average purchasing prices from 2028 (Figure 8.5). Other regions will reach parity between 2030 and 2048, except for MEA where ICE vehicles will keep a lower purchase price throughout our forecast period. The upfront cost of a vehicle is significant for the average household, and lower-income households will probably prioritize a lower purchase price over a lower total cost of ownership – locking more vulnerable households into a cycle of higher ongoing fuel costs (U. Michigan, 2023).

The purchase price of EVs has fallen sharply, but remains higher than ICE vehicles into the 2040s
Relative purchase price of an EV to an ICE

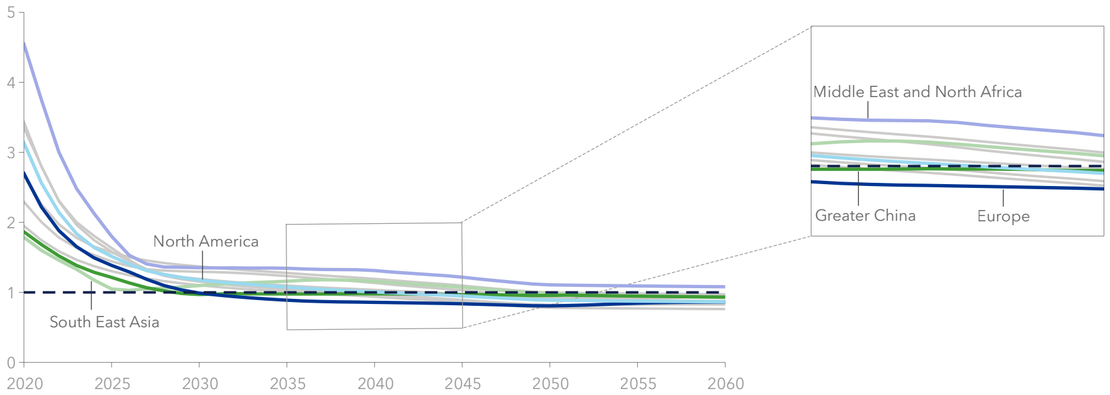


FIGURE 8.5 | The dashed line denotes purchase price parity between EVs and ICE vehicles.

Households directly account for about one third of the world's final energy demand.

Most vehicle owners will change their cars every few years. That is not the case for major home upgrades like installing rooftop or balcony solar PVs or changing out windows and doors for more energy efficient ones. Those steps involve decisions taken only once or twice over a household's lifetime. The right incentives can speed up the pace of such decisions, and it is noteworthy that many governments across the world are shifting beyond general incentives (like feed-in tariffs and production incentives) to providing investment incentives for household participation in the energy transition. Examples abound across almost all world regions, with household incentives arguably most evolved in the EU and in Japan, where the latter pursues an approach that integrates solar, heat pumps, and insulation simultaneously (SurgePV, 2026)

We forecast that energy demand for lights and appliances in residential buildings will grow by 50% between now and 2060, while residential space cooling energy demand will triple over the same period. High-income households will typically invest in energy efficient equipment, while low-income households may lack the means to invest in efficient housing and appliances. Targeted incentives for low-income households to invest in energy efficiency can lead to significant societal benefits (Rocky Mountain Institute, 2025).

Understanding how consumers impact the transition

Electrification is likely to improve overall energy efficiency and reduce energy costs for many consumers. However, its benefits will not be shared equally, nor will household energy bills necessarily decline relative to income (OECD, 2014). Some households will remain unable to invest in efficient end-use equipment without incentives, while in certain use cases natural gas may continue – at least in the medium term – to offer a lower-cost option than electricity, such as gas combi boilers compared with electric resistance heaters.

Understanding the consumer experience of the energy transition remains an important area of enquiry. Households are both affected by the transition and active participants in shaping it. Their technology investments and energy-use behaviours influence demand patterns, while their engagement in local communities and democratic processes helps shape the policies and infrastructure decisions that determine the pace and direction of change. We are devoting increasing attention to these interactions and expect to provide more detailed analysis in future Outlooks, particularly those examining energy transitions at the national level.

Electrification is likely to improve overall energy efficiency and reduce energy costs for many consumers.

9 EMISSIONS AND CLIMATE IMPACT

9.0 Summary 132
9.1 Emissions 133

9.0 SUMMARY

2098
world achieves net-zero emissions under our forecast

2.3°C
forecast global warming by 2100

- **Energy-related CO_2 emissions likely peaked in the last 24 months, and we forecast they will fall 65% to 12 Gt in 2060.** Total carbon dioxide emissions from human activities are comprised of emissions from energy use, industrial processes, and land use changes, minus the CO_2 removed through direct air capture (DAC). These emissions peaked in 2024 at 43 Gt, but we project they will reach net zero only in the final two or three years before the end of the century.
- **Transport is the most-emitting sector in 2060 at 4 Gt, ahead of the power sector at 3 $GtCO_2$.** Power emissions fall 82% over our forecast period compared with 56% for transport.
- **Emissions fall between 2025 and 2060 in nine of the ten ETO regions. Sub-Saharan Africa is the only region where emissions stay flat.** North East Eurasia, Europe, North America, and OECD Pacific all reached peak emissions between 1980 and 2013. Sub-Saharan Africa gets there in 2050. Greater China will likely peak in the mid-2020s and falls by 79% to 2060.
- **Process emissions fall much more slowly than emissions from energy combustion by 2060 – 14% vs 65%.** Process emissions arise from chemical processes rather than energy use. In cement production, for example, calcination releases CO_2 from limestone independently of the kiln fuel. As a result, electrification and fuel switching offer limited opportunities for abating process emissions, contributing to their much slower decline.
- **Direct air capture (DAC) determines when net zero is reached rather than how quickly emissions fall.** We expect DAC removal will be 0.4 $GtCO_2$ in 2060, equivalent to 3% of energy and process emissions that year, and 3.0 Gt in 2100 against remaining emissions of 3.8 Gt.
- **Our forecast reaches net zero in 2098 resulting in end-century warming of 2.3°C.** The 1.5°C budget is exhausted in 2029 and the 2.0°C budget in 2052. Cumulative emissions exceed the 2.0°C carbon budget by 420 Gt. After 2060, cumulative emissions (260 Gt) are 62% of the overshoot.

Total anthropogenic CO_2 by component, against the AR6 carbon budgets
CO_2 emissions ($GtCO_2$/yr) and cumulative emissions from 2020 ($GtCO_2$)

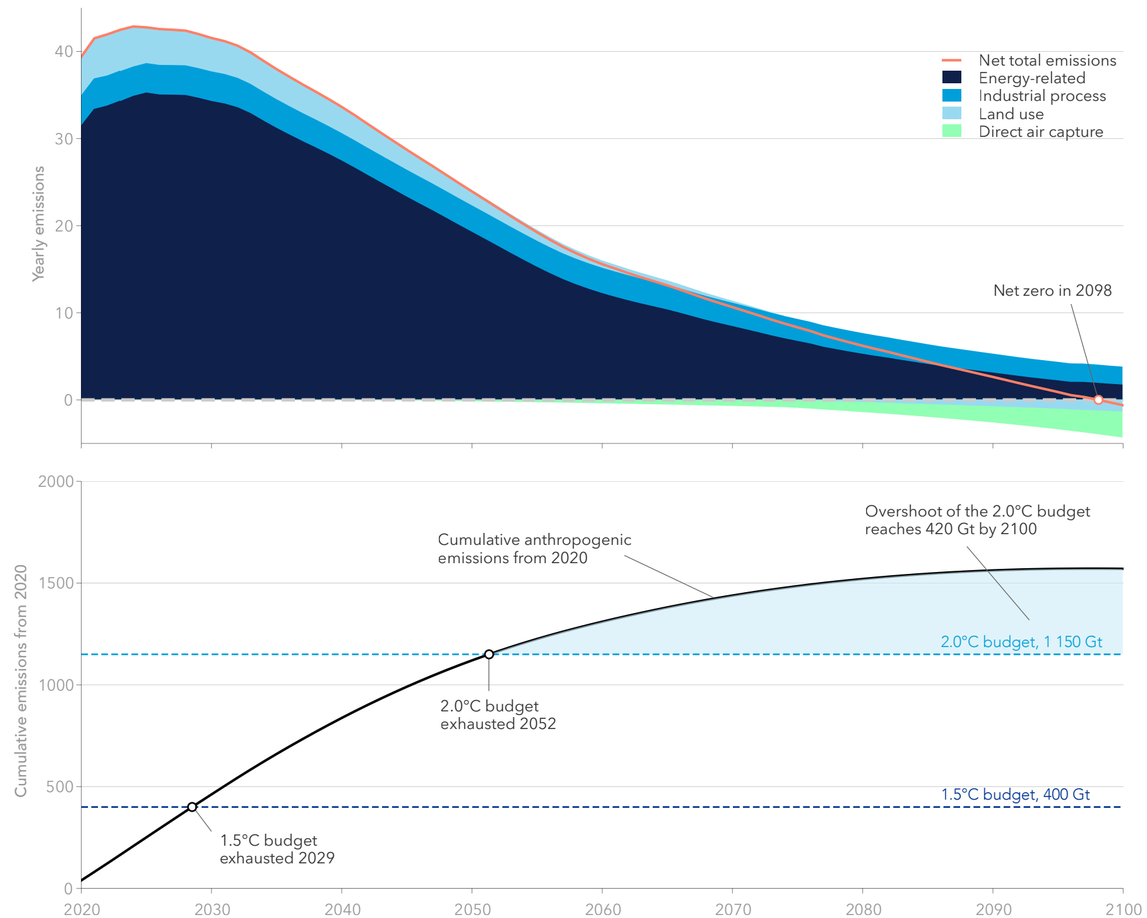


FIGURE 9.1 | Total anthropogenic emissions by component, against the AR6 carbon budgets. Stated at 67% probability and counted from 2020.

9.1 EMISSIONS

Global energy-related CO_2 emissions reached 35 Gt in 2025, the highest level recorded. Industrial process emissions added a further 3.4 Gt. We forecast energy-related emissions falling to 34 Gt in 2030, 27 Gt in 2040, and 12 Gt in 2060. This is a reduction of 65% from the 2025 level. Total CO_2 emissions from human activity are comprised of emissions from energy use, non-energy industrial processes, and land use changes minus CO_2 removed by processes like DAC and bioenergy combined with carbon capture and storage (BECCS). The sum of these emissions likely peaked in 2024 at 42.9 Gt, but will reach net zero only in the last two or three years before the end of this century. That trajectory exceeds the remaining 2°C carbon budget by approximately 420 $GtCO_2$ and implies 2.3°C of warming by 2100.

Global energy-related CO_2 emissions peaked sometime in the last 24 months. We forecast they will fall by 65% by 2060 – a steep reduction, but insufficient to keep cumulative CO_2 from human activity even remotely within the 2°C carbon budget, leading to an overshoot of about 420 Gt by 2100.

The projected overshoot arises from three characteristics of the transition. First, emissions from coal, oil, and gas are plateauing, accumulating at or near historical highs before declining in the 2030s. Second, the rate of emissions decline differs across sectors. It falls fastest where capacity can be substituted, especially in power, and more slowly in hard-to-abate sectors like heavy industry, aviation, and shipping, where abatement depends on replacing long-lived assets, expensive fuel switching to hydrogen, or costly carbon capture installations. Third, residual emissions remain substantial after 2060. Some three-fifths (260 Gt) of the total overshoot accumulates between 2060 and 2100 as emissions continue to reduce but are far from offset by carbon removal or natural land sinks.

Recent data support the forecast that emissions are beginning to plateau and are likely to decline soon. Coal-fired generation in China fell 1.6% in 2025 as record solar and wind additions exceeded demand growth, contributing to the first combined annual decline in coal power across China and India in 52 years (Carbon Brief, 2026; IEA, 2026a). The Global Carbon Project puts total fossil CO_2, which includes cement and other process sources, at a record 38.1 Gt in 2025 (GCP, 2025). However, combining those emissions with land-use emissions, where early estimates indicate a slight fall in 2025, we likely have seen total CO_2 emissions from human activity peak in 2024.

Sector emissions

The fuel mix determines the shape of the decline. Coal accounts for 44% of energy-related emissions in 2025 at 15.5 Gt, oil 30% at 10.7 Gt, and natural gas 25% at 8.9 Gt. By 2060, coal emissions fall 86% to 2.2 Gt, oil 57% to 4.6 Gt, and gas 34% to 5.8 Gt. Coal delivers 57% of the reduction in energy-related CO_2 emissions to 2060, and its share of emissions falls from 44% to 18% (Figure 9.2).

Coal use and emissions plateaued in 2024 and 2025, and estimates point towards similar levels in 2026 (IEA, 2026a). The levelling off in coal-use growth is due to wind and solar additions in the largest coal-burning markets now exceeding the growth in electricity demand. This means incremental generation can be met without additional coal output. China heavily influences the global trend; Chinese coal-fired generation fell 1.6% in 2025 on record solar and wind additions (Carbon Brief, 2026). Coal-fired generation in China falls not because coal capacity is disappearing, but because existing plants are used less. As China's coal fleet is relatively young, most plants remain in the system for decades, but they will run at progressively lower utilization rates.

We forecast oil use and emissions will peak in 2029. The main contributing factor to this decline is the increasing share of EVs; new sales surpass 50% market share on a global basis in 2035.

Energy-related CO_2 emissions by fuel type with peak years within a decade

CO_2 Emissions After CCS ($GtCO_2$/yr)

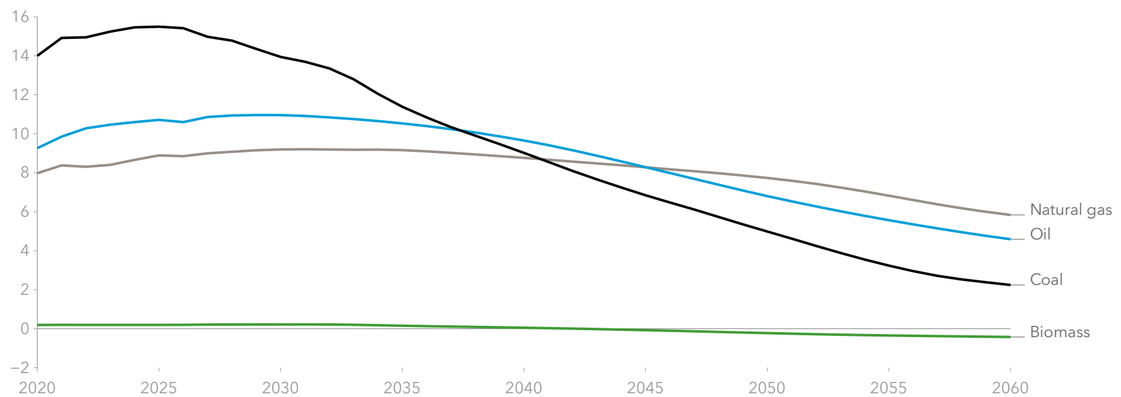


FIGURE 9.2 | Historical data source: IEA WEB (2026)

Gas use and emissions peak in 2034 at 9.2 Gt. Gas substitutes for coal in Asian power generation and heating through the 2020s and early 2030s. This raises gas emissions while lowering total emissions. Electrification of heating then removes the same demand, and gas emissions fall 34% from the 2025 level by 2060.

Emissions from biomass combustion turn negative from 2043 and reach −0.4 Gt in 2060. This represents BECCS, where the captured stream exceeds the fossil carbon released in the supply chain. The volumes are small relative to the three fossil fuels and do not change the shape of the aggregate.

Transport is the largest emitting sector in 2060 at 3.9 Gt, ahead of power at 2.8 Gt

The composition of sector emissions changes over the forecast period. Power is the most-emitting sector in 2025 at 15 Gt (43% of energy-related CO_2), followed by transport at 8.8 Gt (25%), and manufacturing energy use at 6.1 Gt (17%). By 2060 transport is the largest at 3.9 Gt (32%), followed by power at 2.8 Gt (23%), manufacturing energy use at 2.5 Gt (20%), and buildings at 1.9 Gt (16%).

Power decarbonizes fastest, falling 82%, because emissions there are a function of the power generation capacity and that capacity is replaced. Wind and solar take most new capacity additions, unabated fossil plants retire at end of life, and a share of the remaining fossil generation increasingly operates with carbon capture.

Transport CO_2 emissions fall 56%, mainly due to the electrification of road transport. Road emissions decline as EVs replace combustion-engine vehicles across the fleet. Aviation and shipping decarbonize more slowly because batteries are unsuitable for most long-distance operations. In aviation, the supply of sustainable aviation fuels remains constrained by limited production capacity. For shipping, the IMO postponed its proposed global fuel standard and emissions-pricing mechanism, continuing regulatory uncertainty. Decarbonization in hard-to-abate sectors is driven by a combination of regional regulation and private-sector demand.

Manufacturing energy use falls 59%, limited by opportunities for reducing high-temperature process heat emissions. Hydrogen deployment is slower than anticipated and more costly than originally estimated. Electrified high heat is a viable technology, but only reaches commercial deployment late in the forecast period, so the deeper reductions in this sector arrive after 2050.

Buildings CO_2 emissions fall 29%, the smallest reduction of the four largest sectors. Efficiency gains and heat pumps reduce the carbon intensity of space and water heating. However, both floor area and cooling demand grow with population and income, which offsets part of that gain. Data centre and AI electricity demand adds significant demand for new load that is not carbon free. That load is quantified and further described in the AI and Data Centres section on page 46. The net result is that new electricity demand (especially from AI and data centres) slows the fall in power sector emissions without changing its direction.

Emissions fall between 1980 and 2060 in nine of the ten regions

Regional peak years are separated by seven decades (Figure 9.3, left).

In our forecast, regional emissions start to decline when new non-fossil power capacity is large enough to more than cover all growth in electricity demand. This happens earlier in regions where demand grows more slowly. Two further factors contribute. The first is the structure of regional industry. Regions with large cement, steel, and chemicals sectors carry energy and process emissions that peak later. The second is fossil export exposure. Emissions from fossil fuel extraction fall with production volumes rather than with domestic demand. Electricity demand growth therefore provides the principal explanation for differences in peak timing, although it does not account for them in full.

The composition of residual emissions changes as a result. Greater China and North America together account for 49% of energy and process CO_2 in 2025 and 26% in 2060. Over the same period, the Middle East and North Africa, the Indian Subcontinent, and North East Eurasia rise to 49% of the total. The regions that reduce least are those with fossil-based export economies or with demand still growing. These determine the residual that must be abated after 2060.

Regional peak spans seven decades, and the depth of the decline varies more than the timing

Regional evolution of energy and process CO_2 emissions

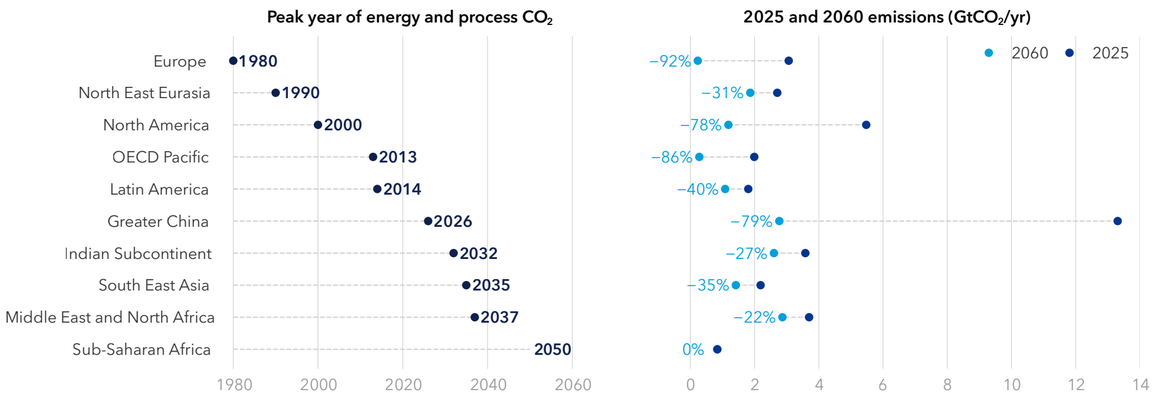


FIGURE 9.3 | Percentage refers to evolution of total regional emissions from 2025 to 2060.

Process emissions fall 14% to 2060 versus 65% for energy combustion

Industrial process emissions were 3.4 Gt in 2025 and will be 2.9 Gt in 2060, a reduction of only 14%. Approximately half of these emissions originate in cement calcination, with the remainder from ammonia, anode consumption, coke ovens, lime, and other chemical processes. We forecast process emissions to be flat to 2030 and then decline gradually as carbon capture is retrofitted and clinker content falls.

The mechanism that abates combustion emissions does not apply here. Calcination releases CO_2 from calcium carbonate as a condition of producing clinker, so the emissions are independent of how the kiln is fired. Electrification and fuel switching therefore reduce only the combustion share of a cement plant's emissions. Clinker substitutes and alternative binders reduce the process share instead, but these are constrained by material standards and by the volume of cement required. That volume grows with construction activity in the regions that are still urbanizing. Implementing carbon capture of CO_2 in cement manufacturing addresses the emissions directly, but adds cost through retrofitting long-lived assets and thus requires relatively high carbon prices.

Energy remains the main contributor of CO_2 emissions
Composition of total CO_2 emissions ($GtCO_2$/yr)

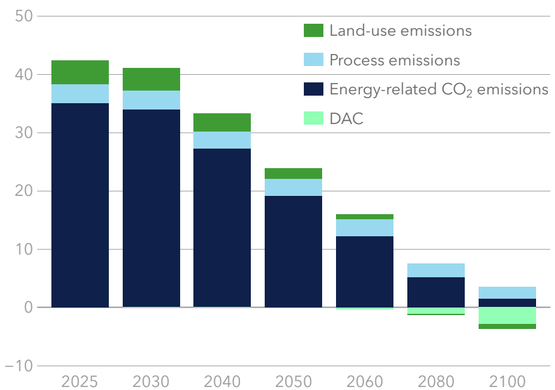


FIGURE 9.4 | Composition of total CO_2 emissions

Process emissions rise from 9% of energy and process CO_2 in 2025 to 19% in 2060 and 55% in 2100, without rising in absolute terms (Figure 9.4). They are the largest single component of the residual that remains after energy combustion has been substantially reduced.

Land-use emissions fall 80% to 0.8 Gt in 2060 – but estimates are uncertain

Emissions from agriculture, forestry, and other land use (AFOLU) are not modelled within the *Energy Transition Outlook*. Instead, we estimate them separately and combine them with modelled energy and industrial process emissions to calculate the warming outcome. For AFOLU, we assume 4.1 $GtCO_2$ in 2025 based on Global Carbon Project (2025) estimates, 3.8 Gt in 2030, 3.0 Gt in 2040, 1.8 Gt in 2050, and 0.8 Gt in 2060. This crosses into net removal in 2075 and reaches a net sink of 1.4 Gt/yr in 2100. Cumulative land-use emissions over 2025 to 2100 are 82 Gt, which represents about 0.05°C of end-century warming and just under 10% of the carbon budget remaining at 2025 for keeping warming below 2°C.

Our estimate for land-use emissions is conditional upon active and successful climate policies being implemented, e.g. results-based forest finance combined with satellite monitoring. The decline between 2030 and 2060 runs at just under 5% a year for three decades, and the observed historical record contains no comparable sustained fall. Net land-use flux held between 4.5 and 4.8 Gt from 2020 to 2024 even as deforestation reduced, because regrowth, degradation, peat drainage, and fire respond to clearing policy on different timescales. Our estimated reduction assumes a steady decline from those levels from around 2030 as forest finance ramps up. The uncertainty around this estimate is wide and asymmetric toward higher emissions, given that fire caused 42% of global tree cover loss in 2025 (WRI, 2026).

However, the base year itself is corroborated by independent estimates. The Global Carbon Project puts net land-use change emissions at 4.1 Gt in 2025, close to 10% less than 2024 on reduced deforestation in South America (GCP, 2025). Tropical primary forest loss fell 36% in 2025 to 4.3 million hectares, from a record 6.7 million in 2024, and Brazil reduced non-fire primary forest loss by 41% to its lowest recorded level (WRI, 2026).

The reason we think reduction in land-use emissions is possible is due to the commitment and progress in results-based forest finance linked to annual satellite monitoring of forest cover. Although the monitoring technology is well established, its proposed use as the basis for predictable payments on a global scale is new. For this mechanism to produce the forecast decline, four conditions must be met: the facility must secure sufficient capital, payments must make forest conservation more valuable than clearing, clearing must decline in response, and net land-use emissions must fall without long delays arising from regrowth and peat dynamics.

None of the four conditions has yet been demonstrated at scale. Announced contributions to the Tropical Forest Forever Facility stood at USD 6.5bn in April 2026, against an interim target of USD 10bn for 2026 and reaching USD 25bn over the coming years with a long-term goal of USD 125bn (TFFF, 2026). We publish the trajectory as our forecast with the condition that the funds disburse and net emissions decline in line with the assumed timescale. The 2020 to 2024 record, in which net flux held flat while deforested area fell, is the strongest evidence against the fourth condition, and continued decline requires the same sustained effort and international cooperation as energy-related emissions.

Direct air capture determines the timing of net zero rather than the rate of decline

Negative emissions technologies (NETs), principally DAC, have little effect on mid-century emissions, but a material effect on the timing of net zero. The technology extracts CO_2 from ambient air for geological storage. Deployment accelerates after 2060 as we expect costs to decline and removing CO_2 becomes more economical than eliminating

some residual emissions directly. DAC is therefore instrumental in balancing remaining emissions and reaching net zero. We expect DAC to remove 0.4 $GtCO_2$ in 2060, which is 3% of energy and process emissions in that year and thus negligible in terms of the mid-century trajectory. Instead, DAC's contribution is concentrated in the final decades: removal rises to 3.0 Gt/yr in 2100, against residual energy and process emissions of 3.8 Gt. Cumulative removal over 2025 to 2100 is 58 Gt (Figure 9.4).

The near-term pipeline for point-source capture argues for caution on the pace of deployment. Cancelled capture capacity in Europe exceeded capacity reaching final investment decision in 2025, and the US reduced public funding (Global CCS Institute, 2025; IEEFA, 2025). Our detailed capture forecast is set out in Section 5.4.

Fossil methane emissions fall by two-thirds between 2025 and 2060

We forecast fossil fuel methane emissions falling from 140 Mt in 2025 to 125 Mt in 2030 and 50 Mt in 2060, primarily driven by lower coal and oil production.

Methane is the second-largest contributor to warming after CO_2, accounting for about 30% of the temperature rise since industrialization (IEA, 2026b). However, most of those emissions come from agriculture. Methane's atmospheric lifetime of roughly a decade makes abatement effective on a short horizon. An error in the methane pathway therefore affects near-term temperature rather than the cumulative CO_2 budget that sets the end-century result. The composition of fossil methane emissions also shifts over time. In our forecast, natural gas declines more slowly than the other sources and therefore accounts for 54% of remaining fossil methane emissions by mid-century compared with 26% for oil and 20% for coal.

Our forecast reaches net zero in 2098, giving end-century warming of 2.3°C

Long-term warming is determined primarily by the cumulative amount of CO_2 released into the atmosphere. Over the relevant range, the relationship between cumulative emissions and temperature rise is approximately linear. Warming therefore depends more on the total quantity of CO_2 emitted than on the timing of those emissions and warming stops only when net-zero emissions is achieved. This relationship, known as the Transient Climate Response to Cumulative Emissions, provides the scientific basis for the carbon budgets assessed in IPCC's *Assessment Report (AR6)*. It also relates a carbon budget expressed in gigatonnes of CO_2 to an expected level of warming.

Atmospheric CO_2 reached a monthly average of 432 ppm at Mauna Loa in May 2026, compared to a pre-industrial 280 ppm (NOAA, 2026). The atmospheric concentration shows how far CO_2 has already risen, while the AR6 budgets indicate how much more can be emitted within a given warming limit. On the AR6 likely budgets at 67% probability, the remaining budget from 2020 is 400 $GtCO_2$ for 1.5°C and 1,150 $GtCO_2$ for 2.0°C, and both already account for non-CO_2 forcing (IPCC, 2021). In our forecast, the 1.5°C budget is exhausted in 2029 and the 2.0°C budget in 2052 (Figure 9.1). Cumulative anthropogenic CO_2 over 2025 to 2060 is 1,100 Gt, and a further 260 Gt accrues between 2061 and 2100 before net-zero emissions are reached in 2098, near the end of the century. We will overshoot the 2°C budget in 2100 by approximately 420 Gt. Applying a transient response of 1,500 $GtCO_2$ per degree at 67% probability, consistent with the basis on which those budgets are stated, gives end-century warming of 2.3°C.

Two qualifications add uncertainty to this estimate. First, the calculation does not fully include carbon-cycle feedbacks such as abrupt permafrost thaw and fire-permafrost interactions, which could reduce the remaining carbon budget. Second, the temperature outcome depends on the assumed climate response. Hansen et al. (2025), argue for an ECS of about 4.5°C per doubling of CO_2, implying greater warming than estimates based on an AR6-central response. However, Boardman et al. (2026) find that the record warm years of 2023 and 2024 do not justify revising the likely climate-sensitivity ranges upward, and their updated TCR estimate remains close to the AR6 assessed range. We continue to observe this research for future updates in our forecast.

Independent climate modelling corroborates 2.3°C

We asked the Center for International Climate Research (CICERO) to evaluate our emissions pathway using the climate assessment procedure applied in AR6. The procedure infills the GHG emissions the ETO does not model, then runs the complete set through MAGICC, the main simplified climate model used in AR6. It gives end-century warming of 2.3°C at the 67% probability level, matching the estimate we derive from the transient climate response to cumulative emissions.

Two independent methods converge on that figure. Running our emissions pathway through the AR6 infilling procedure gives 2.2 to 2.3°C, the lower value where the AR6 land-use estimate replaces ours and the higher where our own emissions are used unaltered. Constraining our pathway between the AR6 scenarios with consistently higher and consistently lower cumulative fossil CO_2 gives a median of 2.3°C. Warming on the central trajectory reaches 2.1°C in 2050 and 2.3°C in 2075.

Emissions we do not model carry the larger remaining uncertainty. Substituting the central two-thirds of AR6 non-CO_2 trajectories moves the 2100 result within 2.3 to 2.4°C, and the sensitivity is asymmetric upward. Our land-use assumption contributes at most 0.1°C, even under substantial deviation from it.

Warming levels off rather than peaks. No central trajectory declines by more than 0.05°C before 2100, so the end-century figure is also the highest warming reached.

10 REGIONAL TRANSITIONS

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Latin America 147
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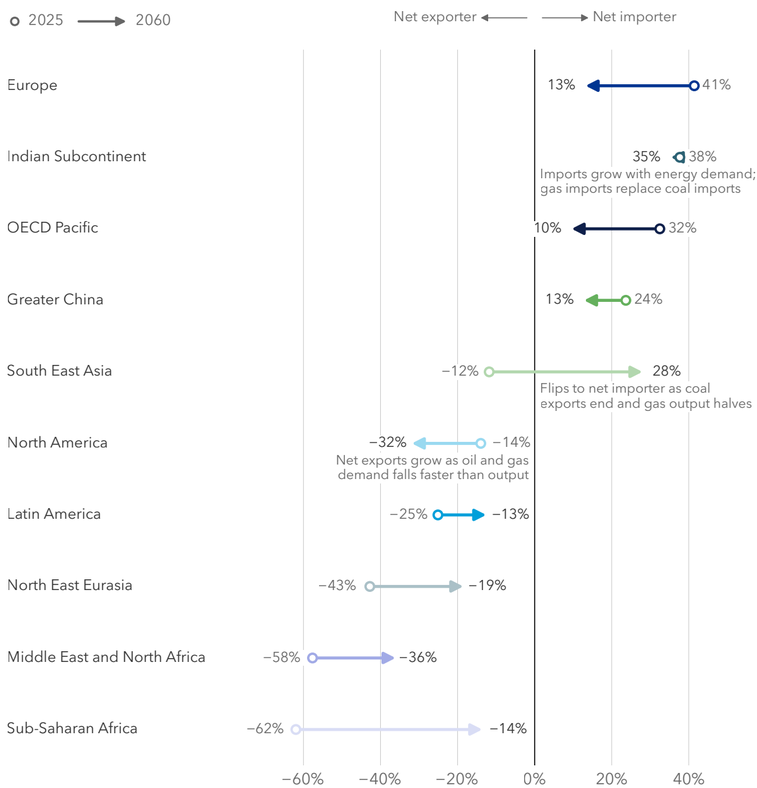
COMPARISON OF REGIONS

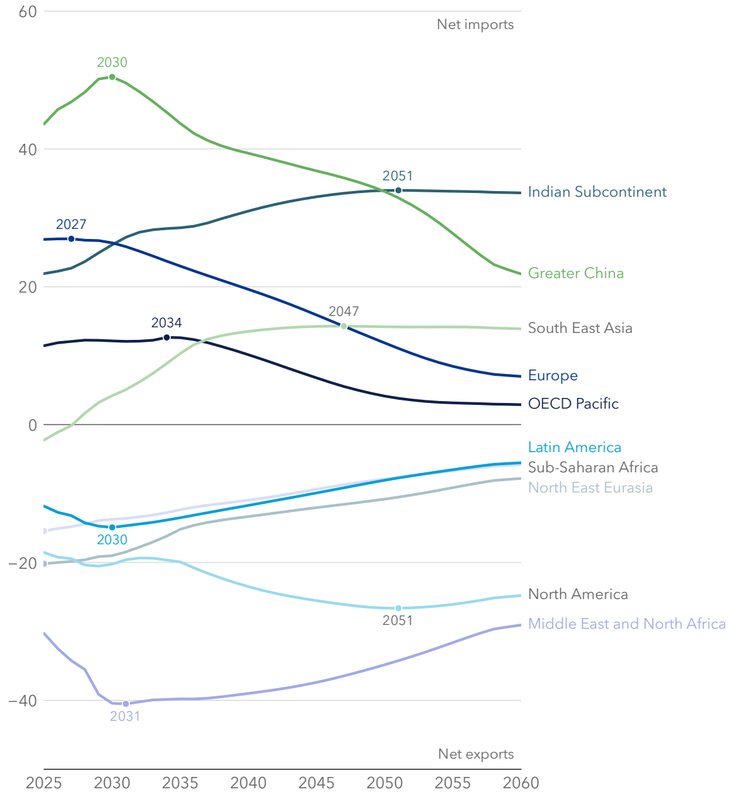
Who reduces import dependence?

Net fossil imports, share of primary energy supply



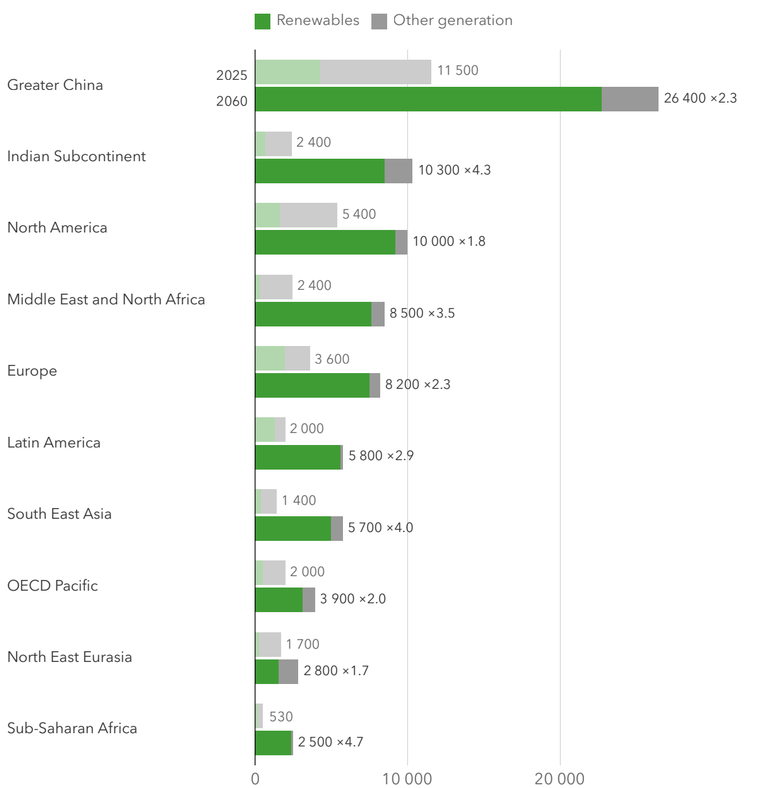
When do net imports peak?

Net fossil imports: oil, gas, and coal (EJ/yr)



How much does power generation grow and green?

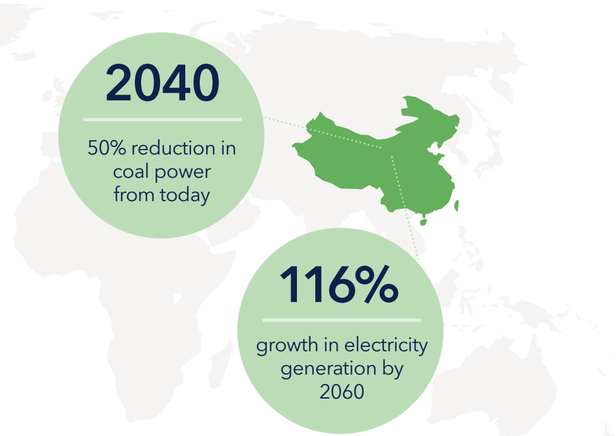
Power generation (TWh/yr)



Net fossil imports: oil, gas, and coal demand minus regional production, in energy terms (negative = net exports). Net import lines: 5-year moving average.
Renewables: solar, wind, hydropower, biomass, and geothermal power generation. ×: growth in power generation 2025–2060.

GREATER CHINA

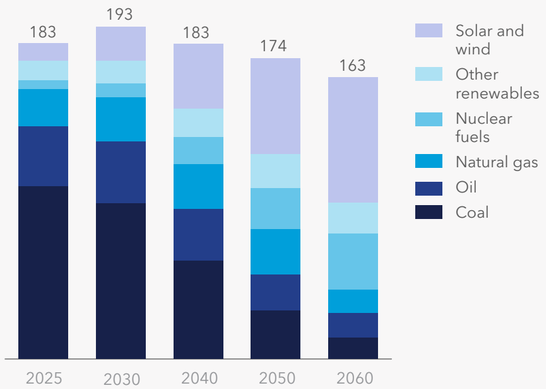
China's *15th Five-Year Plan* frames the next phase of the region's transition and has global implications. Fossil fuels will peak soon as renewables grow. Energy demand peaks in the 2030s as the population shrinks and electrification ensures more efficient energy delivery.



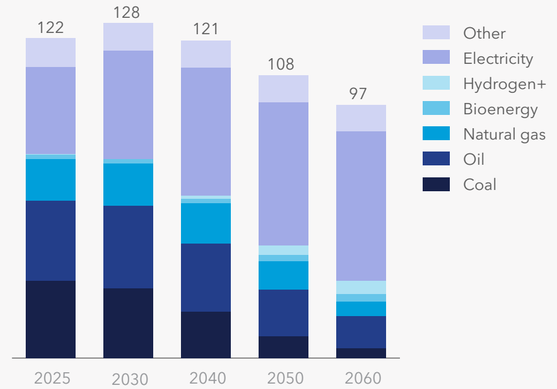
- **Despite GDP per capita more than doubling by 2060, energy demand declines by 20%**. The main reason is a structural shift in China's economy toward less energy-intensive activities. Improving energy efficiency and a 16% population decline further reduce demand. At the same time, electricity increases from 29% to 58% of final energy demand.
- By 2060, China will have net energy-related emissions of 2.8 $GtCO_2$. This is **79% lower than at present,** but does not meet the net-zero emissions goal set for that year.
- **China is leveraging its clean-energy manufacturing strengths** to improve energy security and build competitive low-carbon industries. Leadership in solar, batteries, and electrolysers enables production of green fuels and feedstocks. This, in turn, supports low-carbon steel, aluminium, and other exports while strengthening China's position across the clean-energy supply chain.
- **China's *15th Five-Year Plan* strengthens decarbonization by expanding controls on total emissions**, shifting responsibility for renewable energy use from governments to businesses, and extending carbon markets to new sectors. It also supports cleaner industrial development through investment in transmission infrastructure and by encouraging industry to locate near clean energy resources.

FULL REGIONAL OUTLOOK COMING NOVEMBER 2026

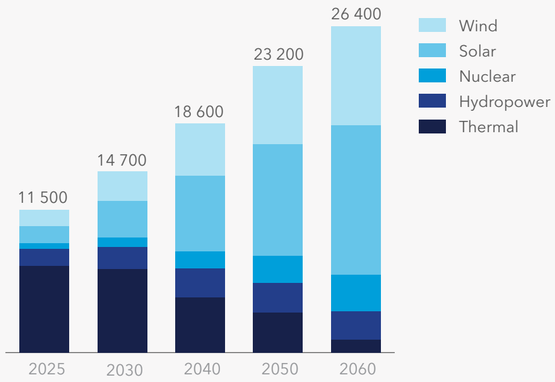
Primary energy
(EJ/yr)



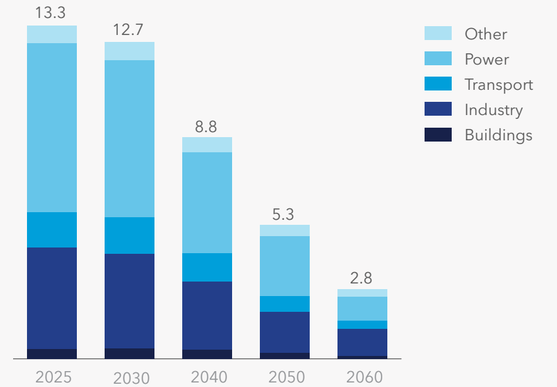
Final energy
(EJ/yr)



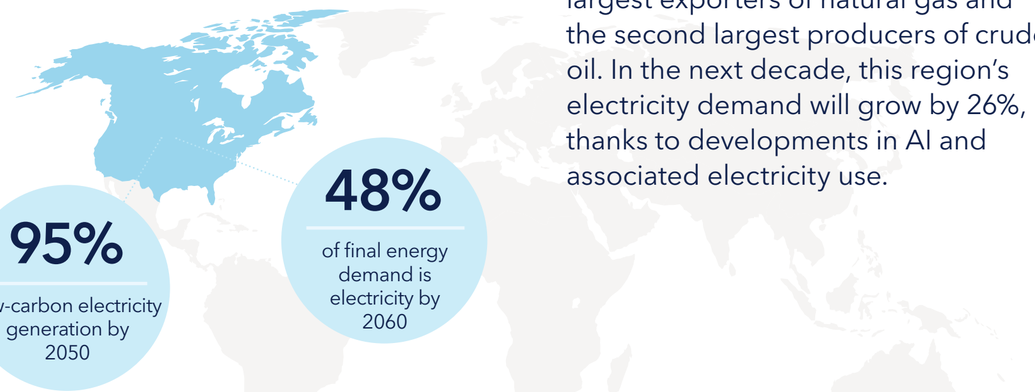
Power generation
(TWh/yr)



CO_2 emissions
($GtCO_2$/yr)



NORTH AMERICA

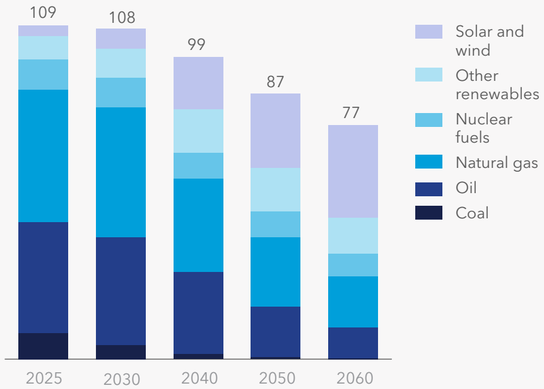


Together, the US and Canada are the largest exporters of natural gas and the second largest producers of crude oil. In the next decade, this region's electricity demand will grow by 26%, thanks to developments in AI and associated electricity use.

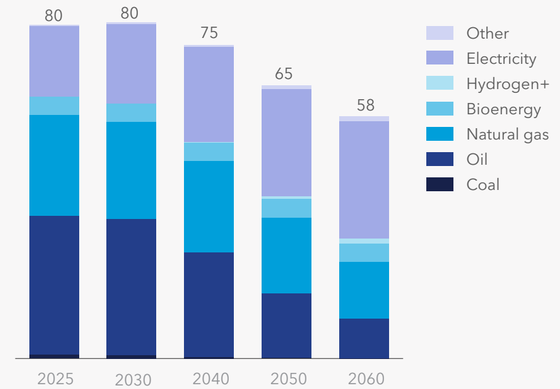
- **Rising electricity demand, driven by AI and data centre demand, and an ageing grid will push up residential power prices and slow electrification in the short term.** Wholesale electricity prices rise from USD 58/MWh in 2026 to USD 67/MWh in 2030, before easing. However, higher network and adequacy costs keep residential prices elevated, slowing electrification uptake.
- **Batteries will begin to displace gas-fired power in the grid during peak hours.** Batteries store abundant midday solar generation and discharge it during evening demand peaks – hours traditionally served by gas-fired peaker plants. This reduces gas utilization, lowers emissions, and enables higher shares of solar PV without compromising system reliability.
- **Renewable deployment remains resilient despite growing policy uncertainty for solar PV and wind.** Falling technology costs, corporate demand for clean electricity, state and provincial policies, and rapidly growing electricity demand ensure that renewables and batteries remain the largest source of new supply capacity across the region throughout our forecast.
- **Energy-related emissions have already peaked and will continue to decline**, despite growing energy demand and policy support for domestic oil and gas. Expanding renewable generation, increasing battery deployment, gradual electrification, and improving energy efficiency more than offset rising electricity consumption and put the region on a sustained downward emissions trajectory.

FULL REGIONAL OUTLOOK COMING JANUARY 2027

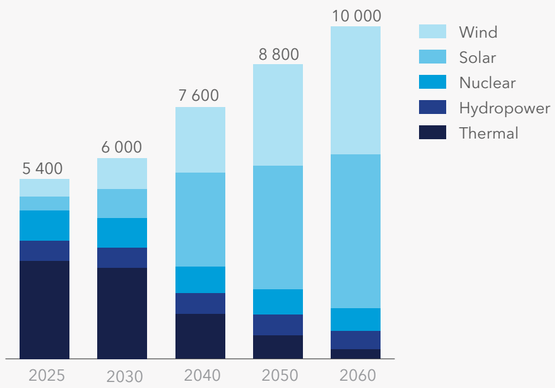
Primary energy
(EJ/yr)



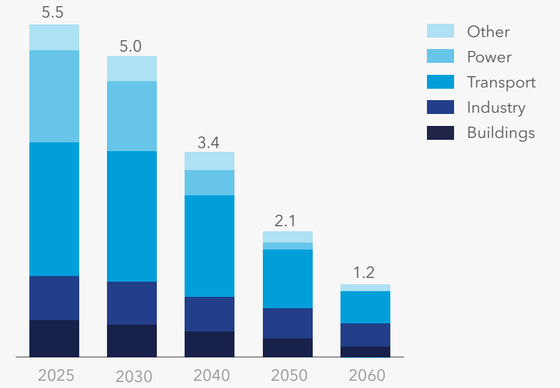
Final energy
(EJ/yr)



Power generation
(TWh/yr)

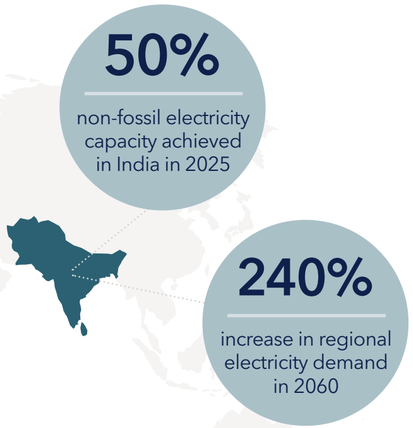


CO_2 emissions
($GtCO_2$/yr)



INDIAN SUBCONTINENT

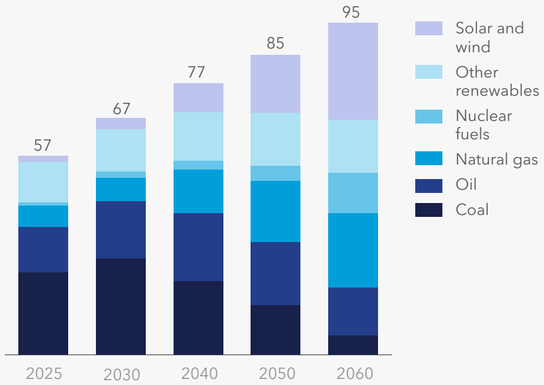
India, Pakistan, and Bangladesh account for 90% of this region's population, GDP, and energy demand. Emissions will decline from 4.2 $GtCO_2$/yr in 2030 to 2.6 $GtCO_2$/yr in 2060 due to expansion of solar and wind.



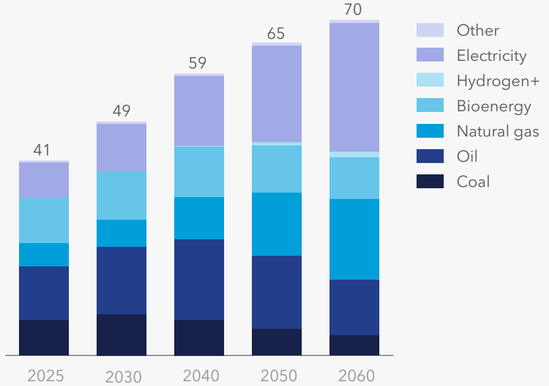
- **India is targeting 500 GW of non-fossil power capacity by 2030 and 100 GW of nuclear capacity by 2047. It achieved 50% non-fossil capacity in 2025**, five years ahead of the 2030 target. We forecast the region will exceed 60% non-fossil capacity (including nuclear and hydro) by 2030.
- **Rising temperatures and GDP per capita will more than double electricity consumption for space cooling**, from 7.4 EJ/yr in 2025 to 25 EJ/yr by 2060. The urban heat island effect, which can make cities 5°C to 10°C warmer, will exacerbate cooling electricity demand for a rapidly urbanizing population.
- **The region will reach peak oil demand in 2042 at 17 EJ/yr then decline, largely led by transport electrification**. The Strait of Hormuz crisis has exposed the region's vulnerability to imported fossil fuels and reinforced the need to transition to diverse, domestic, low-carbon energy sources. Sri Lanka, Maldives, and Bangladesh are particularly vulnerable, with limited domestic energy resources.
- **India's rapid growth in non-fossil generation and electricity demand requires parallel investment in transmission, storage, and flexibility** such as their Green Energy Corridor. The Energy Connectivity in South Asia project aims to support greater cross-border energy integration across the region. This will help to optimize the region's diverse renewable resources, improve flexibility, and support cost-effective decarbonization.

FULL REGIONAL OUTLOOK COMING FEBRUARY 2027

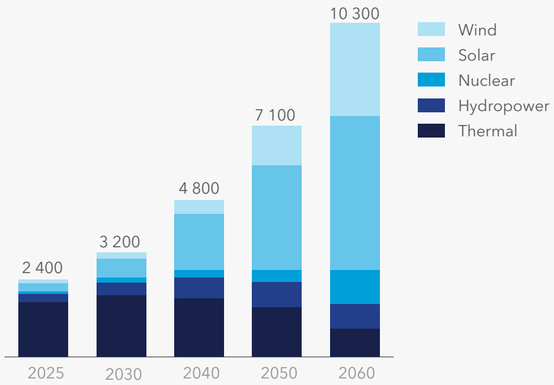
Primary energy
(EJ/yr)



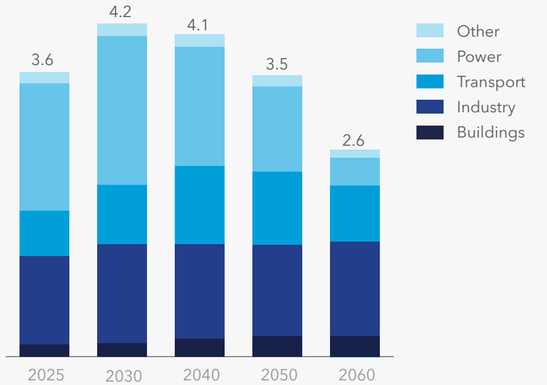
Final energy
(EJ/yr)



Power generation
(TWh/yr)

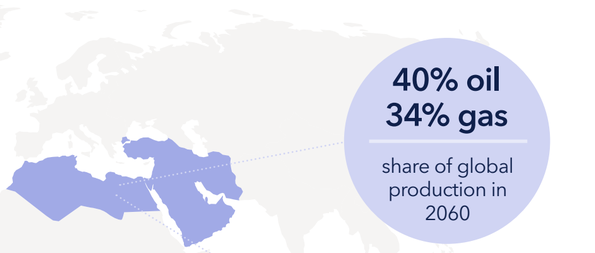


CO_2 emissions
($GtCO_2$/yr)



MIDDLE EAST AND NORTH AFRICA

Despite growing clean-energy investment, the region remains heavily dependent on hydrocarbons for energy exports and domestic power generation. Hydrocarbons will continue to underpin economic activity in the near term, even as renewables, storage, CCS, and hydrogen gradually reshape the energy mix.

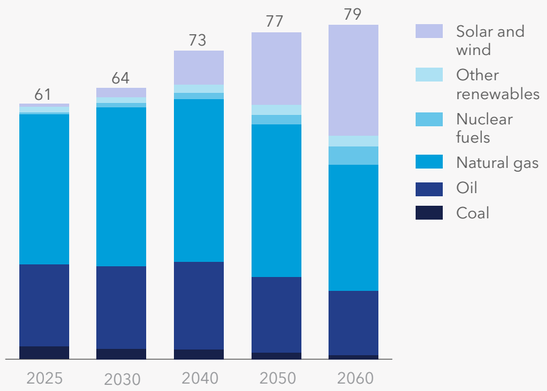


- **The recent disruption has tested the region's supply route security.** With the Strait of Hormuz constrained for months, oil and gas production fell by about 20% and 10% from 2025 levels. Oil and gas investment remains strong, but is increasingly focused on pipelines, storage, ports, and alternative export routes.
- **The region continues to pursue a leading market share in oil and gas supply to Asia**. Its share of global oil production will return to 36% by 2030 and 40% by 2060, with gas reaching 26% by 2030 and 34% by 2060, despite conflict-driven disruption and softer global demand reshaping both volumes and trade routes.

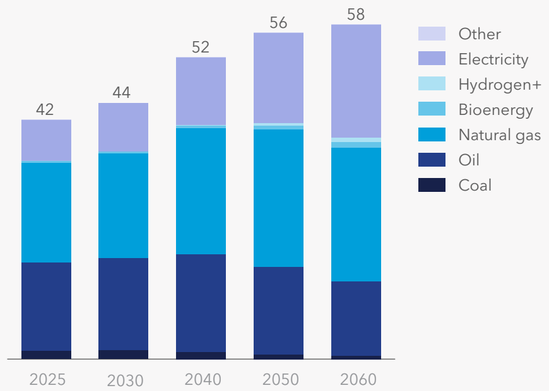
READ THE FULL REGIONAL OUTLOOK HERE

- **Carbon capture, utilization, and storage (CCUS) and hydrogen plans have been delayed or scaled back, but are still growing**. CCUS supports emissions reductions in upstream operations, gas processing, refining, manufacturing, and hydrogen production. These will reach 17 $MtCO_2$/yr in 2030 and 135 $MtCO_2$/yr in 2060. CO_2 removals from BECCS and DAC rise to about 45 $MtCO_2$/yr by 2060.
- **Renewable energy investment will continue to accelerate.** Fossil fuels supply 82% of electricity generation in 2026. However, renewables take the lead by the mid-2030s and reach around 90% of generation by 2060. Led by solar and utility-scale storage, capacity grows ten-fold by 2030.

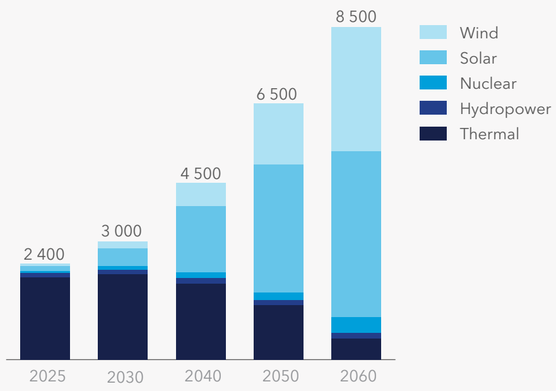
Primary energy
(EJ/yr)



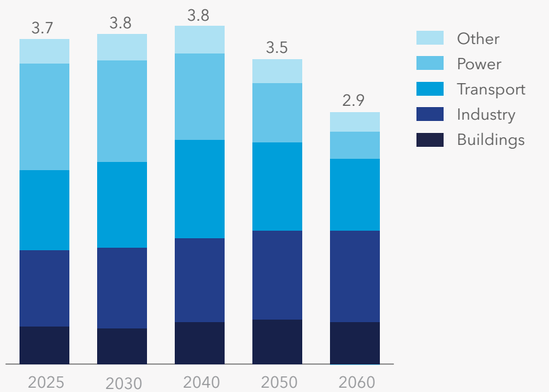
Final energy
(EJ/yr)



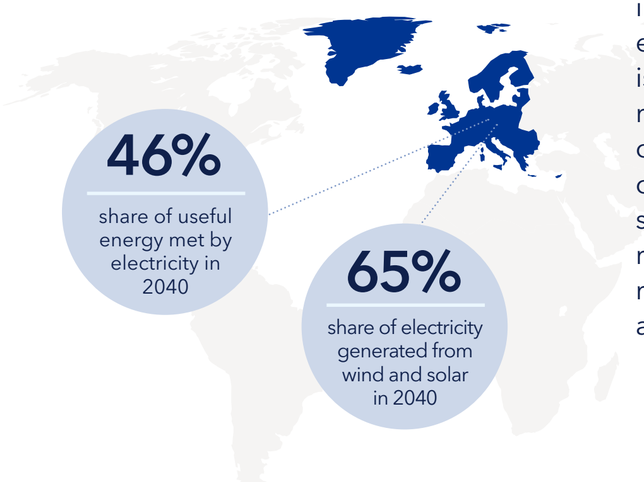
Power generation
(TWh/yr)



CO_2 emissions
($GtCO_2$/yr)



EUROPE



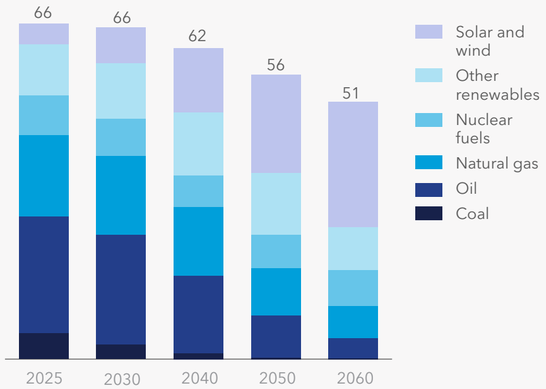
Europe is entering the implementation phase of the energy transition. The challenge is no longer primarily building renewable capacity but rather creating an energy system that can reliably integrate large shares of locally generated renewable electricity while remaining secure, affordable, and competitive.

- **Energy security and affordability have moved up the list of political priorities.** The wars in Ukraine and Iran have exposed Europe's import dependency. As volatile energy prices strain households and undermine industrial competitiveness, policymakers are prioritizing sourcing energy and economic value from regional and domestic supply chains.
- **Electrification through regional renewable generation** is increasingly viewed as the strategic path. The imported share of primary energy will drop from half to a third by 2050. Furthermore, the average power price will decrease and steadily decouple from that of natural gas in the 2030s, as is currently happening in Spain.

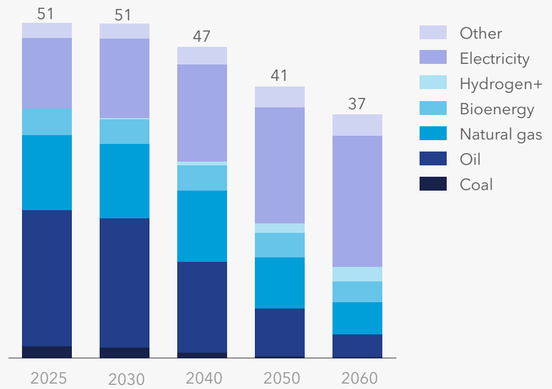
READ THE FULL REGIONAL OUTLOOK HERE

- **Ensuring system adequacy will be the key challenge.** The share of variable renewable grid power will grow from 30% today to 70% in 2050. This will require 10 times as much energy storage alongside grid balancing with flexible demand and rapid grid expansion. Europe is revising its power market to cater to this change.
- **Europe shapes its transition through one of the world's most comprehensive policy toolboxes.** High carbon pricing, the Carbon Border Adjustment Mechanism (CBAM), electrification targets, renewable deployment mechanisms, grid investment programmes, efficiency standards, and sector-specific regulations combine to steer the transition across the energy system. These tools adjust in response to the global market.

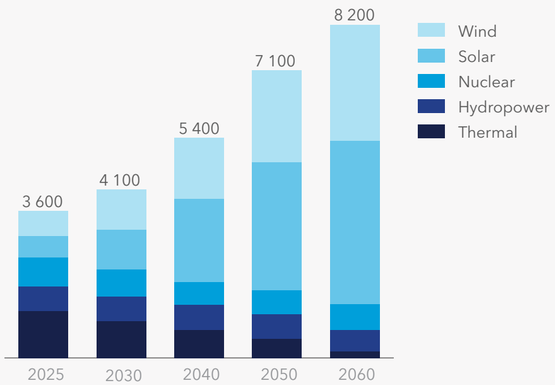
Primary energy
(EJ/yr)



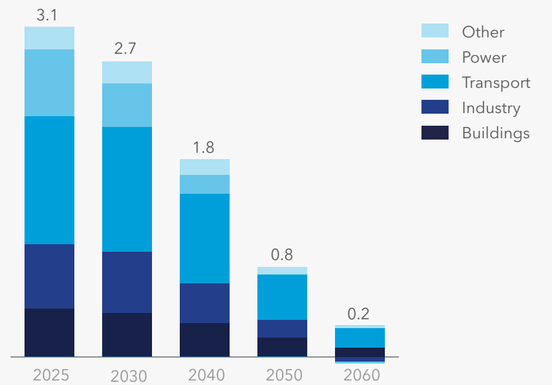
Final energy
(EJ/yr)



Power generation
(TWh/yr)

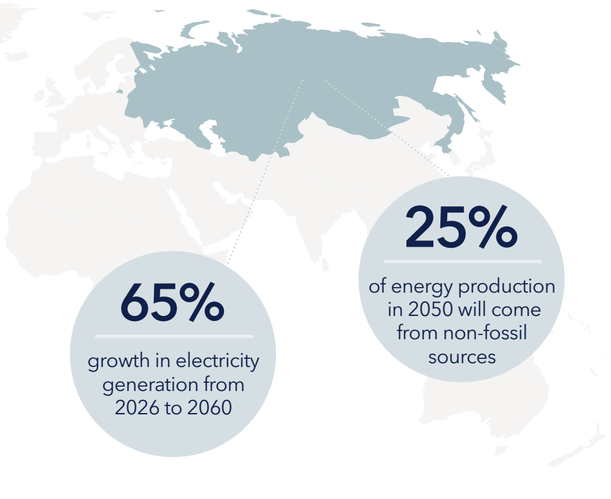


CO_2 emissions
($GtCO_2$/yr)



NORTH EAST EURASIA

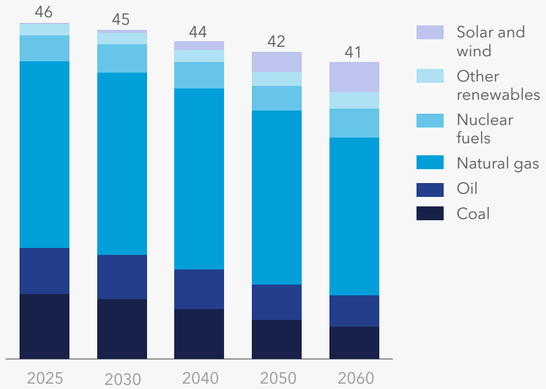
The energy transition in North East Eurasia is splitting into two different trajectories. While Russia, the biggest economy in the region, pushes oil and gas, the other countries commit to converting to green energy. Oil and gas producers continue production for export, while transitioning domestically.



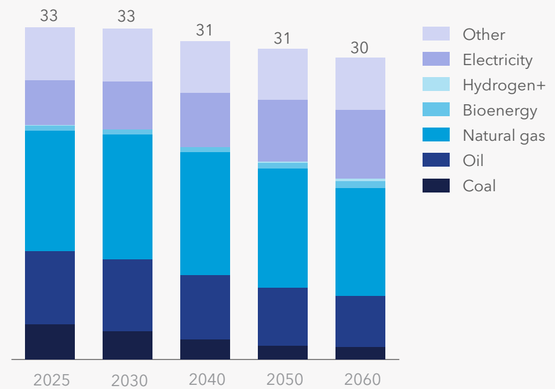
- **Ukrainian strikes on Russian oil production and refinery facilities have caused fuel shortages across Russia.** The outcome depends on whether Russian repair efforts can outpace Ukrainian drone attacks. Any impact is likely to be contained within Russia, although export restrictions continue to impact many countries.
- **Kazakhstan's long-term energy strategy is to replace coal** (70% of today's electricity production) with nuclear power for baseload combined with wind and solar for clean energy growth. Oil and gas production will remain steady for export revenues.
- **Half the Ukrainian power production capacity has been lost in the war.** This has halted the transition to cleaner energy and shifted the focus to energy security. By 2035, Ukraine targets 50% of electricity to come from nuclear (from 49% in 2023) and 50% from wind and solar (from 4% in 2023).
- **We expect around half of the region's installed nuclear capacity to retire from 2035 onwards.** Nevertheless, Russia plans to expand nuclear generation to 2050, and several other countries in the region plan to either increase or begin nuclear power generation starting in the 2030s.

FULL REGIONAL OUTLOOK COMING JANUARY 2027

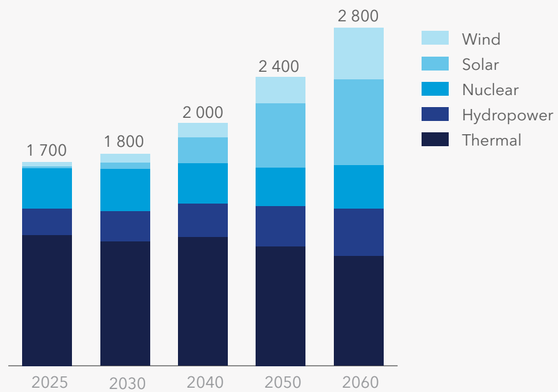
Primary energy
(EJ/yr)



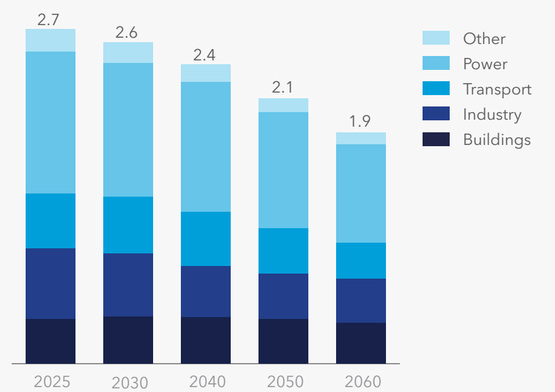
Final energy
(EJ/yr)



Power generation
(TWh/yr)

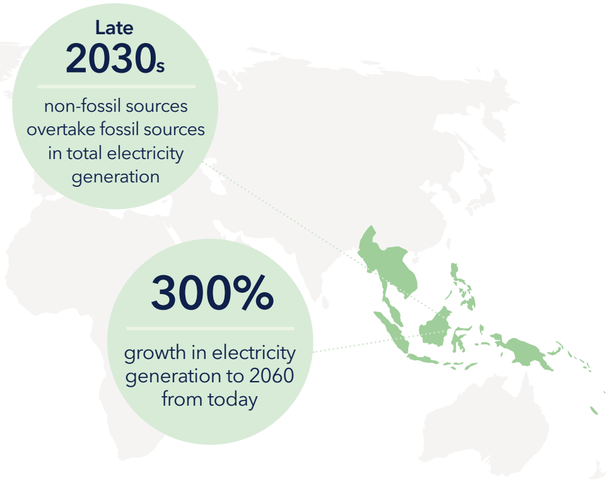


CO_2 emissions
($GtCO_2$/yr)



SOUTH EAST ASIA

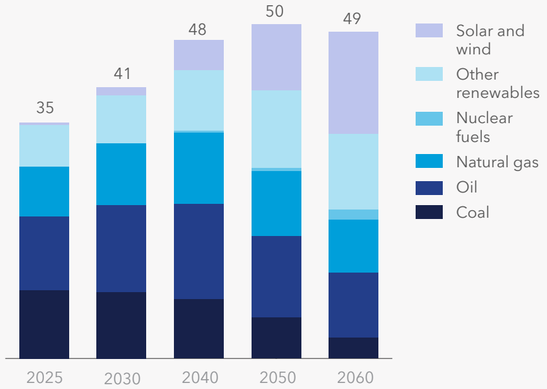
South East Asia is a fast growing, resource rich region which remains heavily dependent on fossil fuels. Rising energy demand, energy security concerns, climate risks, policy reforms, and clean energy investment are accelerating renewables adoption.



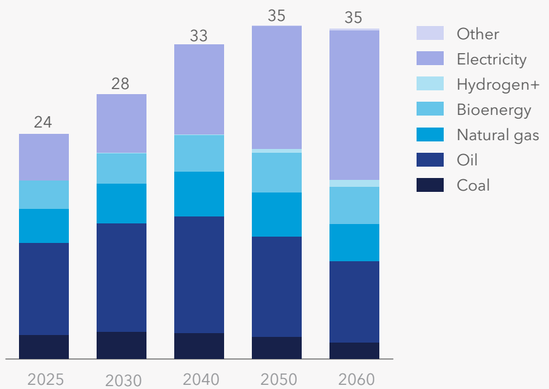
- **The Strait of Hormuz crisis has exposed energy security as a central challenge for the region.** Countries are increasingly looking to diversify their energy supply, which is currently dependent on imported oil. For some countries, a significant portion is shipped through the Strait of Hormuz.
- **The renewable energy buildout is progressing, with solar soon to become the region's largest renewable power source.** Renewables are largely meeting new electricity demand rather than displacing the coal that accounts for nearly half of generation today. Grids and storage remain a challenge: despite growing interest in battery energy storage systems (BESS), market structures and regulatory frameworks are still evolving.
- **Electricity demand is growing rapidly and will nearly double from today to 2040.** The share of electricity in final energy demand today is around 20% and we expect it to reach over 40% in 2060. Residential cooling, EVs, and data centres are the fastest growing demand sectors.
- **The ASEAN Power Grid (APG) remains crucially important for the region's transition.** Interconnected grids and expanded cross-border electricity trade in the region can reduce costs, improve resilience, and support economic development. There is significant potential to expand cross-border transmission through development and expansion of the APG.

READ THE FULL REGIONAL OUTLOOK HERE

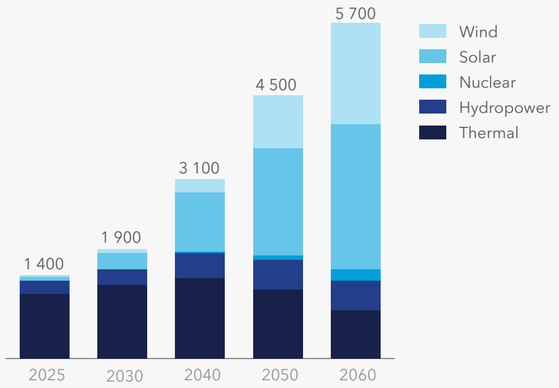
Primary energy
(EJ/yr)



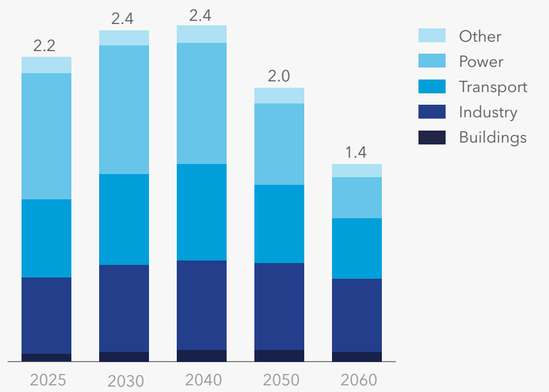
Final energy
(EJ/yr)



Power generation
(TWh/yr)

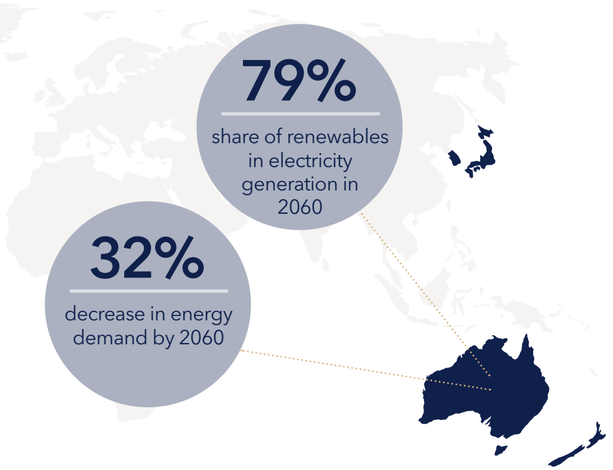


CO_2 emissions
($GtCO_2$/yr)



OECD PACIFIC

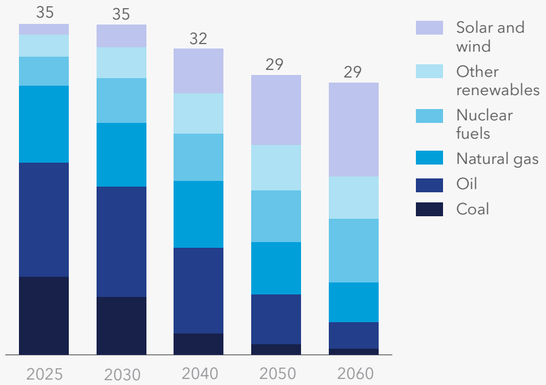
Energy security is accelerating the transition across the OECD Pacific. Rapid solar and battery deployment, renewed nuclear growth, and rising electricity demand are reshaping a region seeking to reduce fossil-fuel imports while maintaining reliable and resilient energy systems.



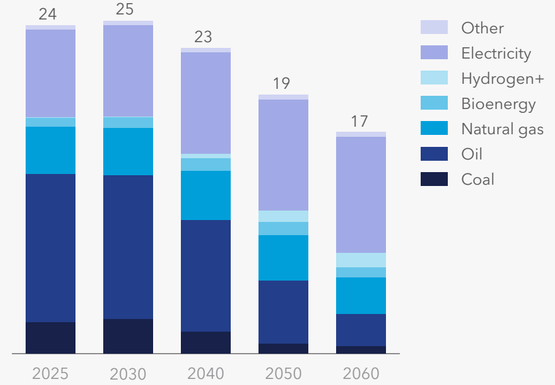
- **The Iran war has reinforced efforts to strengthen energy security** through renewables, storage, and nuclear power. Australia is rapidly expanding solar and storage, while Japan and South Korea are balancing renewables with nuclear power to improve resilience and reduce reliance on imported fossil fuels.
- Population decline, electrification, and broader efficiency improvements reduce total energy demand. At the same time, growing data centre loads and electrification across sectors (especially transport) drive **sustained growth in electricity demand** across the region.
- As solar and wind expand, focus is shifting to grid capacity and flexibility. Australia is leading in battery deployment, while all countries in this region are investing in storage, transmission, and market reforms to maintain system reliability. **Grid bottlenecks and permitting delays continue to slow renewable deployment** across the region.
- **Hydrogen ambitions are becoming more pragmatic.** Proposed projects are shifting from large-scale hydrogen trade towards domestic industrial decarbonization. Australia increasingly targets green iron, steel, and ammonia production. Japan and South Korea target hydrogen deployment in sectors where it delivers the greatest benefits for industrial competitiveness and decarbonization.

FULL REGIONAL OUTLOOK COMING NOVEMBER 2026

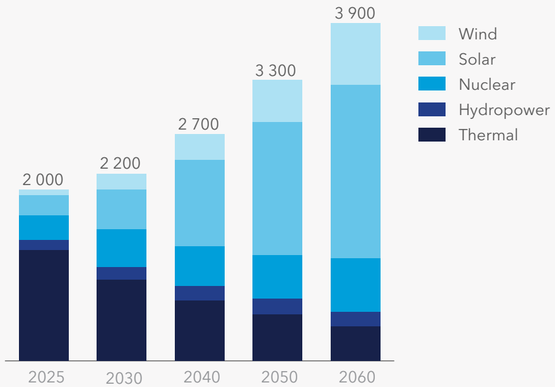
Primary energy
(EJ/yr)



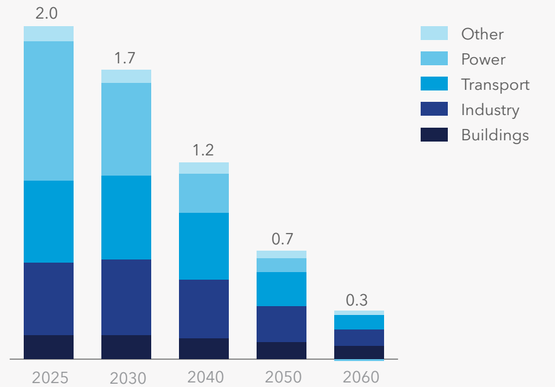
Final energy
(EJ/yr)



Power generation
(TWh/yr)

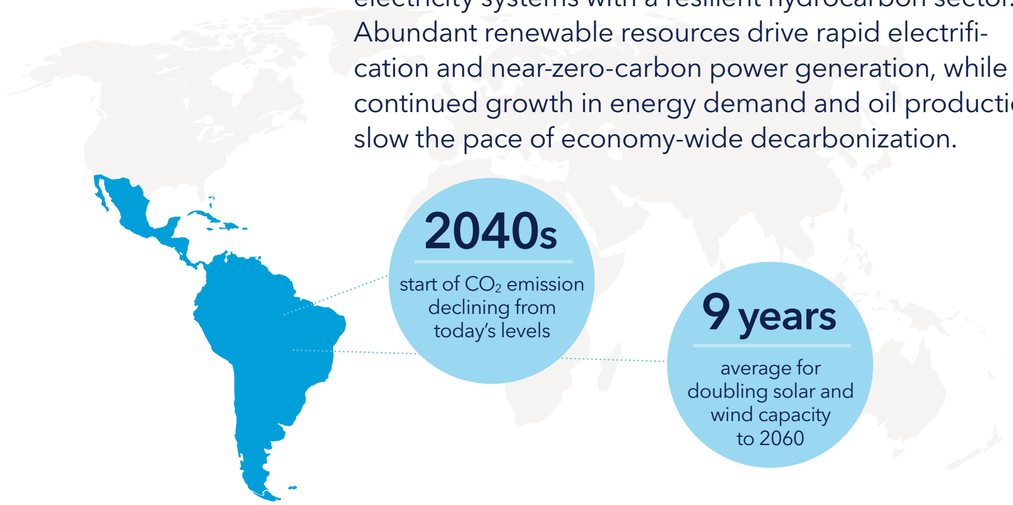


CO_2 emissions
($GtCO_2$/yr)



LATIN AMERICA

Latin America combines some of the world's cleanest electricity systems with a resilient hydrocarbon sector. Abundant renewable resources drive rapid electrification and near-zero-carbon power generation, while continued growth in energy demand and oil production slow the pace of economy-wide decarbonization.

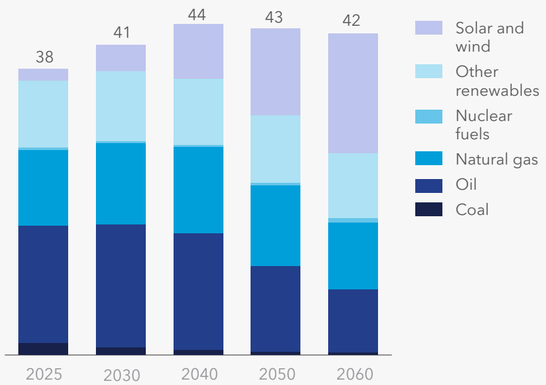


- **Latin America will pursue a dual-track energy transition.** The region combines one of the world's fastest renewable expansions with continued hydrocarbon production. Competitive offshore oil in Brazil and unconventional gas development in Argentina support regional energy supply through the 2040s even as clean electricity rapidly expands.
- **Economic growth slows the pace of emissions reduction.** Population growth, rising incomes, cooling demand, and industrial activity continue to increase energy consumption. As a result, emissions remain broadly stable until 2040 before falling by around half by 2060, far short of net-zero ambitions.

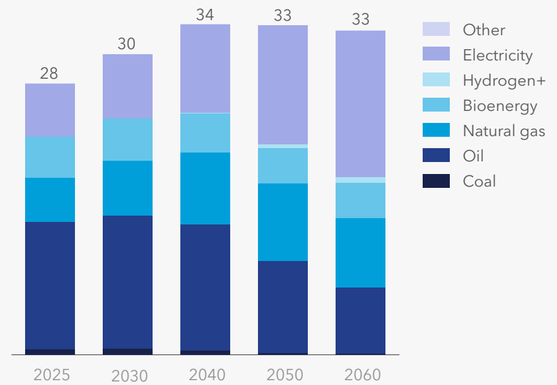
FULL REGIONAL OUTLOOK COMING FEBRUARY 2027

- **Electricity will become the main engine of decarbonization.** Electricity's share of final energy demand rises from about 20% today to almost 45% by 2060. This supports the electrification of transport, buildings, and parts of industry. Renewable electricity increasingly displaces oil after regional energy demand plateaus in the 2040s.
- **Climate resilience is becoming as important as decarbonization.** Hydropower remains the backbone of the power system, but increasing drought risk is driving investment in solar, wind, battery storage, and transmission. By 2060, almost all electricity generation is renewable, with solar and wind becoming the dominant sources.

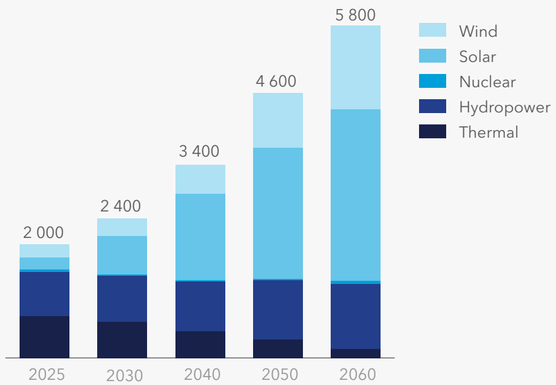
Primary energy
(EJ/yr)



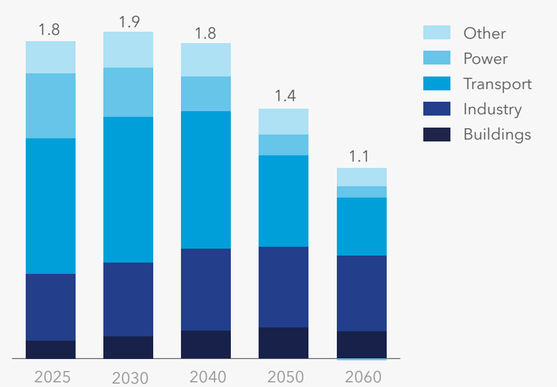
Final energy
(EJ/yr)



Power generation
(TWh/yr)

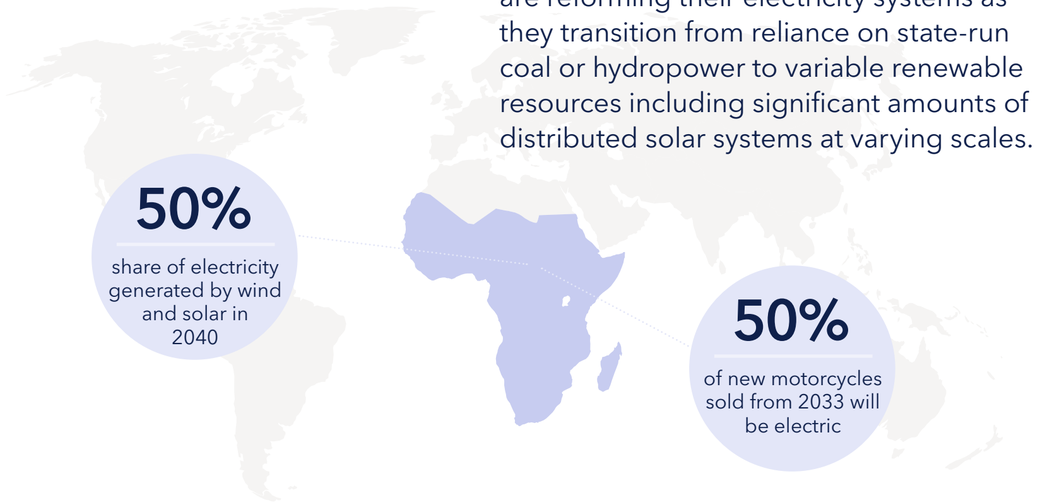


CO_2 emissions
($GtCO_2$/yr)



SUB-SAHARAN AFRICA

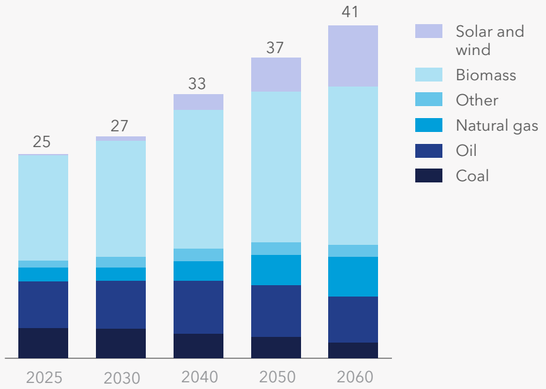
Several countries in Sub-Saharan Africa are reforming their electricity systems as they transition from reliance on state-run coal or hydropower to variable renewable resources including significant amounts of distributed solar systems at varying scales.



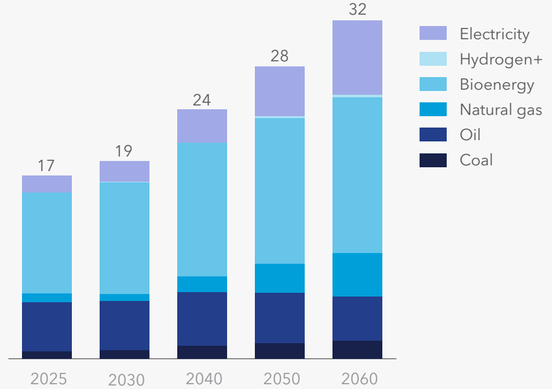
- **Around a dozen countries are pursuing regulatory reform of domestic power systems.** South Africa, which accounts for 45% of the region's total electricity consumption, is using competitive procurement to attract independent power producer investments in solar and wind.
- **This policy shift is happening as electricity demand outgrows ageing coal and hydropower generation infrastructure.** Countries are mostly leapfrogging expensive gas turbines in favour of low-cost solar and wind power, leveraging the region's plentiful renewable resources.
- **Sub-Saharan Africa will remain dependent on traditional biomass** for the majority of energy use, especially for cooking. However, we project electricity demand will grow four-fold by 2060, far outpacing the 50% increase in biomass demand.
- **Electricity demand growth will be driven** by industry and transport electrification and the rising quantity and quality of residential electrification (such as air conditioning and appliances). These are both drivers of, and driven by, GDP growth. Road transport electrification will begin with motorcycles before electric passenger and commercial vehicles become common.

FULL REGIONAL OUTLOOK COMING FEBRUARY 2027

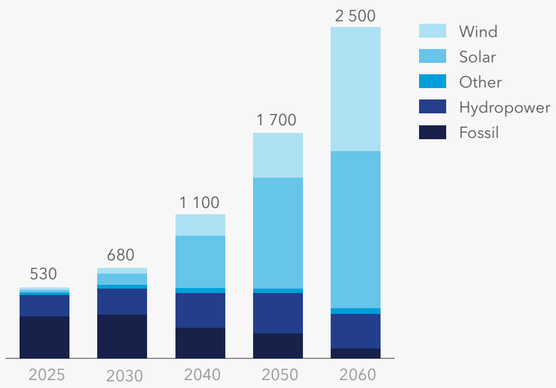
Primary energy
(EJ/yr)



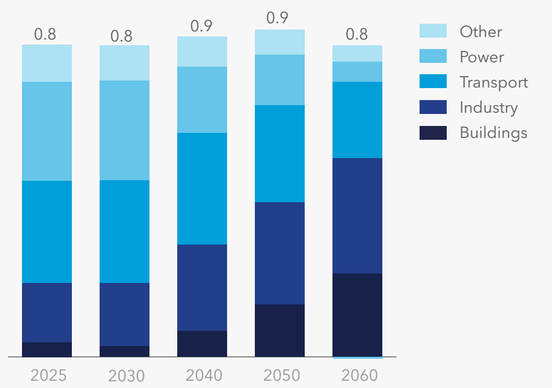
Final energy
(EJ/yr)



Power generation
(TWh/yr)



CO_2 emissions
($GtCO_2$/yr)



11 THE ETO MODEL

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11.1 THE ETO MODEL

The *Energy Transition Outlook* (ETO) is a simulation-based forecast designed to project the most likely pathway for the global energy transition. Rather than presenting an idealized, best-case scenario or a theoretical, cost-optimized solution, it models how the energy system actually behaves. It accounts for how different sectors interact over time, constrained by physical realities and driven by real-world decision-making.

A focus on system response

What makes the ETO approach distinctive is its focus on *system response*. We do not assume that market choices are perfectly rational or seamlessly coordinated. Instead, we model how policies, technologies, and human behaviours continuously influence one another, often generating unexpected feedback loops.

For example, falling battery costs naturally accelerate EV adoption. This rapid uptake increases overall electricity demand, but it also eventually introduces massive vehicle-to-grid (V2G) storage capacity back into the power network. As the energy transition accelerates, these cross-sector interconnections multiply. Dynamics that once occurred in isolation now trigger system-wide ripple effects.

The accompanying system diagram (Figure 11.1) illustrates a selection of the critical feedback loops we formulate within our model. The feedback interactions displayed in the ETO Model diagram generally fall into two categories of feedback processes that dictate the pace of the transition:

– **Reinforcing processes (accelerating growth):** These loops act as an engine for rapid technology deployment. Consider the relationship between carbon capture and storage (CCS) capacity and its underlying costs. The initial build-up of CCS capacity leads to accumulated engineering experience and economies of scale. This naturally drives down the cost of the technology, which in turn makes it more competitive and enables even more capacity to be built. The process continues in a self-reinforcing, upward cycle.

– **Self-balancing processes (natural limits):** These loops act as a natural brake on the system, preventing runaway growth of a single technology. A prime example is electricity generation by type, particularly solar PV. As solar capacity increases relative to other technologies, the grid experiences an oversupply of power during peak daylight hours. Because solar has virtually zero variable operating costs, this oversupply crashes the wholesale electricity price during those hours (a phenomenon known as price cannibalization). This ultimately lowers the revenue generated by solar plants and reduces the economic competitiveness of future solar projects. This self-regulating process naturally caps solar growth once it reaches a critical threshold in the power system unless storage is added to shift supply to higher-value periods.

Simplified model structure and selected key feedback loops in the ETO Model

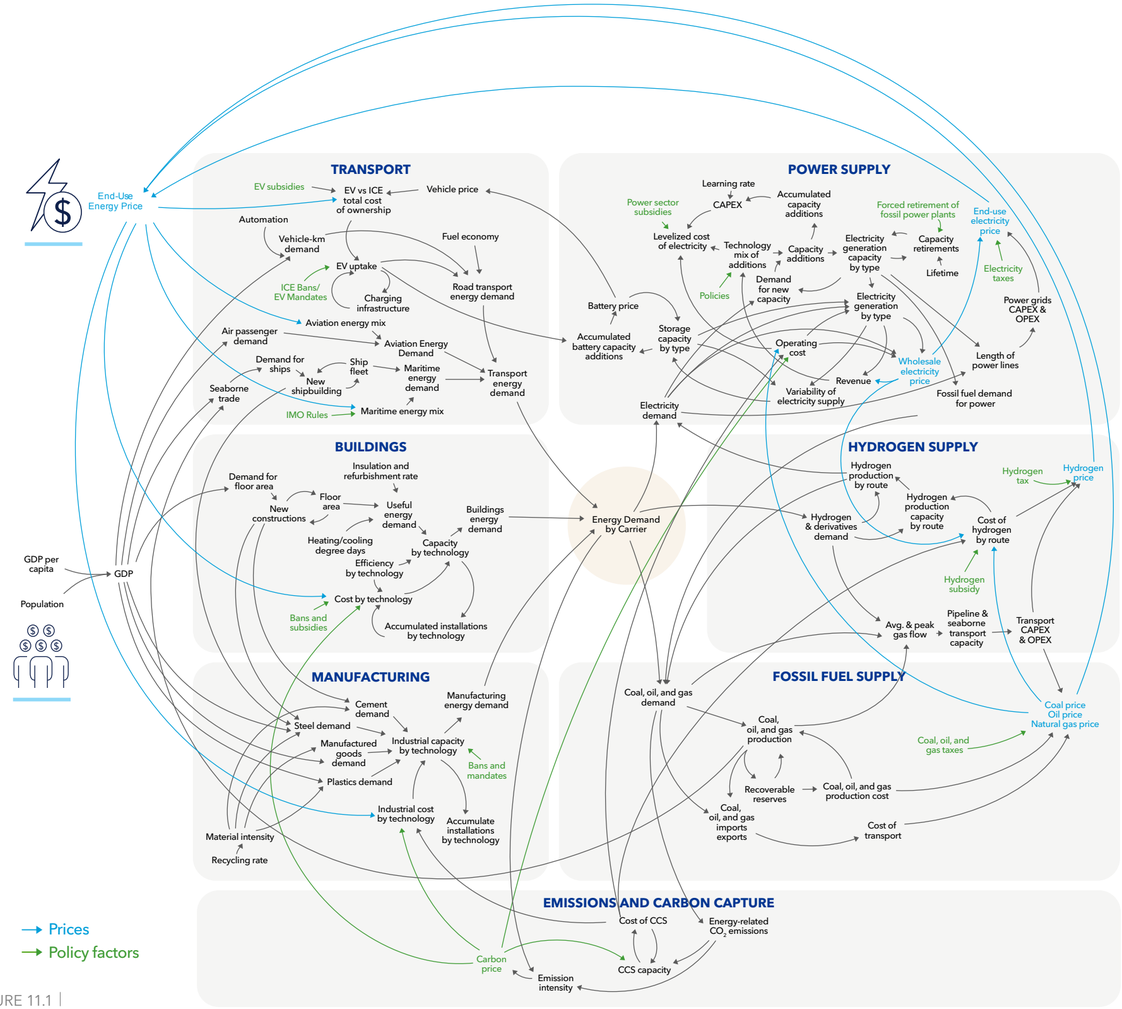


FIGURE 11.1

11.2 HOW THE ETO MODEL OPERATES

A long-term, behaviourally-grounded forecast

The ETO Model simulates the global energy system from 1980 to 2100. This long horizon allows us to represent gradual structural change in technology, demand, supply, and investment. We simulate electricity dispatch hourly to capture grid flexibility and variability; other sectors run on weekly time steps.

Decisions based on available information

We do not assume perfect foresight or centralized planning. Instead, we model how households, companies, and governments act based on what they know, what they value, and the options they face. Investors weigh expected returns. Households care about upfront costs and convenience. Governments pursue a mix of economic, social, and strategic goals. These actors act incrementally, not globally. Their choices are bounded, interdependent, and shape the behaviour of the system itself.

Policy, timing, and constraints

We model policy as it exists in reality – through taxes, subsidies, mandates, and bans. The model includes delays where they matter: in planning, permitting, construction, and learning. It also captures limitations on how fast supply chains can ramp up, how quickly consumers can adopt new technologies, or how long assets stay in use. These frictions, often overlooked in simple models, are essential for understanding transition speed and direction.

Granularity where it matters

The model covers 10 world regions, 12 energy carriers, and over 20 end-use sectors including transport, buildings, and manufacturing (Figure 11.2). This structure allows us to capture fuel switching, technology competition, and regional differences in cost, policy, and behaviour. It also reflects the energy used by whom, for what, and with which technology. This is critical to understanding system interactions.

Data driven and continuously updated

Model parameters are grounded in data from trusted sources such as the IEA, UN, GlobalData, Rystad, and Comtrade. These are supplemented by our expert insights based on the most up-to-date project information and internal research. We regularly update the model to reflect new policy, cost trajectories, and observed market trends. Sensitivity tests are used to test assumptions and improve robustness. The result is a model that reflects how the energy system works – not how we might wish it to.

For further information on our use of systems thinking, you may wish to consult:
From Silos to Systems: The Importance of Whole Energy Systems Thinking. Energy Industry Insights 2025: Special Report, DNV, 2025.

ETO model granularity

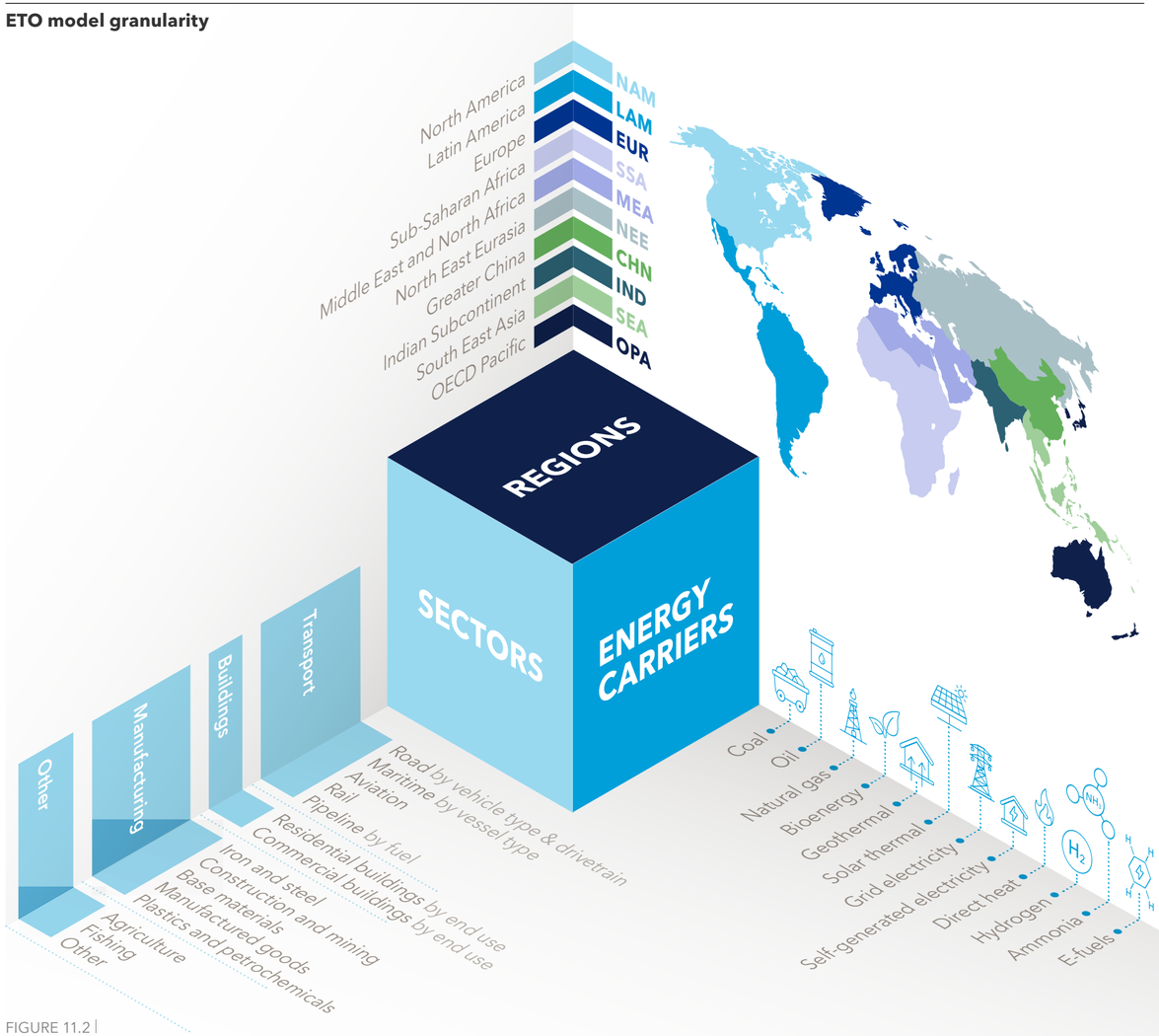


FIGURE 11.2

11.3 WHAT'S NEW IN THE ETO MODEL THIS YEAR

Each year, we refine the ETO Model to reflect the latest data, technology trends, and policy developments. For the 2026 edition (our 10th annual edition), we have made the following prominent updates.

Geopolitical developments

We introduced parameters to reflect the short-term impacts of the war in Iran on energy prices, fossil fuel supply, and energy demand. We have also represented the short-term effects of recent economic developments on cost of capital across regions.

Data centres

We revised our model to distinguish between AI-training and AI-inference-heavy data centres and reformulated demand for AI based on the link between AI capabilities and the sectors of the economy where they can replace labour. We also reflected the impact of data availability as a constraint on AI uptake. We introduced a new formulation for data centre power mix which resolves the competition between grid-electricity and behind-the-meter generation based on cost of investment, cost of backup, the impact of grid bottlenecks on grid connection delays, and connection size.

Hydrogen

The formulation for the hydrogen sector has undergone significant improvement. We have made a distinction between grid-mix electrolysers and electrolysers that use electricity with a power purchase agreement. Based on regional conditions, our formulation finds the economically beneficial configurations endogenously and dynamically.

Electricity generation

We revised the firm capacity requirement in power systems to be stricter. This resulted in more backup conventional capacity. We have made V2G's contribution to the power system variable based on the time of the day. Stability improvements in the formulation for energy requirements that trigger new capacity additions have led to smoother capacity additions.

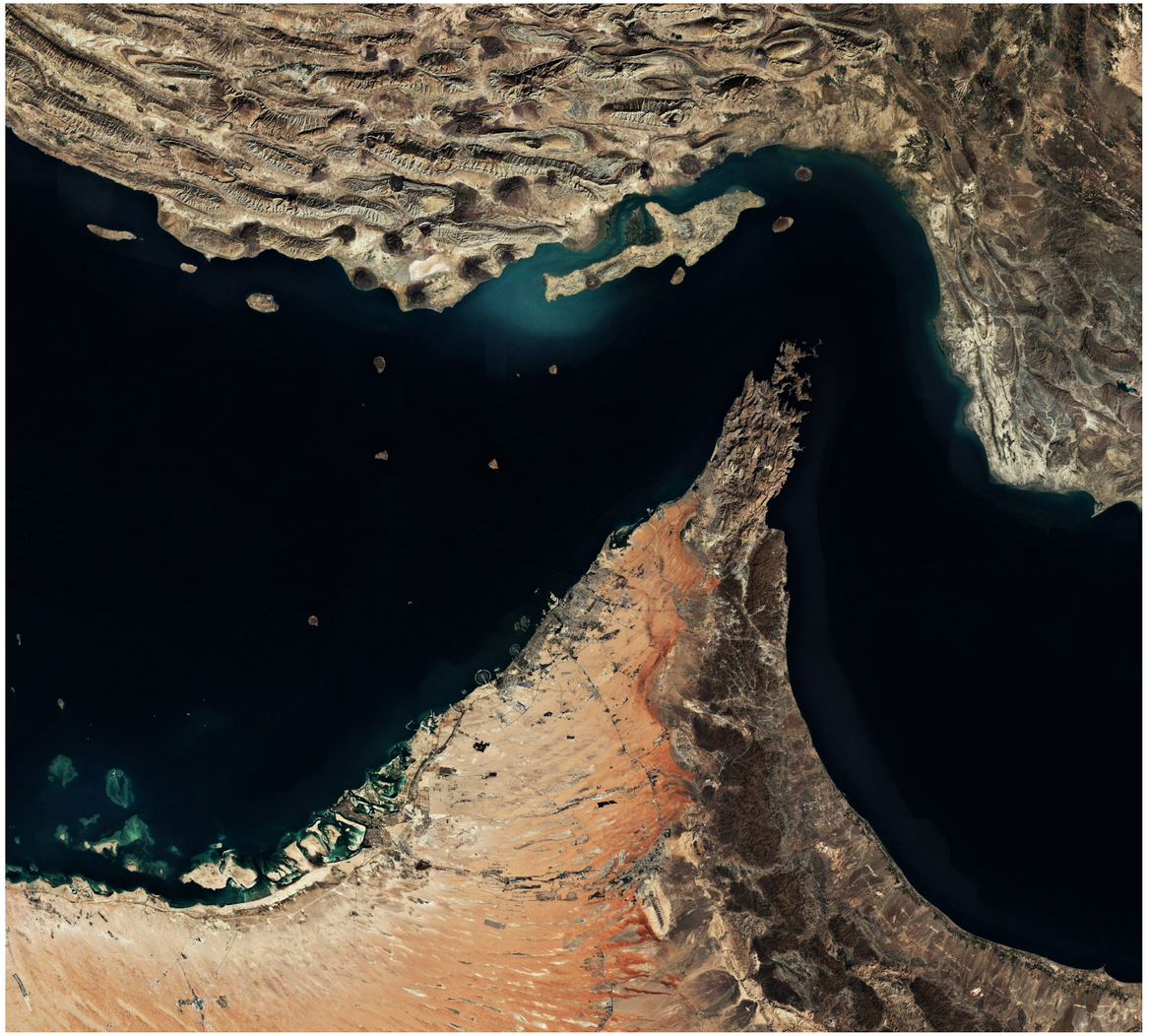
Bioenergy supply

Bioenergy modelling now includes increased granularity in feedstock types and regional supply potential.

Transport

We revised the road vehicle driving distances and fuel economy inputs to better match available data and improved the impact of electrification on final energy demand in rail and aviation.

We introduced parameters to reflect the short-term impacts of the war in Iran on energy prices, fossil fuel supply, and energy demand.



11.4 POLICY FACTORS IN THE ETO MODEL

Our forecast factors in policy measures spanning the entire policy toolbox (Chapter 7).

These affect transition dynamics in three main ways: 1) supporting technology development and deployment, 2) applying technology requirements or standards, and 3) impacting prices. We address nine policy factor areas and their model implementation.

Policy factors included in our Outlook

1. Power

Includes renewable power support from auctions (CfDs, PPAs, capacity payments), investment or production cost support (FiTs, FiPs, tax credits), and behind-the-meter solar PV support for buildings.

We account for fuel and technology import dependency and represent energy security policy via regionally differentiated preferential treatment of domestic sources (solar, wind, coal, gas, nuclear) with incentives/disincentives as a percentage of LCOE.

We incorporate phase-out plans of nuclear, coal, and natural gas.

2. Storage

Includes average regional support for battery storage technologies expressed as a percentage of battery unit costs.

Support levels rise with the share of variable renewables in regional electricity generation, which promote investment in flexibility, while reflecting regional differences in policy ambition and capacity for implementing support.

Includes historical support for behind-the-meter storage for residential and commercial buildings.

3. Zero-emission transport

Road includes average regional purchase subsidies for battery electric and, much less extensively, fuel-cell electric vehicles subsidies, tax exemptions and duty reductions. Support declines over time and is capped by cost disadvantage. We factor in grid constraints limiting charging infrastructure uptake.

Shipping and aviation include fuel-mix shifts driven by carbon price, fuel-blending mandates, investment support, and GHG fuel intensity requirements (DNV, 2026). We assume global agreement on IMO's Net-Zero Framework over the next five years. In aviation, we account for regional clean-fuel policies.

4. Hydrogen

We use a policy-driven capacity pipeline to the early 2030s based on industry databases and expert judgement. Beyond this, support includes CAPEX and OPEX subsidies (whichever is higher), defined separately for low-carbon, renewable, and dedicated nuclear-based production. For grid-connected electrolysers, we include emission threshold requirements and premiums to source renewable electricity (PPA). The model aligns with additionality and hourly temporal matching requirements. CCS in low-carbon H_2 production is driven by carbon prices and tax incentives (NAM). Demand-side support (e.g. CCfDs, quotas) is reflected through explicit support or as policy preference for clean H_2/derivatives in energy/fuel mixes (e.g. maritime and aviation).

5. CCS & DAC

Includes historical and planned CCS capacity through 2032, adjusted for project maturity and risk. These projects are the 'policy-driven' capacity expectations receiving investment and operational support from governments.

We integrate regional policy support beyond this pipeline based on current targets and government funding. Support declines as CCS costs approach carbon prices and we model it as capital subsidies or per-tonne incentives like the US 45-Q.

DAC support reflects government funding to CAPEX.

6. Energy efficiency standards

Building standards for insulation and energy use guide input assumptions, though we do not explicitly quantify policy effects. In Europe, higher retrofit rates reflect subsidies and tax credits for insulation and weatherization.

Vehicles standards are regionally adjusted to real-world fuel use via NEDC correction factors. These guide fuel-efficiency trajectories to 2060 and account for EV adoption.

Shipping includes regulations: *Ship Energy Efficiency Management Plan* (SEEMP), *CII*, *Energy Efficiency Existing Ship Index* (EEXI), and GHG fuel intensity requirements (DNV, 2026).

7. Carbon pricing

CP is reflected as additional costs for fossil fuels in manufacturing, power, hydrogen, ammonia, and methanol production, assuming participation in the same regional schemes, and applies to methane abatement in the energy system. Exemptions and free allowances are included (e.g. EU CBAM policy). Buildings have a separate CP trajectory to reflect the ETS-2 scheme in Europe.

Regional trajectories to 2035 are based on research into policy trends and future CP levels in existing and expected schemes. From 2035 onwards, CPs in the ETO Model are determined endogenously, based on regional emission intensity trends in manufacturing and power sectors resulting from technology transitions, and assuming that CP will be adjusted over time with linkage to decarbonization goals.

8. Taxation

Road transport includes fuel and carbon taxes incorporated in fuel prices, rising with regional carbon prices. Some regions maintain fossil-fuel subsidies. We calculate electricity taxes from the share of charging types, compare their prices with residential prices, and apply them regionally (excluding those with fossil-fuel subsidies).

Buildings and industry include energy taxes that encourage electrification and hydrogen, the latter is also VAT-exempt in industry (all regions).

For grid-connected electrolysers, electricity tariffs reflect regional average tax and support schemes available for hydrogen producers.

9. Pollution

Air pollution prevention includes an air pollution cost proxy added as an operating cost per kWh in power and manufacturing with regionally differentiated implementation (0% to full cost over time) that reflects increasing enforcement of pollution controls.

Plastic pollution intervention includes recycling mandates, taxes, trade restrictions, and EPR, represented by recycling rates and demand reductions via substitution. Europe leads implementation, with other regions following 5 to 15 years later.

11.5 ASSUMPTIONS (GDP, POPULATION)

Some model parameters in the ETO Model are not directly derived from the forecast analysis but are imposed as exogenous factors that drive energy demand. Two such factors that significantly contribute to energy demand are population and economic projections, both globally and for each of the 10 world regions we model.

Population

We base our forecast on population data from the UN Department of Economic and Social Affairs and the Wittgenstein Centre for Demography and Global Human Capital. We use the UN data as the historical reference and the Wittgenstein Centre data, which were last published in 2023 and run to 2100, as a forecast (Wittgenstein, 2024).

The Wittgenstein Centre has developed scenarios that correspond to the five 'Shared Socioeconomic Pathways (SSP)' used by the Intergovernmental Panel on Climate Change. In this Outlook, we follow the central scenario (SSP2) for population, often called the 'Middle of the Road' scenario, and use it as a source of inspiration for other exogenous forecast inputs. Using this approach, we arrive at our 2060 population forecast of 9.9 billion. This is an increase of 19% from the UN's population estimate of 8.3 billion in 2026 (UN, 2024). By 2060, the global population will still be growing, but the rate is reduced to 0.3% per year and Sub-Saharan Africa will be the only region with notable growth, while six of our 10 regions will already be declining.

Our 2060 projection is 1% lower than the latest UN median estimate of 10 billion. The UN median population projection and the Wittgenstein Centre numbers have been converging in recent updates, with some minor regional variations. The main uncertainty in population lies in the long-term (2100 and beyond) forecast where most mainstream forecasts, including the UN's, now indicate global population will peak around 2080 and then decline.

Comparison of Shared Socioeconomic Pathways (SSPs)

SSP2 – 'Middle of the Road'	SSP3 – 'Regional Rivalry: A Rocky Road'
Envisions a balanced economic trajectory with room for global cooperation and gradual improvements.	Reflects a world where economic nationalism, weak institutions, and lack of coordination cause economic stagnation and deepening global inequality.
Narrative summary: – Continuation of historical trends without major disruptions. – No major shifts in policies, governance, or societal behaviour. – Development and income growth proceed unevenly, with some convergence but persistent inequalities.	**Narrative summary:** – A fragmented world characterized by nationalism, regional conflicts, and limited international cooperation. – Nations prioritize security and self-sufficiency over global trade and sustainability. – Development is slow and uneven, especially in poorer countries.

TABLE 11.1

GDP and productivity

GDP per capita is a measure of the standard of living in a country and is a major driver of energy consumption. It is also a good proxy for labour productivity as it reflects the amount of economic output per person. Thus, we also use it as one of the drivers of demand in several places in the ETO Model (vehicle ownership, housing size, etc.).

We base our GDP per capita forecast to 2031 on the GDP per capita growth rates implied by the latest update of the IMF's *World Economic Outlook* (IMF, 2026). Our projections to 2060 are based on the GDP per capita growth rates implied by OECD (2024) and Applied Systems Analysis (IIASA, 2024).

However, due to the current geopolitical landscape and uncertainties in future economic development, we deviate from a pure central scenario (SSP2) for GDP per capita growth. Instead, we combine two pathways, SSP2 and SSP3, and use the average, based on both OECD and IIASA data. This indicates a slightly slower GDP per capita growth trajectory compared with pure SSP2. The implication is that we no longer assume that future economic developments will continue along historical trends. However, we also do not assume that we will transition purely to SSP3 ('Regional Rivalry Pathway') and have therefore chosen a development between these two trajectories (Table 11.1). While we assume a slower overall trajectory, there is a modest 'AI dividend', based on recent IMF short-term forecast (IMF, 2026). Over the long term, our lower-growth pathway already includes a modest AI-related productivity boost in line with other external sources expectations (OECD, 2026). Beyond 2031, the size of this boost is highly uncertain, so we use OECD and IIASA data – a deliberately conservative choice.

We expect the global population to increase 19% and global GDP 94% by 2060.

We expect world GDP to grow from USD 193trn/yr in 2025 to USD 378trn/yr in 2060. This is a near doubling of the economy over the next 35 years. However, the different regions contribute differently, with OECD countries growing much slower compared to developing economies.

The world experienced a 3.1% compound annual GDP growth from 2000 to 2020 (Table 11.2). In the 2040s, this will gradually slow to 2%/yr as population growth slows and ever more national economies become service orientated. Nonetheless, most economies around the world will continue to grow with likely exceptions only in mature economies that are experiencing marked population decline. The fastest growth in GDP per capita between 2025 and 2030 will be in Asia. The Indian Subcontinent will have the highest growth rate at an average of 5.1%/yr, followed by Greater China at 4.8%, and South East Asia at 3.8%/yr.

Compound annual GDP growth rate by region

		2000-2020	2020-2030	2030-2040	2040-2050	2050-2060	2020-2060
NAM	North America	1.7%	2.7%	1.2%	1.0%	1.0%	1.5%
LAM	Latin America	1.8%	3.0%	2.4%	2.0%	1.6%	2.3%
EUR	Europe	1.1%	1.9%	1.1%	0.9%	0.7%	1.2%
SSA	Sub-Saharan Africa	4.2%	4.2%	4.4%	3.8%	3.3%	3.9%
MEA	Middle East and North Africa	3.4%	4.2%	2.8%	2.2%	1.9%	2.7%
NEE	North East Eurasia	3.4%	2.6%	1.6%	1.2%	1.0%	1.6%
CHN	Greater China	8.0%	4.8%	1.9%	1.4%	0.7%	2.2%
IND	Indian Subcontinent	6.0%	6.7%	3.8%	2.7%	2.2%	3.8%
SEA	South East Asia	4.7%	4.6%	3.0%	2.1%	1.6%	2.8%
OPA	OECD Pacific	1.4%	1.4%	1.0%	0.5%	0.2%	0.8%
	World	**3.1%**	**3.6%**	**2.2%**	**1.7%**	**1.4%**	**2.2%**

TABLE 11.2

REFERENCES

0 Highlights

IEA – International Energy Agency (2026). *Energy Technology Perspectives 2026: IEA Supply Chain Risks and Industrial Competitiveness.*

1 Key developments the last 12 months

COP31 Presidency (2026). *COP31 Presidency announces "35% by 2035" electrification target as part of Action Agenda launch*. 9 June.

CREA – Centre for Research on Energy and Clean Air (2026). *China's 15th Five-Year Plan: implications for climate and energy transition.* 6 March.

Deloitte (2025). 2026 *Renewable Energy Industry Outlook.* 29 October.

Direskeneli, H. (2025). 'Russia's Energy Markets: Post-Ukraine War Transformation And 2026 Outlook – Analysis'. *Eurasia review news & analysis.* 29. December.

DNV (2026a). *The growing tail of oil demand destruction.*

DNV (2026b). *The war in Iran and effects on the global energy transition.*

Dunne, D. (2026). 'Santa Marta: Key outcomes from first summit on "transitioning away" from fossil fuels'. *Carbon Brief.* 30 April.

EC – European Commission (2026a). *REPowerEU, four years on.* April.

EC – European Commission (2026b). Nuclear investment needs. *Eighth Nuclear Illustrative Programme (PINC), COM/2026/120.* 10 March.

Evans, S. (2026). 'Iran war: EU strategy sets out 44 actions to limit fossil-fuel price shocks'. *Carbon Brief*. 23 April.

IEA – International Energy Agency (2026a). *World Energy Investment 2026.*

IEA – International Energy Agency (2026b). *Electricity Mid-Year Update 2026.*

IEA – International Energy Agency (2026c). *Energy Technology Perspectives 2026.* Supply chain risks and industrial competitiveness.

IEA – International Energy Agency (2026d). *Oil Market Report, September 2026.*

IMF – International Monetary Fund (2026). *World Economic Outlook Update, July 2026.* Global Economy in Crosscurrents of War and Technology. 8 July.

Kassenova, N. (2026). 'Kazakhstan: An Aspiring Middle Power in the Heart of Eurasia'. *Harvard Kennedy School, Belfer Center,* April.

Kunkel, C. (2025). *Projected data center growth spurs PJM capacity prices by factor of 10.* IEEFA – Institute for Energy Economics and Financial Analysis. 30 July.

Landesmann, M.A. (2026). 'EU Industrial Policy in the Evolving Geopolitical and Geo-economic Environment', in Székely, I.P. and Halmai, P. (eds.) *The Future of European Integration*. Studies in Economic Transition.

Liu, Z. (2026). 'Tracking Chinese Investments in Overseas Industrial Parks'. *Council of Foreign Relations,* 15 July.

MGI – McKinsey Global Institute (2026). *Geopolitics and the geometry of global trade: 2026 update.*19 March.

OECD – Organisation for Economic Co-operation and Development (2026). *OECD Economic Outlook, Volume 2026 Issue 1.* June.

Pao, J. (2026). 'US polysilicon tariffs to move solar makers to domestic materials'. *Asia Times.* 6 August.

Paullin, C. (2026). *Virginia regulators approve new Dominion rates, assign more costs to data centers.* Inside Climate News. 7 January.

S&P Global (2025). *Foreign direct investment trends in Sub-Saharan Africa.* 27 March.

SDG News (2026). *Dilma Rousseff's Push to Expand the Global South's Financial Power.* 14 August.

Shamim, S. (2026). 'How shipping insurance rates are rising, as Hormuz, Bab al-Mandeb shut down'. *Al Jazeera.* 23 July.

Spektor, M. (2025). 'Rise of the nonaligned. Who wins in a multipolar world?'. *Foreign Affairs.* January/February.

TimesTech (2026). *India on track for record 50 GW solar year as ALMM-II mandate reshapes domestic supply chain.* 13 August.

UNCTAD – UN Trade and Development (2026). *Hormuz disruption deepens global economic strain across trade, prices and finance.* 1 April.

UNFCCC – UN Framework Convention on Climate Change (2025). Global Mutirão: Uniting humanity in a global mobilization against climate change. 21 November version.

United Nations (2026). General Assembly Strengthens Fight against *Climate Change, Adopting Contentious, Yet Broadly Supported Draft among Several Texts on Myriad Topics.* 20 May.

USTR – Office of the United States Trade Representative (2026). *Fact Sheet: President Trump Directs USTR Section 301 Action in Response to Brazil's Unreasonable Acts, Policies, and Practices.* July.

WEF – World Economic Forum (2026a). *Energy Transition Index 2026. Energy security.* 18 June.

WEF – World Economic Forum (2026b). *Davos 2026: Special address by Mark Carney, Prime Minister of Canada.* 20 January.

Xue, X. and Larsen, M. (2025). *China's Green Leap Outward: The rapid scale-up of overseas Chinese clean-tech manufacturing investments*. Net Zero Industrial Policy Lab. 9 September.

2 The DNV forecast

BCG – Boston Consulting Group (2025). *Delivering the Energy Transition Will Come Down to the Wires.*

DNV, 2025. *From Silos to Systems: The Importance of Whole Energy Systems Thinking.* Energy Industry Insights 2025 Special Report.

IEA – International Energy Agency (2024). *Building the Future Transmission Grid: Strategies to Navigate Supply Chain Challenges.* IEA Publications.

IEA WEB (2026). *World Energy Balances,* IEA, Paris

3 Energy demand

AFFO – U.S. Department of Energy Alternative Fuels and Feedstocks Office (2026). *Sustainable Aviation Fuels.*

ALG (2024). 'Modal competition and substitution: high-speed rail versus air transport'. *ALG-Global.* June.

ATAG – Air Transport Action Group (2025). 'Fact sheet 3: Tracking aviation efficiency'. *Aviation Benefits Beyond Borders.* 7 January.

Auto Remarketing (2024). *Survey shows most EV drivers can charge at home, but still rely on public chargers.* 1 March.

Becker, J., Rush, N., Barnes, E., and Rein, D. (2025). 'Measuring the impact of early-2025 AI on experienced open-source developer productivity'. *METR*.

Bernard, M.R. (2023). *European Union Alternative Fuel Infrastructure Regulation (AFIR).* International Council on Clean Transportation (ICCT).

Chevron (2026). *Chevron signs 20-year power agreement with Microsoft for West Texas data center.* 22 June.

Clarksons (2026). *Shipping Intelligence Network.* Accessed: 1 September 2026.

C-SAF – Canadian Council for Sustainable Aviation Fuels (2023). *Canadian Council for Sustainable Aviation Fuels launches Canada's first roadmap for SAF.* 5 June.

Curran, I. (2026). 'Data centres account for almost one quarter of Irish electricity usage in 2025'. *The Irish Times*. 7 July.

DGMT – Directorate-General for Mobility and Transport (2025). 'Commission launches plan to accelerate high-speed rail across Europe'. *European Commission.* 5 November.

DiGangi, D. (2025). 'Some load forecasts using "unrealistically high load factors": Grid Strategies'. *Utility Dive.* 19 November.

DNV (2026a). *Maritime Forecast to 2050,* 2026 Edition.

DNV (2026b). *Alternative Fuels Insight.* (Accessed: 16 July 2026).

DNV (2025). *Energy Transition Outlook 2025: A global and regional forecast to 2060.*

EASA – European Union Aviation Safety Agency (2026). *2025 Aviation Fuels Reference Prices for ReFuelEU Aviation.* 26 February.

EC – European Commission (2023). *Renewable Energy Directive*.

EC – European Commission (2026). *Heavy-duty vehicles: Council adopts targeted flexibility for manufacturers to comply with CO_2 targets,* Consilium.

Economist (2026a). *The geopolitics of air-conditioning.* July 23.

Economist (2026b). *Can America retrain workers before AI leaves them behind?* August 2.

EIA – U.S. Energy Information Administration (2026). *Petroleum markets responded to disruptions in the Middle East in the second quarter.* 15 July.

EIA (2025). *Amid regional conflict, the Strait of Hormuz remains critical oil chokepoint*.

Fischer, I. (2026) *Behind-the-Meter Power Solutions: The Data Center Industry's New Reality.* 19 February. Behind-the-Meter Power Solutions: The Data Center Industry's New Reality – datacenterHawk.

Freemark, B.Y. (2026). 'Transit Project Openings in 2026: A Global Review'. *The Transport Politic.* January 20.

IATA – International Air Transport Association (2026a). *Jet Fuel Price Monitor.* (Accessed: 30 September 2026).

IATA (2026b). *Global Outlook for Air Transport Energy in Crisis*. June.

IATA (2025). *The global commercial aircraft fleet.*

ICS – International Chamber of Shipping (2026). *ICS Maritime Barometer Report 2025-2026*.

IEA – International Energy Agency (2026a). 2026 *Energy Crisis Policy Response Tracker – Data Tools.*

IEA (2026b). *Executive summary – Electric Car Markets in a Time of Uncertainty.*

IEA (2026c). *Electric vehicle charging – Global EV Outlook 2026*.

IEA (2026d). *Vehicle-to-grid technology – Analysis. IEA*.

IEA (2026e). *Electricity 2026: demand.*

IEA (2026f) *Key questions on energy and AI.* World Energy Outlook Special Report.

IEA (2026g). *World energy investment 2026*. 28 May.

IEA (2019). *The Future of Rail Opportunities for energy and the environment.*

IEA WEB (2026). *World Energy Balances,* IEA, Paris.

Jiying, G. (2026). 'China to Require Heavy Truck Charging Stations on Highways to Boost EV Truck Use, Cut Emissions'. *Yicai Global,* 16 June.

JLL (2026). *2026 Global data center market outlook.* 5 January.

Jones, J.M. (2026). *Americans Oppose AI Data Centers in Their Area.* May 13.

Kwa, T., et al. (2025). 'Measuring AI ability to complete long tasks'. *METR.* arXiv:2503.14499.

Lago, C. (2026). *Pantheon eyes large-scale behind-the-meter data center campus in Croatia.* 4 May.

Massenkoff, M., et al. (2026). 'Anthropic Economic Index report: cadences'. *Anthropic*. 26 June.

McKay, J. (2026) EV Battery Evolution | Compare the Market AU.

Mercure, J.-F. et al. (2025). 'Evidence of a cascading positive tipping point towards electric vehicles,' *Nature Communications,* 17(1), p. 240.

Natal, J.M. and Sadikov, A. (2026). 'The Oil Market Absorbed the War Shock, but Buffers Are Running Low'. *IMF.* 15 July.

Pei, M. and Zhou, O. (2026). 'China focuses on building voluntary demand for SAF over issuing mandates'. *S&P Global.* 16 July.

PIB – Press Information Bureau (2026). *Ethanol Blending in India.*

Plucinska, J. and Hepher, T. (2026). 'European airlines likely beat 2% green jet fuel target last year, sources say' *Reuters.* 30 March.

Railway Pro (2026). 'China's high-speed rail tightens global lead'. *Railway Pro.* April 2.

Sagaria, S., van der Kam, M., Boström, T. (2025). 'Vehicle-to-grid impact on battery degradation and estimation of V2G economic compensation'. *Applied Energy,* 377, p. 124546.

Srivathsan, B., Sorel, M., Sachdeva, P. (2024). 'AI power: expanding data center capacity to meet growing demand'. *McKinsey & Company.* 29 October.

UNDP – United Nations Development Programme (2025). *2025 Global Multidimensional Poverty Index (MPI): Overlapping Hardships: Poverty and Climate Hazards.* New York.

US EPA – U.S. Environmental Protection Agency (2026). *Final Renewable Fuel Standards for 2026 and 2027.*

Wei, H., Struyven, D., and Dart, S. (2026). 'US data center power demand projected to double by 2027'. *Goldman Sachs.* 20 May.

Willems, A. (2026). *Seattle enacts year-long ban on new AI datacenters.* 10 June.

4 Electricity

ACER – Agency for the Cooperation of Energy Regulators (2025) *Getting the Signals Right: Electricity Network Tariff Methodologies in Europe. ACER Report on Network Tariff Practices.*

Berkeley Lab (2025). *Land-Based Wind Energy Technology Update: 2025 Edition.* Published 08/2025.

Bloomberg (2026). *China's Wasting Too Much Renewable Power as Curtailments Rise.* 8 April.

BNetzA – Bundesnetzagentur (2026) *AgNes (Allgemeine Netzentgeltsystematik Strom).* (Accessed: 30 September 2026)

Bundesnetzagentur (2024). *Grid Expansion (Netzausbau) in Germany.* Federal Network Agency.

Carnegie – Carnegie Endowment for International Peace (2026). *Beyond the Hype: Assessing Hyperscaler Nuclear Commitments Against US Energy Realities.*

Civillini, M. (2026). 'China's carbon emissions rise again as more clean power is wasted'. *Climate Change News.* 4 June.

CSIS – Center for Strategic and International Studies (2026). *China's Nuclear Energy Priorities Under Its 15th Five-Year Plan.*

DOE – Department of Energy (2026). *National Transmission Needs Study.* U.S. Department of Energy.

Down to Earth (2024). *As climate change snuffs life out of hydroelectric sources, Zambia feels need to diversify energy mix.* 29 April.

Economist (2026). *The geopolitics of air-conditioning.*

EC – European Commission. (2023). *Electricity Grids and Interconnections in Europe*.

ECRB – Energy Community Regulatory Board (2026) *ECRB Best Practice Report on Tariff Methodologies in the Energy Community*. Energy Community Regulatory Board, March 2026.

EDF (2025). *EDF shares its forecasted cost estimate of the EPR2 programme for EUR 72.8bn*. Press release.

EIA (2025) *Wholesale Electricity and Natural Gas Markets Data*. U.S. Energy Information Administration.

Energy Connects (2025). *UAE breaks ground on world's first gigascale round-the-clock renewable energy project.* 24 October.

ENTSO-E (2024). *Ten-Year Network Development Plan (TYNDP) 2024.* European Network of Transmission System Operators for Electricity.

ENTSO-E, T&D Europe, DSO Entity, Europacable (2025). *Electrifying Europe: Industry and grid operators discuss the skilled workforce gap.*

ESA – Euratom Supply Agency (2025). *Euratom Supply Agency Annual Report 2024.*

Euronews (2026). In *Finland, the world's first facility to bury nuclear waste is set to begin operations.* 9 April.

Fervo Energy (2026). *Enhanced Geothermal Has Been Proven at Scale. Here's What Two Years of Production Data Show.*

Framatome (2026). *EPR2 program.*

Gallup (2025). *Nuclear energy support near record high.*

Galvin, Ray (2026). 'Free midday electricity in a Solar Sharer scheme: Should Germany follow Australia's lead?'. *Renewable Energy 267*.

GlobalData (2025). *Power Database.* GlobalData.

GlobalData (2026). *Power Database.* GlobalData.

Good Energy Collective (2022). *State of play: siting of nuclear projects.*

Haneklaus, N. et al. (2025). 'Dependencies of the European Union and the world on Russian nuclear fuel cycle services, and how to reduce them'. *Energy Strategy Reviews,* 62, 101923.

Harder, Amy (2026). *What it's like inside the data centers powering the AI boom.*

IAEA – International Atomic Energy Agency (2023). *Nuclear safety, security and safeguards in Ukraine.*

IEA (2026a). *Cooling a hotter world: El Niño meets strong growth in global electricity demand.*

IEA (2026b). *Global Energy Review 2026: Nuclear.*

IEA (2026c). World Energy Investment 2026. IEA, Paris.

IEA (2026d). *Securing Clean Energy Technology Supply Chains.* IEA, Paris Securing Clean Energy Technology Supply Chains – Analysis – IEA

IEA (2026e). *Global EV Outlook.* IEA, Paris.

IEA (2025a). *The Path to a New Era for Nuclear Energy.*

IEA (2025b). *World Energy Employment 2025.*

IEA (2025c). *The battery industry has entered a new phase.* The battery industry has entered a new phase – Analysis – IEA.

IEA (2025d). *China's share in global PV manufacturing capacity, 2024 and 2030.*

IEA (2025e). *Building the Future Transmission Grid: Strategies to Navigate Supply Chain Challenges.*

IEA (2024). *Batteries and Secure Energy Transitions.*

IEA (2023). *Electricity Grids and Secure Energy Transitions.*

IEA (2022). *Nuclear Power and Secure Energy Transitions: From Today's Challenges to Tomorrow's Clean Energy Systems.*

IEA WEB (2026). *World Energy Balances,* IEA, Paris.

Infrastructure Investor (2025). *The gas turbine crunch: Why supply won't meet demand.*

IHA (2026). *2026 World Hydropower Outlook.* International Hydropower Association.

Laha, M. (2025). 'Chinese Export Controls on Cathode Technology: The EU Must Increase Insight into Supply Chains'. *German Council on Foreign Relations – DGAP.* 18 December.

Lovering, J., Swain, M., Blomqvist, L., Hernandez, R.R. (2022). *Land-use intensity of electricity production and tomorrow's energy landscape*. PLOS ONE, 17(7), e0270155.

Myllyvirta, L. (2026). 'Analysis: China's CO2 climbs 2% in early 2026 due to 'wasted' wind and solar'. *Carbon Brief.* 4 June.

NAO – National Audit Office (2026). *Sizewell C.* HC 33, Session 2026-27. https://www.nao.org.uk/wp-content/uploads/2026/05/sizewell-c.pdf

NAO (2017). *Hinkley Point C.* HC 40, Session 2017-18.

NEA – Nuclear Energy Agency and OECD (2026). *Nuclear Energy Outlook: Global Installed Capacity to 2050 and Beyond.*

NEA and OECD (2021). *Small Modular Reactors: Challenges and Opportunities.* NEA No. 7560.

NESO – National Energy System Operator (2025). *Strategic Spatial Energy Plan and Electricity Network Planning.* https://www.neso.energy

Nord Pool (2025). *Transition to 15-minute Market Time Unit (MTU).*

NRC – US Nuclear Regulatory Commission (2024). *Backgrounder on uranium import ban.*

Nuclearеurope (2020). 'Foratom sounds alarm over nuclear skills shortage in Europe'. *ANS Nuclear Newswire.* Published under the organization's former name, Foratom.

Nuclear Scaling (2026). *2026 Landscape of US Domestic Advanced Nuclear Energy Supply Chain.* March.

Nuclear Skills Plan (2026). *GBP 65.6 million doctoral focal award to strengthen UK nuclear skills, energy security and sovereign defence capability.* March.

NucNet (2026). *Poll shows over 90% support for nuclear power in Poland.*

Ofgem (2024). *RIIO-ET3: Economic Lives of Electricity Transmission Network Assets.* Ofgem.

Pew – Pew Research Center (2025). *Support for expanding nuclear power is up in both parties since 2020.* 16 October.

PIB – Press Information Bureau, Government of India (2025a). *Nuclear mission: 100 GW of nuclear energy by 2047.* Department of Atomic Energy.

PIB (2025b). *SHANTI Bill 2025*.

Regeringskansliet (2026). *Ansökan om etablering av ny kärnkraft.*

Rystad Energy (2026) *Renewables Market Analysis.*

Shale Magazine (2026). *The $2.7bn plan to cut ties with Russian nuclear fuel.*

smr.intel (2026). *Every nuclear-powered data center deal.*

Spector, J. (2026). 'Suddenly, the US manufactures a ton of grid batteries'. *Canary Media.* 26 March.

Squassoni, S. (2024). 'New Nuclear Energy: Assessing the National Security Risks'. *Institute for International Science and Technology Policy,* George Washington University.

Star Energy Geothermal (2026). I*ndonesia's leading geothermal producer.*

Stringer, D. (2026). 'Iran Oil Shock Pushes Asia and Europe to Adopt Renewable Energy'. *Bloomberg*. 5 June.

The White House (2025). *Executive orders on nuclear energy*, May 2025 (400 GW by 2050; 18-month NRC deadlines).

Think Power Solutions (2025). *Modernizing Transmission and Distribution: Grid Hardening at Every Level.*

Thunder Said Energy (2025). *Costs of nuclear power.*

US DOE – US Department of Energy (2022). *Investigating Benefits and Challenges of Converting Retiring Coal Plants into Nuclear Plants.* Office of Nuclear Energy with Argonne, Idaho and Oak Ridge National Laboratories.

WindEurope (2026). *Wind energy dis- and misinformation: Undermining Europe's Security and Competitiveness.* April.

WNA – World Nuclear Association (2026). *Nuclear Power in India.* 15 July.

WNA (2024). Nuclear Power and Energy Security. Updated December 2024.

WNN – World Nuclear News (2026a). *Finland looks to reform nuclear energy legislation.* 19 March.

WNN (2026b). *US Department of Energy closes loan for Duane Arnold restart*. 8 September.

WNN (2025). *What is the budget for Canada's first SMR project?*.

Woods, B. (2025). 'Nuclear power's radioactive waste problem'. *CNBC*. 9 November.

5 Fossil fuels

Agnolucci, P. and Makarenko, N. (2026). 'Natural gas market: LNG trade disruptions and market rebalancing'. *World Bank Blogs.* 4 June.

Bloomberg (2026). *Saudi Oil Logistics Roiled Again by Houthis' Red Sea Threat,* 23 August.

DNV (2025). *Energy Transition Outlook: CCS to 2050.*

Hernandez, A. (2026). 'Natural gas spending to hit 10-year high in 2026 as oil investment falls, IEA says'. *Reuters*. 28 May.

IEA – International Energy Agency (2026a). *World Energy Investment 2026.* 28 May.

IEA (2026b). *Oil Market Report – July 2026.* 10 July.

IEA (2025). *World Energy Investment 2025.* 5 June.

IEA WEB (2026). *World Energy Balances,* IEA, Paris.

Martinsen, A. and Satwani, K. (2026). 'The cost of war: Gulf energy infrastructure left facing a $25 billion repair bill'. *Rystad Energy.* 25 March.

Russell, C. (2026). 'China did use crude stockpiles to ease Iran shock, but not that much'. *Reuters*. 17 June.

Samonte, J.K. (2026). 'Coal Holds Steady as China Cuts Import Forecast'. *Trading Economics.* 10 February.

Talukdar, T. and Torres Diaz, C. (2026). 'Asia's coal demand to rise by 100 million tonnes in wake of Middle East conflict'. *Rystad Energy.* 8 June.

Young, J. and Manning-Pickett, T. (2026). 'U.S. natural gas exports to grow nearly 30% by 2027 as LNG facilities ramp up'. *U.S. Energy Information Administration.* 16 April.

Zeitlin, M. (2026). 'How the Iran War Is Fueling a Coal Comeback'. *Heatmap News.* 27 March.

GCCSI – Global CCS Institute (2026). *CCS Quarterly Update.* Q1 2026. April 2026. Members only content.

Patel, S. (2025). 'Google Commits to First U.S. Gas-Fired Power Plant with Integrated CCS for Data Centers'. *POWER Magazine.* 24 June.

Chevron (2026). *Chevron Signs 20-Year Power Agreement with Microsoft for West Texas Data Center.* Chevron Newsroom. 22 July.

6 Clean molecules & thermal supply

NRCan (2025). *Zibi District Thermal System.* (Accessed: 9 July 2026).

New York State Senate (2022). *Senate Bill S9422.* (Accessed: 14 July 2026).

Massachusetts DPU (2024). *DPU Establishes Network Geothermal Guidelines.* (Accessed: 14 July 2026).

BBSR (2024). *Heat transition in Germany: Status quo of municipal heat planning.* (Accessed: 14 July 2026).

DNV (2022). *Hydrogen Forecast to 2050.*

Guo, X. et al (2023). *High-Quality Development of China's Geothermal Industry.*

IEA – International Energy Agency (2026). *Renewables in District Energy.*

IEA (2025a). *Outlook for Biogas and Biomethane.*

IEA (2025b). *Opportunities for district heating in the changing energy landscape.* (Accessed: 9 July 2026).

JASPERS (2024). *JASPERS guide to decarbonisation of district heating systems.*

Die Bundesregierung (2023). *The public sector set to become a role model.*

Regjeringen (2025). *The data centre industry – a sustainable industry of the future for the digital Norway.*

7 Policy

ACEEE – American Council for an Energy-Efficient Economy (2025). *2025 International Energy Efficiency Scorecard.* October. Authors: Jasmine Mah, Alex Aquino, Daivie Ghosh, and Andrew Hoffmeister. Washington, DC.

ACER – European Union Agency for the Cooperation of Energy Regulators (2025). *Security of EU electricity supply. 2025 Monitoring Report.* 20 November.

Asmelash, H. (2026). 'Fuel subsidies are an easy fix for the Iran war's energy price shock — and the wrong one'. *Carnegie Endowment for International Peace.* 5 May.

Aurora Energy Research (2025). *Capacity remuneration mechanisms in Europe.* January.

BloombergNEF (2026). *Energy Transition Investment Trends 2026.*

CGTN (2026). *China sets three-year drive to cut energy use, carbon emissions in key industries.* 15 June.

Changing Transport (2025). *Transport in the new NDCs 3.0.*

EC – European Commission (2026a). *Commission boosts Europe's competitiveness, decarbonisation and independence with Electrification Action Plan and ETS review.* 17 July.

EC (2026b). *Directive of the European Parliament and of the Council amending directive 2003/87/ec and decision (eu) 2015/1814 as regards driving competitiveness and cost-effective decarbonisation, COM(2026) 616 final.* 17 July.

EC (2025). *Commission proposes upgrade of the EU's energy infrastructure to lower bills and boost independence.* 10 December.

EC (2023). *COM(2023) 757 final Grids, the missing link – An EU Action Plan for Grids.*

GBFR – Global Banking & Finance Review (2026). *EU plans tax changes to reduce electricity bills, draft shows.* 8 June.

GEM – Global Energy Monitor(2025). *Global Coal Plant Tracker.* July.

Jaeger, Joel (2025). 'These Countries Are Adopting Electric Vehicles the Fastest'. *Systems Change Lab,* 5 December.

ICAP – International Carbon Action Partnership (2026). *Emissions Trading Worldwide: Status Report 2026.* April.

IEA – International Energy Agency (2026a). *State of Energy Policy 2026.*

IEA (2026b). *World energy Investment 2026*. May.

IEA (2025a). *Energy Efficiency 2025 – Buildings.*

IEA (2025b). *Global Hydrogen Review 2025.*

IMF – International Monetary Fund (2026). *World Economic Outlook UPDATE: Global Economy in Crosscurrents of War and Technology.* 2026 July.

Krugman, P. (2026). *Renewable Energy and National Security – The wind and the sun don't need to transit the Strait of Hormuz.*

McKibben, B. (2026). *Sunlight travels 93 million miles to reach the earth – None of them through the Strait of Hormuz.*

Ministry of Finance (2024). *National Electricity Plan (Transmission) launched by Cabinet Minister for Power and Housing & Urban Affairs.* 14 October.

OECD (2025). OECD *Inventory of Support Measures for Fossil Fuels 2025: Policy Trends.* OECD Publishing, Paris. December. https://doi.org/10.1787/6fff0cb7-en.

Public First (2026). P*owering Up: Business perspectives on electrification across 18 countries.* June. Research commissioned by: We Mean Business Coalition, E3G, Global Renewables Alliance.

REN21(2026). *Renewables-based Economy (RBE) Tracker 2026*.

Rushton, H. and Patonia, A. (2026). 'Bankability of Hydrogen Projects: Key Risks, Financing Challenges and Mitigation Solutions'. *OIES Paper:* ET 52. ©2026 Oxford Institute for Energy Studies.

Slocat (2025). *Transport in a crisis era: Seizing opportunities for just, sustainable transformation.*

Stott, M. (2026). 'How has India dealt with the economic hit from the Iran war?'. *Financial Times.* 16 June.

Systems Change Lab (2026). *Scale up technological carbon removal.*

UNIDO and EELA (2025). *Harmonisation of energy efficiency policies and implementation mechanisms in the EAC, ECOWAS and SADC regions.* December. © 2025 United Nations Industrial Development Organization.

World Bank (2026). *State and Trends of Carbon Pricing 2026.* https://www.worldbank.org/en/publication/state-and-trends-of-carbon-pricing

World Bank (2025). *State and Trends of Carbon Pricing 2025.*

Xin, Zheng (2026). 'China to invest 5 trillion yuan in power grid over next 5 years'. *China Daily,* 2 February.

8 Finance outlook

BloombergNEF (2026) *Energy Transition Investment Trends.*

EC – European Commission (2026). *Financing Building Renovations.*

IEA – International Energy Agency (2025a). *World Energy Investment.*

IEA – (2025b). *Cost of Capital Observatory.*

IEA – (2025c). *Energy Efficiency 2025.*

IEA – (2023). *Electricity Grids and Secure Energy Transitions.*

Koefoed, A. and Selvakkumaran, S. (2025). 'Costly capital: Money for green megawatts in Sub-Saharan Africa'. *DNV.* 4 October.

OECD – Organisation for Economic Co-operation and Development (2014). *Determinants of Households' Investment in Energy Efficiency and Renewables.*

Rocky Mountain Institute (2025). *Low-Income Energy Efficiency.* Updated June 1, 2026.

SurgePV (2026). Solar incentives in Japan 2026: *Market Guide and Incentives.* Written by: Keyur Rakholiya and Rainer Neumann. 30 January.

University of Michigan (2023). *EV transition will benefit most US vehicle owners, but lowest-income Americans could get left behind.* 11 January.

9 Emissions and climate impact

Boardman et al. (2026). 'Emergent constraints on climate sensitivity and recent record-breaking warm years'. *Earth System Dynamics,* 17.

Carbon Brief (2026). *Coal power drops in China and India for first time in 52 years after clean-energy records.*

GCP – Global Carbon Project (2025). 'Global Carbon Budget 2025'. *Earth System Science Data.*

Global CCS Institute (2025). *Global Status of CCS 2025.*

Hansen, J. et al. (2025). *Global warming acceleration and climate sensitivity.*

IEA – International Energy Agency (2026a). *Global Energy Review 2026: CO_2 emissions.*

IEA (2026b). *Global Methane Tracker 2026: Key findings.*

IEA WEB (2026). *World Energy Balances,* IEA, Paris.

IEEFA – Institute for Energy Economics and Financial Analysis (2025). *Europe's potential carbon capture and storage project pipeline is losing steam.*

IPCC – Intergovernmental Panel on Climate Change (2021). *Climate Change 2021: The Physical Science Basis. Working Group I contribution to the Sixth Assessment Report.*

NOAA – National Oceanic and Atmospheric Administration (2026). *Trends in atmospheric carbon dioxide.*

TFFF – Tropical Forest Forever Facility (2026). *Tropical Forest Forever Facility demonstrates rapid progress at IMF/World Bank Spring Meetings.* April 2026.

WRI – World Resources Institute (2026). *Tropical rainforest loss drops 36% in 2025, but fires threaten global progress*'. April 2026.

11 The ETO Model

DNV (2026). *Maritime Forecast to 2050. 2026 Edition.*

DNV (2025). *From Silos to Systems: The Importance of Whole Energy Systems Thinking.* Energy Industry Insights 2025 Special Report.

IIASA – International Institute for Applied Systems Analysis (2024). *IIASA GDP Data prepared by Jesus Crespo Cuaresma and Jakob Gregor Zellmann.*

IMF – International Monetary Fund (2026). *World Economic Outlook Update, April 2026. Global Economy in the Shadow of War. 8 July.*

OECD (2026). *Foundations for Growth and Competitiveness 2026.* OECD Publishing, Paris. 9 April.

OECD (2024). *GDP projections by the OECD Data prepared by Rob Dellink, Jean Chateau, Laura Atarody, Ruben Bibas, and Elisa Lanzi.*

UN (2024). *United Nations World Population Prospects 2024.*

Wittgenstein (2024).

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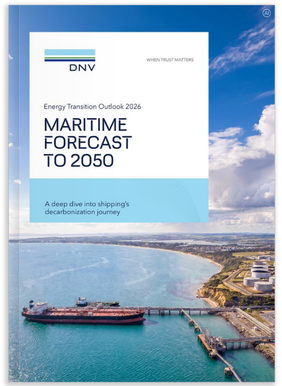
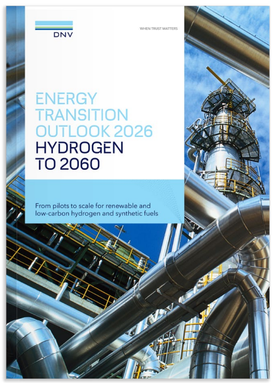


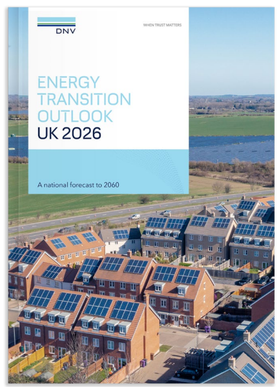
Maritime forecast to 2050
The 2026 edition of our Maritime Forecast to 2050 uncovers key trends shaping the future of shipping. Now in its 10th edition, our Maritime Forecast explores how shipping can navigate an increasingly complex and uncertain energy transition. As ships ordered today will operate well beyond 2050, decisions taken now must balance near-term compliance with the long-term flexibility needed to respond to evolving regulations, fuel markets, and technologies.

The report explores four regulatory scenarios, ranging from the imminent adoption of the Net Zero Framework (NZF) to prolonged gridlock, and their implications for shipping. We explore how stronger regulation leads to more efficiency – up to 25% lower fleet energy demand by 2050 under stronger global regulatory signals. More specifically, fuel demand depends on regulatory pathways: Low-GHG fuel demand could reach 22 Mtoe by 2030 and 185 Mtoe by 2050 under stronger regulatory scenarios.



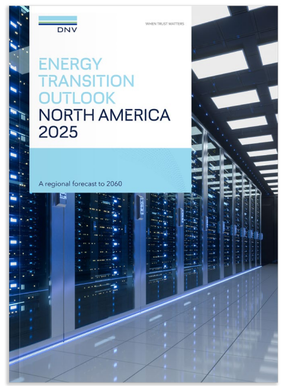
Hydrogen to 2060
DNV's Energy Transition Outlook: Hydrogen to 2060 revisits our hydrogen forecast four years after the previous edition, drawing on lessons from an industry that has moved from early optimism to the realities of project development and deployment.

As decarbonization ambitions collide with high costs, policy uncertainty, and energy-security concerns, the report provides a realistic assessment of hydrogen's future role in the energy system. Based on DNV's independent modelling and industry insight, it examines where hydrogen is most likely to scale, where deployment will remain limited, and what is needed to accelerate investment. The report tracks hydrogen demand, investment, trade, and regional development through to 2060, identifying the conditions required to move from ambition to industrial scale.



Energy Transition Outlook: UK to 2060
Our ETO UK 2026 provides DNV's independent forecast of the UK energy system to 2060 and assesses progress towards the country's 2030, 2035, and 2050 climate goals.

While the UK is making strong progress, with wind capacity set to double and solar to triple by 2030, it remains off track to meet its key climate commitments. Natural gas will still play an important role in power generation in 2030, while slow progress in buildings and transport leaves emissions reductions well short of the 2035 target. By 2050, emissions fall substantially but do not reach net zero. Despite these challenges, the long-term outlook is positive, with electrification, efficiency gains, and low-carbon energy delivering a cleaner, more resilient, and more affordable energy system.



Energy Transition Outlook: North America to 2060
Since we published our first ETO North America in 2023, consumers have sent a clear signal that they want energy that is affordable, reliable, and secure. Driving that shift are changes in US policy and trade, geopolitical tensions, supply-chain constraints, and the rapid expansion of energy-intensive industries. These pressures were already evident as this report was being prepared in 2025 and have only intensified amid recent global events. The report examines how these forces are reshaping North America's energy future, from the continued role of fossil fuels and natural gas to the long-term growth of solar and wind, the pace of decarbonization, and the implications for electricity prices, energy security, and climate goals.

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This report has been prepared by DNV as a cross-disciplinary exercise between the DNV Group and our Energy Systems and Maritime business areas across 20 countries. The core model development and research has been carried out by a dedicated team in our Energy Transition Outlook research unit, part of the Group Research & Development division. In addition, we have been assisted by internal and external experts, with the core names listed below:

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